



GE Healthcare

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Book 1 of 2
Pages 1 - 124

LightSpeed 5.X Plus Installation Manual

Table of Contents, Chapters 1-3 (Mechanical Installation)

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

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IMPORTANT PRECAUTIONS

LANGUAGE

WARNING

- THIS SERVICE MANUAL IS AVAILABLE IN ENGLISH ONLY.
- IF A CUSTOMER'S SERVICE PROVIDER REQUIRES A LANGUAGE OTHER THAN ENGLISH, IT IS THE CUSTOMER'S RESPONSIBILITY TO PROVIDE TRANSLATION SERVICES.
- DO NOT ATTEMPT TO SERVICE THE EQUIPMENT UNLESS THIS SERVICE MANUAL HAS BEEN CONSULTED AND IS UNDERSTOOD.
- FAILURE TO HEED THIS WARNING MAY RESULT IN INJURY TO THE SERVICE PROVIDER, OPERATOR OR PATIENT FROM ELECTRIC SHOCK, MECHANICAL OR OTHER HAZARDS.

AVERTISSEMENT

- CE MANUEL DE MAINTENANCE N'EST DISPONIBLE QU'EN ANGLAIS.
- SI LE TECHNICIEN DU CLIENT A BESOIN DE CE MANUEL DANS UNE AUTRE LANGUE QUE L'ANGLAIS, C'EST AU CLIENT QU'IL INCOMBE DE LE FAIRE TRADUIRE.
- NE PAS TENTER D'INTERVENTION SUR LES ÉQUIPEMENTS TANT QUE LE MANUEL SERVICE N'A PAS ÉTÉ CONSULTÉ ET COMPRIS.
- LE NON-RESPECT DE CET AVERTISSEMENT PEUT ENTRAÎNER CHEZ LE TECHNICIEN, L'OPÉRATEUR OU LE PATIENT DES BLESSURES DUES À DES DANGERS ÉLECTRIQUES, MÉCANIQUES OU AUTRES.

WARNUNG

- DIESES KUNDENDIENST-HANDBUCH EXISTIERT NUR IN ENGLISCHER SPRACHE.
- FALLS EIN FREMDER KUNDENDIENST EINE ANDERE SPRACHE BENÖTIGT, IST ES AUFGABE DES KUNDEN FÜR EINE ENTSPRECHENDE ÜBERSETZUNG ZU SORGEN.
- VERSUCHEN SIE NICHT, DAS GERÄT ZU REPARIEREN, BEVOR DIESES KUNDENDIENST-HANDBUCH ZU RATE GEZOGEN UND VERSTANDEN WURDE.
- WIRD DIESE WARNUNG NICHT BEACHTET, SO KANN ES ZU VERLETZUNGEN DES KUNDENDIENSTTECHNIKERS, DES BEDIENERS ODER DES PATIENTEN DURCH ELEKTRISCHE SCHLÄGE, MECHANISCHE ODER SONSTIGE GEFAHREN KOMMEN.

AVISO

- ESTE MANUAL DE SERVICIO SÓ LO EXISTE EN INGLÉS.
- SI ALGÚN PROVEEDOR DE SERVICIOS AJENO A GEMS SOLICITA UN IDIOMA QUE NO SEA EL INGLÉS, ES RESPONSABILIDAD DEL CLIENTE OFRECER UN SERVICIO DE TRADUCCIÓN.
- NO SE DEBERÁ DAR SERVICIO TÉCNICO AL EQUIPO, SIN HABER CONSULTADO Y COMPRENDIDO ESTE MANUAL DE SERVICIO.
- LA NO OBSERVANCIA DEL PRESENTE AVISO PUEDE DAR LUGAR A QUE EL PROVEEDOR DE SERVICIOS, EL OPERADOR O EL PACIENTE SUFRAN LESIONES PROVOCADAS POR CAUSAS ELÉCTRICAS, MECÁNICAS O DE OTRA NATURALEZA.

ATENÇÃO

- ESTE MANUAL DE ASSISTÊNCIA TÉCNICA SÓ SE ENCONTRA DISPONÍVEL EM INGLÊS.
- SE QUALQUER OUTRO SERVIÇO DE ASSISTÊNCIA TÉCNICA, QUE NÃO A GEMS, SOLICITAR ESTES MANUAIS NOUTRO IDIOMA, É DA RESPONSABILIDADE DO CLIENTE FORNECER OS SERVIÇOS DE TRADUÇÃO.
- NÃO TENHA TENTAR REPARAR O EQUIPAMENTO SEM TER CONSULTADO E COMPREENDIDO ESTE MANUAL DE ASSISTÊNCIA TÉCNICA.
- O NÃO CUMPRIMENTO DESTA AVISO PODE POR EM PERIGO A SEGURANÇA DO TÉCNICO, OPERADOR OU PACIENTE DEVIDO A CHOQUES ELÉTRICOS, MECÂNICOS OU OUTROS.

AVVERTENZA

- IL PRESENTE MANUALE DI MANUTENZIONE È DISPONIBILE SOLTANTO IN INGLESE.
- SE UN ADDETTO ALLA MANUTENZIONE ESTERNO ALLA GEMS RICHIEDE IL MANUALE IN UNA LINGUA DIVERSA, IL CLIENTE È TENUTO A PROVVEDERE DIRETTAMENTE ALLA TRADUZIONE.
- SI PROCEDA ALLA MANUTENZIONE DELL'APPARECCHIATURA SOLO DOPO AVER CONSULTATO IL PRESENTE MANUALE ED AVERNE COMPRESO IL CONTENUTO.
- NON TENERE CONTO DELLA PRESENTE AVVERTENZA POTREBBE FAR COMPIERE OPERAZIONI DA CUI DERIVINO LESIONI ALL'ADDETTO ALLA MANUTENZIONE, ALL'UTILIZZATORE ED AL PAZIENTE PER FOLGORAZIONE ELETTRICA, PER URTI MECCANICI OD ALTRI RISCHI.

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このサービスマニュアルには英語版しかありません。

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注意:

本维修手册仅存有英文本。

非 GEMS 公司的维修员要求非英文本的维修手册时，

客户需自行负责翻译。

未详细阅读和完全了解本手册之前，不得进行维修。

忽略本注意事项会对维修员，操作员或病人造成触

电，机械伤害或其他伤害。

DAMAGE IN TRANSPORTATION

All packages should be closely examined at time of delivery. If damage is apparent write "Damage In Shipment" on ALL copies of the freight or express bill BEFORE delivery is accepted or "signed for" by a GE representative or hospital receiving agent. Whether noted or concealed, damage MUST be reported to the carrier immediately upon discovery, or in any event, within 14 days after receipt, and the contents and containers held for inspection by the carrier. A transportation company will not pay a claim for damage if an inspection is not requested within this 14 day period.

Call the Service Center, People's Republic of China, (86) 800-810-8188 immediately after damage is found. At this time be ready to supply name of carrier, delivery date, consignee name, freight or express bill number, item damaged and extent of damage.

Complete instructions regarding claim procedure are found in Section S of the Policy And Procedures Bulletins.

14 July 1993

CERTIFIED ELECTRICAL CONTRACTOR STATEMENT

All electrical Installations that are preliminary to positioning of the equipment at the site prepared for the equipment shall be performed by licensed electrical contractors. In addition, electrical feeds into the Power Distribution Unit shall be performed by licensed electrical contractors. Other connections between pieces of electrical equipment, calibrations and testing shall be performed by qualified GE Medical personnel. The products involved (and the accompanying electrical installations) are highly sophisticated, and special engineering competence is required. In performing all electrical work on these products, GE will use its own specially trained field engineers. All of GE's electrical work on these products will comply with the requirements of the applicable electrical codes.

The purchaser of GE equipment shall only utilize qualified personnel (i.e., GE's field engineers, personnel of third-party service companies with equivalent training, or licensed electricians) to perform electrical servicing on the equipment.

IMPORTANT...X-RAY PROTECTION

X-ray equipment if not properly used may cause injury. Accordingly, the instructions herein contained should be thoroughly read and understood by everyone who will use the equipment before you attempt to place this equipment in operation. The General Electric Company, Medical Systems Group, will be glad to assist and cooperate in placing this equipment in use.

Although this apparatus incorporates a high degree of protection against x-radiation other than the useful beam, no practical design of equipment can provide complete protection. Nor can any practical design compel the operator to take adequate precautions to prevent the possibility of any persons carelessly exposing themselves or others to radiation.

It is important that anyone having anything to do with x-radiation be properly trained and fully acquainted with the recommendations of the National Council on Radiation Protection and Measurements as published in NCRP Reports available from NCRP Publications, 7910 Woodmont Avenue, Room 1016, Bethesda, Maryland 20814, and of the International Commission on Radiation Protection, and take adequate steps to protect against injury.

The equipment is sold with the understanding that the General Electric Company, Medical Systems Group, its agents, and representatives have no responsibility for injury or damage which may result from improper use of the equipment.

Various protective materials and devices are available. It is urged that such materials or devices be used.

LITHIUM BATTERY CAUTIONARY STATEMENTS



CAUTION Risk of Explosion

Danger of explosion if battery is incorrectly replaced. Replace only with the same or equivalent type recommended by the manufacturer. Discard used batteries according to the manufacturer's instructions.



ATTENTION Danger d'Explosion

Il y a danger d'explosion s'il y a remplacement incorrect de la batterie. Remplacer uniquement avec une batterie du même type ou d'un type recommandé par le constructeur. Mettre au rebut les batteries usagées conformément aux instructions du fabricant.

OMISSIONS & ERRORS

Customers, please contact your GE Sales or Service representatives. GE personnel, please use the GEMS CQA Process to report all omissions, errors, and defects in this publication.

Revision History

Revision	Date	Reason for change
1	07/15/04	Initial release document.
2	01/10/05	Chapter 1, Updated Floor Anchor information Chapter 2, Sec 8.0 System Ground Connections: Corrected figure 2-31. Added Chapter 8 - Customer Touch Leakage Test.
3	06/30/05	Chapter 1, Updated Gantry & Table Alignment Bar & Hole Locations in 3.5 - Install Alignment Bars for PQR13038013. Chapter 1, Updated diagram for Table Base Gantry/Table Level Locations A and B in 3.6 - Level the Table for PQR13038013. Chapter 1, Added diagram for alternate anchoring holes on gantries in 3.11 - Gantry & Table Redundant Anchor Holes PQR13038013. Chapter 1, Updated Table Base Anchor Assembly. Chapter 5, Updated the intercom adjustment method for console intercom pots and for gantry intercom pots. Chapter 5, Revised procedures of Autovoice/Intercom Check for PQR13038084. Chapter 5, Deleted procedures for monitor setup for CRT and LCD, referring instead to product documentation, according to PQR13038082. Chapter 5, Updated Interference Test. Chapter 6, Deleted "CT number adjustment" from flowchart in introduction, according to PQR13038090. Chapter 6, Revised directions for reference procedures in image quality calibration process. Chapter 6, Deleted manual scan for 20cm QA Phantom Image Series Image Performance Verification. Chapter 6, Deleted manual scan parameters for System Functional Test, according to PQR13038092.
4	10/12/05	PQR 13044488: Updated Top Cover info in Appendix A. PSR 13050792: Added Section 4.4 to Chapter 5 Chapter 5: - Added Sections 4.2, 4.5, 4.6, 4.7, 4.8 and 4.9 - Updated Section 4.1 Chapter 9: Updated Section 3.0 (Form 4879)

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Preface

Publication Conventions

Purpose: This section means to inform the reader on publication conventions used. So that the reader can identify safety and general material that is considered important by its format. This includes the interpretation of computer screen text as either input or output. There are a number of specific text and paragraph styles/conventions used within this section to accomplish this task. Please become familiar with the conventions used within this publication before proceeding.

Section 1.0 Safety & Hazard Information

1.1 Text and Character Representation

Within this publication, different paragraph and character styles have been used to indicate potential hazards. Paragraph prefixes, such as hazard, caution, danger and warning, are used to identify important safety information. Text (Hazard) styles are applied to the paragraph contents that are applicable to each specific safety statement. Words describe the type of potential hazard that may be encountered and are placed immediately before the paragraph it modifies. Safety information will normally include:

- Type of potential hazard
- Nature of potential injury
- Causative condition
- How to avoid or correct the causative condition

EXAMPLES OF HAZARD STATEMENTS USED

A few examples are provided that have been adapted from GEMS' global document standard (2119696-100). They include paragraph prefixes and modified text styles.



CAUTION
Pinch Points
Loss of Data
Sharp Objects

Caution is used when a hazard exists that can or could cause minor injury to self or others if instructions are ignored. They include for example:

- Loss of critical patient data
- Crush or pinch points
- Sharp objects



DANGER
EXCESSIVE
VOLTAGE
CRUSH
POINT

DANGER IS USED WHEN A HAZARD EXISTS THAT WILL CAUSE SEVERE PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- ELECTROCUTION
- CRUSHING
- RADIATION



**WARNING
ROTATING
EQUIPMENT
BARE WIRES**

WARNING IS USED WHEN A HAZARD EXISTS WHICH COULD OR CAN CAUSE SERIOUS PERSONAL INJURY OR DEATH IF INSTRUCTIONS ARE IGNORED. THEY CAN INCLUDE:

- Potential for shock
- Exposed wires
- Failure to Tag and lockout system power could allow for un-command motion.



**NOTICE
Equipment
Damage
Possible**

Notice is used when a hazard is present that can cause property damage but has absolutely no personal injury risk. They can include:

- Disk drive will crash
- Internal mechanical damage, such as to the x-ray tube
- Coasting the rotor through resonance.

It's important that the reader not ignore hazard statements in this document.

1.2 Graphical Representation

Important information will always be preceded by the exclamation point  contained within a triangle, as seen throughout this chapter. In addition to text, several different graphical icons (symbols) may be used to make you aware of specific types of hazards that could possibly cause harm.

ELECTRICAL



LASER



MECHANICAL



HEAT



RADIATION



PINCH



Some others make you aware of specific procedures that should be followed.

AVOID STATIC ELECTRICITY



TAG AND LOCK OUT



WEAR EYE PROTECTION



Section 2.0 Publication Conventions

2.1 General Paragraph and Character Styles

Prefixes are used to highlight important non-safety related information. Paragraph prefixes (such as Purpose, Example, Comment and Note) are used to identify important but non-safety related information. Text styles are also applied to text within each paragraph modified by the specific prefix.

EXAMPLES OF PREFIXES USED FOR GENERAL INFORMATION

- Purpose:** Introduces and provides meaning as to the information contained within the chapter, section or subsection, such as used at the beginning this chapter for example.
- Note:** Conveys information that should be considered important to the reader.
- Example:** Used to make the reader aware that the paragraph(s) that follow are examples of information possibly stated previously.
- Comment:** Represents “additional” information that may or may not be relevant.

2.2 Page Layout

Publication Part Number & Revision Number

Publication Title

Headers and footers in this publication are designed to allow you to quickly identify your location. The document's part number and revision number appears in every header on every page. Odd

numbered page footers indicate the current chapter, its title, and current page number. Even page footers show the current section and its title, as well current page number.

2.3 Computer Screen Output/Input Character Styles

Within this publication different character styles are used to indicate computer input and output text. Character (input, output, and variable) styles are used and applied to the text within a paragraph so as to indicate directions. Computer screen output and input is also formatted using mono (fixed width) spaced fonts.

Example: This paragraph denotes computer screen fixed output. It's output is fixed
Fixed Output from the sense that it does not vary from application to application. It's the most commonly used style used to indicate filenames, paths, and text.

Example: *This paragraph denotes computer screen output that is variable. Its output
Variable Output varies from application to application. Variable output is sometimes found placed between greater than and lesser than operators. For example: <variable_ouput>*

Example: This paragraph denotes fixed input. It's typed input that will not vary
Fixed Input from application to application. Fixed text the user is required to supply as input.

Example: *This paragraph denotes computer input that can vary from application to
Variable Input application. Variable text the user is required to supply as input. Variable input sometimes is placed between greater than and lesser than operators. For example: <variable_input>. In these cases, the (<>) operators are dropped prior to input. Exceptions are noted in the text.*

2.4 Buttons, Switches and Keyboard Inputs (Hard & Soft Keys)

Different character styles are used to indicate actions requiring the reader to press either a hard or soft button, switch, or key. Physical hardware, such as buttons and switches, are called hard keys because they are hard wired or mechanical in nature. A keyboard or on/off switch would be a hard key. Software or computer generated buttons are called soft keys because they are software generated. Software driven menu buttons are an example of such keys. Soft and hard keys are represented differently in this publication.

Example: A power switch **ON/OFF** or a keyboard key like **ENTER** is indicated by applying a character style
Hard Keys that uses both over and under-lined bold text that is bold. This is a hard key.

Example: Whereas the computer MENU button that you would click with your mouse or touch with your hand
Soft Keys uses over and under-lined regular text. This is a soft key.

Chapter 1

Position Subsystems



NOTICE Potential for Data Loss and/or Equipment Damage.

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system. Any other revisions of this manual may not exactly match your system.

Section 1.0

Install Table/Gantry Introduction

Note: This chapter describes how to mount, position, and level the CT Scanner subsystems. Before you start the installation, make sure the site preparation complies with conditions and instructions found in the pre-installation manual. Failure to comply will result in excessive installation delay and potential increased, unrecoverable installation costs. This product is designed to meet specific mechanical installation standards, which should be reviewed prior to installing this system.

1.1 Skill Set Required



NOTICE All Mechanical Installers must meet these minimum requirements.

Only people experienced and trained in the installation process of a GE CT Scanner should perform the work described in this document. A person doing the installation should have:

- Installed a GOC 1 or GOC 2 console.
- Completed three LightSpeed Ultra 4 or 8 or 16 CT installations as a helper, prior to leading an installation. Installation leader must have completed project management, leadership training, and completed all required training prior to becoming a project leader.
- Reviewed the LightSpeed Installation Video, Safety Video, Installation Training CD, and successfully passed all related tests. (Refer to [Table 1-1](#).) New installation training materials to be available in 4Q03.

Table 1-1 Installation and Training Media

MEDIA	PART NUMBER
LightSpeed Installation CD	2289471-100
LightSpeed Installation Video	2289468-100
LightSpeed Installation Training CD	2289779-100
GE Safety CD	2290198-100
GE Safety Video	2290199-100

- A working understanding of the mechanical installation process, prior to beginning.
- Completed environmental health and safety (EHS) training, understand and apply lockout/tagout processes, completed Hazcom training, completed Eye & Personal Protection Safety, and have taken Blood Borne Pathogens training.
- All mechanical vendor training and records must be verifiable and available to GEMS.

- Mechanical vendors shall be responsible for training all employees involved in the installation of GEMS CT products.
- It is the mechanical vendors responsibility to require and monitor that all of the above safety requirements are maintained and met on installation sites.
- Mechanical vendor installation teams must have basic measurement skills, system layout skills, and the ability to read and understand blue prints. Must have professional communication skills needed to communicate with customer any changes or additional requirements needed to complete the installation project.

1.2 Floor and Room Preparation

1.2.1 Preparation

Consult your local GE Sales and service representative about your specific needs. It is the purchaser's (buyer) responsibility to provide an approved support structure and an approved method of mounting. General Electric is not responsible for any failure of the support structure or method of anchoring.

The LightSpeed 5.X system has a total floor load of approximately 6500 lbs (2950 kg). About 5175 lbs (2350 kg), including patient (450 lbs (204 kg)), is concentrated in the table-gantry assembly. Refer to the *Pre-installation Manual* for more information.

1.2.2 Flooring

Do not place the scanner on any resilient flooring. Resilient tile or carpeting may slowly yield over a period of time and disturb the alignment of the table to the gantry. Refer to the floor template to determine locations where resilient flooring material should be removed.

Limitations include:

- No part of the floor surface within the table, gantry, or the two interface areas between table and gantry, should be higher than the support areas for the table and gantry.
- The floor structure must withstand the occupied weight of table and gantry, as well as the individual contact area loading of these components.
- The method and placement of anchors, or through bolts, must not reduce the structural strength of the floor.

If you have to remove the gantry covers in order to move the gantry into the room, refer to [Appendix A](#), for the cover removal procedure. Please read the caution statement on [page 97](#) before removing the gantry covers.

1.3 Overview

Procedures in this chapter provide detailed instructions to position, level, and anchor the gantry and table securely for operation. The LightSpeed 5.X system uses adjustable leveling pads to support the gantry and table. The gantry has four (4) primary leveling pads located on the gantry base. The table has five (5) pads used for leveling it.

The process you will be following is:

- 1.) Use the room-layout template to determine the general position of the gantry and table.
- 2.) Move the gantry into position.
- 3.) Level gantry.
- 4.) Use the three alignment bars to position the table relative to the gantry.
- 5.) Level the table to the gantry, and anchor the system.

Use the template to position the system; however, use the gantry and table to locate and drill the anchor holes. Drill the anchor holes with the system in place. Refer to [Section 3.0 on page 29](#) and

Figure 1-16, on page 37, for an example of this procedure. This CT system installation procedure requires the items listed in Section 1.5 on page 23 and Section 1.6 on page 24.

1.4 Pre-Installation Template

Always use the room-layout template (2363851) during installation. The gantry and table will not be properly aligned if existing holes are used. The template shows the location of the gantry and table anchor holes.

This template is shipped with the system. It may also be ordered via the Web.

1.5 Installation Alignment Tool Kit

The installation personnel provide the installation tool kit, P/N 2110953. The tool kit contains the following special tools (See Figure 1-1). **This installation alignment toolkit requires yearly calibration and must have a current calibration sticker.**

- **Guide Pin (Bolt) Tool** included in the tool kit (not needed).
- **Long Alignment Bar** (in two pieces), P/N 2106369: use to position the table, relative to gantry
- **Two Short Alignment Bars**, P/N 2106368: use to position the table, relative to the gantry
- **Adjuster Tool**, P/N 2107863: use to turn the adjusters, which in turn level the system.
- **½" Drill Bushing**, P/N 2106205: use to center and guide the bit when drilling the anchor holes
- **Vacuum Attachment (funnel)**, P/N 2110944: use to vacuum debris out of the anchor holes
- **Spanner Wrench**, P/N 2110003: if necessary, use to turn the adjusters after anchoring system
- **Anchor Seating Tool**, P/N 2247965: used to properly seat anchors

Figure 1-1 Installation Alignment Toolkit Accessories

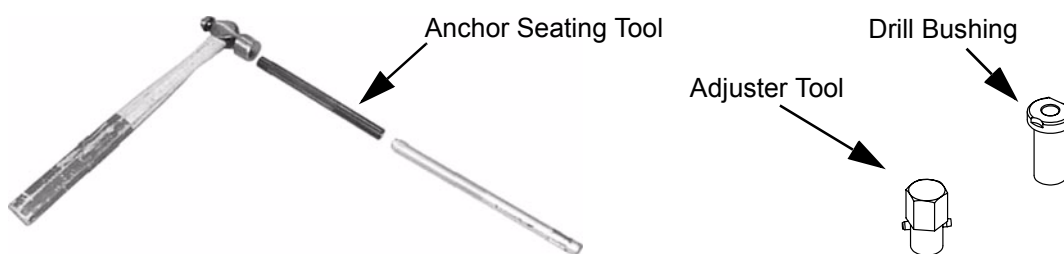


Figure 1-2 Gantry/Table Alignment Toolkit



1.6 Required Common Tools and Supplies

The following tools and supplies are required for installation of the scanner. Please refer to [Appendix B on page 117](#), for pictorial descriptions of the tools and supplies.

WRENCHES

- Standard and Metric combination wrench sets
- Standard and Metric Hex Key (Allen wrench) sets
- ½" and 3/8" drive torque wrench: 0-100 N-m (0-100 ft.-lb.) Must be calibrated yearly.

SOCKETS AND EXTENSIONS

- 3/8" and ½" drive ratchet wrenches
- ½" drive 3" & two 6" long extensions
- 3/8" drive 12" long extension
- Standard & Metric 3/8" drive socket sets
- ¾" deep well socket 3/8"
- 1", 1-1/8", 1-¼" & 1-½" sockets for ½" drive
- 3/8" drive universal joint
- Metric hex bit set ¼" or 3/8" drive, including:
 - 14mm hex bit 3/8" or ½" drive (14mm ball hex helpful)
 - 10mm hex bit 3/8" drive

SCREW DRIVERS

- Phillips screwdriver set (small, medium, and large)
- Straight blade screwdriver set (small, medium, and large)

DRILL BITS

- Complete set of standard (U.S.) drill bits
- Metric tap set
- ½" masonry bit, min. 8" long USA – 18" optional (for rear table hole)
- 13mm masonry bit, min. 203mm long – 457mm optional (for rear table hole)
- 3" (76mm) hole saw with 1/4" (6mm) masonry bit (to remove flooring)

POWER TOOLS

- 3/8" or 1/2" drill, cordless or electric
- Reciprocating Saw (Sawzall or equivalent) and assorted blades.
- Hammer Drill & Bit (8" min, 12" max)
- Sears 17740 Shop vacuum or equivalent, with "HEPA" or dry wall dust filter (Sears part number 17918) or equivalent
- 25' Extension power cords

HAND TOOLS

- Ball-Peen Hammer (1lb or 2lb)
- Tongue & Groove Pliers (large)
- Diagonal Cutting Pliers, Large (to cut 1/0 ground)
- Framing Square (e.g., Empire 16" x 24" aluminum square)
- Diagonal Cutting Pliers, Small
- Large pry bar
- 4', 2' & 9" torpedo levels (see [Table 1-2: Recommended Levels](#))
- Laser level (see [Table 1-2: Recommended Levels](#))

Table 1-2 Recommended Levels

9"	Johnson Magnetic Level, model 7500M*
	Johnson Magnetic Level, model 4500
	Stanley Magnetic Level
2'	Johnson Professional Box Beam Level, model 9624*
	Empire Titan Professional Box Beam Level, model 900 series
4' (nominal)	48" Johnson Professional Box Beam Level*
	42" Stanley Contractor Grade Level
	48" Empire Titan Professional Box Beam Level, model 900 series
	48" Stabila Aluminum Box Beam Level, Kit 24816
Laser**	Sears Laser Level Tool (\$40)

* Preferred levels

** Laser level is required for table cradle alignment. Using this tool reduces the alignment procedure by 1 man-hour. New tables have a cradle center line to be used with the laser level.

ELECTRICAL TOOLS

- DVM
- Continuity tester

PERSONAL SAFETY EQUIPMENT

- Safety shoes*
- Safety glasses*
- Gloves
- LOTO Kit (supplied)*
- Hearing Protection*
- 6' & 8' Step ladder

* These PPE items are absolutely required for every installation job, with NO exceptions.

Note: A box labeled Installation Support Kit is shipped with each system. It contains paint, masking tape, cleaners, towels, and other materials needed to install this CT scanner.

Section 2.0

Delivery and Inventory Procedure

2.1 Delivery Procedure

2.1.1 System Transportation - Temperature Extremes

When transporting the CT system, ensure that the system is not exposed to temperatures or humidity outside the following specifications.

Temperature: 0° to +120° F (-18° to +49° C)

Humidity: 20% to 80%

NOTICE **Component Freezing occurs if CT system is exposed to temperatures below 0° F (-18° C) for a period longer than two days.**

Allow a minimum of 12 hours for the CT system to adjust to ambient room temperature, prior to installation.

2.1.2 Working with the Mover

Follow the instructions provided by your installation specialist regarding working with equipment movers. Help direct movers on where to place equipment and which items you need first.

Generally movers should move all equipment into the customer room. Door removal and other site changes to move equipment should be done only as directed by the install specialist.

For component sizes and weights, refer to the *LightSpeed 5.X Pre-Installation* manual.

2.1.3 Floor Protection

It is suggested that the movers use floor protection. Most equipment movers can provide floor protection during the equipment delivery. Installers should provide floor protection for the room.

2.1.4 Equipment Delivery Route

Prior to equipment delivery, review the delivery route with the movers. Refer to the installation specialist for any additional delivery instructions.

2.1.5 Removing Gantry Dollies and Covers

Refer to [Appendix A](#), for the dolly and cover removal procedures. Please read the caution statement on [page 97](#) before removing the gantry covers.

2.1.6 Check for Shipping Damage

Check for the following damage, which may have occurred during shipping:

- Equipment damage: paint scrapes and cover damage.
- Damage to hospital property: floors, door frames and walls.

If damage is found—or items are missing in shipment—notify the appropriate service personnel:

- For item(s) missing in shipment or short shipped, contact the install specialist to have the item(s) shipped.
- Report damaged items to:

- (800) 548-3366, Option 8 — Bill Kennedy, or
- (262) 544-3744 — John Carini, CT Manufacturing, John.Carini@med.ge.com.

For further details, see [“DAMAGE IN TRANSPORTATION,” on page 7](#).

2.1.7 A1 Breaker

Lock-out and tag-out the A1 breaker now.

Figure 1-3 Sample A1 Breaker

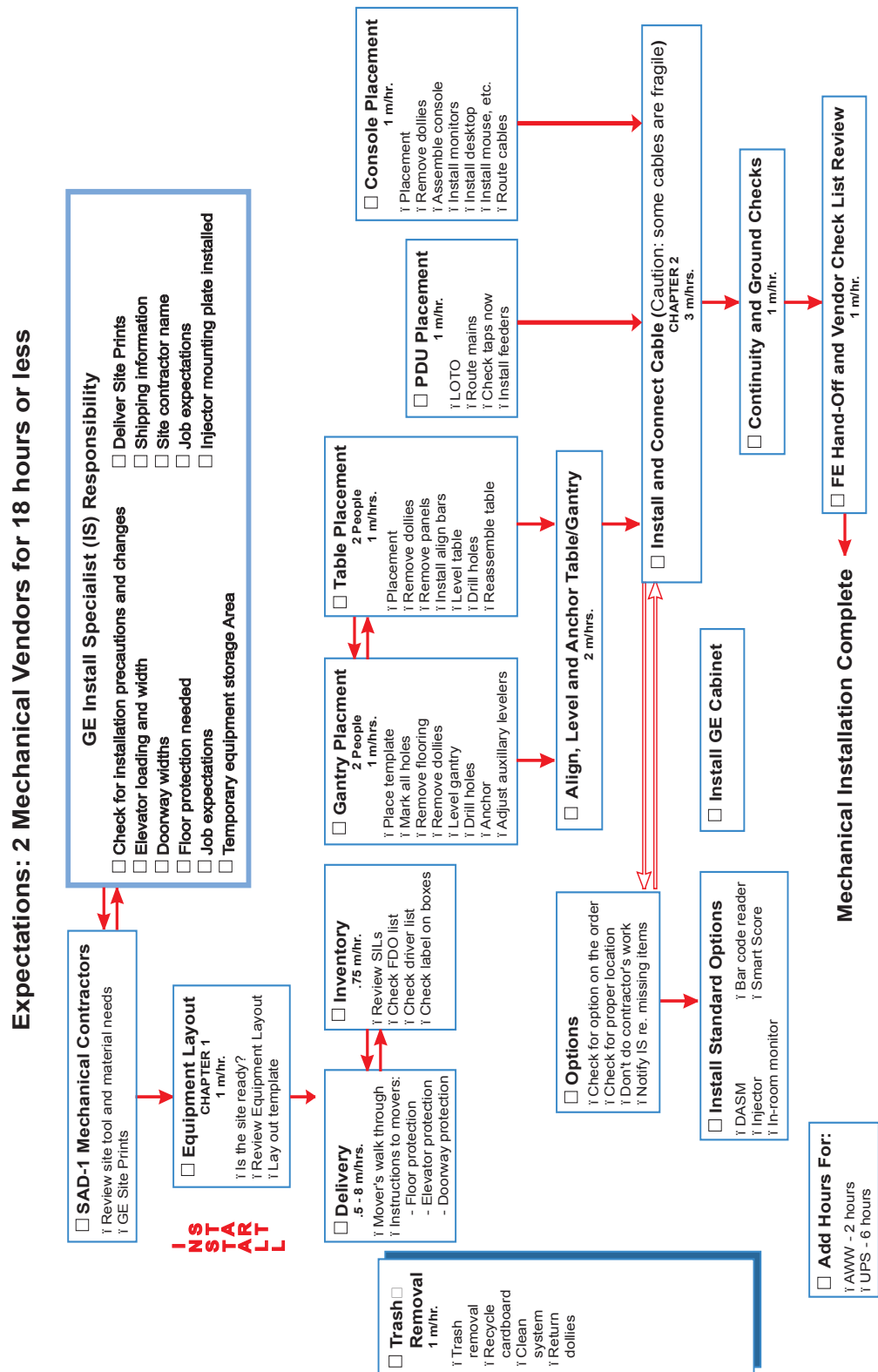


2.1.8 Installation Support Kits

An Installation Support Kit is shipped with every system. Locate this box now and open it. All included materials are to be used during the installation process. These items are to be left ON SITE, for future service needs.

2.2 Mechanical Block Diagram

Figure 1-4 Mechanical Installation Block Diagram



Section 3.0

Install and Level Table/Gantry

3.1 Establish the Room Layout

Using the GE print (developed for your site) to establish the room layout, make sure all the operating and service clearances shown on the print are observed. Using the supplied template, locate the anchor holes. Make sure they clear structural interferences in the floor.

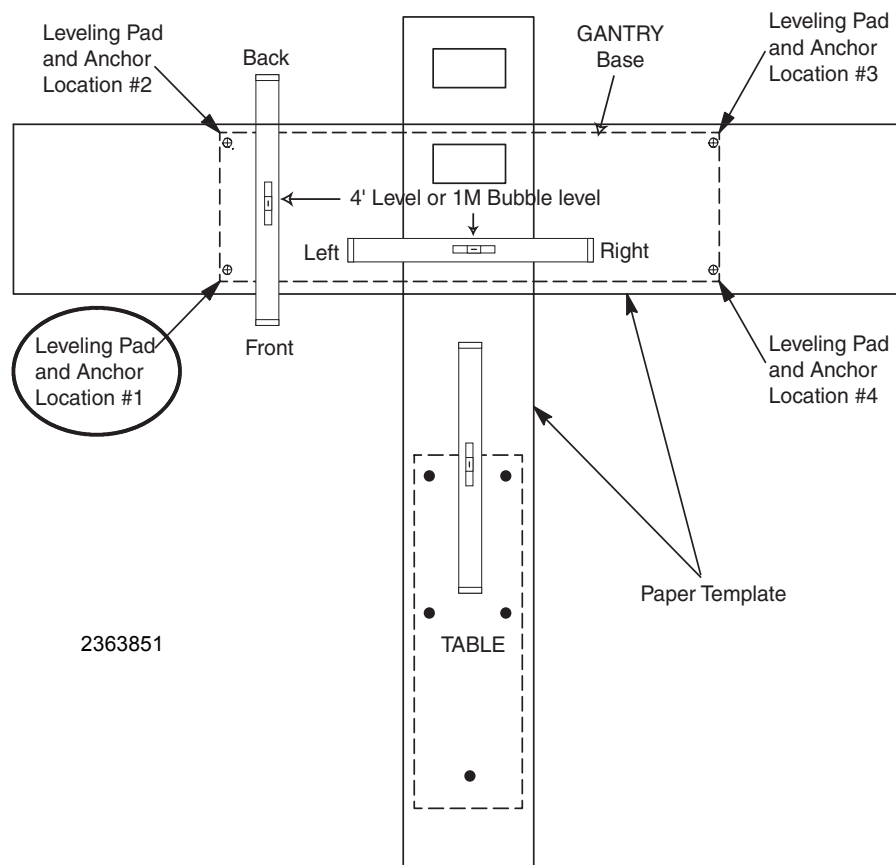
Clean the area. Free the mounting surface of any material that may interfere with the positioning and leveling of the system.

- 1.) Lay out the pieces of the floor template (P/N: 2363851).
- 2.) Start with the Gantry template—align per the GE print.
- 3.) Place the table template over the top of the Gantry template. Align the scan and table center-lines and secure the templates to the floor. Make sure there are no potential clearance issues.
- 4.) Now, check the level of the floor (See [Figure 1-5](#)) across the templates.

Note:
Clearances
differ by model.

LightSpeed 5.X service clearance areas are different than LightSpeed 1.X and earlier scanners.

Figure 1-5 Hole Locations



- 5.) Scribe a mark (e.g., use a center punch) at each of the gantry's mounting hole locations (there are four (4) of these).
- 6.) Using a center punch, mark the three (3) table mounting hole and two (2) leveler locations.



NOTICE

Positioning requires cutting 15 holes in the floor.

Before you drill or cut any flooring, make sure the appropriate hospital personnel have approved the location of the table/gantry.

- 7.) Cut tiles (or other resilient flooring) around all holes punched in the template for the gantry and table. Use a utility knife with a heat gun, a 3" hole saw with a 1/4" masonry bit or other adequate tools to cut the flooring.
- 8.) Some sites may require sealing of the floor penetrations after the flooring is removed.

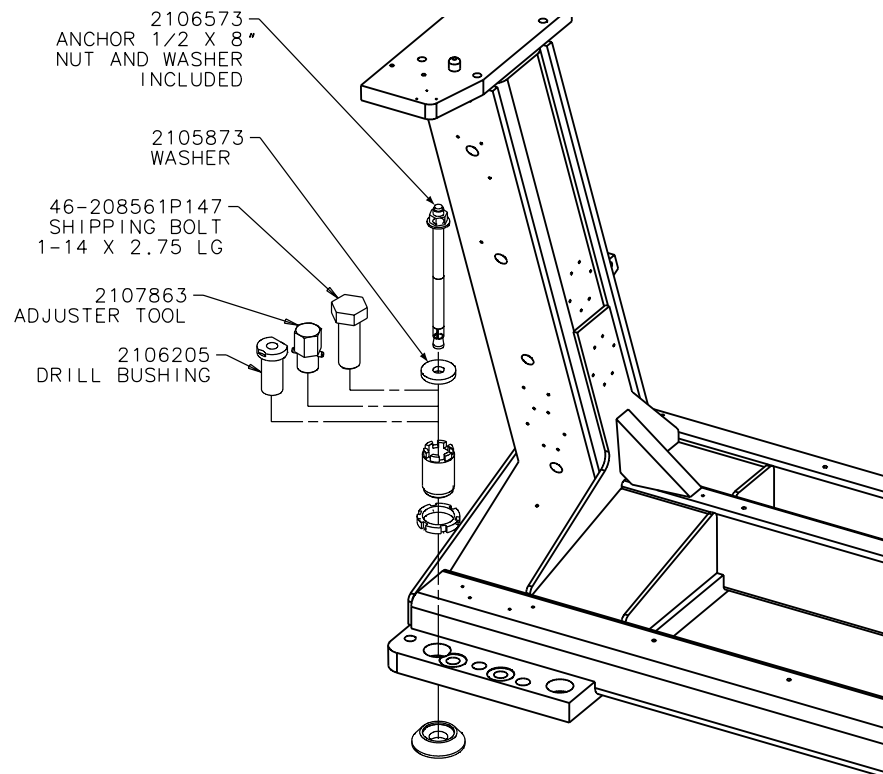
Note:

- Use RTV or other sealant to seal the floor covering, as necessary.
- All documentation in this manual is based on mounting the table / gantry on a concrete floor only.

3.2 Position the Gantry

- 1.) Remove all the transportation packaging, except for dollies, from the gantry.

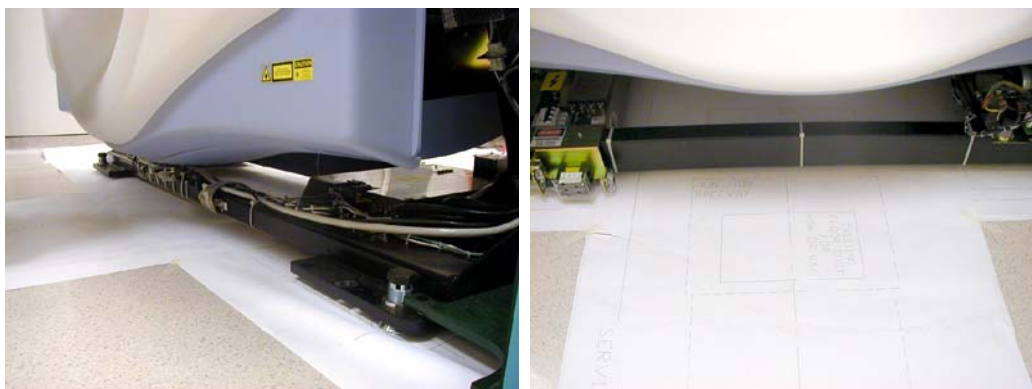
Figure 1-6 Table Base Installation Hardware



Note: Adjusters are used at each anchor location. Anchor hole ID is 1" (2.5 cm). Void between adjuster and anchor must be filled according to local building codes for seismic application.

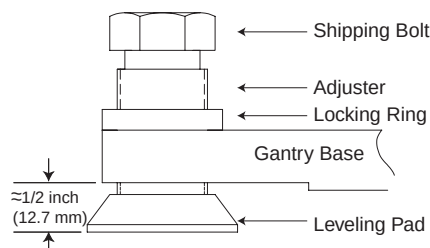
- 2.) Position the gantry over the template appropriately.

Figure 1-7 Gantry and Table Paper Templates



- a.) Locate the four (4) leveling pads, and position each of them beneath its associated adjuster.
- b.) Use the dollies to evenly lower the gantry, until it is just off of the floor (approximately $\frac{1}{2}$ " or 12.7 mm). Use a $\frac{1}{2}$ " ratchet to raise and lower the dollies.
- c.) Carefully rotate the gantry into the correct position over the template.

Figure 1-8 Gantry and Table Base Leveling Pads (Starting Positions)



- 3.) Remove the paper templates from the floor and discard of properly.
- 4.) Loosen the locking rings and shipping bolts so you can fine tune the leveling pads to compensate for slight variations in the floor surface.
- 5.) Position the gantry so that the adjusters are centered over their respective holes scribed earlier into the floor.

IMPORTANT:

Make certain to route the gantry power cord under the two rear gantry rails, before removing the gantry shipping dollies.

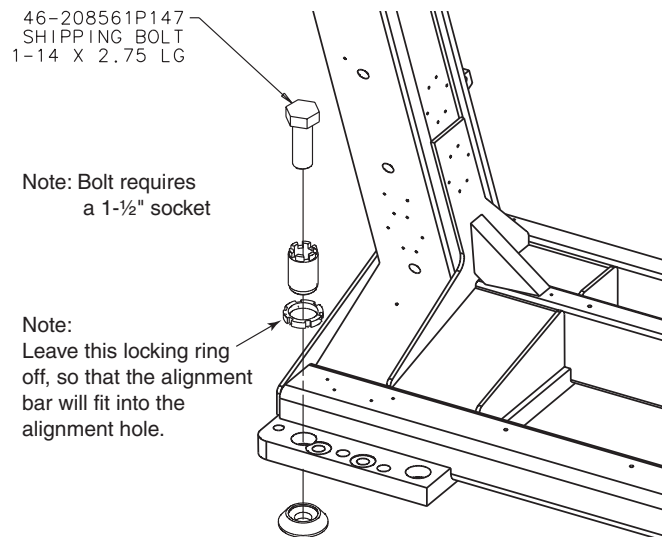
- 6.) Using a $\frac{1}{2}$ " ratchet, gently lower the gantry until it rests on the floor, over the marked areas.
- 7.) Using a 14mm hex socket, remove the dollies from the gantry by removing the three dolly bolts found at both ends of the gantry (Figure 1-9).

Figure 1-9 Gantry Dolly Bolts



- 8.) Remove the (4) gantry shipping bolts, using a 1½" socket.

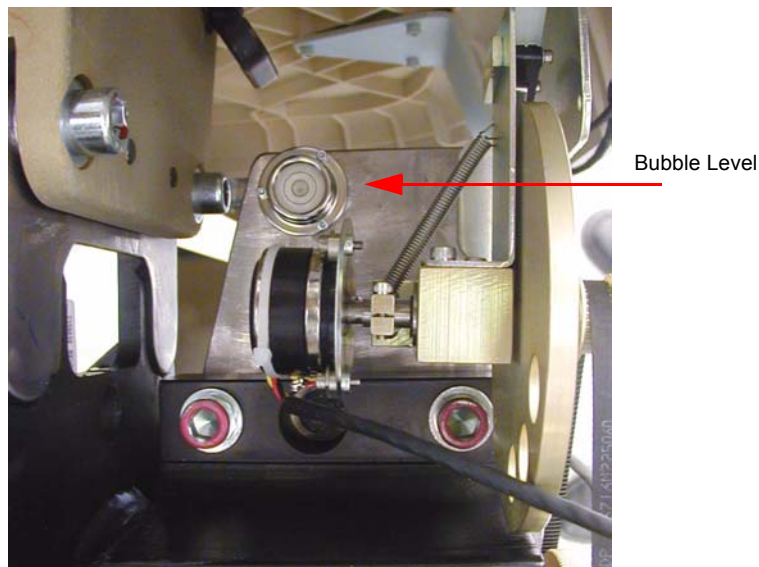
Figure 1-10 Gantry Shipping Bolts



3.3 Level the Gantry

The gantry uses 2 bubble levels that are permanently mounted to machined surfaces on the stationary base to tell when it is level.

Figure 1-11 Gantry Bubble Level



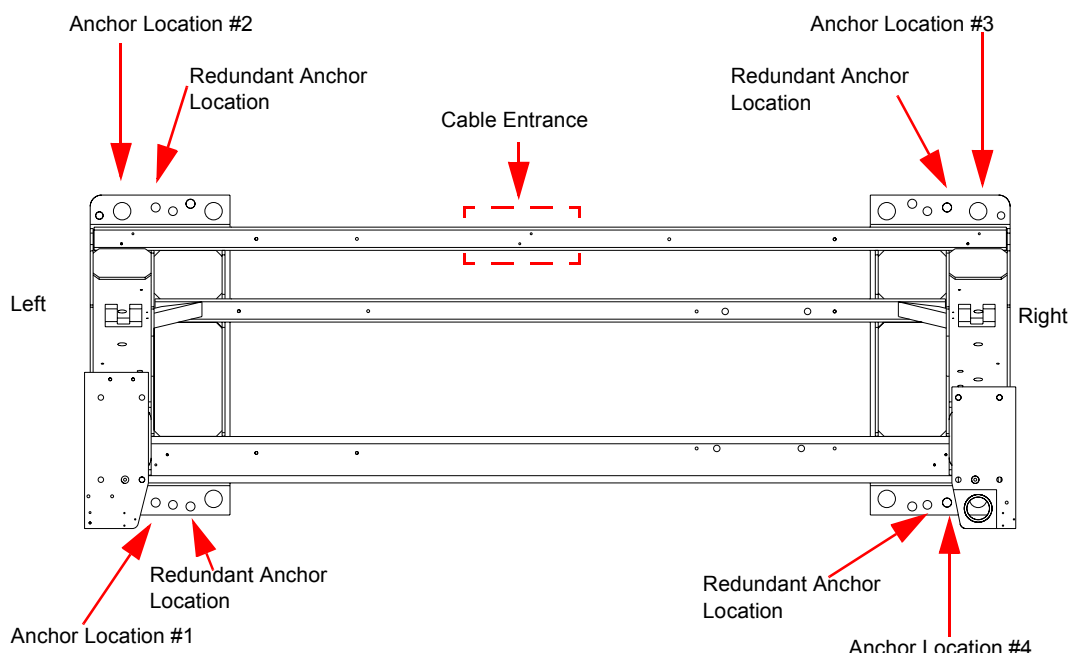
Bubble levels are located on both ends of the gantry stationary base. They're located on the stationary base near a point where the rotating structure pivots mount to the base structure. (See [Figure 1-11](#).) The gantry is properly leveled when the bubble is centered. (See [Figure 1-13](#), on page 34.)

- 1.) **Remove adjuster #1 lock ring** and only loosen all the other (#2, #3 and #4) adjuster lock rings. This can be done by using a spanner wrench 2110003 (the spanner wrench can be found in the alignment toolkit). Check to make sure auxiliary levelers do not cause interference.
- 2.) Systematically turn each of the gantry's adjusters (locations 1, 2, 3 and 4 in [Figure 1-12](#) or [Figure](#)) until both bubble levels are centered left to right, and front to back.
 - Begin by turning each adjuster no more than 1 turn at a time.
 - Use the adjuster tool, 1¹/₈" socket, and the ½" drive ratchet to turn each adjuster. (Refer to [Figure 1-6](#), on page 30.)

Systematic Procedure for Leveling gantry follows:

- 1.) Level the left side from front to back by turning adjusters #1 and #2.
- 2.) Level the right side from front to back by turning adjusters #3 and #4.
- 3.) Level the side (right or left) that is higher with respect to the other side. Turn both adjusters on a side equally until that side is level. The side should now also be level.

Figure 1-12 Gantry Base "Adjuster" Locations — Top View

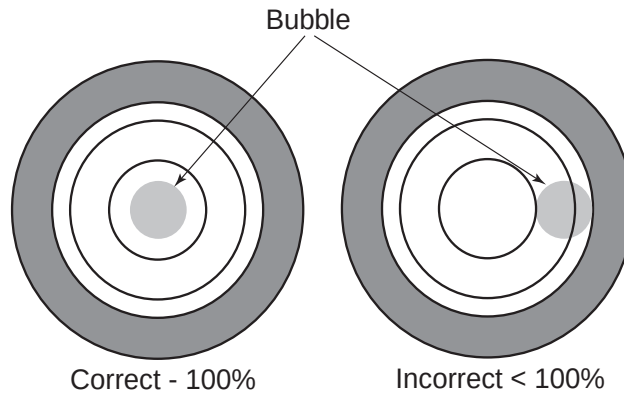


- 3.) When the bubble levels are centered ([Figure 1-13](#)), each of the four (4) leveling pads should be carrying a portion of the gantry weight. Distribution of the gantry weight prevents the base frame from rocking during normal operation. **DO NOT leave any adjuster un-loaded or floating.**

Figure 1-13 Bubble Level Centering

Correct level is 100% of bubble within small circle

Incorrect Level is less than 100% of bubble within small circle



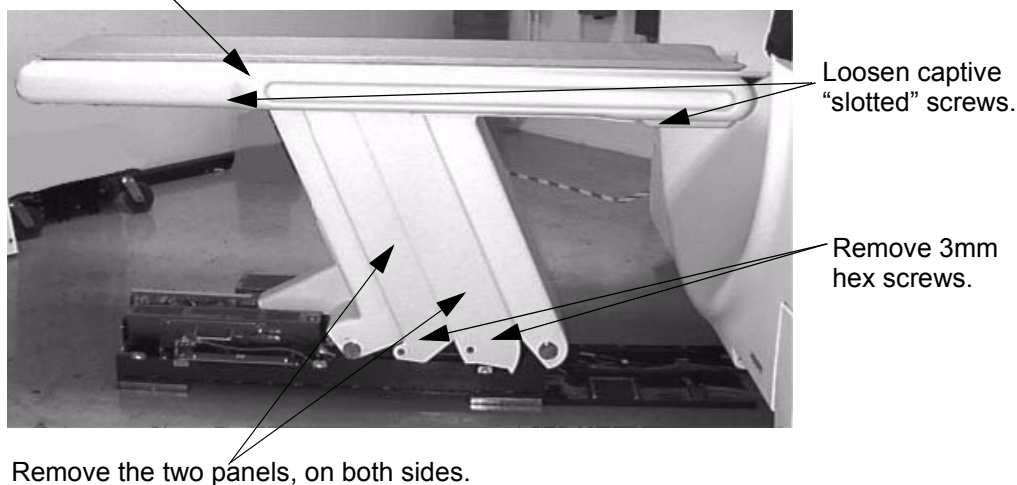
3.4 Position the Table

- 1.) Remove all the transportation packaging and boxes, except dollies, from the table.
- 2.) Wheel the table to its approximate position relative to the gantry, using the marks made earlier.
 - a.) Locate the table leveling pads and position them against the base of the table, using the adjusters with a 1½" socket and ½" ratchet.
 - b.) Use the dollies to evenly lower the table until it rests on the leveling pads using an ½" ratchet.
- 3.) **Loosen adjuster shipping bolts and locking rings “one turn” before you lower the table to the floor.** Loosen the locking rings and shipping bolts so you can **fine tune** the leveling pads to compensate for slight variations in the floor surface. Use a 1½" socket with 6" extension to loosen the shipping bolts.
- 4.) Remove the table left and right upper side covers (refer to [Figure 1-14](#)).

Note: The captive screws have a tendency to fall out occasionally.

Figure 1-14 Table Cover Removal

Remove left and right side covers.



- 5.) Remove the four side panels, using a 5mm hex key. Without disconnecting the ground straps at the z-channel, carefully lay the side panels on top of the padded cradle.



NOTICE

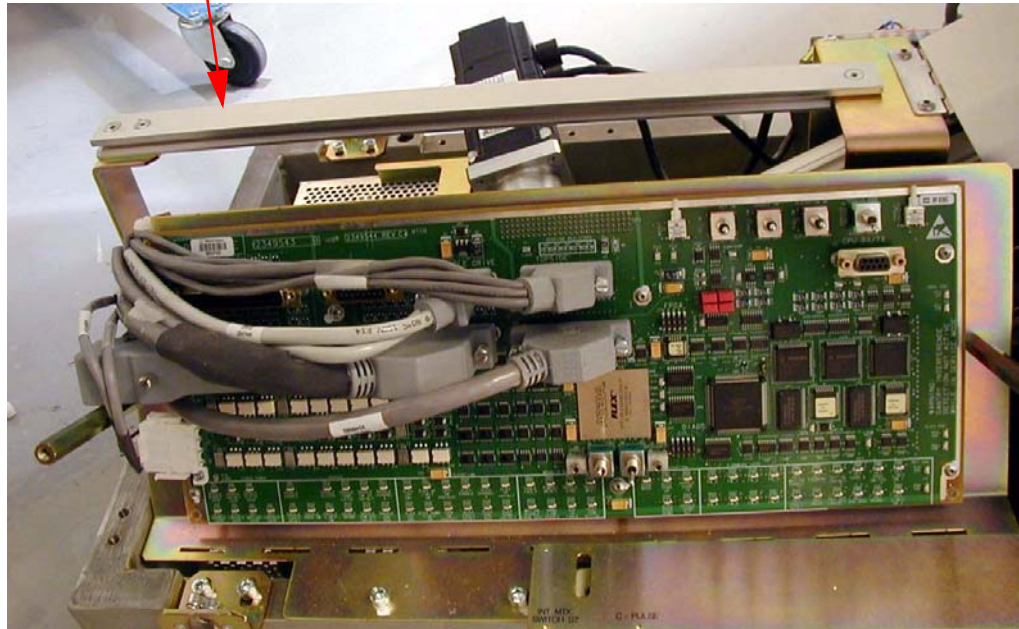
Potential for Equipment Damage.

Take extreme care to prevent damaging the MTCB printed wire assembly. Use some packing material to protect these components.

- 6.) Remove the table base cover center support bar. (See [Figure 1-15.](#))

Figure 1-15 Center Support Bar

Remove this bar.

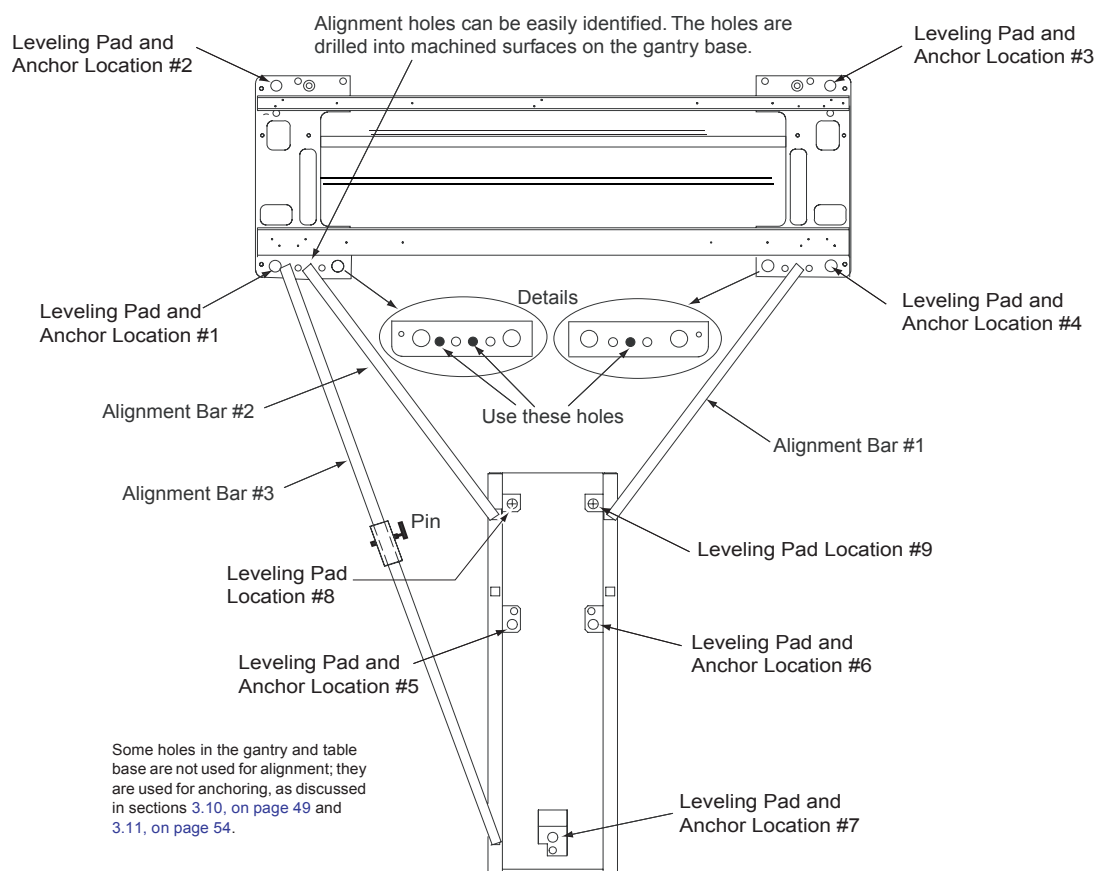


3.5 Install Alignment Bars

- 1.) Assemble the two (2) halves of the long gantry/table alignment bar, #3:
Adjuster locking ring at location #1 (see [Figure 1-16](#)) must be removed to properly position the alignment bars. Also, observe the following:
 - **Bars should be at room temperature ($70^{\circ}\text{F} \pm 1\text{ degree}$ or $21^{\circ}\text{C} \pm 1\text{ degree}$).**
 - **The alignment bar must be calibrated. They must have a valid calibration sticker to be accurate.**
 - Do not use or trust bars that have a missing locking pin or loose end cams.
- a.) Insert the pin and rotate the handle to tighten the joint. If the joint remains loose, tighten the tip of the expanding pin to increase its diameter and try again.
- b.) Check the serial numbers on both halves of the alignment bar; make sure the serial numbers match.

- 2.) The short alignment bars, #1 and #2, are identical and interchangeable. Refer to [Figure 1-16](#):
 - a.) Install the short alignment bars #1 & #2 first. Bars should fit snugly.
 - b.) Pivot the far end of the table until the long alignment bar #3 fits into place.
 - c.) Adjust the position of the table until the three alignment bars fit into the $\frac{3}{4}$ " alignment holes in the gantry base and the $\frac{1}{2}$ " alignment holes in the table base.
 - d.) Fully seat all three bars on their mating surfaces.

Figure 1-16 Gantry & Table Alignment Bar & Hole Locations



- 3.) After installing the alignment bars, lower the table to the floor and remove the transportation dollies ([Figure 1-17](#)).

Figure 1-17 Table transportation dollies



- a.) Temporarily remove alignment bar #1, to remove the front table dolly.
- b.) Replace alignment bar #1 after you remove the front dolly.
- 4.) Remove the five adjuster shipping bolts and loosen the lock rings.
- 5.) Use the 1½" socket and 6" extension to remove the shipping bolts.

3.6 Level the Table

3.6.1 Before You Begin

- Only adjusters should be used to level the table.
- Use the adjuster tool, 1½" socket, and ½" drive ratchet to turn each adjuster.
- Place a mark on the adjuster to aid in counting adjuster rotation.

3.6.2 Procedure

- 1.) Loosen and unload adjusters #5 and #6. (Refer to [Figure 1-18](#).) Do not remove shipping bolts.
- 2.) Using a 4' level, measure the levelness of the table at locations A, B and C.
 - Place the level on the machined surfaces of the alignment bars.
 - Do NOT rest the level on the radiused surface on alignment bar #3. (See [Figure 1-16](#).)Because the leveling pads rest against the bottom of the base, you have to raise the table from its lowest position to get it leveled.
- 3.) Map out the adjustments needed to level all 3 locations.
- 4.) Remove the shipping bolts from adjusters 7, 8 and 9 only.

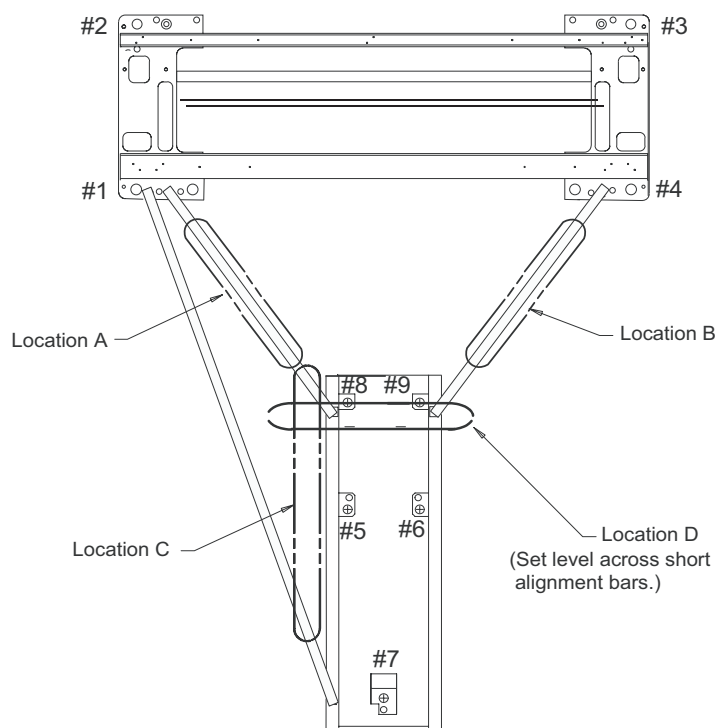
- 5.) Systematically level locations A, B, and C (refer to [Figure 1-18](#)) by turning each adjuster. We suggest a maximum of 1 turn at a time.

NOTICE

Always use the adjuster tool to raise or lower the table.

- Use adjuster #8 to level Location A.
- Use adjuster #9 to level Location B.
- Use adjuster #7 to level Location C.
- Use adjuster #8 and 9 to level Location D.

Figure 1-18 Table Base Gantry/Table Level Locations



- 6.) Continue to adjust until Locations A, B, and C are centered at the same time.
- 7.) Check location D. Fine tune only.
- 8.) Screw down the adjusters at table Locations #5 and #6, until each rests tightly against its leveling pad. Tighten each adjuster a little more than hand tight, but take care not to unload any of the other table adjusters. It's important that locations #5 - #9 ([Figure 1-18](#)) all bear part of the table's load.

- 9.) The table base covers need a minimum of $27\text{mm} \pm 3\text{mm}$ between the bottom of the table base and the floor for the covers to fit correctly. Check the distance between the table base and the floor, to ensure the covers will fit properly. See [Figure 1-19](#) and [Figure 1-20](#).

Figure 1-19 Typical Table Base Clearance

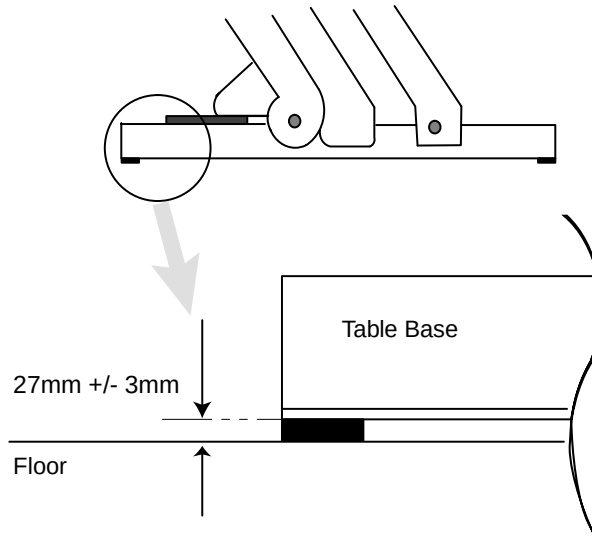


Figure 1-20 Excessive Table Base Cover Clearance



Note: If excessive clearance cannot be avoided, fill the gap. For example, you can use 0.5" round pipe insulation to fill the gap.

3.7 Table-Gantry Mechanical Alignment

3.7.1 Required Tools

- 2' Carpenter's Square
- 2' Level
- Masking Tape
- Flat-blade Screwdriver
- Craftsman Laser Trac™ Level with Carrying Case and Laser Enhancing Glasses
Sears item #00948247000
Mfr. model #48247
\$39.99
- 4' Level
- 9" Level
- Metric Tape Measure
- 1' Pry Bar

3.7.2 Alignment Conditions

- 1.) Complete all sections of this procedure before drilling any anchor holes.
- 2.) Alignment tool was used to align and level the table gantry.
- 3.) Alignment tool MUST have a current calibration sticker.

3.7.3 Alignment Specification

Mechanical alignment must be within ± 3 mm
Table cradle travel must be parallel
Table must be perpendicular to the gantry

3.7.4 Procedure

NOTICE Use caution while removing the gantry scan window.



- 1.) Remove gantry scan window. (Refer to [Figure 1-21](#).)

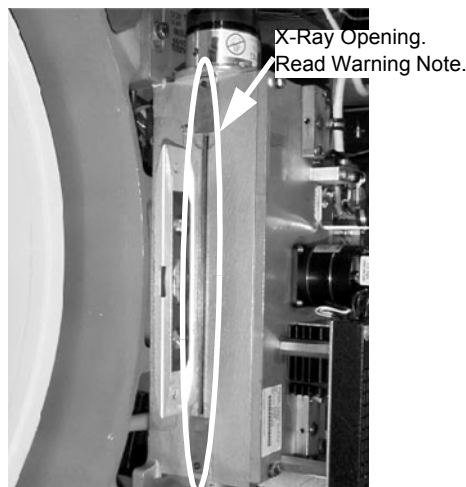
Figure 1-21 Removing gantry scan (mylar) window



- 2.) Do not remove the alignment bars unless told to do so in the procedure

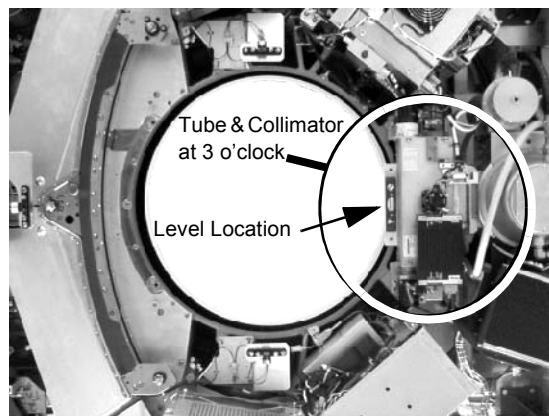
- 3.) Do not place any object near or into the x-ray opening. This slot must not be nicked or marred (refer to [Figure 1-22](#)).

Figure 1-22 X-Ray Opening



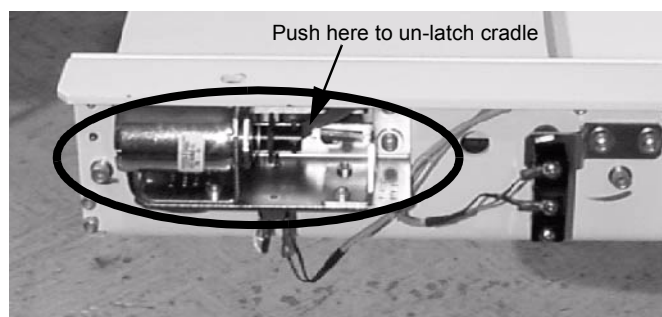
- 4.) Rotate gantry by hand until collimator face plate is at 3 o'clock (refer to [Figure 1-23](#)). Use 9-inch level to verify plate is vertical. With power off, the gantry movement is tight.

Figure 1-23 Three o'clock position



- 5.) Remove the cradle shipping bolt, cradle right side cover. Cradle shipping bolt is located under the table/cradle assembly at the foot end of the table. It has a RED TAG attached to it.
- 6.) Push the plunger on the right rear of the table to unlatch the cradle (refer to [Figure 1-24](#)).

Figure 1-24 Unlatch Cradle



3.7.4.1 Cradle Center Procedure

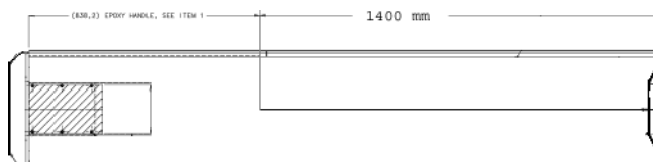
- 1.) Center the cradle on the drive rollers assembly by pushing the cradle into the gantry to its maximum position (1400 mm) and back to just before the latch position six times. The cradle should be centered as shown in [Figure 1-25](#).

Figure 1-25 Centered Cradle



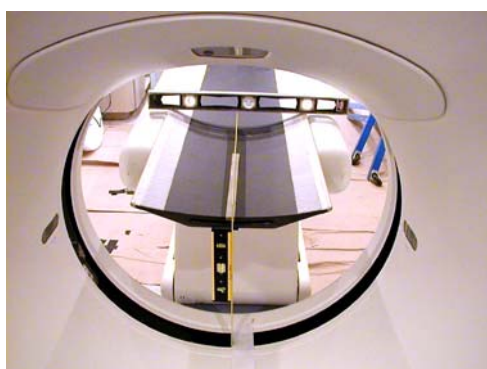
- 2.) Adjust cradle as needed, using the six bolts located on the back of the cradle ([Figure 1-26](#)).

Figure 1-26 Use bolts to adjust cradle



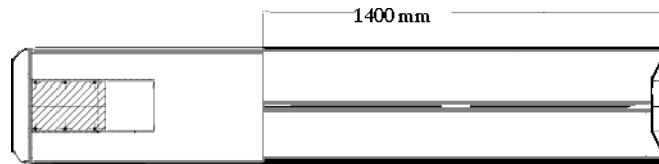
- 3.) Unlatch and move the cradle towards the gantry a few inches until the front of the cradle is near the gantry opening.
- 4.) Use a 2ft level to check cradle level from side to side. Level the cradle using the table front level adjusters #8 & #9. (See [Figure 1-27](#).)

Figure 1-27 Place level across cradle



- 5.) Place the level in the center of the cradle and check for level (see [Figure 1-28](#)). The width and the length of the cradle must be level when finished. Re-tighten all adjuster lock rings.

Figure 1-28 Place level along the length of the cradle



- 6.) Find the table center line marked on the front edge of the cradle. If your system had no center line mark, complete the following procedure to make a cradle center line.
- 7.) Using a metric tape measure or other suitable measuring device:
 - a.) Using supplied masking tape, place a 380mm (15") strip on the front of the cradle. See [Figure 1-29](#).

Figure 1-29 Masking tape on the front of the cradle

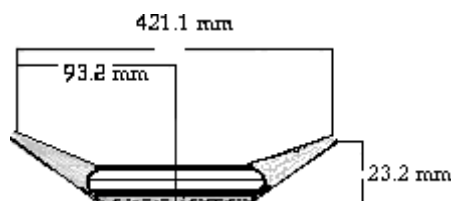


- b.) Determine the cradle center-line by measuring the cradle width 421.1mm (210.5mm center) Use a square or straight edge to determine the center.

Note:

The white edges on the cradle may not always parallel and a second measurement is needed. See [Figure 1-30](#).

Figure 1-30 Front of Cradle



- c.) Find the center of the inside slot. Width of the inside slot is 186.4 mm (93.2 mm center.) Use a square or straight edge to determine the center. See [Figure 1-31](#).

Figure 1-31 Determine slot center



- d.) Use these two dimensions to determine the table center line by connecting the two marks.

Measure back 300mm, then measure across the cradle and place a second cradle center mark on the tape. Use a square or straight edge to connect the two marks. See [Figure 1-32](#).

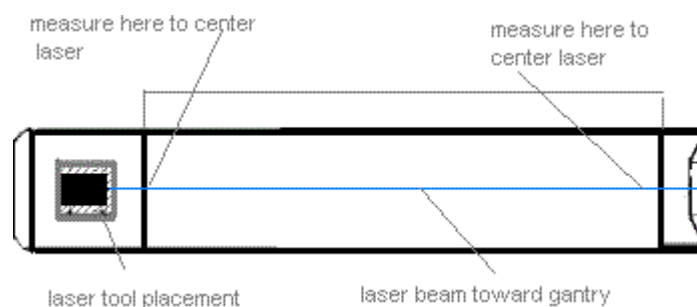
Figure 1-32 Use a square to connect marks



3.7.4.2 Laser Cradle Center Procedure

- 1.) Assemble the laser tool
 - a.) Mount the laser to the magnetic base unit
 - b.) Note the CL alignment pin on the magnetic base unit
- 2.) Level the laser tool
 - a.) Place the laser tool on the rear of the cradle with the laser pointing toward the gantry. See [Figure 1-33](#).

Figure 1-33 Position Laser Tool



- b.) Using the three adjustment screws level the laser. See [Figure 1-34](#).

Figure 1-34 Laser Level Adjustment Screws



- 3.) Center the laser on the cradle

POSSIBLE EYE INJURY.

LASER BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

WHEN OPERATING THE LASER LEVEL, DO NOT STARE AT THE LASER BEAM.

- a.) Turn on the laser. See [Figure 1-35](#).

Figure 1-35 Turn on the Laser



- b.) Place a metric tape across the cradle in front of the laser. See [Figure 1-36](#).

Figure 1-36 Place Metric Tape in Front of Laser



WARNING

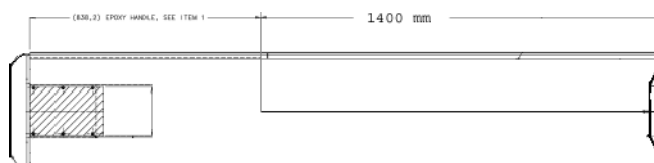


- c.) Adjust the laser tool so that the beam is centered at ~ (210mm)
- d.) Move the metric tape to the front edge of the cradle and adjust beam center (210mm)
- 4.) Using a metric tape carefully measure from cradle center line to collimator faceplate. The laser beam should pin point the exact measurement. Record this measurement as A on the worksheet.
- 5.) Carefully rotate the gantry by hand until the tube and collimator is in the nine o'clock position.
- 6.) Use a bubble level to precisely position the collimator faceplate at the nine o'clock position.
- 7.) Carefully measure from cradle center mark to collimator faceplate.
- 8.) Record this measurement as B on the worksheet.
- 9.) Perform the equation in the SPEC column to determine if the table should be moved. If so move the table half the distance to correct the error.
- 10.) The table and gantry are in alignment when both measurements meet the specification.
- 11.) If A and B are not equal, move the table and repeat until measurements meet the specification (± 1 mm).

3.7.4.3 Cradle/Table Parallel Movement

- 1.) Extend the cradle into the gantry 1400 mm (see [Figure 1-28, on page 44](#)).

Figure 1-37 Extend Table Cradle



- 2.) Re-measure the collimator face plate to the cradle center distance at 9 o'clock and confirm that the distance measured in table 1 is the same. Repeat at the 3 o'clock position.
- 3.) Rotate the gantry and re-measure each side until the front measurements match the extended measurement. (+/- 1 mm)
- 4.) Move the table using an appropriate tool until all measurements are equal +/- 1 mm. Note: Alignment bars must be on during these adjustment
- 5.) Recheck cradle level (front to back and across) and re-level as required.
- 6.) Visually inspect the table / cradle for perpendicular movement. Final perpendicular movement will be determined later with x-ray.

Table 1-3 Alignment Worksheet

FRONT	A 3 o'clock	B 9 o'clock	SPEC A-B ± 1 mm
Front measurement			
Back			
#1			
#2			
Extended Measurement			
# 1			
Final Measurement			
# 1			

3.8 Tighten the Lock Rings

- 1.) Re-check bubble levels and loading first. Refer to [Figure 1-18, on page 39](#).
 - a.) Re-check that bubbles levels at locations, A through D on the table.
 - b.) Re-check that bubbles levels at both ends of the gantry.
 - c.) Re-check that adjusters at Locations 1, 2, 3, 4, 7, 8 and 9 carry a portion of the load.
 - d.) Re-check that gantry “inside” levelers carry a portion of the load.

CAUTION



Eye protection is required when using a hammer and chisel.

- 2.) Tighten the lock rings at Locations #1 through #9 with the spanner where possible. Use a hammer and chisel to tighten the lock rings only where you can not use the spanner.

3.9 Drill the Anchor Holes

WARNING



POTENTIAL FOR PATIENT INJURY.

IMPROPERLY SECURED TABLE MAY TIP, DISLODGING PATIENT.

PROPER ANCHORING IS KEY TO MAINTAINING PATIENT SAFETY DURING SYSTEM OPERATION.

3.9.1 Notes to Mechanical Installers

GE provided floor anchors are designed for use ONLY on concrete floors that meet the 4-inch concrete floor requirement. Supplied floor anchors must be installed by a trained contractor, and shall be set to a minimum depth of 3-inches at each anchor point. ANY anchors having more than 1-inch of thread showing above the nut, when torque is set to 55 lb.-ft, shall have a second anchor installed in the closest adjacent hole. this is because the minimum anchor engagement length in the concrete was not met. The second anchor shall be installed to the standard depth and torque specification. **Do not cut anchor bolts that extend longer than the 1-inch limit.**

3.9.1.1 Note 1: Alternate Anchoring

If at least four anchors cannot be set for the gantry, and at least three anchors for the table, then the installer must inform GE that the minimum anchoring cannot be met. Additionally, a structural engineering contractor must be engaged to determine the anchoring method and to certify that their anchoring meets the stated GE minimum load requirement and torque specification.

3.9.1.2 Note 2: Non-Concrete Floors

All other anchoring methods — on floor types other than the concrete minimum — must be determined at the customer's expense, by a structural engineering contractor. The anchoring method must be certified to meet the stated GE minimum load requirement and torque specification.

3.9.1.3 Note 3: GE Notification

It is not the role of the mechanical contractor or installer to determine acceptable methods to install or anchor equipment on non-4-inch concrete floors. The appropriate GE contact person shall be notified that the facility's floor type DOES NOT MEET the installation mounting requirement for the installation procedure (described in this Installation Manual), and therefore the table-gantry mounting process CANNOT continue.

3.9.2 Drilling Procedure

- 1.) Make certain that the gantry scan window is in place, to prevent dust from entering the gantry.

Figure 1-38 Gantry Scan Window

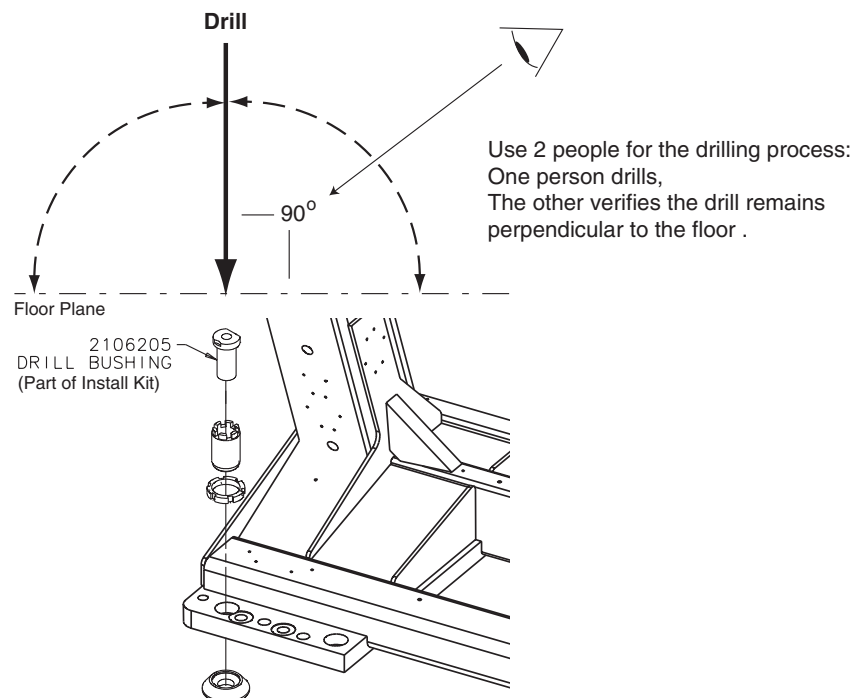


- 2.) Use a piece of tape to mark the drill bit depth of 190 mm ($7\frac{11}{16}$ ") from the tip of the 13mm ($\frac{1}{2}$ ") masonry drill bit.
- 3.) Use the 13mm ($\frac{1}{2}$ ") bit to drill all seven (7) anchor holes (refer to [Figure 1-41](#) for locations.)
 - Review [Figure 1-39](#), prior to drilling.

NOTICE
Potential for
Equipment
Damage from
Dust

You must cover all electronic assemblies in the table base prior to drilling to prevent damage due to the dust created during drilling.

Figure 1-39 Drilling Position



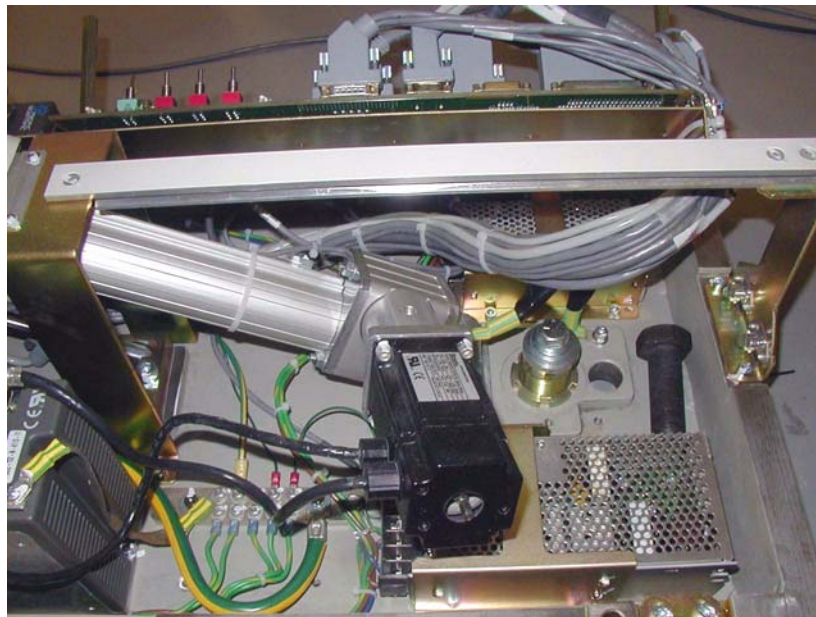
- **Important** - Drill the holes perpendicular to the floor. While one person drills the holes,

position a second person to watch the relationship between the drill bit and floor. Make sure the bit remains absolutely perpendicular to the floor throughout the drilling operation. Always use the mechanical guide supplied with the alignment kit.

- Stop drilling every 15 or 20 seconds and clear the hole of debris. This helps to prevent binding of the drill bit.
- a.) Place appropriate protection to prevent damage and dust contamination to electronic assemblies. Refer to [Figure 1-15, on page 36](#), for placement of dust protection materials.
- b.) Drill each hole until the mark on the drill bit is even with the top of the drill bushing. (195mm)
- c.) Place the drill bushing inside each adjuster, to keep the hole vertical, and centered within the adjuster.
 - * Use the drill bushing to center the anchor holes in adjuster Location #5, #6, and #7, to provide maximum lateral alignment capacity when you center the cradle on isocenter during subsequent system testing.
 - * Take care not to injure yourself on the gantry cover brackets.
 - * Take care to avoid the actuator motor lead when you drill the Location #7 anchor hole. Remove the lead if necessary.

Note: An 18" bit will make drilling anchor location #7 easy, and will prevent component damage (see [Figure 1-40](#)).

Figure 1-40 Table Assembly Showing Actuator Motor and Anchor Location #7

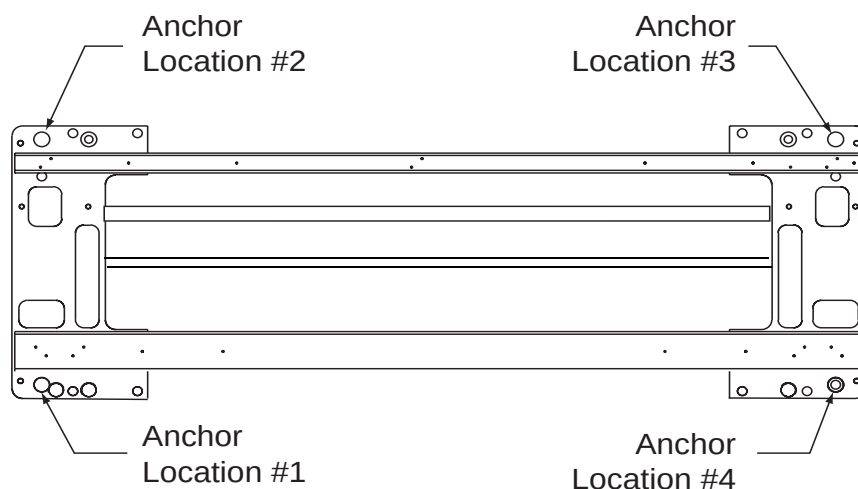


- d.) Stop the drill every 15 or 20 seconds and clear the hole of debris to prevent binding of the drill bit.
- 4.) Vacuum all debris from the inside of the hole:
- Note: **A drywall dust filter should be used on the vacuum.**
- a.) Place the funnel tip inside the hole; place the vacuum hose in the funnel.
 - b.) Continue to vacuum while you drill, if you can, to keep gantry and table as free of dust contamination as possible.
 - c.) When you finish clearing the anchor hole, vacuum the debris from the surrounding area.
 - d.) Stop drilling frequently to let the drill bit cool.
 - e.) All holes must be a minimum of 108 mm deep (see [Figure 1-43, on page 53](#)). Recheck all

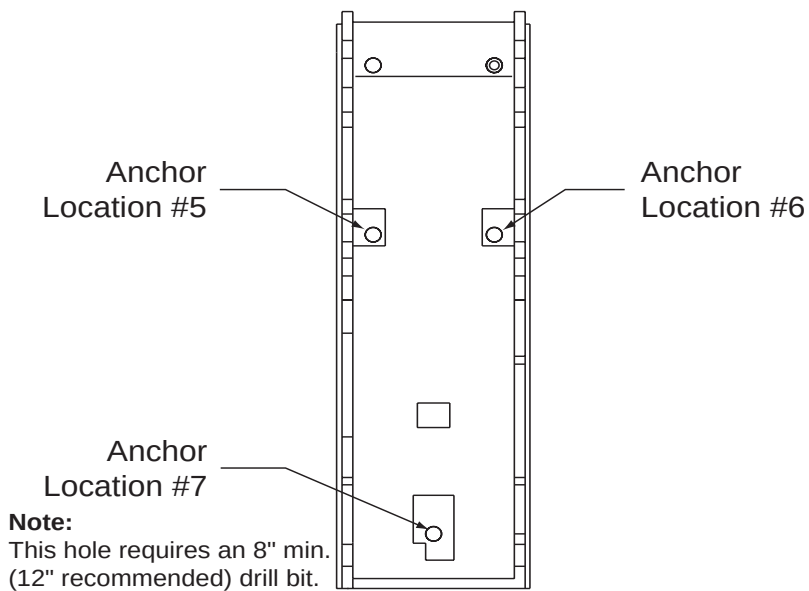
holes.

- 5.) Do not drill anchor holes at Location #8 and #9. These locations help support the weight, but do not anchor the table into place.

Figure 1-41 Anchor Locations, Standard



Note: Anchor holes in base are 7/8" (2.3 cm).



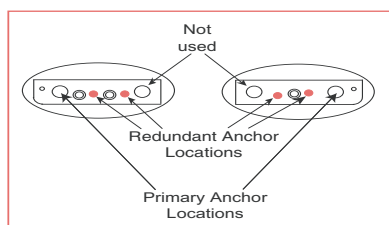
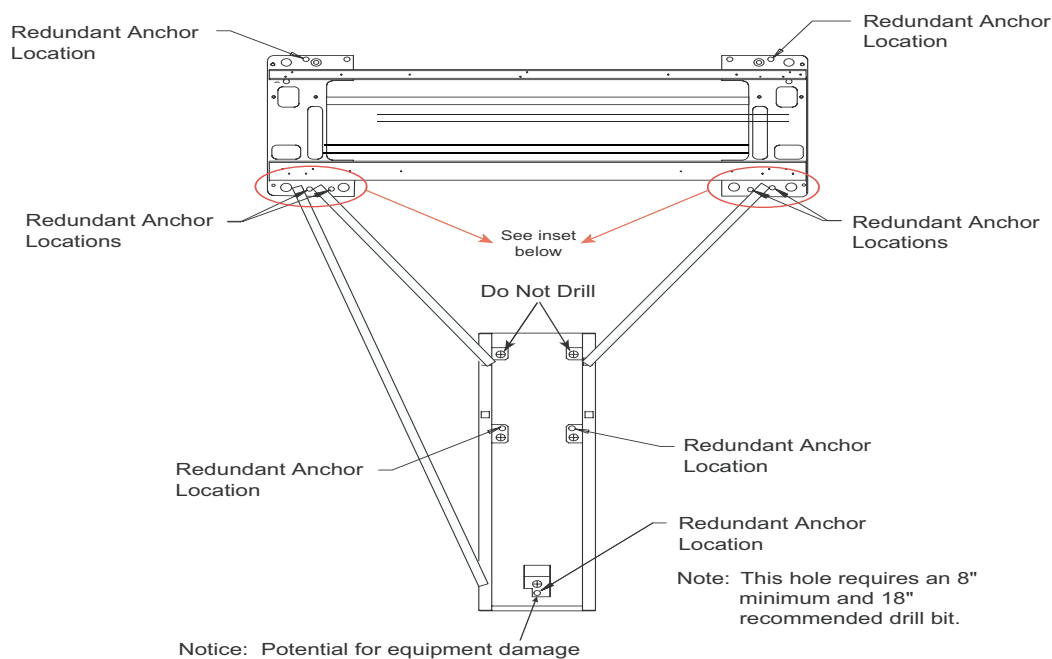
3.10 Gantry & Table Redundant Anchor Holes

If you cannot use one of the adjuster anchor holes due to structural interference, such as reinforcement bars in the concrete, you must use one of the redundant anchor locations, as shown in [Figure 1-42](#).

- The gantry requires a minimum of four (4) anchors, one (1) in each corner.
- The table requires a minimum of three (3) anchors, one (1) at location #5, #6 and #7.

If you must use a redundant anchor hole in the gantry, you must remove the gantry covers. See [Appendix A](#), for gantry cover removal.

Figure 1-42 Table Base Redundant Anchor Locations



Mark the 13mm ($\frac{1}{2}$ ") drill bit with a piece of tape 170 mm ($6\frac{11}{16}$ ") from the tip of the bit, and follow the instructions in [Section 3.9 on page 48](#).

3.11 Install the Anchors - 4-inch Concrete Floors Only

NOTICE These instructions apply to installing anchors in 4-inch (or greater) concrete floors **ONLY**. (Refer to the notes in Section 3.9.1 on page 48.)

Recommended - Use "Hilti Kwik-Bolt II" anchors for this procedure. P/N 2106573 (½" dia. by 8" long).

- 1.) Remove the three alignment bars and repack the install kit.
- 2.) Refer to Figure 1-43. Assemble the anchors before you install them:
 - a.) Remove the nut and washer from the anchor.
 - b.) Add a ¼" thick washer (PN 2105873) under the regular anchor washer
 - c.) Reassemble the anchor washer and nut and position nut so top is flush with threads of anchor before hammering into hole (refer to Figure 1-43).

Figure 1-43 Table Base Anchor Assembly

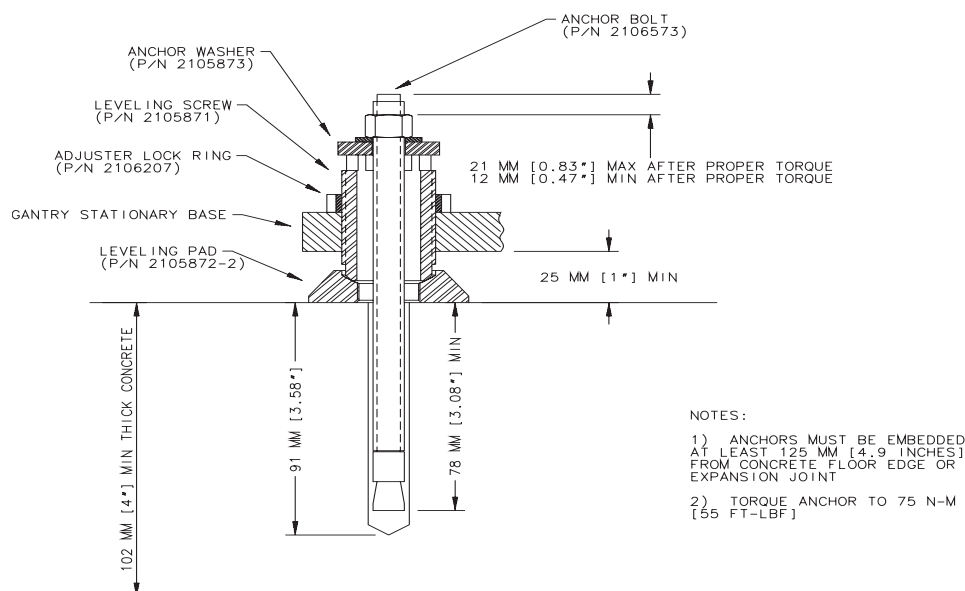
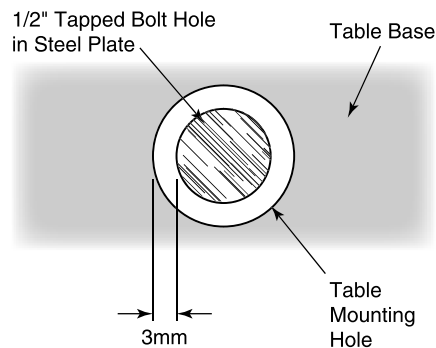


Figure 1-44 Measure Table Base Anchor Thread



- 3.) Fixed site Installations must be centered under mounting holes in table base, with 3mm of space visible surrounding the tapped bolt hole. [Figure 1-45](#).

Figure 1-45 Center tapped holes under mounting holes in table base



Bolt centering is important to provide $\pm 3\text{mm}$ of adjustment for electrical alignment. Always use the drilling centering tool when drilling all bolt holes.

CAUTION



Eye protection is required when using a hammer.

- 4.) Refer to [Figure 1-43](#) for instruction on inserting the anchors in each hole. Using the anchor seating tool, hammer the anchor downward to *completely insert the anchor in the hole*.
- 5.) Repeat Sections [3.7.4.2](#), and [3.7.4.3](#).
- This procedure is designed to mechanically center the table cradle within the gantry. All steps are important to achieve final electrical x-ray alignment of 1 mm. Alignment bars are still needed to level and position the gantry. Alignment bars must be calibrated yearly.
- 6.) Once alignment has been verified, torque all mounting bolts.
Tighten the Location #1 through #7 anchors and torque to $75 \pm 6 \text{ N-m}$ ($55 \pm 5 \text{ ft.-lb.}$).
- 7.) Replace all the table parts you removed during installation, including covers, ground strap and the center support bar. Check the motor lead.

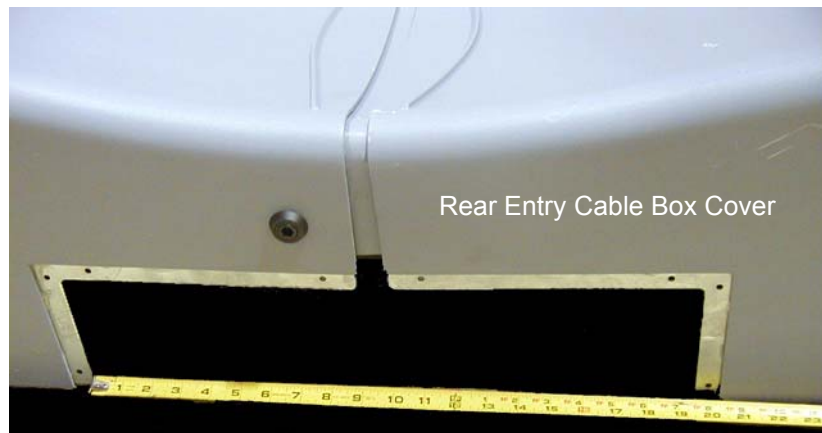
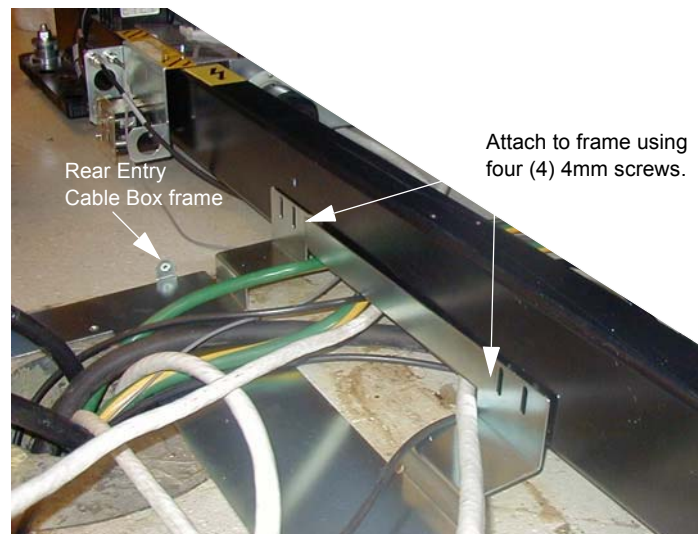
Note: If you cannot replace the lower table cover because the floor interferes, adjust all of the table and gantry levelers by $\frac{1}{2}$ turn increments to raise the table/gantry until the lower table covers clear the floor. (Refer to [Figure 1-20, on page 40](#), for table base cover clearances.) Return to Sections [3.3 on page 32](#), [3.6 on page 38](#) and [3.8 on page 48](#) to level the gantry, level the table, and tighten the locking rings, respectively.

Section 4.0

Rear Entry Cable Box

A rear entry cable box (B7850RC) is used when the cables to the gantry cannot be brought-up inside the gantry base. The box is not supplied with the system and must be ordered separately. The rear entry cable box frame attaches to the gantry base. See [Figure 1-46](#).

Figure 1-46 Rear Entry Cable Box



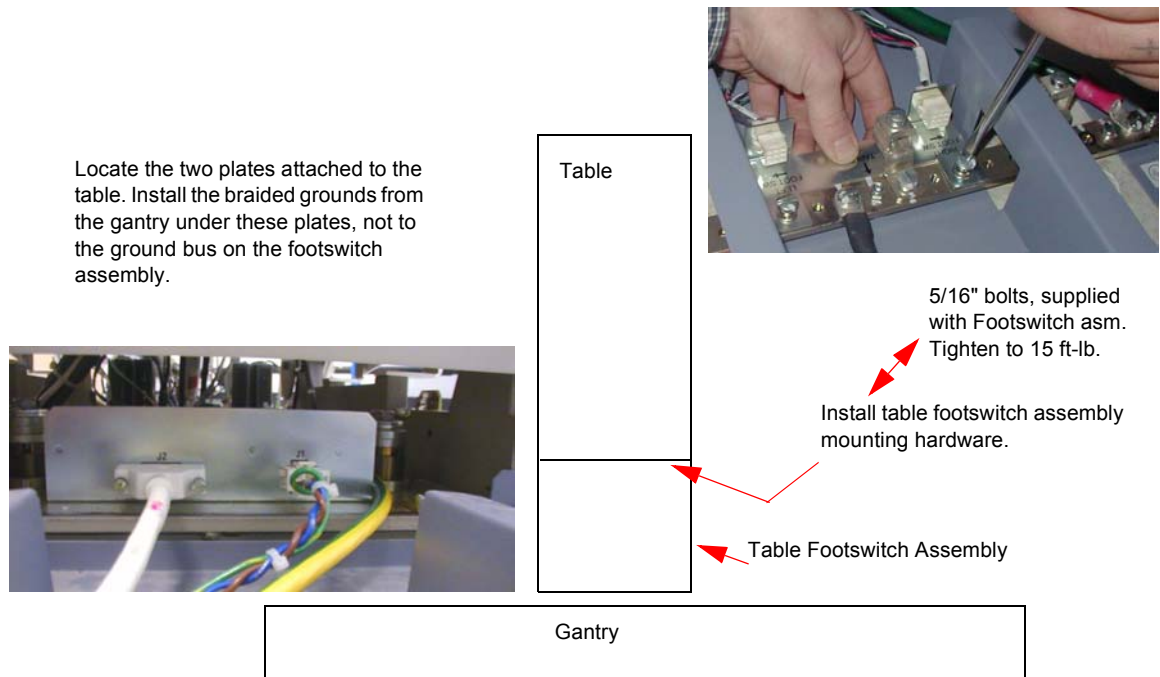
The filler plate may need to be cut in order to fit the rear cable box into the cover opening. Use the supplied white molding around the filler plate.

Section 5.0

Install Table Footswitch Assembly

Install the table foot-switch assembly as shown in [Figure 1-47](#), along with filler strip.

Figure 1-47 Install Table Foot-switch Assembly

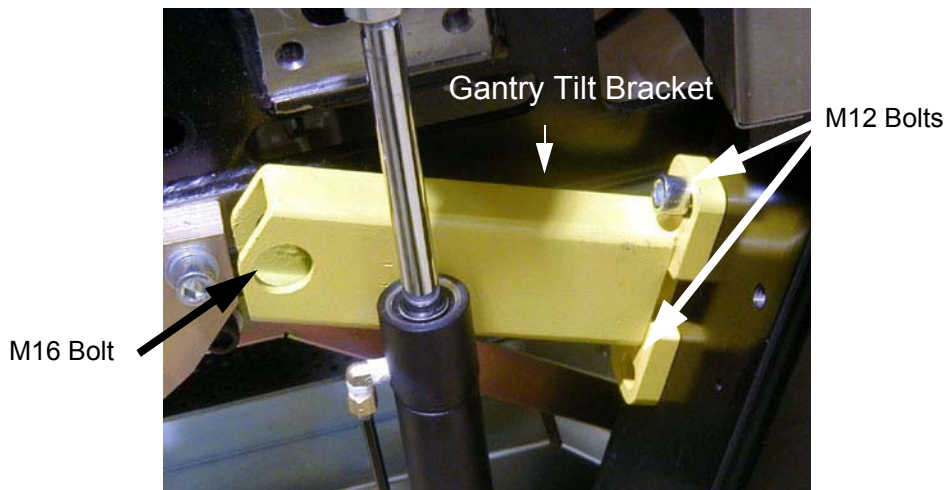


- 1.) Remove foot switch assembly, and locate hardware.
- 2.) Loosen six (6) hex head screws and remove footswitch assembly.
- 3.) Place footswitch cover in a secure place.
- 4.) Locate the supplied 2' velcro in the install support kit, and apply two strips to the bottom of the footswitch assembly.
- 5.) Using the two 5/16" bolts and blue loctite (242), attach the footswitch assembly.
- 6.) Remove the 2" braided strip and gantry cable that are secured to gantry base, and place in the cable tray.
- 7.) Remove three (3) bolts that secure the footswitch bar.
- 8.) Take off the footswitch bar and discard.
- 9.) Locate the new footswitch bar from the install support kit.
- 10.) Install footswitch bar on cover, using the same three screws.

Section 6.0

Remove Gantry Tilt Bracket

Figure 1-48 Gantry Tilt Bracket Removal



- 1.) Refer to [Figure 1-48](#). Remove the M12 bolts using a 10 mm Hex wrench.
- 2.) Loosen the M16 bolt 1-2 turns and check the Gantry tilt bracket, it should be loose to the touch. If loose continue with step 4.
- 3.) Check the hydraulic connections for leaks or lack of fluid. You will have to wait until the system can be energized to use the tilt controls to relieve the load on the tilt bracket prior to removal. Do not use force to remove the bracket.
- 4.) If the bracket feels loose, remove the M16 bolt using a 14 mm Hex wrench.
- 5.) Remove the bracket.
- 6.) Close the gantry covers and reinstall the scan window.
- 7.) Store brackets in the gantry base.

CAUTION



If tilt bracket is not loose, stop and put the M12 bolts back in and tighten tilt bracket back in place. If there is a load on the tilt bracket, removal may cause the gantry to suddenly tilt all the way back due to a possible lack of hydraulic pressure.

Section 7.0

Position the Power Distribution Unit



- LOCKOUT/TAGOUT IS REQUIRED BEFORE PERFORMING THIS TASK.
USE THE SUPPLIED LOTO KIT.
- 1.) Roll the PDU into position on its permanently mounted casters. Leave at least 15.5 cm (6") between the PDU and back wall to allow cooling air to circulate.
- Note: Connecting the primary incoming power (steps 2 through 5, below) is performed by the customer’s electrical contractor.

Table 1-4 Contractor Connections

Connection or Wall Box	AWG #	Connection From	Connection To PDU	Installed & Checked
TS1	#1	PDB-A	TS1-1	
	#1	PDB-B	TS2-2	
	#1	PDB-C	TS3-3	
	#1/0	GND	N/G (Do NOT connect anything to neutral point.)	

- 2.) Run the main input power conductors and ground though flexible metal conduit (attached between the PDU chassis and room duct-work) so you can move the PDU away from the wall during service.

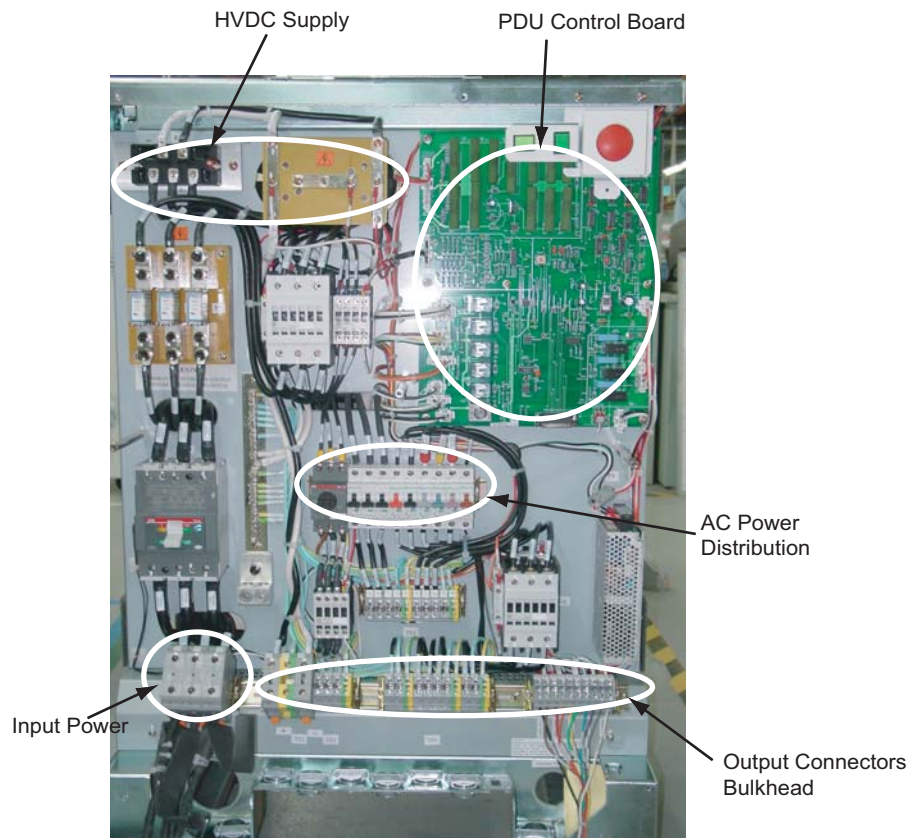
Figure 1-49 Flexible Conduit for PDU Power



- 3.) Locate the hole cover plate in Box 1 and attach the flexible metal conduit to the PDU.
- 4.) Cut the three phase and 1/0 ground wire to size.

5.) Attach cables as shown in [Figure 1-49](#) and [Figure 1-50](#).

Figure 1-50 PDU Area Locations



Section 8.0

Install Operator Console

8.1 Unpack Console

Figure 1-51 Console boxed on skid



- 1.) Remove all items from the console.
- 2.) Remove all packing materials and discard.
- 3.) Place the step-board under the front edge of the skid and step on it to raise the front edge of the skid as in [Figure 1-52](#).

Figure 1-52 Step-board used to raise front edge of skid



- 4.) Remove the two front cushions from the bottom of the skid. Refer to [Figure 1-53](#)

Figure 1-53 Cushion on bottom of skid



- 5.) Remove the Seismic Brackets from each side of the console. Refer to [Figure 1-54](#). Save brackets for use later if you need to mount the console to the floor.

Figure 1-54 Removing the Seismic Brackets



- 6.) Lift up on the strap on the front of the step-board ([Figure 1-52](#)) to lower the skid. Remove the step-board.
- 7.) Ensure the console stabilizers are in line with the notched portion at the front of the skid. This will allow enough clearance to smoothly roll the console down the ramps. Refer to [Figure 1-55](#).

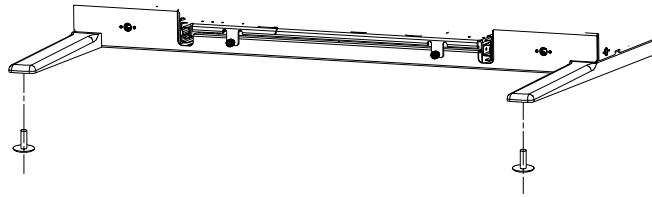
Figure 1-55 Console ready to be unloaded



- 8.) Move console to installation location.

- 9.) Attach the two pads to the console stabilizers. Refer to [Figure 1-56](#).

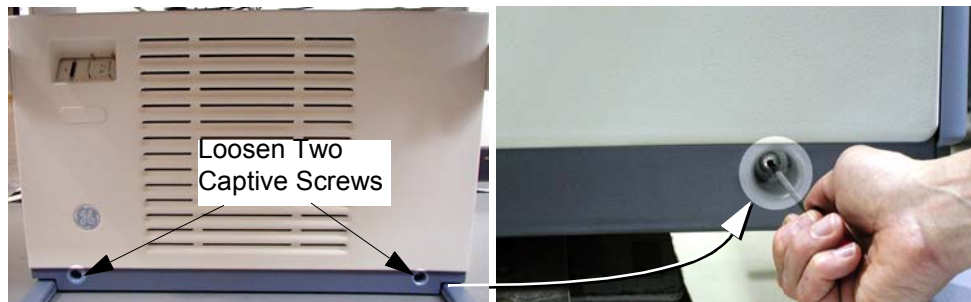
Figure 1-56 Console Stabilizer Pads



- 10.) Screw the pads all the way into the legs

8.2 Remove Console Covers

Figure 1-57 Front Cover Removal - Global Consoles



- 1.) From the service desktop, shut down the system
- 2.) Turn off console power.
- 3.) Loosen two captive screws at bottom of console.
- 4.) Rotate bottom of cover outward and upward until it may be lifted free of the console at the top.

8.3 Adjust Table Top Height and Position Console

- 1.) Refer to [Figure 1-58](#). The console normally arrives with the bottom hole on the monitor top aligned with the fourth bolt from the bottom.

Figure 1-58 Global Console tabletop adjustments



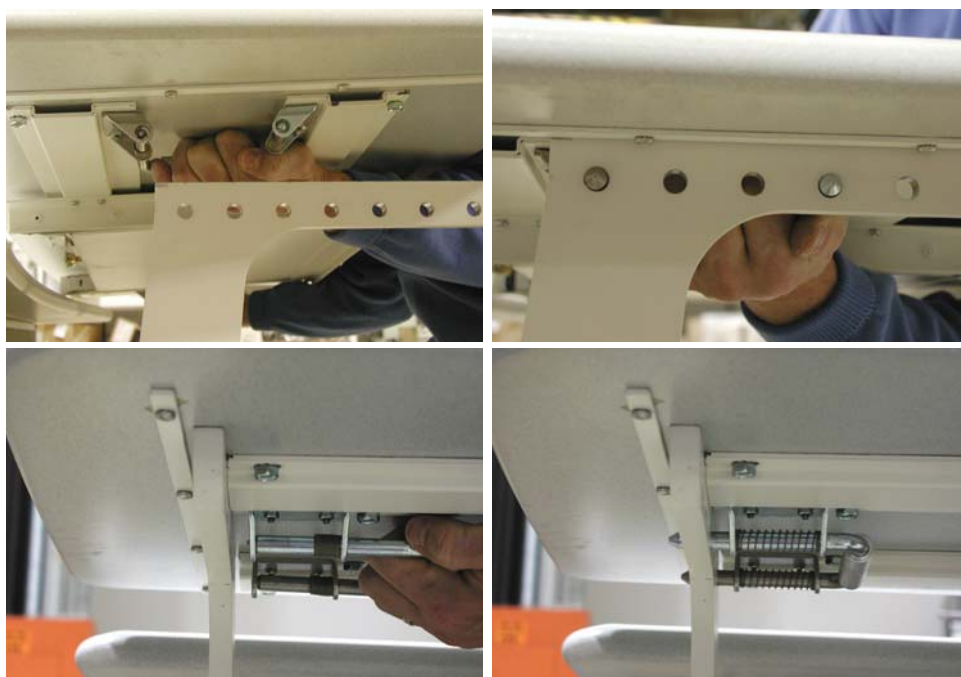
- 2.) For optimum operator comfort, align the bottom hole of the keyboard table top with the fifth hole from the bottom on the console.

Your site may have different requirements and you may have to adjust the monitor top and/or keyboard table top up or down from this position.

- Always select a table top height that permits the operator to see the patient on the table.
 - Keep the one bolt-hole relationship between the monitor top and the keyboard table top. Fasten the keyboard table top one (1) hole lower than the console table top ([Figure 1-58](#)).
- 3.) Install the console side covers.
Side covers are left and right specific.
 - 4.) Move the console into its final position, maintaining a 6" (15 cm) minimum distance between the console and the wall.

8.4 Attach Keyboard Table Top

Figure 1-59 Keyboard Table Top Placement



- 1.) Locate the spring loaded latches at either end of the underside of the keyboard table top.
- 2.) Pull latches inward (toward center of table top).
- 3.) Position table on brackets so that latches align with holes.
- 4.) Release latches, making certain that the pins fully engage the bracket holes (pins should protrude beyond outside edge of bracket).

8.5 Monitor and SCSI Tower Placement

- 1.) Locate and unpack the two monitors. Be careful as each monitor weighs approximately 80 pounds (36 Kg) each.
- 2.) Carefully place the monitors on the console desktop. **This step requires two people.**
- 3.) Locate and unpack the SCSI drive tower.
- 4.) Place the SCSI drive tower on the console desktop. A suggested placement is illustrated in [Figure 1-60](#).

Figure 1-60 Global Console tabletop component placement



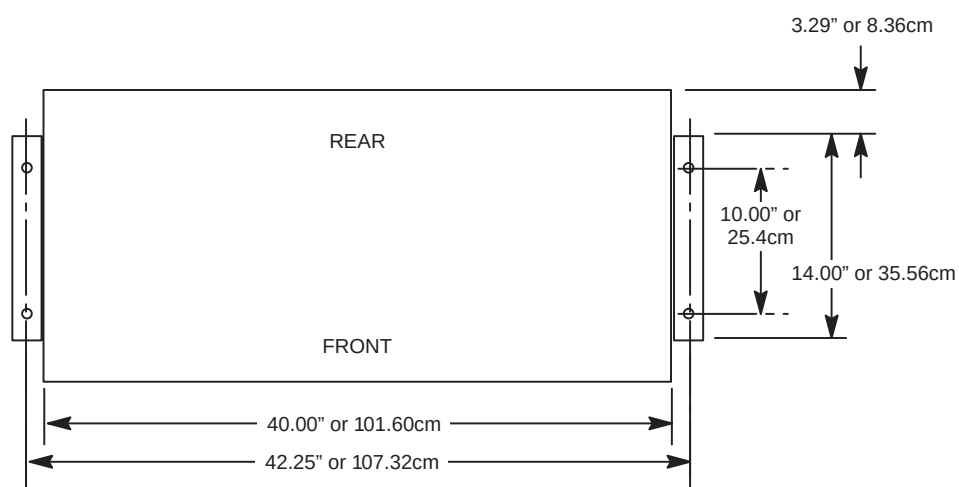
Section 9.0 Seismic Mounting

The Seismic Kit is part number R4390JC.
A seismic kit is also available for the SCSI Tower.

9.1 Console

If site specifications require seismic mounting, use ½" bolts to mount the brackets to the floor. Refer to [Figure 1-61](#) for hole placement. The console seismic brackets are shipped with the console. They are used on the console shipping skid.

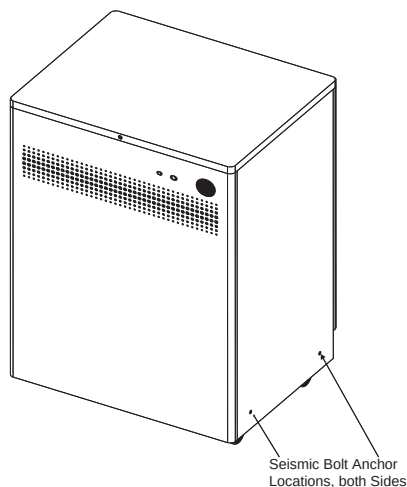
Figure 1-61 Seismic Console Mounting Hole Locations



9.2 Power Distribution Unit

If site specifications require seismic mounting, use the PDU seismic brackets that were shipped with the seismic kit. Refer to [Figure 1-62](#) for hole placement.

Figure 1-62 Seismic PDU Mounting Hole Locations



Chapter 2

Power, Ground & Interconnect Cables



NOTICE Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Site use of conduit, floor duct, wall duct, or a raised computer floor, as well as the individual component layout determines the system cable sequence. If your site has floor or wall ducts that will interfere with placement of the table/gantry, it may be important to have the movers unload the cable boxes (8 & 9) first and run those cables while others unload the subsystems.

- Try to run the system cables after the contractor completes the contractor supplied wiring.
- All ground wires and other contractor wiring should be complete to the point of equipment placement.

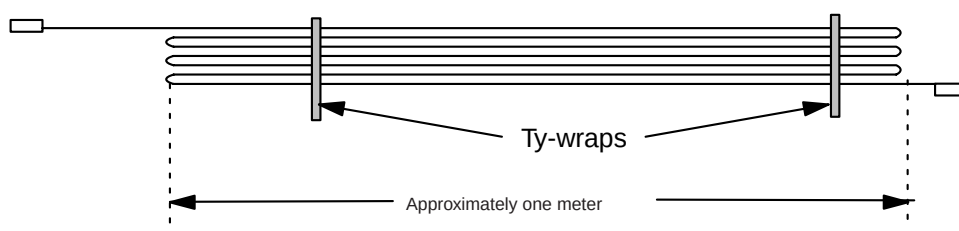
NOTICE Potential for Equipment Damage



Do not store excess cable in the bottom of the PDU or Gantry.

When possible, store excess cable length in a serpentine configuration, approximately one meter long. (**Do NOT coil excess cable.**)

Figure 2-1 Excess Cable Storage Configuration



- Keep signal and control cables away from power cables and power wiring. When you lay cables in a raceway, locate the signal cables in a separate section of the raceway, or a separate conduit.
- Check all connections for tightness.
 - Use suitable tools and judgment.
 - Check all visible connections, especially ground connections.
- Check for reasonable cable routing.
 - Take into consideration necessary take-up distances for equipment maintenance, etc.
 - Try to complete as neat a job as possible.

1.1 System Component Identification

Identify all system cables by the system component designators listed in Table 2-1 on page 68. Each end of a system cable has a label, and may have a color near the connector, (refer to Table 2-2 on page 68) to indicate the component and the jack identifier of the component.

Table 2-1 System Component Identifiers

DESIGNATOR	SYSTEM COMPONENT
CT2	Gantry
CT1	Patient Table
PM	Power Distribution Unit
OC	Operator Console (Console Computer)
WL	X-Ray ON Warning Light

1.2 Cable Color Identifiers

The ends of the cables may be marked with a piece of blue, yellow, red, or orange colored tape to help with the cable installation. Table 2-2 on page 68 lists the subcomponent, and corresponding color.

Table 2-2 Cable Color Identifiers

SUBCOMPONENT	COLOR
Gantry	Blue
Table	Yellow
PDU	Red
Console Computer	Orange

Table 2-3 LightSpeed 5.X System Interconnect Cables

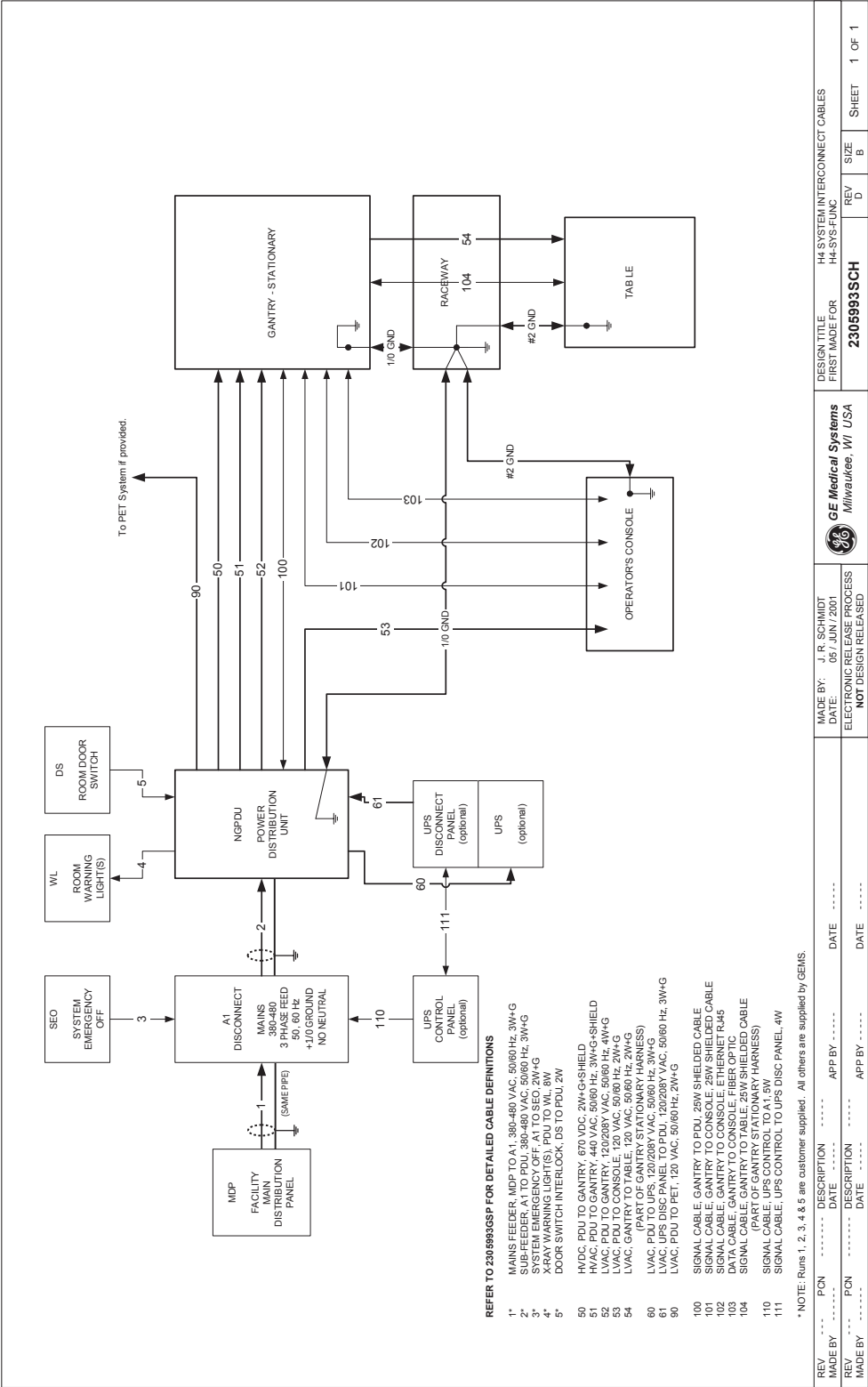
RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
1	Facility MDP to Room Disconnect (A1)	cust. supplied	cust. supplied
2	Room Disconnect (A1) to PDU	cust. supplied	cust. supplied
3	Room Disconnect (A1) to System E-Off	cust. supplied	cust. supplied
4	PDU to Room Warning Light(s)	cust. supplied	cust. supplied
5	PDU to Scan Room Door Switch	cust. supplied	cust. supplied
50	HVDC Power Cable - PDU to Gantry	2343529	2343529-2
51	HVAC Power Cable - PDU to Gantry	2343530	2343530-2
52	LVAC Power Cable - PDU to Gantry	2343528	2343528-2
53	LVAC Power Cable - PDU To Operator's Console	2343531	2343531-2
54	LVAC Power Cable - Gantry to Table	n/a	n/a
55	Ground, PDU to Raceway	2371450	2371450-2
56	Ground, Raceway to Console	2371450-3	2371450-4

Table 2-3 LightSpeed 5.X System Interconnect Cables (Continued)

RUN NO.	DESCRIPTION	PART NUMBER	
		LONG CABLES (KIT 2281840-4)	SHORT CABLES (KIT 2281840-5)
60	LVAC Power Cable - PDU to Optional UPS	-	-
61	LVAC Power Cable - UPS Disconnect Panel to PDU	-	-
90	LVAC Power Cable - PDU to PET	-	-
100	Signal Cable - Gantry to PDU	2333152	2333152-2
101	Signal Cable - Gantry to Console	2333150	2333150-2
102	Signal Cable (Ethernet) - Gantry to Console	2352714-2	2352714-3
103	Data Cable (Fiber Optic) - Gantry to Console	2117848-2	2117848-7
104	Signal Cable - Gantry to Table	n/a	n/a
110	Signal Cable - UPS Control to Room Disconnect (A1)	-	-
111	Signal Cable - UPS Control to UPS Disconnect Panel	-	-

Section 2.0
System Interconnect Diagram

Figure 2-2 System Interconnect Diagram



Section 3.0

Contractor Connections

Table 2-4 Contractor PDU Connections

CONNECTION OR WALL BOX	AWG #	CONNECTION FROM	CONNECTION TO PDU	INSTALLED AND CHECKED
A1	#1	Load - T1	TS-1 L1	
	#1	Load - T2	TS-1 L2	
	#1	Load - T3	TS-1 L3	
	#1/0	GND	TS-1 GND (Do NOT connect anything to neutral point.)	
WL (Warning light) See Figure 2-28 , on page 87.	#14	LV Source -1	TS6 1	
	#14	LV Source -2	TS6 2	
	#14	X-Ray ON Light -1	TS6 3	
	#14	X-Ray ON Light -2	TS6 4	
	#14	Sys-ON Light -1	TS6 5	
	#14	Sys-ON Light -2	TS6 6	
	#14	Ready Light -1	TS6 7	
DS (Scan Room Door Switch)	#14	Door SW-1	TS6 9	
	#14	Door SW-2	TS6 10	

Note:
Add #2 ground
wire.

IMPORTANT: Add AWG #2 ground wire from Table frame to Table/Gantry raceway ground bar (as shown in [Figure 2-2](#)).



WARNING



WORK WITH THE ELECTRICAL CONTRACTOR TO BE SURE EXTERNAL POWER SOURCE IS TURNED OFF.

Section 4.0 Console Connections

4.1 SCIM, Keyboard, Trackball & Mouse Installation

- 1.) Route the keyboard cable under the SCIM, as shown in [Figure 2-3](#).

Figure 2-3 SCIM control with keyboard cable routed through SCIM



NOTICE

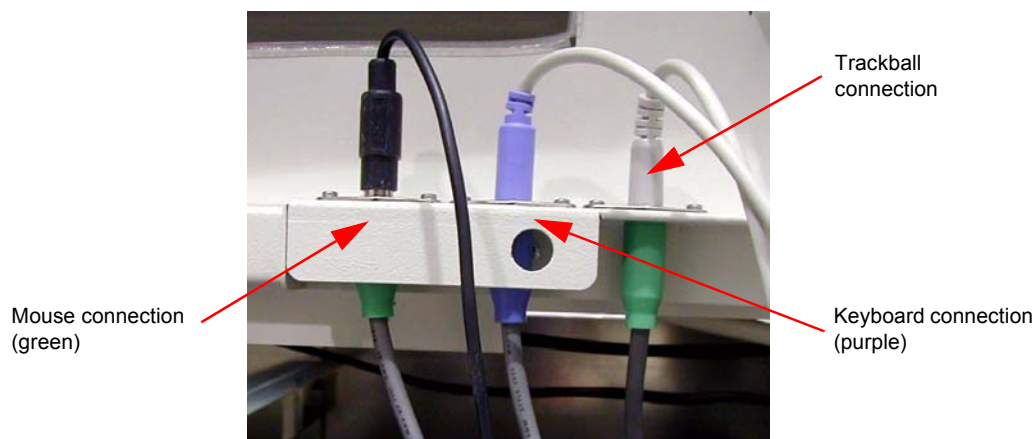


Potential for equipment damage

Never connect a mouse or keyboard with the host computer powered “ON”. Doing so can destroy components within the host computer.

- 2.) The keyboard and mouse connect to the ports on the top of the console front bulkhead. The keyboard connects to the port on the right, the mouse connects to the left port ([Figure 2-4](#)).

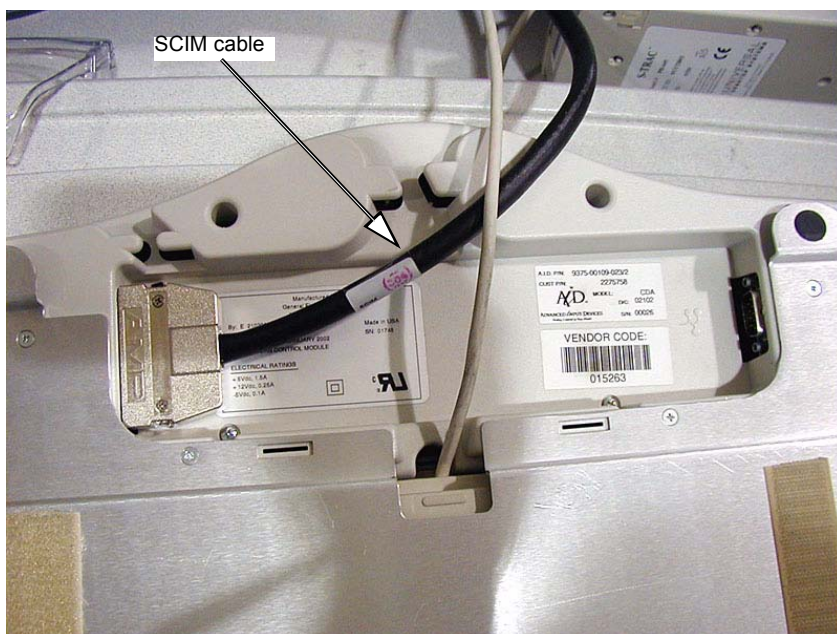
Figure 2-4 Global Console front bulkhead, showing keyboard and mouse connections



- 3.) Connect the SCIM cable to the SCIM as shown in [Figure 2-5](#) (note the cable routing).

Note: Make sure the SCIM connector fits snug. Some molding may need to be removed to allow the cable to fit snug.

Figure 2-5 SCIM bottom, showing cables and keyboard mounting bracket



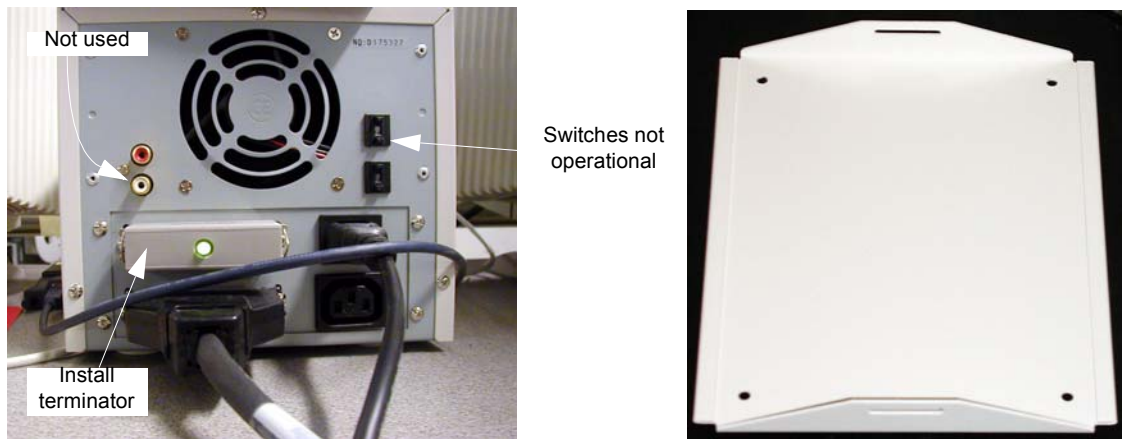
- 4.) Select and install the proper overlay for your system: (1) with Tilt or (2) without tilt. Verify that none of the buttons get caught and stuck under the overlay. Pay close attention to the prescribed tilt button on systems with the tilt feature.
- 5.) The keyboard should attach to the SCIM using the supplied Velcro strip and fit snugly against the SCIM when finished, as shown in [Figure 2-6](#).

Figure 2-6 SCIM connected to the keyboard with the US English tilt overlay installed



4.2 Connecting the SCSI Tower

Figure 2-7 SCSI tower: Connections (left) and Optional Seismic Mounting Bracket (right)



- 1.) Connect the SCSI cable to the rear of the SCSI tower.

Note: The cable is included in the SCSI Cable Kit and is the same as the DASM cable.

- 2.) Connect the SCSI terminator to the rear of the SCSI tower.
- 3.) Connect the power cable to the rear of the SCSI tower.

4.3 Connecting the Monitor

- 1.) With the monitor and computer switched off, connect the video signal cable to the monitor's video input (see [Figure 2-8](#)).

NOTICE

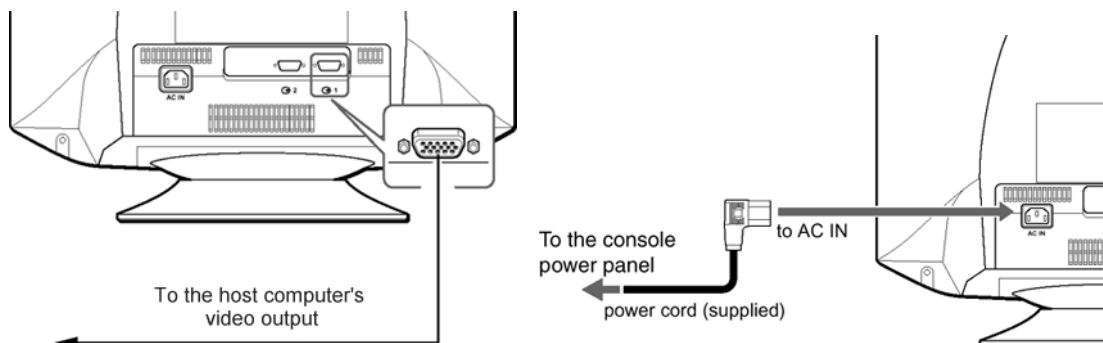


Equipment Damage Possible

Do not touch the video signal cable connector pins as this might bend them. When connecting the video signal cable, check the alignment of the HD15 connector. Do not force the connector in the wrong way, or the pins might bend.

- 2.) Connect the power cord to the monitor, then connect it to console power panel outlet J1 or J7.

Figure 2-8 Monitor Connections (Sony Model GMD-F520 -- GE #2333243)



4.4 Power Panel Connections

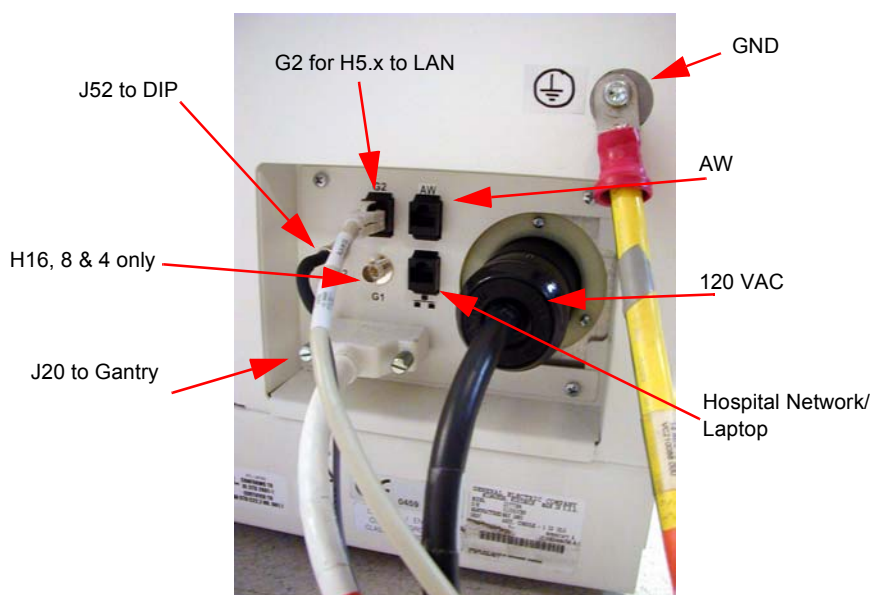
Note: Console power is single phase power. Outlet assigned is not critical.

- 1.) Connect the console power cable to the console power panel.
- 2.) Connect console component power cords as listed in [Table 2-5](#). ("J numbers" increment from top to bottom, left to right)

Table 2-5 Power Panel Outlet Assignments - GC-GRE

J#	DEVICE	J#	DEVICE
J1	Video Splitter	J9	Host Computer
J2	Remote Monitor	J10	Power Monitor
J3	Modem	J11	DARC
J4	IG 3	J12	Power Monitor
J5	IG 1	J13	
J6	IG 2	J14	Power Media Conv
J7	SDDA	J15	Left Fan
J8	DASM	J16	Right Fan

Figure 2-9 Power Connections

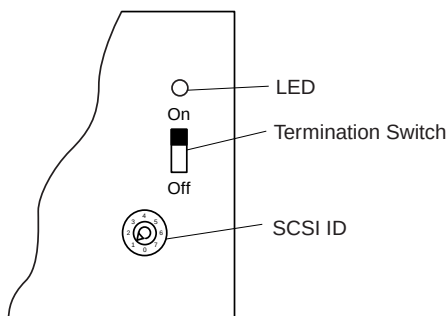


4.5 Installing a DASM (Hardware Option)

If you have a digital DASM to install, complete the following steps to connect it to the computer.

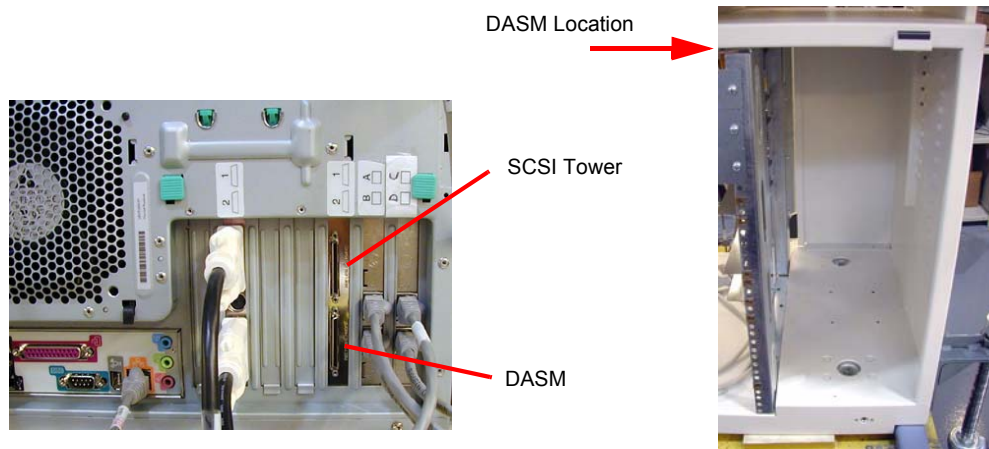
- 1.) Turn the DASM over so you are looking at the bottom.
- 2.) Set the DASM SCSI ID to 1.
- 3.) Set the DASM termination to ON.

Figure 2-10 DASM Bottom



- 4.) Set the RS232/RS422 switch to the RS232 position, for a digital DASM.
- 5.) Refer to [Figure 2-11](#). Attach the SCSI cable supplied with DASM kit to the back of the DASM.

Figure 2-11 Rear view of HP xw800, showing DASM SCSI cable connection



- 6.) Attach the free end of the SCSI cable to the rear of the Host Computer (see [Figure 2-11](#)).
- 7.) Plug DASM power cord into J10 (or any open slot).
- 8.) Attach remaining cable(s) from camera to DASM and install the DASM to the right of the slide-out tray.

Note: See your *Software Load Procedures* manual for instructions on configuring cameras.

4.6 Modem Option

If you have a modem to install, do it now. Place the modem on top of the outlet box and attach it with velcro to the power supply.

Hook up the power, phone, and serial line as shown in drawing 2378862. The drawing is located in the back cover of the console.

Figure 2-12 Modem

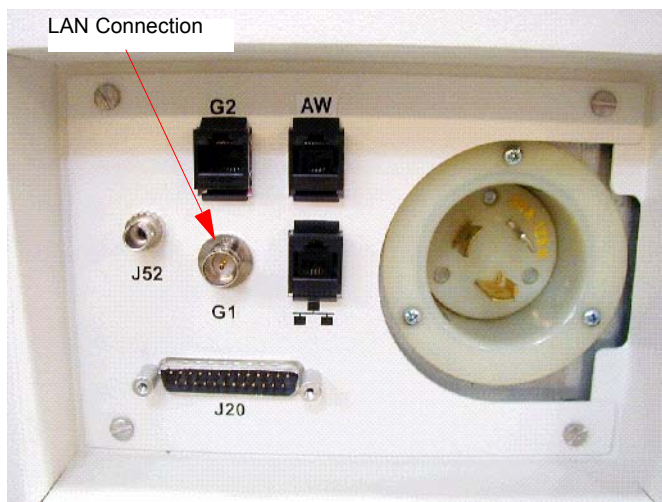


4.7 LAN Connections

Plug LAN cable (2266887) into G1 of rear bulkhead on console.

Note: G2 is used for LightSpeed Pro¹⁶ only.

Figure 2-13 LAN Connection



Section 5.0 Gantry Cable Connections

Please refer to [Figure 2-2](#) for complete system interconnect details.

Table 2-6 Gantry Cable Connections

TO	FROM	CABLE DESCRIPTION
Gantry Power Pan	PDU	HVDC
Gantry Power Pan	PDU	440VAC
Gantry Power Pan	PDU	120VAC
Gantry Power Pan	Console	Fiber - <i>Take extreme care when you install the fiber optic DAS data cable. Do not step on, kink, or sharply bend this fragile DAS cable.</i>
Gantry Power Pan	Console	LAN
MSUB (J9)	Console	Control
MSUB (J11)	PDU	Control

NOTICE



- 1.) If using a rear cable entry box (B7850RC), install it now, before routing cables to gantry.

Potential for equipment damage.

Observe correct polarity when connecting the high voltage DC power. Reversing these leads will result in serious equipment damage. The HVDC positive conductors have red insulation and are labeled “ONE.” The HVDC negative conductors have black insulation and are labeled “TWO.” Lead “ONE” must be connected to lead “ONE,” and lead “TWO” must be connected to lead “TWO.”

Observe correct phase rotation when connecting the axial motor power. Phases one, two and three should be connected top to bottom.

- 2.) Install the cables to the gantry power pan. The power pan is located on the rear of the gantry at its base. See [Figure 2-15](#), [Figure 2-16](#), and [Figure 2-17](#) for connections.

Note: The gantry 120VAC cable may not fit under the gantry frame. Install this cable before gantry placement—or remove the power plug—to route it under the gantry.

Figure 2-15 Gantry Power Pan Connections

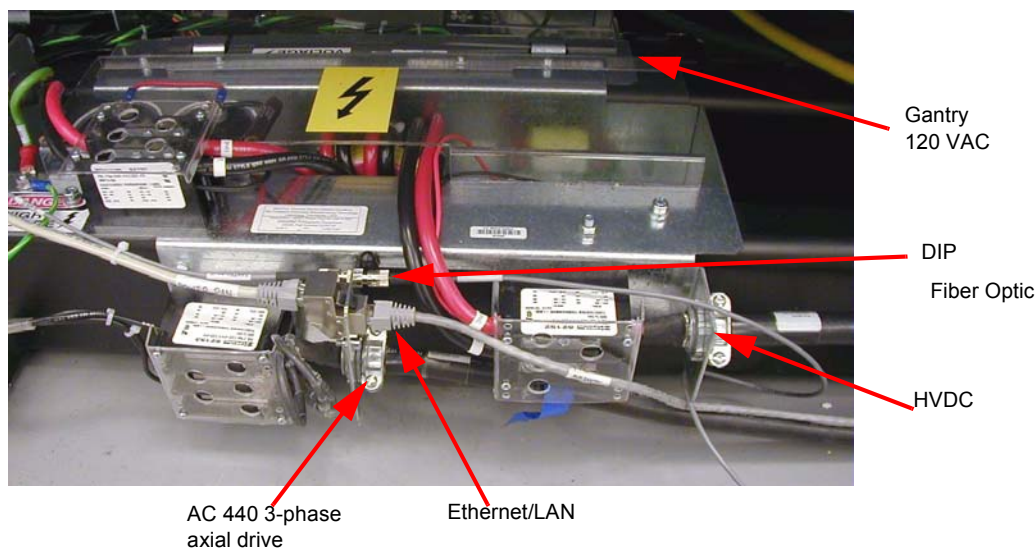


Figure 2-16 Gantry Power Pan

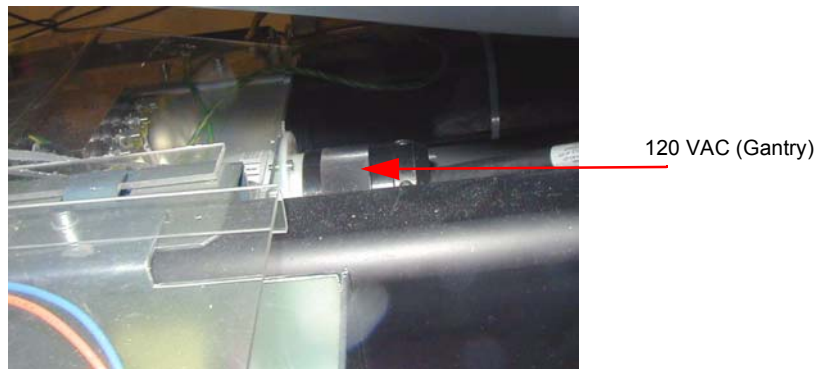
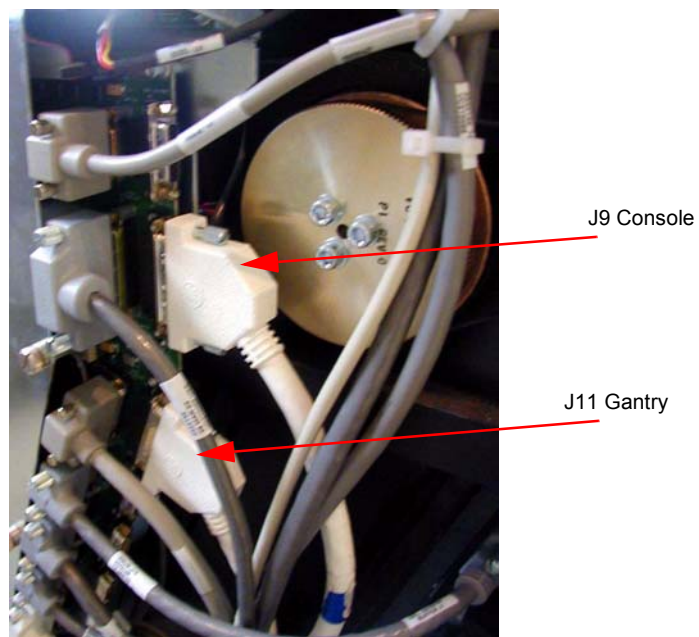


Figure 2-17 MSUB Connections



- 3.) Install cables to the gantry MSUB.

Section 6.0

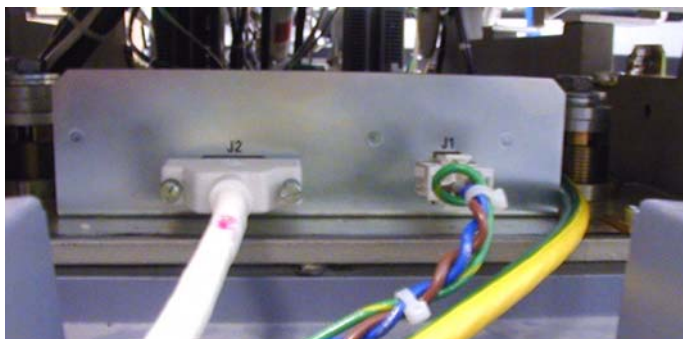
Table Connections

Pull and connect the following cables:

Table 2-7 Cables Connected to Table

J#	CABLE DESCRIPTION
J1	120 VAC
J2	Signal Cable

Figure 2-18 Table Bulkhead Connections



Section 7.0

PDU Cable Connections & Configuration



CAUTION

Do not work in an energized PDU. When working on the PDU, follow this simple rule: Always tag and lock out power to the PDU at the “main” disconnect. Failure to do so can result in electrocution or death.

Do apply power to the PDU until all work has been completed and all PDU covers are in their proper place.

7.1 NGPDU

As seen in [Figure 2-19](#), a number of cables must be installed throughout the PDU. Specific details on each connection can be found in the sub-sections that follow. Use [Figure 2-19](#) for reference. The PDU has been designed to have cables routed into the PDU from its behind and/or from beneath it.

Figure 2-19 PDU Cable Connections - Front

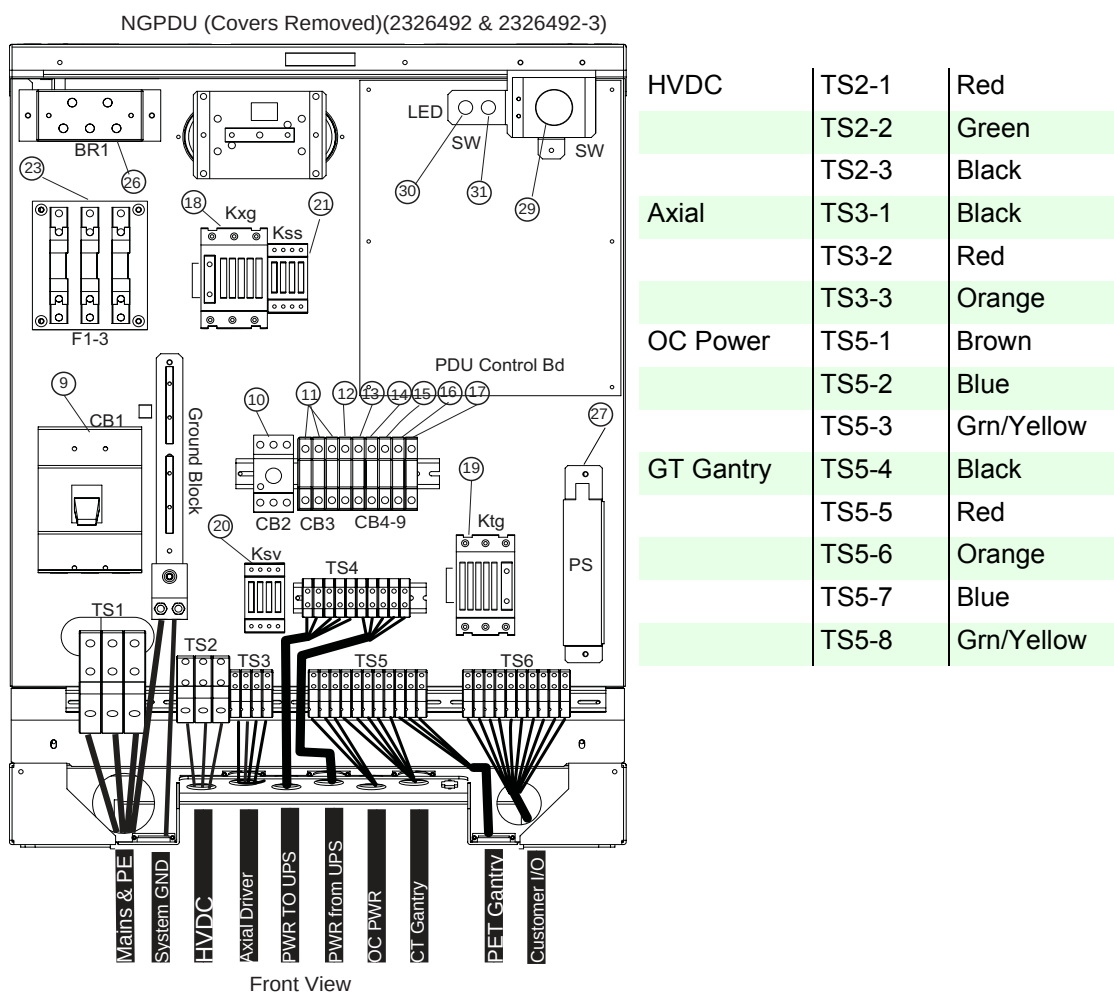
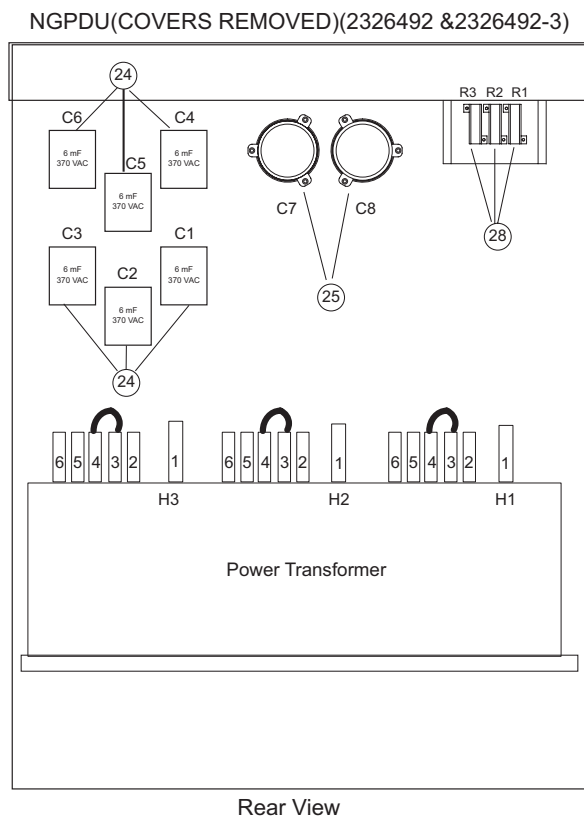


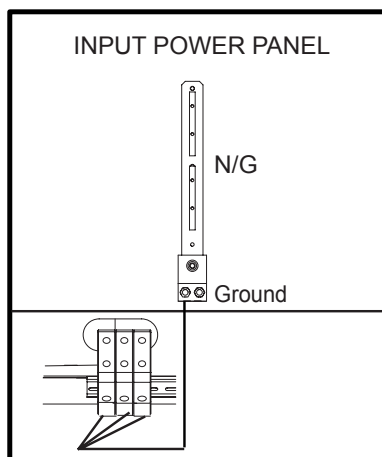
Figure 2-20 PDU Cable Connections - rear



7.1.1 Panel - 380 - 480VAC Mains “TS1” Input Power Connection

- 1.) Remove the TS3 panel front cover.
- 2.) Strip the wires to fit securely on the power block.
- 3.) Observe incoming phases (L1, L2 and L3) and insert bare leads into power block.
- 4.) Insert “vault” ground into PDU “vault” ground lug.
- 5.) Tighten all fasteners securely and replace the TS3 front panel.

Figure 2-21 Input Power Panel Connections

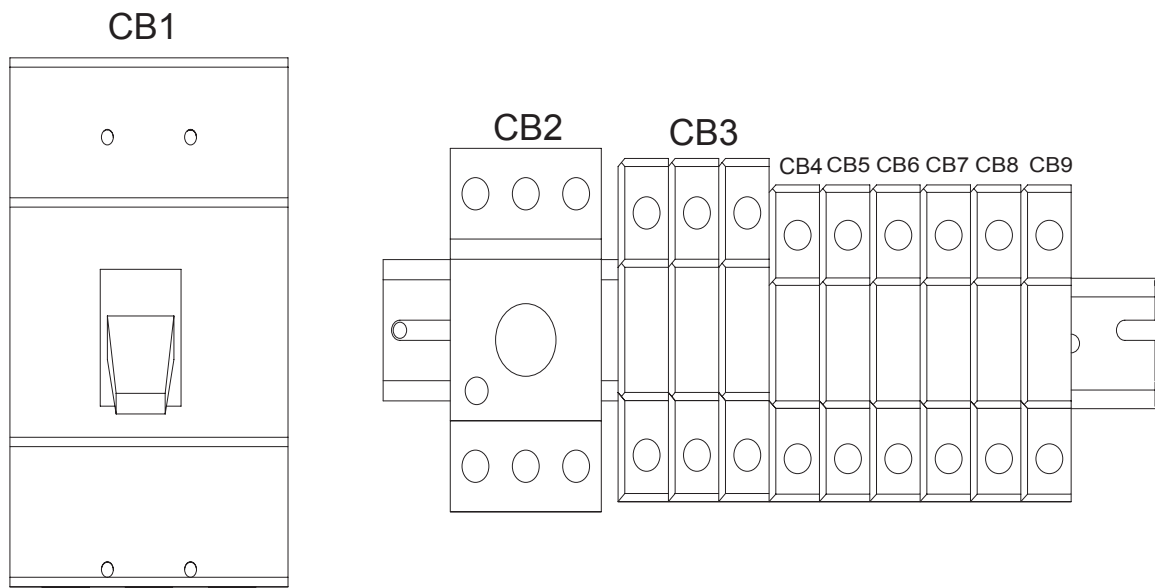


When Mains power is available to the PDU, the “TS1” power light will be illuminated.
(See [Figure 2-19.](#))

7.1.2 Panel - Circuit Breakers

Place the circuit breakers in the “off/down” position during installation, even with Mains incoming power tagged and locked out. After you have completed work on the PDU, you may return the circuit breakers to the “ON” positions.

Figure 2-22 Circuit Breaker Panel



By design, when CB3 is in the “OFF” position, circuit breakers 4, 5, 6, and 7 are switched “OFF”. CB3 is essentially in series with these breakers.

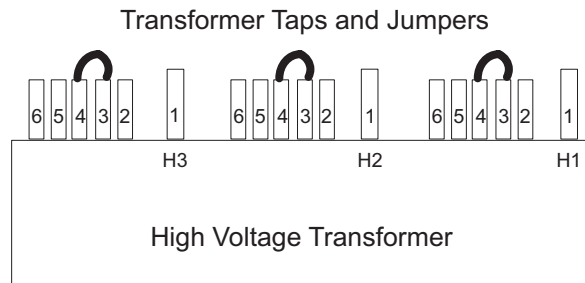
Table 2-8 Panel Circuit Breaker Descriptions

CIRCUIT BREAKER	DESCRIPTION
CB3	Fully Winding Protection (Master power of CB 4, 5, 6, and 7)
CB4	CT Gantry Service Outlets
CB5	CT Gantry rotating loads
CB6	Table & CT Gantry Station Loads
CB7	Operator Console
CB8	PET Gantry
CB9	NGPDU Control Power Supply

7.1.3 Transformer (480VAC) Taps

Verify that the transformer taps are set properly. The transformer taps are set to 480 VAC operation at the factory. The taps should be set as shown in [Figure 2-23](#).

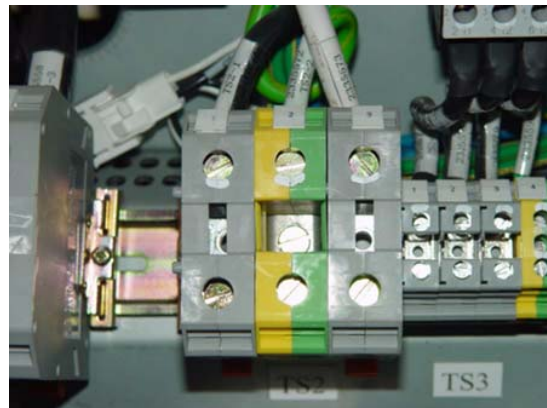
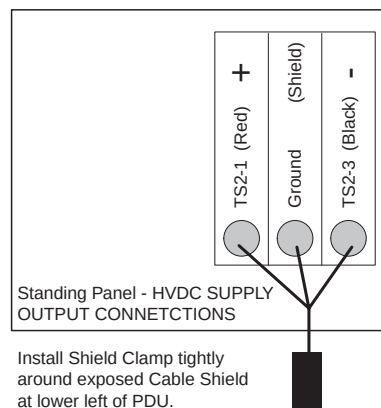
Figure 2-23 PDU Tap Positions for 480 volt operation



7.1.4 HVDC Connection

Connect the internally shielded HVDC cable to TS2 on the standing panel. See [Figure 2-19](#) for the location of the connector and [Figure 2-24](#) for details. Observe polarities and grounds.

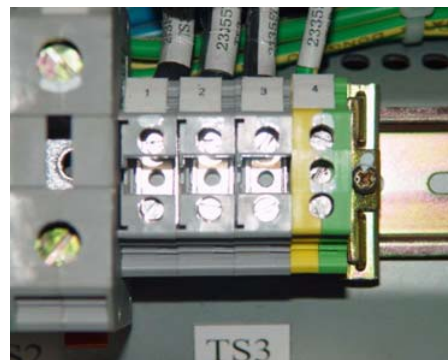
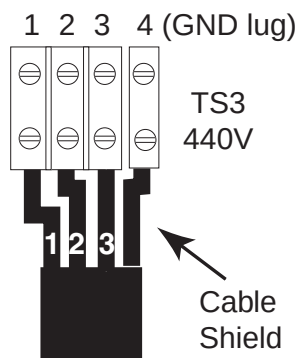
Figure 2-24 HVDC Connection



7.1.5 440V Connection

Connect the internally shielded 440V cable from the gantry to TS3 on the panel. See [Figure 2-19](#) for the location of the connector and [Figure 2-25](#) for details. Observe the labels on the cable leads for proper identification and orientation.

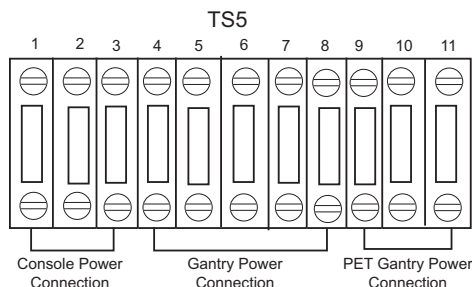
Figure 2-25 440VAC Connection



7.1.6 Gantry & Console Power Connections (120V)

Both Gantry and Console power cables come pre-terminated. Simply plug the Gantry power cable into “J4” and the Console power cable into “J5” as shown in [Figure 2-26](#).

Figure 2-26 Gantry & Console Power Connections



7.1.7 Console Power Cable Re-termination

Power cable re-termination should be used as a last result. Short and long cables are available.

- 1.) Carefully remove the power plug and **record the color of the wires** in [Table 2-9](#). The terminals are labelled X, W and G on the plug.

Table 2-9 Console Power Cable Termination

TERMINAL	X	W	G
Description	Hot	Neutral	Ground
Color			

- 2.) Cut the cable to desired length and dress ends.
- 3.) Re-install the power plug, according to the orientations recorded in [Table 2-10](#).
- 4.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 2-10 Resistance Verification Points

FROM PDU PLUG	TO CONSOLE PLUG END	
CB1 -11 (Black) (J5 Phase X (Brown))	Phase X (Brown)	☐ Check box when complete
A3 Neutral Buss Bar (Blue) (J5 -13 Phase W)	Phase W (Blue) Neutral	☐ Check box when complete
A3 Ground Buss Bar (J5 -22 Ground Green)	Ground (Green) Ground Green screw	☐ Check box when complete

7.1.8 PDU Control Cable

The PDU control cable comes pre-terminated and should not be re-terminated in the field. Excess cable length must be stored. Simply plug the cable into “J2” on the “A4” panel. Secure it by using the fasteners intergrated into cable’s connector shell.

7.1.9 System Ground Connection

Connect the ground wire (green with a yellow strip) from the Table/Gantry raceway ground bus to the system ground lug in the PDU. See [Figure 2-19, on page 82](#), and [Figure 2-27](#), below.

Figure 2-27 PDU System Ground Connection

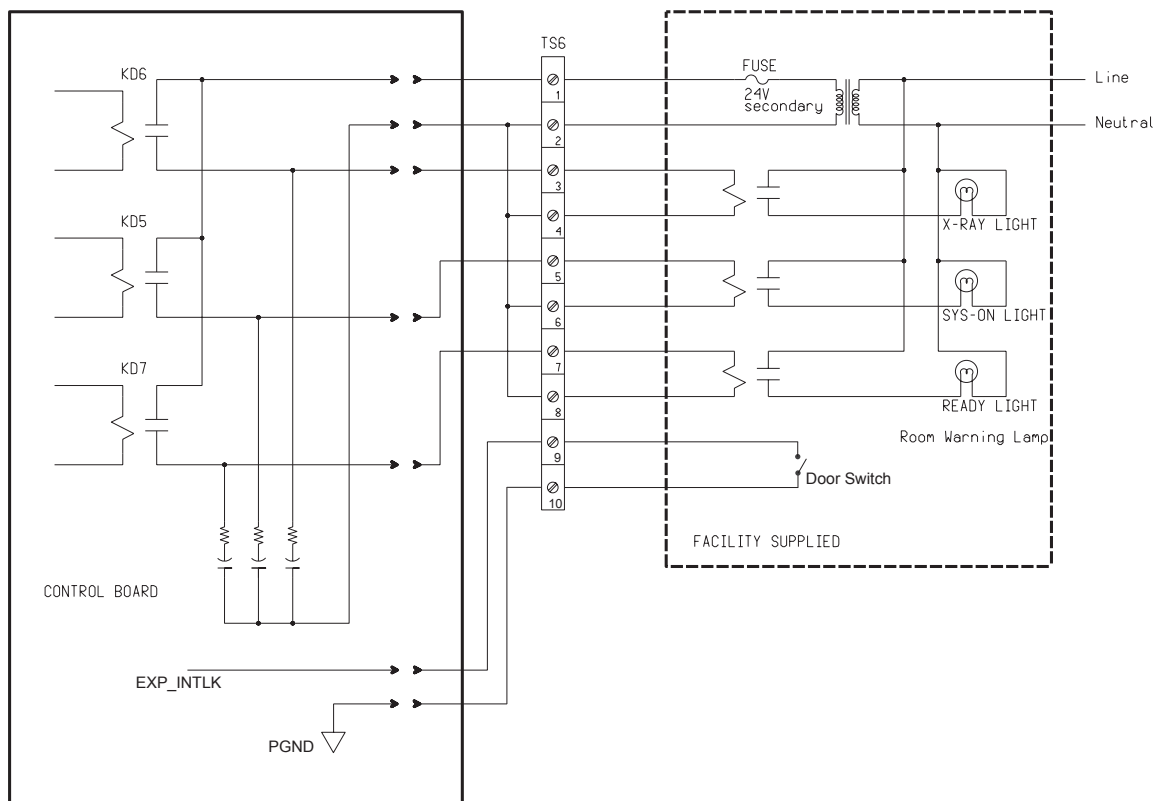


7.1.10 Warning Light & Door Interlock Connections

7.1.10.1 Warning Light Configuration & Connection

- 1.) Warning Light is controlled by signals from the system.
- 2.) This step is site specific. The PDU by default is configured for “no” external warning light connection. If you have external warning lights, see [Figure 2-28](#) for proper connection.

Figure 2-28 Typical TS6 Warning Light & Door Interlock Connections



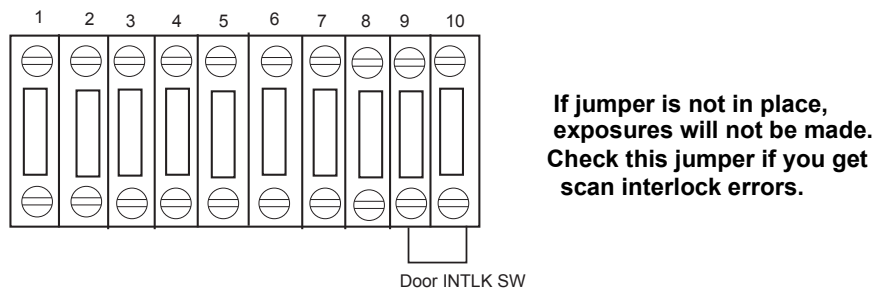
It is recommended that you use the four (4) wire method of adding a X-ray warning light to a room, as shown in [Figure 2-28](#). When using this method, you:

- Minimize EMC interference.
- Increase contact life of the relay used in the PDU.

7.1.10.2 Door Interlock Connections

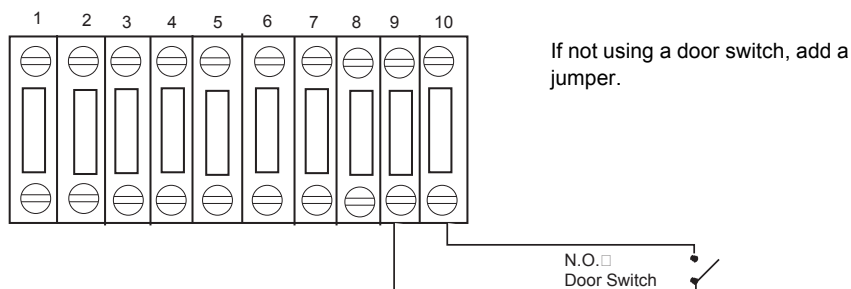
Door interlocks are used to prevent X-Rays from being generated when the scan room door is open. The Door Interlock circuitry in the PDU is shipped from the factory engaged. This means the system cannot generate X-ray until disengaged. A short must exist between pins 9 & 10 for X-ray to be generated. Using a small piece of wire, short pins 1 and 2 together. See [Figure 2-29](#).

Figure 2-29 Without a Door Interlock



To use the system with a door interlock, wire a normally open switch between pins 1 & 2 that is attached to the interlock.

Figure 2-30 With a Door Interlock

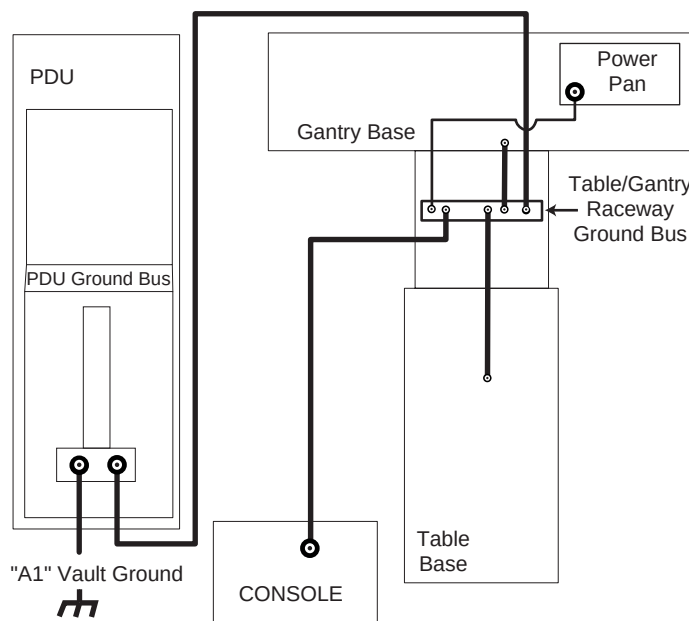


Section 8.0

System Ground Connections

As seen in [Figure 2-31](#), the Table/Gantry raceway ground bus is used to centralize all system grounding. The system ground is tied to vault ground at the PDU, through its chassis.

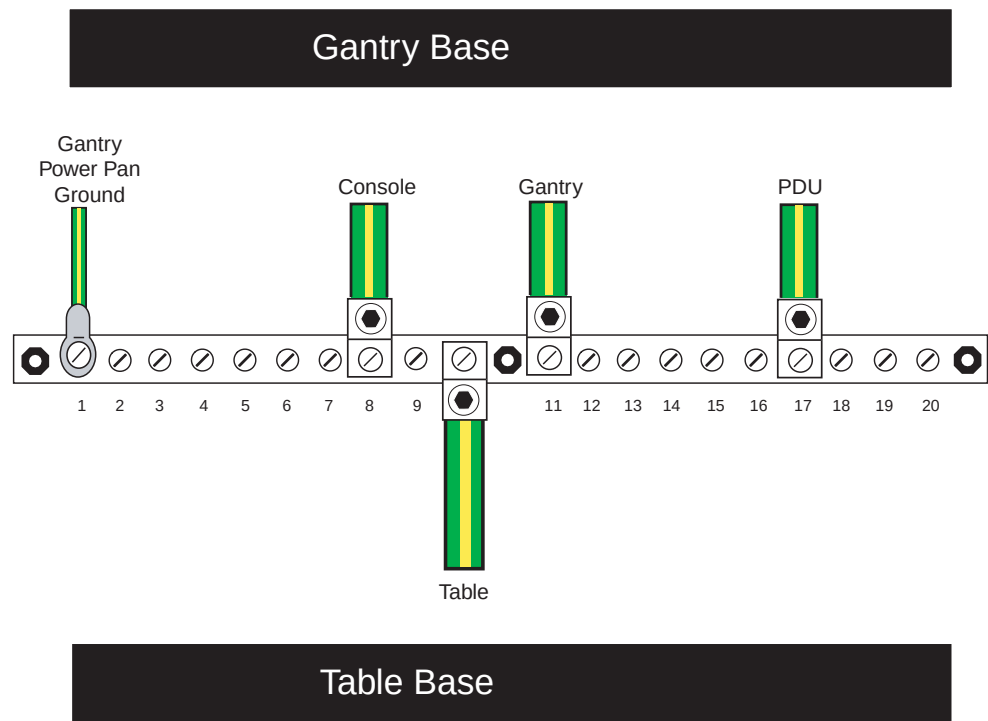
Figure 2-31 CT System Ground Connections



The gantry is tied to system ground at a number of points. It is important that all of these ground connections are securely made. See [Figure 2-32](#).

Check that the terminals will not be loosened, by moving or swinging the cables rather strongly by hand.

Figure 2-32 Table/Gantry Raceway Bus - Grounds



Various types and sizes of wire are used to ground the system. Please use the type and sizes specified in [Table 2-11](#), below.

Table 2-11 System Ground Connections

AWG #	CONNECTION TO	CONNECTION TO
#1/0	PDU	Power Main
#1/0	Gantry (Power Pan)	Raceway
#2	Console	Raceway
#1/0	Gantry	Raceway
#2	Table (frame)	Raceway
#1/0	PDU	Raceway

Chapter 3

System Continuity & Ground Checks



NOTICE

Potential for Data Loss and/or Equipment Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

System Continuity & Ground Checks (Mechanical Contractor)

Use this section to check cable and ground connections.

1.1 Tools Required

- Digital VOM
- 30 ft of #18 wire
- 600 VAC meter leads

1.2 Procedure

Reference [Figure 3-1: Front View of NGPDU, with Covers Removed on page 93](#) and [Figure 3-2: Gantry Power Pan on page 93](#).



WARNING



USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES; LOCK OUT WALL POWER.

- 1.) Remove all System Power at the A1 Mains Disconnect Panel. Follow Lockout/Tagout procedures.
- 2.) Put the UPS in the Service Position.
- 3.) Remove the PDU A3 input power panel cover.
- 4.) Verify, with a voltmeter, that mains power is disconnected.
- 5.) Verify that less than 1 ohm of resistance exists between the following ground connections:

Table 3-1 Mains Connections to PDU

FROM	TO	
Wall ground connection	PDU Cabinet	☐ Check box when complete

6.) Verify that less than 1 ohm of resistance exists between the following connections:

Table 3-2 Resistance Verification Points

FROM	SIGNAL NAME (COLOR)	TO	
PDU TS2-1	+HVDC (Red)	Gantry HV Power Pan TS1-1	☐ Check box when complete
PDU TS2-2	HVDC Ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU TS2-3	-HVDC (Black)	Gantry HV Power Pan TS1-2	☐ Check box when complete
PDU Ground Bus	HVDC shield	Gantry HVDC cable shield	☐ Check box when complete
PDU TS3-1	Axial drive 440vac (Black)	Gantry HV Power Pan TS2-1	☐ Check box when complete
PDU TS3-2	Axial drive 440vac (Red)	Gantry HV Power Pan TS2-2	☐ Check box when complete
PDU TS3-3	Axial drive 440vac (Orange)	Gantry HV Power Pan TS2-3	☐ Check box when complete
PDU TS3-4	Axial drive ground (Green/Yellow)	Gantry Power Pan Chassis	☐ Check box when complete
PDU Ground Bus	Axial drive shield	Gantry 440 VAC cable shield	☐ Check box when complete
PDU TS5-1	120vac Phase A (Brown)	Console J?	☐ Check box when complete
PDU TS5-2	120vac Neutral (Light Blue)	Console J?	☐ Check box when complete
PDU TS5-3	Ground (Green/Yellow)	Console J?	☐ Check box when complete
PDU TS5-4	120vac Phase A (Black)	Gantry LV Power Pan A1J1 Filter - L1	☐ Check box when complete
PDU TS5-5	120vac Phase B (Red)	Gantry LV Power Pan A1J1 Filter - L2	☐ Check box when complete
PDU TS5-6	120vac Phase C (Orange)	Gantry LV Power Pan A1J1 Filter - L3	☐ Check box when complete
PDU TS5-7	120vac Neutral (Light Blue)	Gantry LV Power Pan A1J1 Filter - N	☐ Check box when complete
PDU TS5-8	Ground (Green/Yellow)	Gantry Power Pan Chassis A1J1 Filter Ground Stud	☐ Check box when complete

Figure 3-1 Front View of NGPDU, with Covers Removed

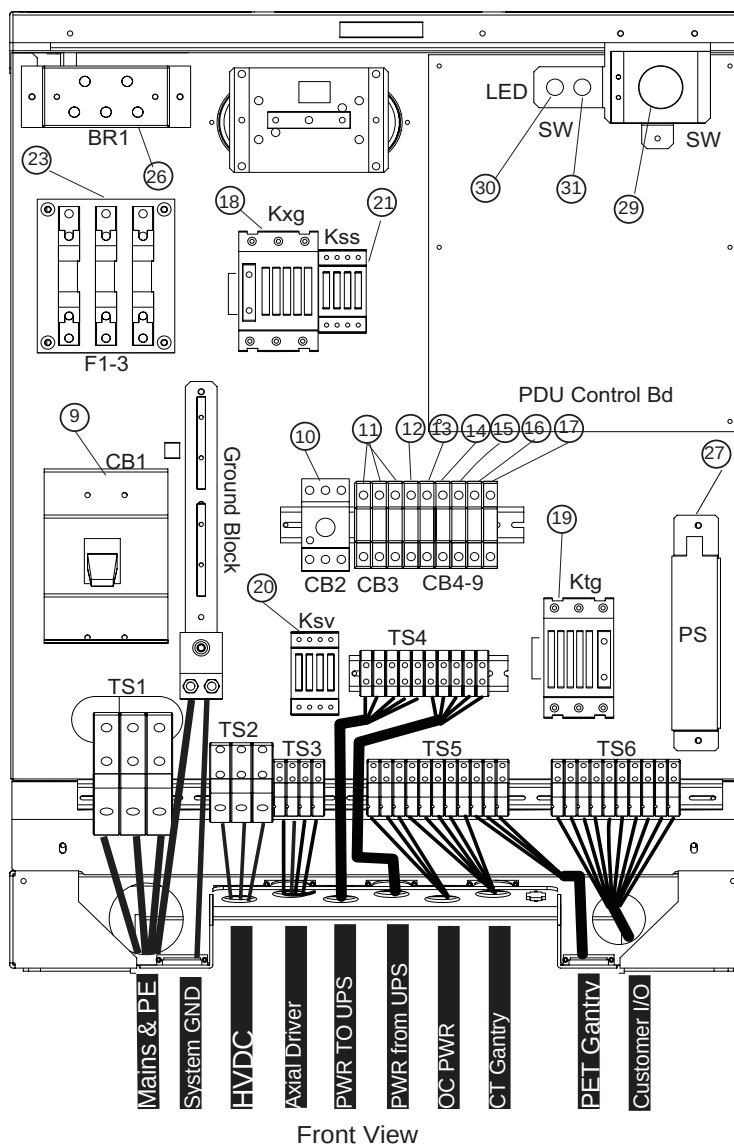
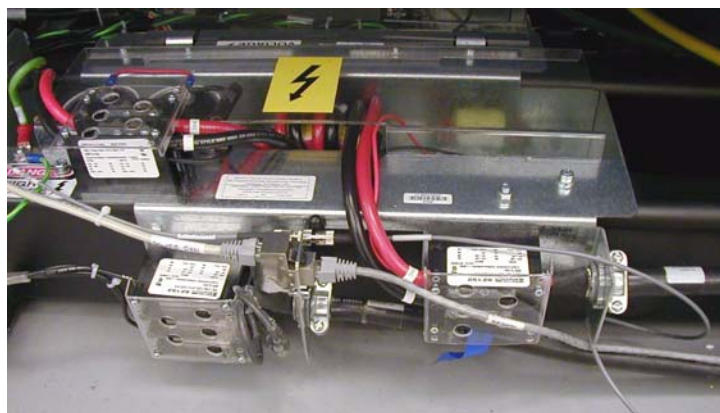


Figure 3-2 Gantry Power Pan





WARNING



TURN OFF ALL PDU CIRCUIT BREAKERS.

- 7.) Set an ohmmeter to the lowest scale. Check between the following points for shorts to ground. Verify no continuity exists between the following points:

Table 3-3 No Continuity Verification Points

FROM PDU	TO A1 BREAKER BOX	
TS2-1 (+HVDC) (Red)	vault ground	☐ Check box when complete
TS2-3 (-HVDC) (Black)	vault ground	☐ Check box when complete
TS3-1 (440vac output) (Black)	vault ground	☐ Check box when complete
TS3-2 (440vac output) (Red)	vault ground	☐ Check box when complete
TS3-3 (440vac output) (Orange)	vault ground	☐ Check box when complete

- 8.) Leave the metal cover off the PDU A3 input power panel until you complete the checks in the next section.

Section 2.0 Site Ground Continuity Check

Use an ohmmeter to verify the presence of **less than 1.0 ohm of resistance** between each of the following points:

Table 3-4 Resistance Verification - Site Ground

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

Section 3.0

Mechanical Installation Completion Section

Complete the Mechanical Vendor checklist:

3.1 Clean Up

- ☐ Move box labeled “Customer” to one corner.
- ☐ Move Field Engineer Material (phantoms, hoist, kits, tools) to another corner or counter top.
- ☐ Put all documentation (incl. software, HHS doc) in the purple box.

3.2 Remove Trash

- ☐ Recycle cardboard.
- ☐ Remove skids and wood.

3.3 Return Dollies

- ☐ Check the packing list found in the purple box.

3.4 Install Covers

- ☐ Recheck Gantry and Table anchor for 75ft-lb.
- ☐ PDU covers.
- ☐ Table foot-switch assembly (refer to [Figure A-30, on page 115](#)).
- ☐ Console covers.
- ☐ Gantry base covers (refer to [Figure A-29, on page 114](#)).

3.5 Clean the Equipment

- ☐ Use the contents of the cleaning kit to clean the equipment.

3.6 Touch Up Nicks

- ☐ Use the paint found in the cleaning kit to touch up nicks in the equipment.

3.7 Complete Paperwork

- ☐ Collect all Product Locators.
- ☐ Complete the Field Engineer sign-off.

Appendix A

Removal & Installation of Covers

Section 1.0 Gantry Side Covers

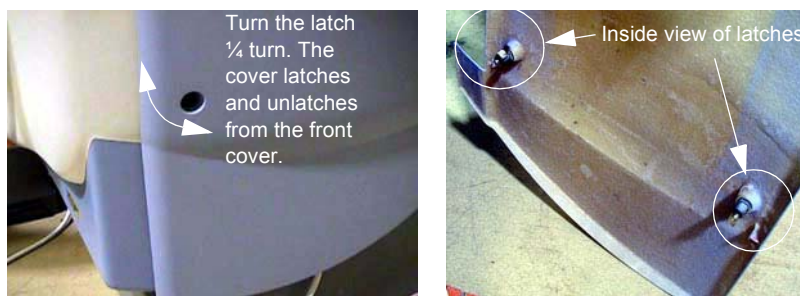
1.1 Side Cover Removal

CAUTION



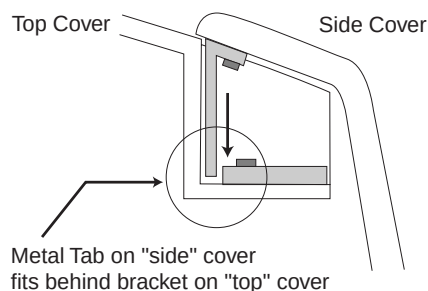
- 1.) Lower table to home (lowest) position.
Potential for injury if covers removed and power is left "ON".
Always remove the right side cover first, and turn OFF power at the Service Switch Panel.
- 2.) Use an 8mm Hex wrench to unlatch the side cover from the front cover. See [Figure A-1](#).

Figure A-1 Side Cover Latches



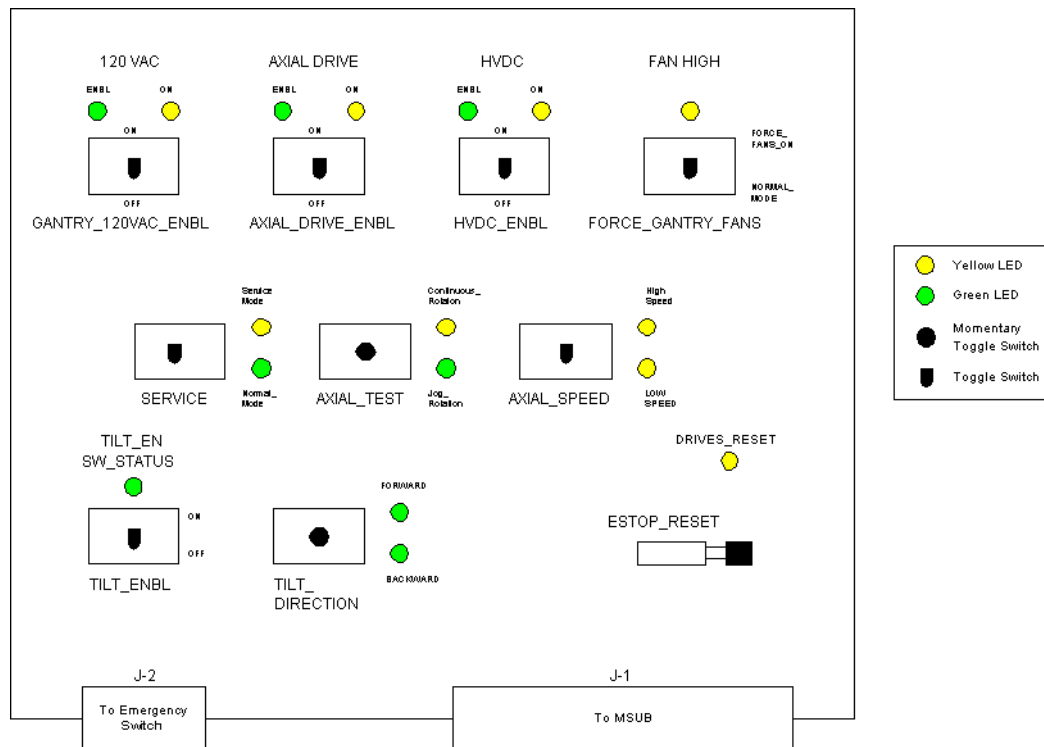
- 3.) Remove the right side cover by lifting it upward to release the two (2) latches, located on the top edge of the cover. Once removed, the MSUB/TGP left side should be exposed.

Figure A-2 Side and Top Cover Clasp



- 4.) Turn OFF the **120 VAC**, **AXIAL DRIVE** and **HVDC** power switches on the gantry service switch panel (see [Figure A-3](#)).

Figure A-3 Gantry Service Switches



- 5.) Repeat steps 1-3 for the left side cover.

1.2 Side Cover Installation

- 1.) To install a side cover, place it over the top cover and let the two (2) side cover latches slide behind the metal tabs, located on the top cover. See [Figure A-2](#).
- 2.) Use Hex wrench to secure the side cover to front cover by turning the bolts a quarter turn. See [Figure A-1](#).

Section 2.0 Gantry Top Covers

2.1 Top Cover Removal

CAUTION Potential for Shock.



Voltage may be present. Potential for injury if covers removed and power is left ON. Always remove the right side cover first, and turn OFF power at the service switches.

NOTICE



Always turn OFF the HVDC before the 120 VAC. Turning OFF 120 VAC power before HVDC power can result in equipment damage.

- 1.) Remove the gantry right side cover (see Gantry Side Covers Removal and Re-install).
 - 2.) Turn OFF the three (3) main power switches (HVDC, 120VAC, and Axial Drive) on the Service Switch Panel (SSP). See [Figure A-3](#).
 - 3.) Disconnect the power cable that connects the fans on the top cover to the gantry.
- Note: The cooling fans are now mounted on the top covers.
- 4.) Take the end of the top cover nearest to the side cover and tilt upwards.
 - 5.) Slide the cover down to disengage the tab from the mounting bracket. See [Figure A-4](#).

Figure A-4 Top Cover Tabs and Bracket



- 6.) Lift the cover clear and repeat the above steps for the other cover.

2.2 Top Cover Installation

The top cover consists of two (2) pieces. Install the front and rear gantry covers, if not already installed. See [Section 3.0 on page 100](#), and [Section 4.0 on page 108](#).

- 1.) Take one of the top covers and align the tabs on the cover with its associated bracket. Lift and slide the cover into place.
- 2.) Take the other top cover and align the tabs on the cover with its associated bracket. Lift and slide the cover into place.
- 3.) Connect the cable from the fans to the gantry.

NOTICE



Always turn ON the 120 VAC before the HVDC. Turning ON HVDC power before 120 VAC power can result in equipment damage.

- 4.) Turn on the three (3) power switches.
- 5.) Ensure fans work properly.
- 6.) Re-install the gantry side covers.

Section 3.0

Gantry Front Cover



NOTICE

Potential for cover damage.

Front and rear cover removal and installation can be safely accomplished by one (1) person using the dollies provided with the system. Failure to use these dollies will significantly increase the likelihood of damage to the covers. Do not lean covers against walls.

3.1 Front Cover Dolly Setup

DANGER



DO NOT USE DOLLIES ON UNEVEN SURFACES SUCH AS STEPS OR ELEVATOR THRESHOLDS. THE DOLLIES ARE DESIGNED TO BE USED ON FLAT LEVEL FLOORS WITHIN THE SCANNING SUITE ONLY. MISUSE CAN RESULT IN PERSONAL INJURY OR DAMAGE TO COVERS OR OTHER FACILITY ITEMS.

CAUTION



Rotating arms on the stand are supposed to be stiff. If they fall freely, tighten the tensioning nuts. Loose rotating arms will reduce the stability of the dollies when supporting the front cover. Do not lubricate.

- 1.) Arrange Dolly sections for assembly. The base and post can be assembled only one way.
Refer to [Figure A-5](#) and [Figure A-6](#).
 - The base uses two (2) palm screws to clamp the four (4) legs in the open or usage mode.
 - The base also uses the same palm screws to prevent the legs from falling in storage mode.
 - The top post can be inserted in either base and is keyed for proper engagement.
 - The top post locking pin prevents the sections from separating during usage.

Figure A-5 Front Cover Dolly in Storage Mode

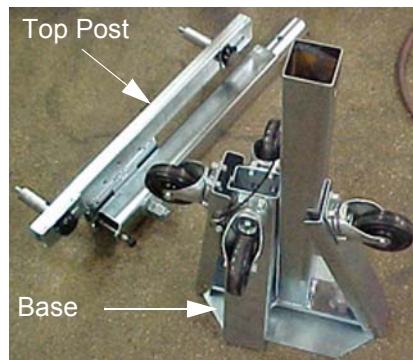
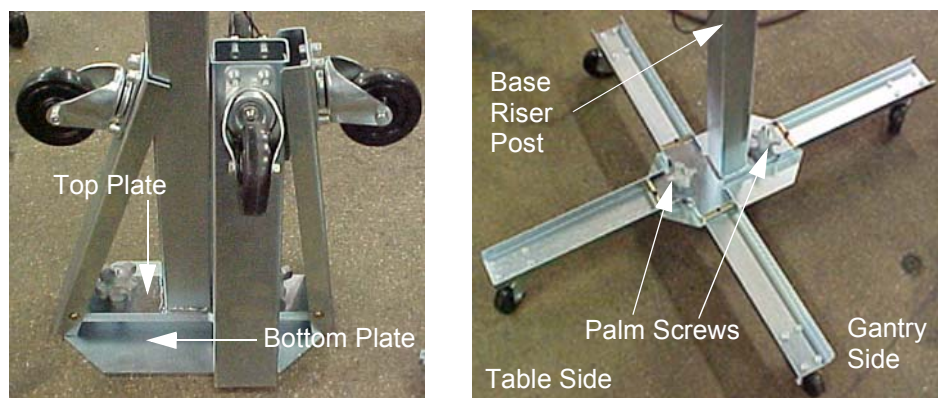


Figure A-6 Front Cover Dolly Base Assembly



- 2.) Unfold the base legs by loosening both palm screws to the top of their travel.
- 3.) Carefully unfold the legs so that the castors touch the floor.
- 4.) Tighten the palm screws to clamp the legs between the base top and bottom plates.

Note: Lifting the base by the riser post while leaving the castors on the floor will ease palm screw tightening. Reference [Figure A-6](#).

WARNING



ENSURE BOTH PALM SCREWS ARE TIGHTENED SECURELY AND THE LEGS ARE CLAMPED TIGHTLY BETWEEN THE BASE TOP AND BOTTOM PLATES. FAILURE TO DO SO WILL RESULT IN INSTABILITY DURING FRONT COVER HANDLING.

- 5.) Insert top post into the base riser post. Align the key for complete engagement.
- 6.) Insert top post locking pin to secure both top and bottom sections.
- 7.) Reverse above steps to disassemble.

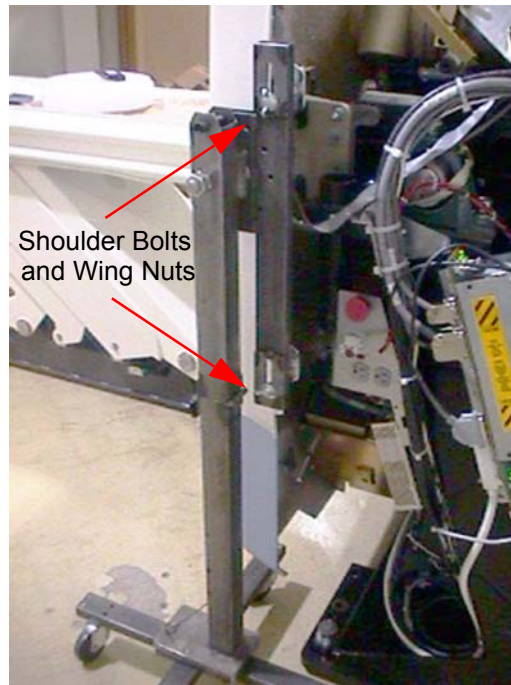
Note: For base storage only one (1) palm screw needs to be tightened. This will engage the bottom base plate and the leg ends preventing the legs from unfolding during transport and storage.

3.2 Removal

- 1.) Position the table at its lowest position.
- 2.) Remove gantry side and top covers, if you have not already done so. See [Section 1.0 on page 97](#). Make sure that the three (3) power switches have been turned off. See [Figure A-3](#).

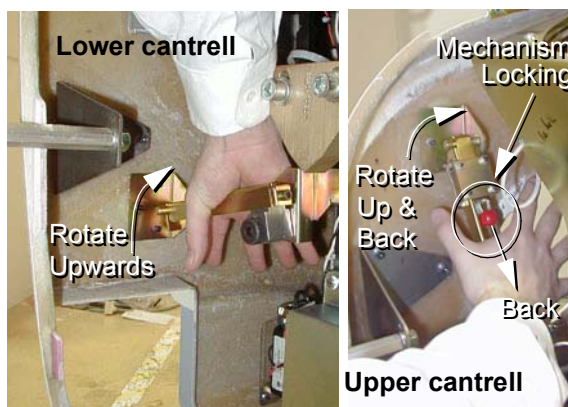
- 3.) Assemble the front cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the gantry securely. This will make cover installation easier. See [Figure A-7](#).

Figure A-7 Front Side Dolly



- b.) Attach side dolly to the shoulder bolts and secure assembly with two (2) wing nuts.
 - c.) Repeat steps a and b to assemble the other side dolly.
- 4.) Detach front cover J3 and J2 and front cover BKHD J1 cables.
- 5.) Remove front cover
 - a.) Disengage upper and lower cantrell brackets on both sides of the cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-8](#).

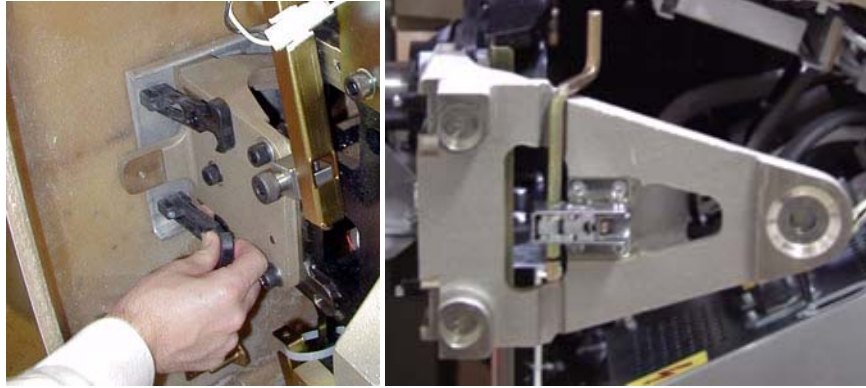
Figure A-8 Releasing cover brackets



- 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.

- b.) Disengage the rubber retaining straps on both sides. You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.
- c.) Also lift and rotate cover locking arm to unlocked position.

Figure A-9 Rubber retaining straps and Cover Locking mechanism



- 6.) Rotate front cover away from gantry.
 - a.) Move front cover away from gantry giving enough space (about 5 feet) between front cover and gantry.
 - b.) Pull the locking pin and rotate front cover away from gantry. Place locking pin in one of the side dolly perforations. See [Figure A-10](#).

Figure A-10 releasing Front Cover Dolly Hinge



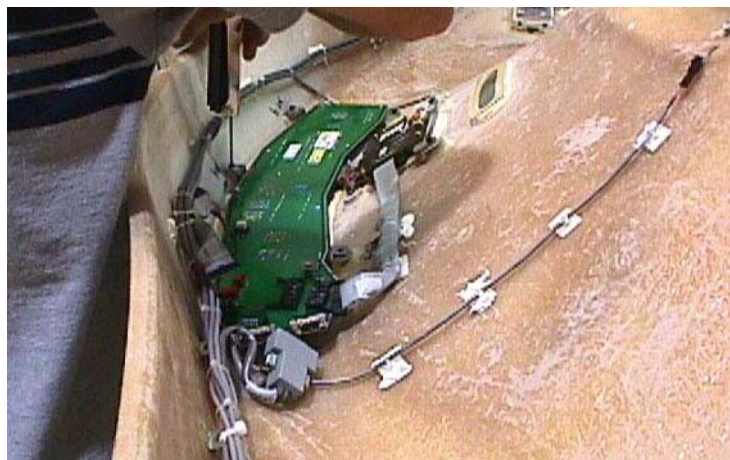
Figure A-11 Front Cover Removal Sequence



1. Upper Left>Rotate cover to make room for step 2.
2. Upper Right>Rotate cover to clear the table. Roll the cover to foot end of table.
3. Lower Left> Rotate the cover upside down to provide clear work area.
Note: Remove Gantry Display and Control Panel in position 2.

- 7.) Rotate the cover horizontally and move it back and over the table to a safe location. Once in a safe location, you may over-rotate the cover full vertically but upside down.
- 8.) Remove the gantry display and one (1) of the cover's control assemblies, and place them into the service positions.
 - a.) Remove the gantry display and place it into its service position.
 - * The gantry display is held in place with (5) thumb screws. Use a flat-blade screwdriver to remove the Display. Reference [Figure A-12](#).

Figure A-12 Gantry Display Removal



- * Place the Display in the bracket on the right side of the gantry. See [Figure A-13](#).

Figure A-13 Gantry Display Service Mounting Location



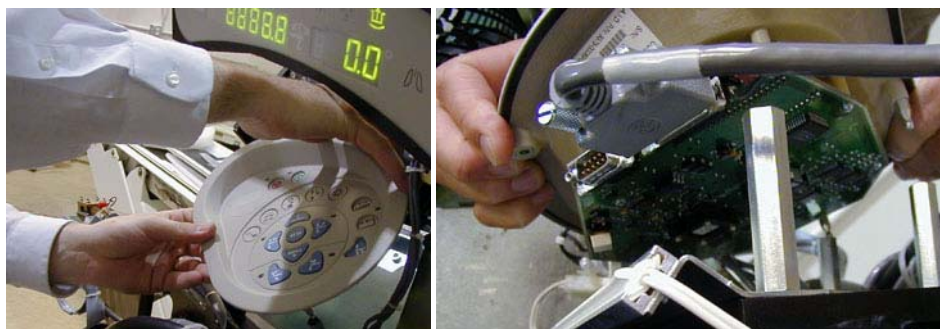
- * Disconnect the cabling at the right rear gantry cover. Only (1) cable will connect to the Gantry Display. Connect the cable taken from the rear cover to the display.
- b.) Remove a gantry control and place it into its service position.
- 1.) Press on each ball stud until the panel is released. Keep one hand on the control panel at all times to prevent it from dropping to the floor.

Figure A-14 Gantry control panel removal



- 2.) Align the ball studs with their associated receivers and snap into place.

Figure A-15 Control panel service position



- Note:
- 3.) Connect cable to terminator located on the cantrell arm. Reference [Figure A-16](#).
There are 3 cables, each of which is unique. The ribbon cable is not used in the Service configuration. The other 2 cables will only fit in the terminator or the control panel, not both.

Figure A-16 Gantry Service Mode Cable Terminator



3.3 Installation

- 1.) Remove the gantry display and control assembly from their service positions and re-attach them to the gantry cover.
 - a.) Disconnect cables from Display and Gantry Control Panels.
 - b.) Install Gantry Display in front cover. Secure the 5 thumbscrews. With a flat-blade screw driver gently tighten past finger tight.
 - c.) Install the gantry control panel making sure the ball studs are secure within the receivers.
 - d.) Re-attach cables.

- 2.) Rotate gantry back to its vertical position.

NOTICE



Potential for front cover damage.

When you put rotate the gantry back to its vertical position, make sure not to scratch the front cover with the edge of the table cradle.

- 3.) Attach the front cover.
 - a.) Align the studs on both sides of the front cover with each associated receiver. Receiver is located on the gantry frame.

Figure A-17 Cover stud and Mounting bracket receiver



- b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

You may find it helpful to lift "up" on the cover to align the stud while attaching the rubber retaining straps.

- 4.) Re-attach upper and lower cantrell brackets on both sides.
 - a.) Remove upper Cantrell brackets from service position and rotate them into position over their associated retaining pins. See [Figure A-18](#).

Figure A-18 Service position of upper and lower cantrell brackets

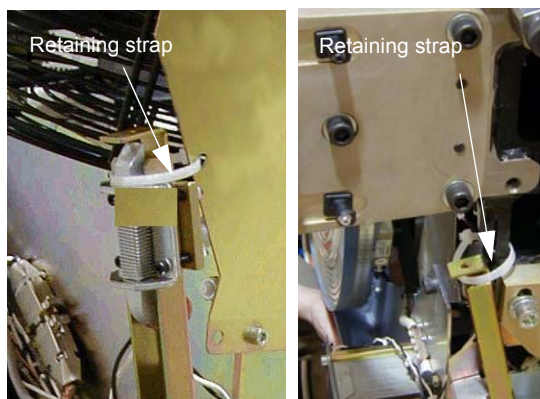
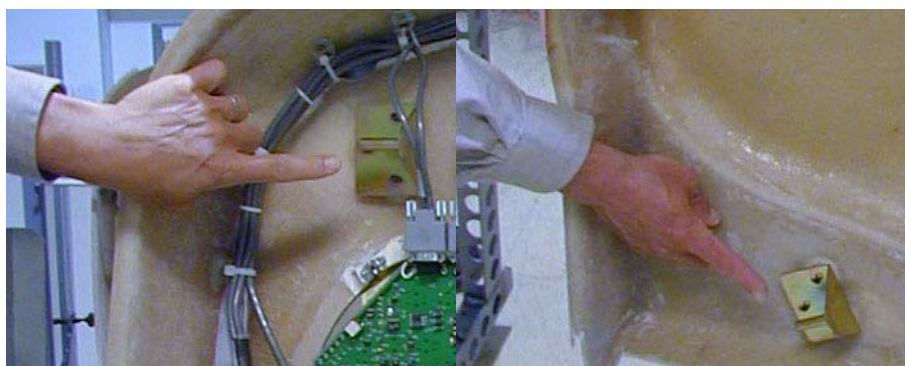


Figure A-19 Cover retaining pins (top and bottom)



Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides. See [Figure A-20](#).

Figure A-20 Locking the cover brackets into place



- b.) Remove lower cantrell brackets from service position (see [Figure A-18](#)), and rotate them into position over their associated retaining pins. Press down firmly on the bracket and

- Note: snap it into place. See [Figure A-20](#).
- Mis-adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not require adjustment during normal use.
- 5.) Remove dolly, disassemble and store safely away for later use.
 - 6.) Re-attach cables to cover.

Section 4.0 Gantry Rear Cover

4.1 Removal

- 1.) Assemble the rear cover dolly.
 - a.) Tighten the two (2) shoulder bolts to the rear cover.

Figure A-21 One side of the Rear cover dolly



- b.) Fit side dolly through the shoulder bolts and secure assembly with two (2) wing nuts. See [Figure A-21](#).
 - c.) Repeat steps a and b for the other side dolly.

CAUTION



Potential for injury if covers removed and power is left "ON".

- 2.) Disconnect cables on the right side of the rear cover.
- 3.) Remove rear cover.
 - a.) Disengage upper and lower cantrell brackets on both sides of the rear cover.
 - 1.) Using steady but firm pressure, lift each of the lower cantrell brackets from their associated retainers. See [Figure A-8](#).
 - 2.) Disengage the locking mechanism on the upper cantrell brackets by using your thumb to slide the trigger (red lever) back. This will release the locking mechanism and allow the cantrell to be rotated upwards with steady and firm pressure.
 - b.) Disengage the rubber retaining straps on both sides.

4.2 Installation

- 1.) Position cover in back of gantry
- 2.) Attach the rear cover
 - a.) Align the studs on both sides of the rear cover with the receivers located on the gantry frame.
 - b.) Insert the stud on one side into its associated receiver and attach the rubber retaining straps. Then insert the stud on the other side into its associated receiver and attach its rubber retaining straps.

Note: You may find it helpful to lift “up” on the cover to align the stud while attaching the rubber retaining straps.

- 3.) Reattach upper and lower cantrell brackets on both sides.
 - a.) Remove upper cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place. The locking mechanism on each upper bracket should lock the bracket securely into place. Do this on both sides.
 - b.) Remove lower cantrell brackets from service position and rotate them into position over their associated retaining pins. Press down firmly on the bracket and snap it into place.

Note: Adjustment of the cantrell brackets can cause misalignment of the top and side covers. The upper and lower cantrell brackets do not re-quire adjust during normal use.

- 4.) Remove dolly, disassemble and store safely away.
- 5.) Re-attach cables to cover.
- 6.) Re-install the mylar (scan) window. Carefully bend the scan window and place it into the channel (groove) provided in the covers.

Figure A-22 Installing the mylar window



Section 5.0

Gantry Cover Alignment Adjustment Guide

5.1 Overview

This section explains the adjustment capability available for the Gantry covers and guidelines for performing adjustments.

5.2 Tools Required

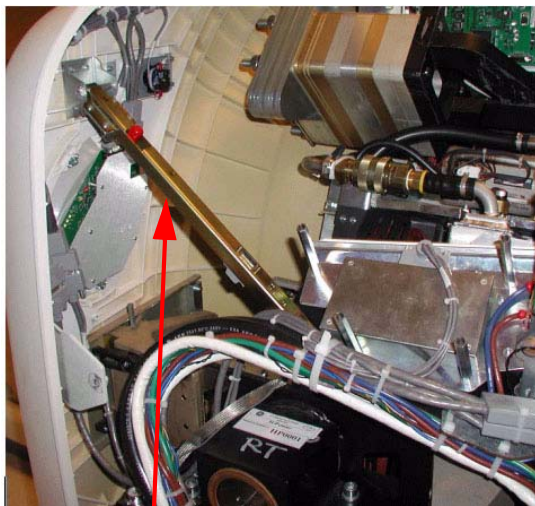
3mm, 5mm, 8mm, Hex wrenches

5.3 Adjustment Guide

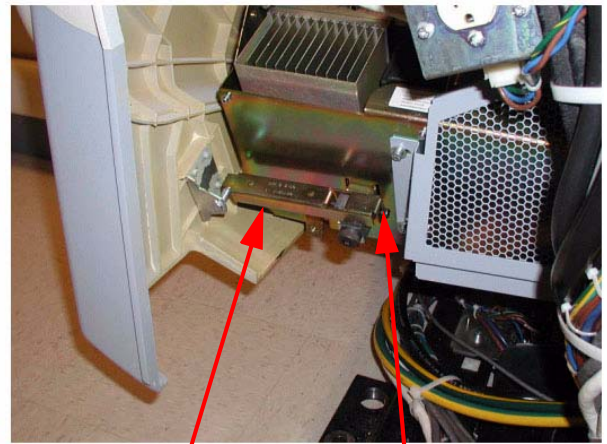
5.3.1 Front/Rear Cover adjustment

The front and rear covers are held in a fixed vertical position by the cover locking mechanisms at the vertical middle of each cover on each side. The only adjustment capability is found on each of the cover cantrells (see [Figure A-23](#), one near each corner of the front and rear covers. Adjustments are best done at zero tilt.

Figure A-23 Cantrells



Top Cantrell same
for all top latches



Bottom Cantrell same
for all bottom latches

Adjustment screw on
end of all brackets

- 1.) Remove the Gantry side covers using a 8mm hex wrench to release the bottom latches. Refer to [Section 1.0 on page 97](#), for further details.
- 2.) Disable gantry rotation, 120VAC and HVDC via the service switch panel.
- 3.) Disconnect the fan power via the white molex connector (one for each top cover). Remove the top covers by lifting up and pulling to the side of the gantry.
- 4.) Remove the scan window to avoid creases or kinks during cover movements.
- 5.) Using a 5mm hex wrench, adjust the bolt on the gantry end of the cover cantrells (see [Figure A-23](#)) to lengthen or shorten the cantrell arm that will push or pull the cover corner. The distance

between the inside edges of the front and rear covers will be typically 27" + 1/8" (689mm). Distance should remain the same as measured along the inside edges from top to bottom with both covers hanging level as seen with a level on the inside vertical edge.

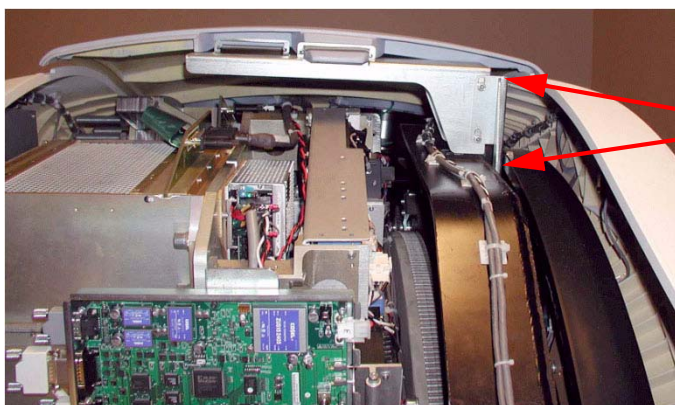
5.3.2 Scan Window

Position the scan window in the opening to verify that it seats all the way around with no gaps. The scan window should seat easily if the front and rear covers have been adjusted properly. If not, go back and readjust such that the scan window seats properly. There is no easy solution other than adjusting and looking for fit.

5.3.3 Top Cover Adjustment

The top covers rest in a top bracket that is mounted to the stationary gantry frame and subsequently just rest on top of the front/rear covers. There are also two alignment fingers on the outer edge of the top covers that need to sit in brackets on the front/rear covers.

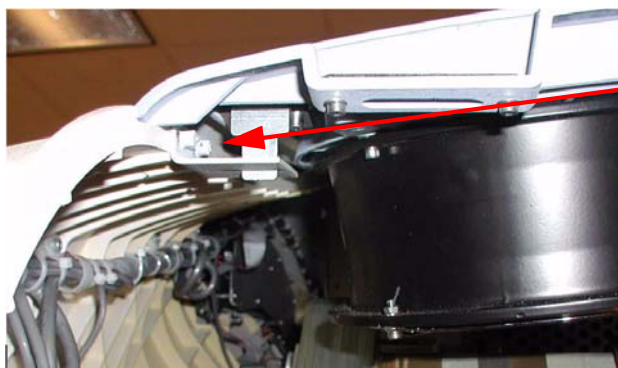
Figure A-24 Top Cover Bracket



Screws on both sides allow adjustment up/down only

- 1.) Set the top covers back on the gantry and look for proper fit. The top covers should rest on the front/rear covers all along the edges. If they do then skip the rest of this section.
- 2.) If the top covers do not rest on the gantry at the top then lift them back off and use a 5mm hex wrench to loosen the bolts that hold the top cover bracket for adjusting vertical alignment to allow the top covers to rest on the front and rear covers.
- 3.) The brackets on the front/rear covers for the fingers on the outer edge of the top covers are adjustable using a 3mm hex wrench and sliding the bracket till it lines up with the top cover finger. (See [Figure A-25](#))

Figure A-25 Top Cover Fingers



Adjustment bolts for aligning bracket to top cover finger

- 4.) Verify the top cover tapered edges fit over the lip along the top edge of the front/rear covers. If

not, go back to Section 5.3.1 and readjust the cantrells as necessary.

- 5.) When done adjusting the top covers, reconnect the fan power molex connections.

5.3.4 Side Covers

The side covers just hang from the top covers by two fingers that are not adjustable. When set into the top covers, the side covers should hang straight down with the tapered edges fitting over top of the lip of the front/rear covers.

5.3.5 Finish

- 1.) Re-Enable gantry rotation via the Axial Enable switch on the service switch panel.
- 2.) Replace the side covers and latch using a 8mm hex wrench on the side cover bottom latches.
- 3.) Run an axial scan to ensure proper system operation prior to turning system back over to the customer.

Section 6.0 Gantry Scan Window

6.1 Remove Scan Window

- 1.) Grab the window at the top and pull firmly downward.
- 2.) Continue to pull until the top of the scan window makes contact with the bottom portion of the scan window.
- 3.) Hold the top and bottom portions of the scan window together, grasp both sides of the scan window, move them together and lightly pull upward, until you can free the window from between the front and rear covers.

Figure A-26 Scan Window Removal



6.2 Install Scan Window

- 1.) Install the front and rear covers.
- 2.) Deform the scan window, as shown in [Figure A-27](#), and nest the scan window at the bottom of the opening between the front and rear covers, ([Figure A-28](#)) with the rivets in the 6 o'clock position installation position. Remember the rivets must be in the 12 o'clock position when the mylar window is fully installed.
- 3.) After you complete the initial seating of scan window, let the window slowly unfold, and work both sides of the window into position, starting at the bottom and finishing at the top.
- 4.) Make sure you position the window with the rivets at the 12 o'clock position, and the mylar window slit at either the 3 or 9 o'clock position.

Figure A-27 Install Scan Window

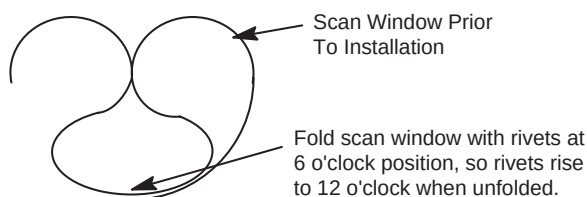
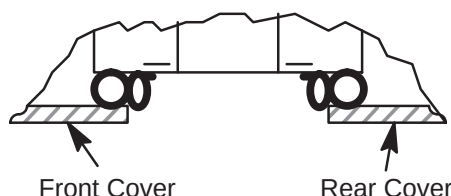


Figure A-28 Scan Window Nested Between Front and Rear Cover



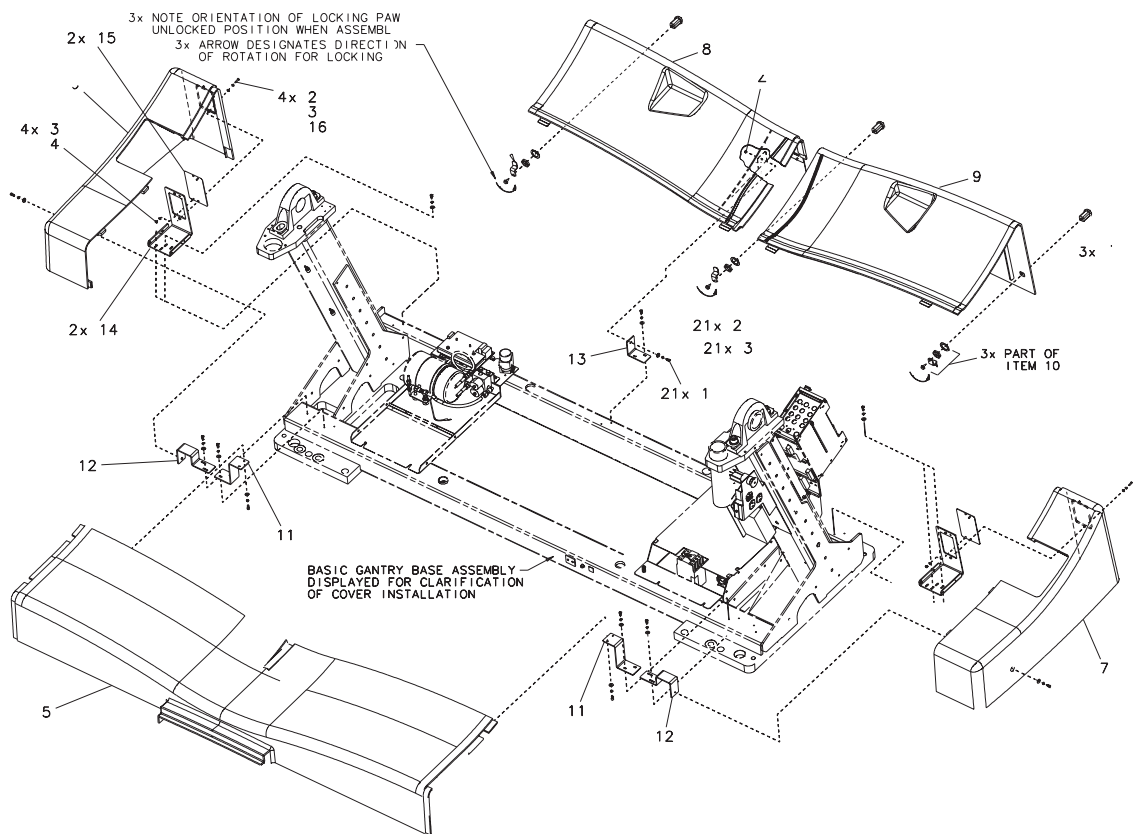
Section 7.0 Gantry Base Covers

Refer to [Figure A-29](#) for the following assembly sequence.

Note: Tighten means torque to 2.3 Nm

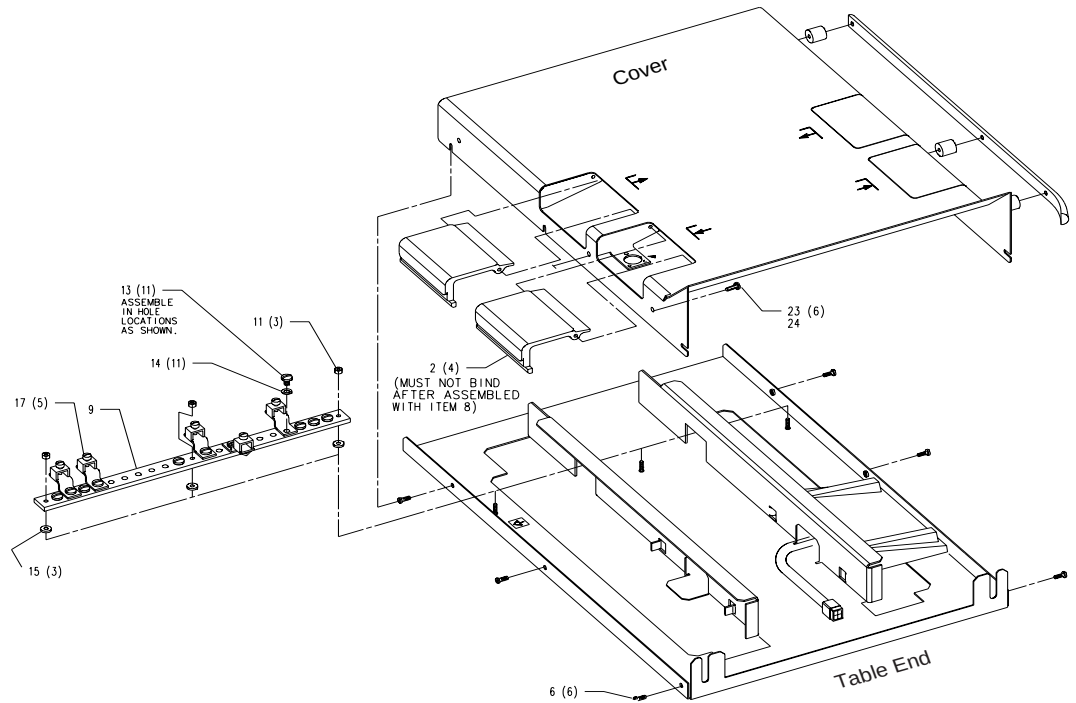
- 1.) Assemble two (2) brackets (item 11) to front cover (item 5), using four (4) hardware items 1, 2 and 3. Align leg of brackets parallel to front edge of cover and tighten. Position cover on gantry base, with bracket slots aligned to gantry holes. Center cover left to right, and attach with four (4) hardware items 1, 2 and 3, as shown, and tighten.
- 2.) Assemble two (2) brackets (item 12) and two (2) brackets (item 13) to gantry base, using eight (8) hardware items 1, 2 and 3. Finger tighten hardware with brackets moved outward to end of slots. Install side covers (items 6 & 7) on base, pushing brackets (items 12 & 13) inward until properly aligned with front cover. Remove side covers, tighten fasteners and replace side covers, using two (2) hardware items 1, 2 and 3 and one item 4 on each cover, and tighten.
- 3.) Assemble last bracket (item 13) loosely to gantry base with two (2) hardware items 1, 2 and 3. Assemble latch (item 10) to rear cover (item 8), oriented as shown. Install rear cover (item 8) to base, properly aligned to side cover (item 6). Attach rear cover to bracket with hardware items 1, 2, 3 and 4, tightening all fasteners. Lock latch (item 10).
- 4.) Assemble two (2) latches (item 10) to rear cover (item 9), oriented as shown. Place cover on gantry base, aligned to covers 7 and 8. Lock both latches.

Figure A-29 Gantry Base Covers (except RT)



Section 8.0 Footswitch Covers

Figure A-30 Table Footswitch Assembly



Appendix B

Pictorial Representation of Required Tools

Use the following guide as a reference, if you are unsure of a tool listed in [Section 1.6, on page 24](#).

Table B-1 Required Tools

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Adapter		Sears Industrial: $\frac{3}{8}$ " to $\frac{1}{2}$ " (9-4258)
Ball-Peen Hammer		Sears Industrial: 1lb/2lb (9-38465)
Canned Air		Miller Stephenson: Aero Duster (MS-222N)
Clamp on Amp Meter		Sears Industrial: 9-WTAD105
Combination Wrench Set		Sears Industrial: U.S. Standard & Metric (9-44048)
Cordless Screwdriver		Sears Industrial: 9-MU65401
Deep Well Socket		Sears Industrial: $\frac{3}{4}$ " X $\frac{3}{8}$ " (included with 9-34496)
Dental Pick		
Diagonal Cutting Pliers		Sears Industrial: Small (9-45077)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Drill		Sears Industrial: $\frac{3}{8}$ " or $\frac{1}{2}$ " (9-27859)
Drill Adapter		Sears Industrial: 3" X $\frac{3}{8}$ " (9-APSZ24)
Drill Bit Set		Sears Industrial: U.S. Standard (9-66084)
DVM		Sears Industrial: 9-82028 Sears Industrial: 9-FL873
Extension for Ratchet Wrench		Sears Industrial: 3" X $\frac{1}{2}$ " (9-44133)
Gloves		Sears Industrial: Large (9-40502)
Hammer Drill		Sears Industrial: $\frac{1}{2}$ " (9-27205)
Hex Bit Set		Sears Industrial: $\frac{1}{4}$ " (9-SK45508)
Hex Key (Allen Wrench) Set		Sears Industrial: U.S. Standard (9-46284)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Level		Sears Industrial: 4' (9-39856)
Masonry Bit		
Open-End Wrench (Thin or Standard Tappet)		Snap-on: 10mm (SRSM10) & 21mm (LTAM2124)
Ratchet Wrench		Sears Industrial: $\frac{3}{8}$ " (9-43175)
Reciprocating Saw with Blades		Sears Industrial: 9-MU650921
Safety Glasses		Sears Industrial: 9-18650
Safety Shoes		
Screwdriver Set		Sears Industrial: Phillips & Straight (9-41505)
Socket Set		Sears Industrial: Standard $\frac{3}{8}$ " (9-34496)
Sockets		Sears Industrial: 1 $\frac{1}{8}$ " X $\frac{1}{2}$ " (9-47516)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Table B-1 Required Tools (Continued)

TOOL NAME	PICTURE	EXAMPLE PART NUMBER*
Step Ladder		Sears Industrial: 6' (9-WN6006)
Tongue & Groove Pliers		Sears Industrial: Large (9-CL440)
Torpedo Level		Sears Industrial: 9" (9-39829)
Torque Wrench		Sears Industrial: $\frac{3}{8}$ " (9-WR3470)
Universal Joint		Sears Industrial: $\frac{3}{8}$ " (9-4435)
Vacuum Cleaner		Sears Industrial: 8 Gal (9-17780)

* Part Numbers given for reference only. GEMs does not endorse any tool brand name.

Appendix C

Option Cables

Run all option cables that connect to the Table, Gantry or Console.
For example the remote monitor (through the splitter) and SmartScore.
For more information, refer to “LightSpeed Linux CRT & LCD Monitor Option”

Figure 3-3 Remote Monitor Splitter



Figure 3-4 SmartScore





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Book 2 of 2
Pages 1 - 136

LightSpeed 5.X Plus Installation Manual

Chapters 4 to End
Calibration, Integration & Testing

Retain this installation manual permanently with your CT System. Any other revisions of this installation manual may not exactly match your system.

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Chapter 4

Electrical Introduction



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss and equipment damage, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 Introduction

Use the continuity and ground checks to verify the system power connections have not shorted to ground and that the ground and neutral connections are intact.

Section 2.0 Review Mechanical Hand Off Material

Complete the Mechanical Hand Off checklist:

- ☐ All options were installed. If not, contact your install specialist.
- ☐ Check for short ships.
- ☐ Review cable connections with mechanical team.
- ☐ Complete mechanical vendor evaluation.

Section 3.0 Training

This product requires a trained FE to proceed with the calibrations in this section.

Section 4.0

Required FE Common Tools and Supplies

4.1 FE Calibration and Service Tool List

Note: Items with a “check” (✓) are included in the Install Support Kit.
Special CT Tools used for mechanical alignments:

- | | |
|---|---|
| ✓ 5mm hex bit for 3/8" drive 6" long
(Snap-On FAML5E or equivalent) | ✓ 10mm open-end thin wrench
(Snap-On SRSM10 or equivalent) |
| ✓ 10mm hex bit for ratchet wrench (ball end)
(Snap-On FABM10E or equivalent) | ✓ 21mm open-end thin wrench
(Snap-On LTAM2124 or equivalent) |
| ✓ 2.5mm Allen hex bit for 1/4" | ✓ 14mm ball hex socket bit for 3/8" drive |
| • Standard FE Tool Kit | ✓ 14 mm hex socket bit for 1/2" drive |
| • Torque Wrench Kit | |

4.2 Electrical Tools

These tools must be calibrated yearly.

- Fluke 87 DVM or equivalent
- Clamp on amp meter

4.3 Image Quality Calibration Tools

- GE Performance Phantom
- QA Phantom (2206352)
 - 24cm (2144721)
 - 48cm (2144721-2)
 - Phantom Holder

4.4 Detector Service Tools

A DAS/Detector Service Kit (PN 2344539) is included in the Install Support Kit.

- Aero Duster
- AMAX Contact Cleaner

4.5 Optional Tools

Needed for warranty recalibration:

- GE HV Bleeder
- Scope with 10X probes
- Beckman CT231 clamp-on Amp probe: 46-194427P228 (Fits up to #2 size wire)
- Beckman CT232 clamp-on Amp probe: 46-194427P270 (Fits #1 size wire and larger)
- Scope Probe-to-bleeder Cable 46-219921G1 (Belden #8422 cable, 30 ft)
- Polaroid type 52 film and developer
- Radial dial indicator (mm or inches) and mounting bracket
- Caliper Dial indicators (mm or inches)
- Ground Rod Tester Clamp on Amp Probe

4.6 Safety Materials

Note: Items with “checks” (√) are included in the Install Support Kit

- √ Lockout/Tagout kit, or equivalent
- √ Nitrile Rubber Gloves
- Safety Glasses

4.7 Cleanliness

Any dirt on the surface increases leakage current on the filter or converter cards and causes the DAS to fail the drift spec. Wear Nitrile gloves (part number 2207303-6 [large] and 2207303-7 [extra large]) when you handle the DAS, because fingerprints on the board can cause problems during humid conditions. Use only clean, new Nitrile gloves. Do not use hospital grade gloves.



NOTICE NEVER USE AN ERASER TO CLEAN ANY PART OF THE DAS.
DUST COVERS REQUIRED for installations on construction sites.

Section 5.0

Requirements/Assumptions

- This procedure will be performed by an appropriately trained engineer.
- All stations in a suite must have the same SUITE ID.
- You need the name of all hosts in the suite and their corresponding Internet/Ethernet numbers.
- You need the Internet (IP) addresses the first time you execute a reconfig on the system.
 - When you connect the system to a network, contact the system administrator to obtain the IP addresses for all the computers in the suite.
 - You also need an IP address for each gateway (second) ethernet board in any OC or IC.
 - You can use the default internet number on stand-alone systems (stand-alone = not connected to any network).
- For more detailed information and instructions regarding Network Integration, see [Chapter 7 on page 69](#).

Chapter 5

Electrical Integration and Safety Verifications



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Electrical Power ON & Ground Checks



WARNING



**THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES.
USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.**

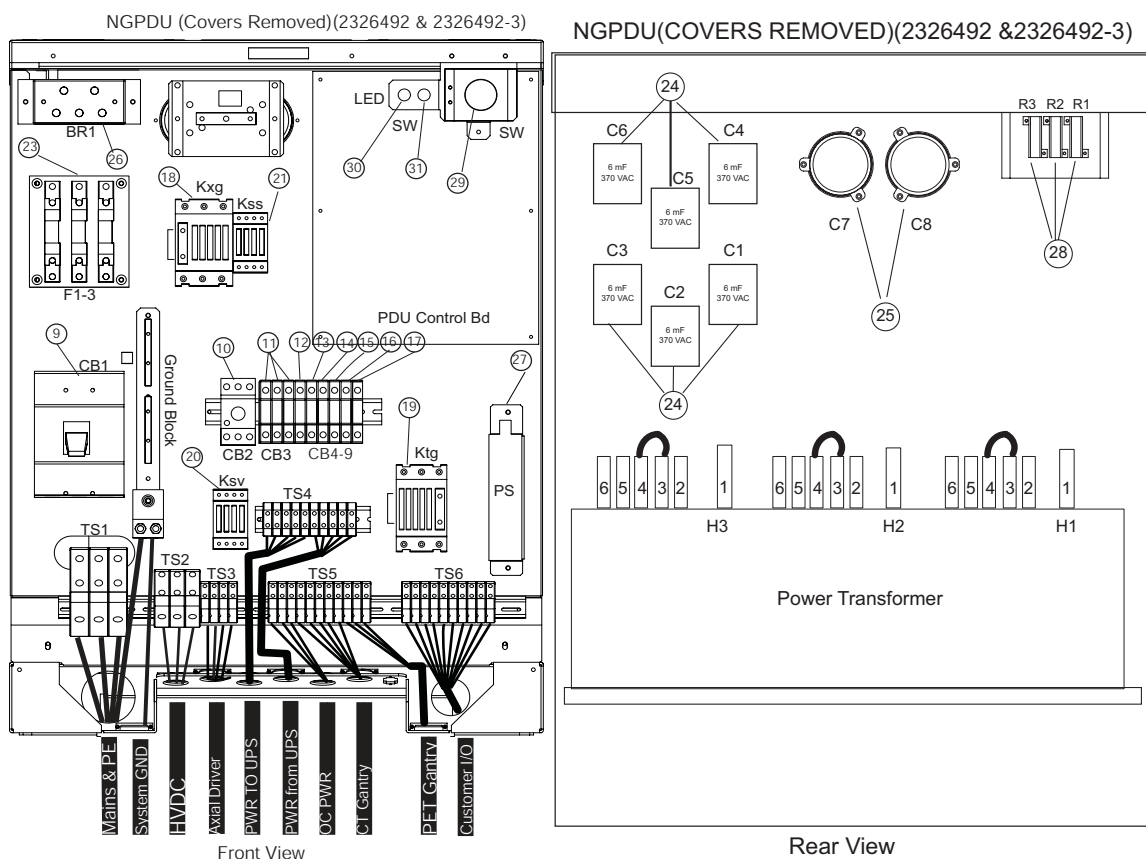


Figure 5-1 NGPDU

1.1 Required Tools

- Multimeter with a rating of at least 1000 volts
- Multimeter leads with a rating of at least 1000 volts

1.2 Initial PDU Configuration



WARNING



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

1.2.1 Circuit Breakers

Set all circuit breakers to OFF

1.2.2 Relay Board

- 1.) Set SW to the normal position.
- 2.) When system is already, three lamps are both lighting (refer to [Figure 5-2](#)).

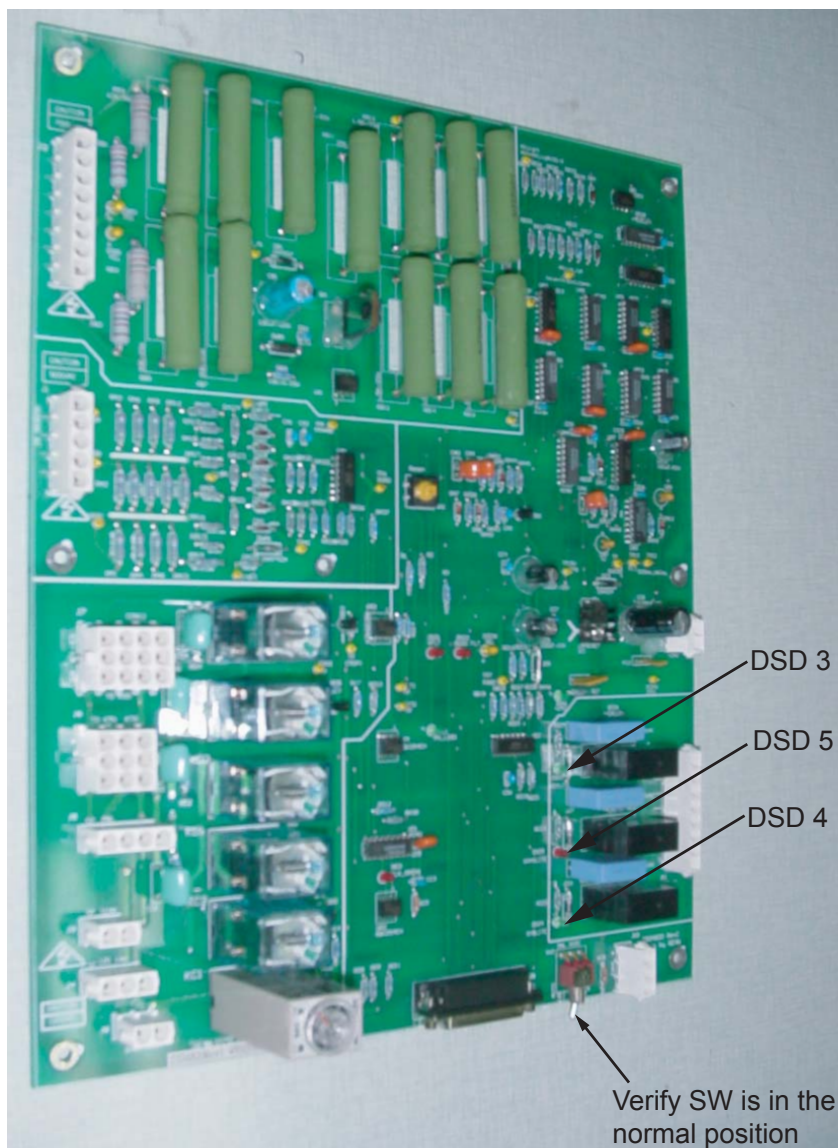


Figure 5-2 NGPDU Control Board

1.2.3 Power Switches

Turn OFF all power switches on all subsystems

- Gantry
- Table
- DAS
- Console

1.2.4 Hardware and Connection Check

Use this step to check mechanical connections and tighten anything that may have shaken loose during shipment. Verify **all** hardware and connections in the PDU are securely fastened.

1.2.5 Covers

Install, or verify the presence of, all the lexan safety covers.

1.3 Suite Emergency Off Checks



WARNING

VERIFY ALL PERSONNEL HAVE CLEARED THE SYSTEM BEFORE YOU TURN ON WALL POWER.

- 1.) Turn wall power ON to the PDU.
- 2.) **Press the suite emergency off button and verify it turns off wall power to the PDU.**
[Typically, this red palm button is located on the wall close to the console, within the scan suite.]
- 3.) Verify that all “Emergency Off” button are working properly.
- 4.) Leave power “OFF”

1.4 Line Transformer Settings

1.4.1 Requirements

- 1.) The PDU is shipped configured for 480VAC.
- 2.) Complete only if your site uses a voltage other than 480VAC.
- 3.) If PDU is configured for 480VAC, go to [1.5](#). Otherwise, proceed to [Section 1.4.2](#).



WARNING

MAKE SURE YOU TURNED OFF, TAGGED AND LOCKED THE MAIN WALL POWER BEFORE YOU CHANGE TAPS. FAILURE TO DISCONNECT POWER AT MAIN INPUT MAY RESULT IN ELECTROCUTION. TURN OFF WALL POWER TO CONNECT OR MOVE METER LEADS, OR TO REMOVE OR INSTALL COVERS.

1.4.2 Line Input Conditions

- 1.) Monitor the No Load Line to Line Voltage at L1, L2, L3, during the workday. Do not record this data during “brown out” conditions.
- 2.) After you determine the nearest nominal line, verify the tap connections match (refer to [Table 5-1](#) and [Figure 5-3](#) for tap locations).

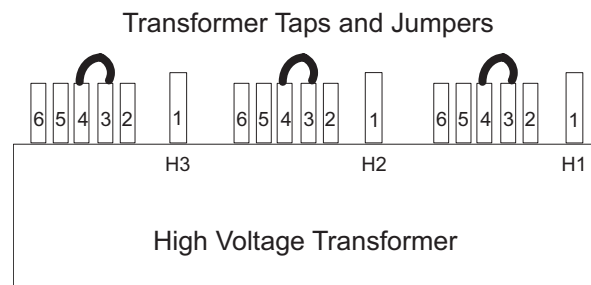


Figure 5-3 PDU Tap Positions (Rear)

Note: **Taps should be shipped as shown for 480 VAC only. For all others, you must move the taps. The tap check should be completed by the mechanical installer.**

- 3.) Verify that the No Load Line to Line Voltage never falls outside the corresponding minimum

and maximum values listed in [Table 5-1](#).

- 4.) Use a 0-750 AC voltmeter of 3/4% accuracy to measure the line-to-line voltages at L1, L2, and L3.
 - Verify the highest line-to-line voltage does not exceed 1.02 times the lowest voltage.
 - **Example:** If the lowest voltage equals 474, the highest voltage should not exceed $474 \times 1.02 = 483.5$ volts.



THIS PROCEDURE MEASURES POTENTIALLY HAZARDOUS VOLTAGES. USE AND FOLLOW LOCKOUT/TAGOUT PROCEDURES.

NO LOAD Line to Line Voltages		TAP CONNECTIONS (All 3 phases must have same the configuration)		
Nominal	Maximum Range (10%)	Phase A Connection	Phase B Connection	Phase C Connection
480V*	432 to 528*	3-4*	3-4*	3-4*
460V	414 to 506	3-5	3-5	3-5
440V	396 to 484	3-6	3-6	3-6
420V	378 to 462	2-4	2-4	2-4
400V	360 to 440	2-5	2-5	2-5
380V	342 to 418	2-6	2-6	2-6
240V**	216 to 264**	1-4**	1-4**	1-4**
220V**	198 to 242**	1-5**	1-5**	1-5**
200V**	180 to 220**	1-6**	1-6**	1-6**
* Factory Default				
** 2326492-3 PDU only				

Table 5-1 PDU Line Tap Connections

1.5 System Power Up



CAUTION Verify all personnel have cleared the system before you turn on wall power.

- 1.) Turn ON A1 breaker panel.
- 2.) Turn ON all system power switches and breakers (table, gantry, PDU, console).

SUB-SYSTEM POWER-UP

- 1.) Turn ON switch S3 in the table (120vac 24hr power).
- 2.) Turn ON the Gantry **120VAC**. (Light should turn on.)
- 3.) Turn **AXIAL DRIVE ENABLE** ON. (Light should turn on.)
- 4.) Turn **HV DC ENABLE** ON. (Light should turn on.)
- 5.) Push the Service Switch Panel reset button (see [Figure 5-4](#)).

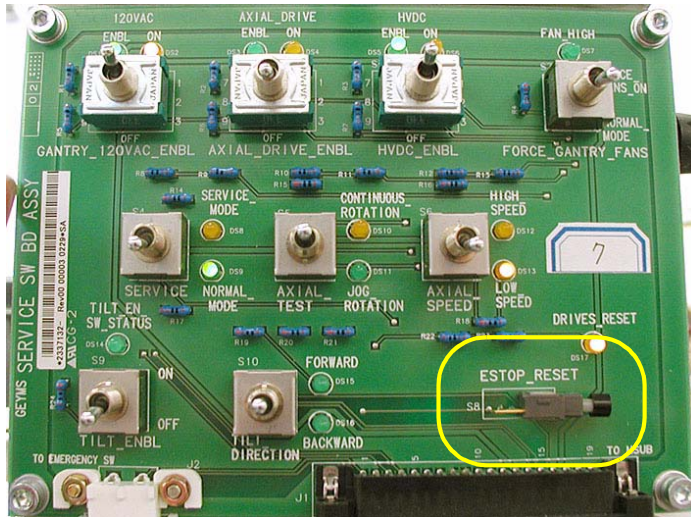


Figure 5-4 Service Switch Panel

AXIAL ENABLE SWITCH TEST

- 1.) Turn OFF axial drive enable switch AXIAL_DRIVE on the Service Switch Panel.
- Note: For the initial condition, do not leave the tube at the 2:30 position.
- 2.) Clear the gantry area for rotation.
- 3.) Press the alignment light push button.
- 4.) Verify that the gantry did not rotate.
- 5.) Turn ON axial drive enable switch AXIAL_DRIVE on the Service Switch Panel.

ROTATION SAFETY CHECKLIST

- 1.) Turn OFF axial drive enable switch AXIAL_DRIVE.
- 2.) Turn OFF HVDC enable switch.
- 3.) Press red E-STOP button.
- 4.) Manually rotate the gantry 360 degrees. (Keep one finger on the Gantry button.)
- Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
 - Listen for any loose parts. (Tighten parts as needed.)
- 5.) Turn ON axial drive enable switch AXIAL_DRIVE.

⚠ WARNING

MAKE SURE THERE ARE NO OBSTRUCTIONS AROUND THE GANTRY. PRESSING THE ALIGNMENT LIGHT PUSHBUTTON WILL CAUSE THE GANTRY TO ROTATE.

- 6.) Press the alignment light push button.
- 7.) Verify that the gantry rotates.
- 8.) Perform a 4 second X-ray OFF scan.

⚠ NOTICE

During the scan, it may be necessary to enter the scan room, to obtain a better listening position. If so, keep a finger on one of the four E-STOP buttons (on the gantry), to quickly stop the gantry, if necessary.



- a.) From the console, click on the SERVICE DESKTOP icon.
- b.) Select DIAGNOSTICS.
- c.) Select DIAGNOSTIC DATA COLLECTION
- d.) Set the scan time to 4.00 seconds
- e.) Leave the door open. (This makes it easier to hear any loose or interfering parts.)
 - * Listen for any interference between the rotating and stationary parts. (Correct any interference problems.)
 - * Listen for any loose parts. (Tighten parts as needed.)
- 9.) After completing the 4 second scan, repeat steps 9a through 9e, with the following scan times:
 - 2.0 second scans
 - 1.0 second scans
 - 0.7 second scans
 - 0.5 second scans

1.6 Emergency Stop Check

- 1.) Use the gantry push-buttons to advance the cradle about 0.5m (2ft) from the home position.
- 2.) Press one of the E-STOP buttons on the gantry.
- 3.) Make sure the TABLE POWER shuts off, and the green LED flashes.
- 4.) Depress one of the table elevation buttons, to verify the emergency stop disabled table elevation.
- 5.) Depress one of the cradle drive buttons, to verify the emergency stop disabled the cradle drive.
- 6.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER. (120 VAC LED stops flashing.)
- 7.) Press the other E-STOP button on the gantry.
 - a.) Make sure the Table Power shuts off.
 - b.) Manually move the cradle to the home position to make sure the cradle clutch released.
 - c.) Make sure the cradle latches securely in the home position.
- 8.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER.
- 9.) Press one of the four table tape switches to make sure the table down motion stops. Repeat with the three remaining table tape switches.
- 10.) Press the console emergency stop switch; make sure the Table Power shuts off.
- 11.) Press one of the RESET buttons to turn on X-RAY DRIVES POWER. (See [Figure 5-4](#)).



Figure 5-5 Reset buttons on Gantry and Service Switch bank.

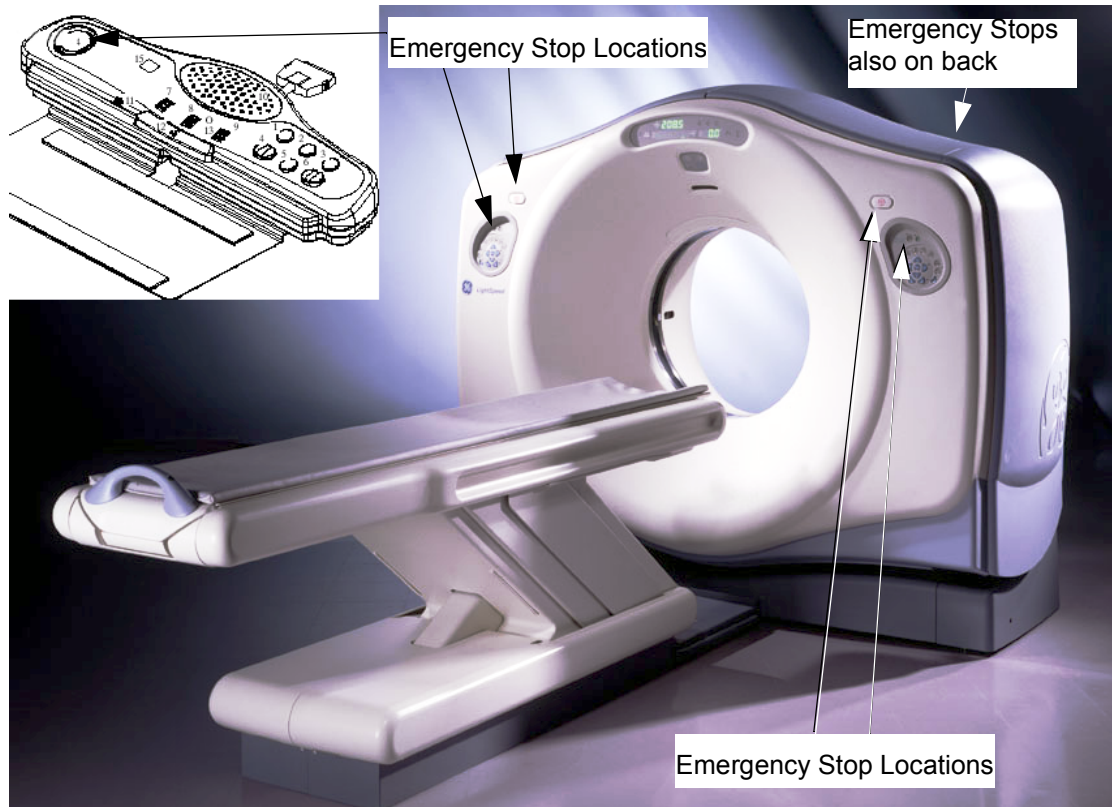


Figure 5-6 Emergency Stop button positions

Note: Emergency Stop buttons are located on the front and rear of the gantry (8 in all). They are also located on both sides of the table base (4 in all) as noted in [Figure 5-6](#).

Section 2.0 Computer Integration

2.1 Introduction

This Section describes the reconfiguration, system state restore, options, and monitor adjustment procedures.

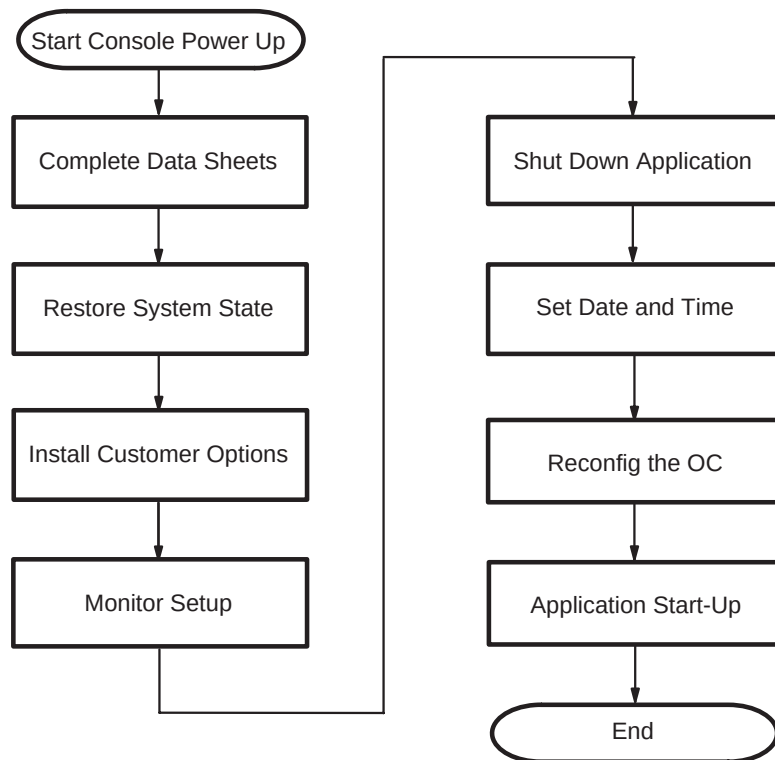


Figure 5-7 Computer Integration Process Overview

2.2 System Configuration Data Sheets

For convenient removal and use during installation, System Configuration Data Sheets appear in Appendix D. Please locate and complete to them at this point during installation.

2.3 Restore System State

Tools Required: None

Your system should have a system state DVD, located in the software collector box. The system state DVD contains:

- Collimator Characterization
- Phantom Calibrations
- Gen Cal
- Other Data

The installation process uses all the system state files. At this time, use the system state DVD to restore all files.

If you cannot locate an existing system state DVD, you must recalibrate your system.



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the PM icon.

3.) Select SYSTEM STATE.

4.) Insert the DVD in the DVD drive.

5.) Select CHARACT.

6.) Select CALS.

7.) Select RESTORE to restore the system characterization and phantom calibration files to the system.

Note: **Restore State can take as long as ten minutes, although typical times average about three minutes. When Restore State completes, dismiss the tool, and proceed to the next section.**

If any error should occur during the restore process, see the Software Load Procedure manual (Load From Cold) for information regarding possible error messages and their recovery.

8.) Click NO for Reset Scan Hardware popup message.

9.) Select DISMISS.

2.4 Install Customer Options

Tools Required: None

2.4.1 Software

Note: **Your system may have a DVD that contains customer purchased options. If your system has an options DVD, install it at this time--otherwise skip this section.**

Ensure that the options DVD is NOT write protected at this time. The initial install requires that the DVD be write enabled; subsequent installs can be done with the DVD write protected.



1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



2.) Click on the CONFIGURATION icon.

3.) Select INSTALL OPTIONS.

An Options Window appears (Figure 5-8):

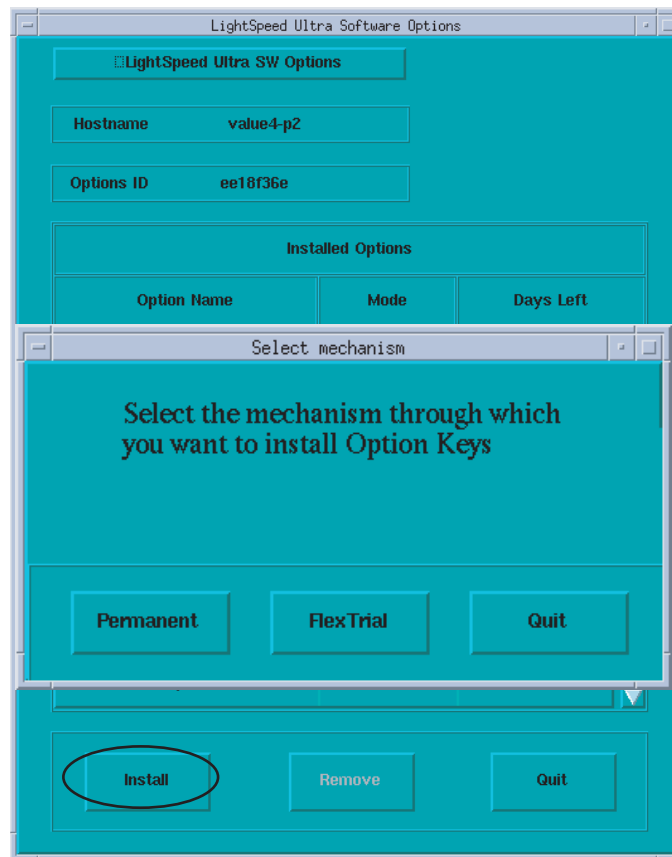


Figure 5-8 Options Window when First Selected

- Check the FDO to see what options were ordered.
- Compare FDO options to those on the Option DVD.
- If different contact your local sales representative.

- 4.) Insert the options DVD into the DVD drive and click on OK. (If you do not have an options DVD, click on OK anyway, wait for the abort pop up, then abort the process.)

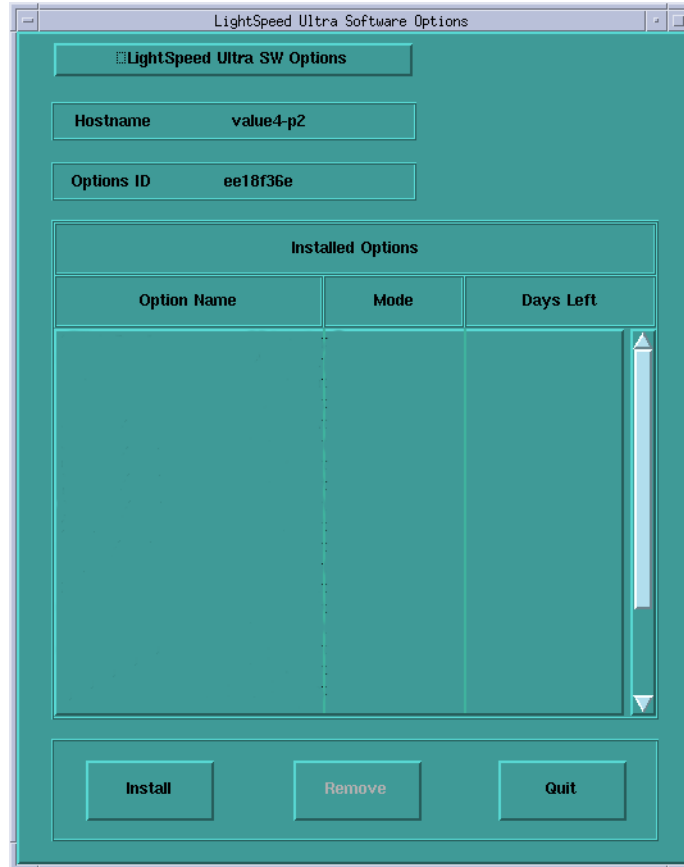


Figure 5-9 Example: Options Window

- 5.) Select all of the options in the left-hand column to install the corresponding software.
- 6.) Select INSTALL. A box may appear while the options are loading. When an option is displayed in the Installed Options list, then installation of that option is complete. Note that some options take a fraction of a second to install, while options like 3D may take a half minute (due to the fact that they are installing software).
- 7.) After the options are installed, select QUIT.
- 8.) Select OK.
- 9.) Remove the DVD and write protect the side with options.
- 10.) When the system prompts to Reboot, click YES, and reboot the system to complete the installation.

2.4.2 Camera

Tools Required:

- Small flat blade screw driver
- Data collected from data sheets (See “Camera Application Configuration” on page 249.)
- Software Load Procedures manual
- System Service manual.

Note: If a DASM is required, the DASM hardware must be installed before proceeding.
For details on camera configuration, refer to the Software Load Procedures manual.
For details on troubleshooting the camera, refer to the System Service manual.

- 1.) Click on the SERVICE DESKTOP icon.



- 2.) Select CONFIGURATION icon.
- 3.) Select INSTALL CAMERA.
- 4.) Select ADD.
- 5.) Select DASM or DICOM.
- 6.) Follow procedures on the screen.
- 7.) Return to Home Page



2.5 Monitor Setup

The LightSpeed 5.X systems are shipped with LCD monitor. For information on LCD monitor color setup, please refer to the documentation shipped with the monitor.

2.6 Contrast/Brightness Black & White Adjustment Procedure

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the IMAGE QUALITY icon.
- 3.) Select INSTALL SMPTE IMAGE and wait approximately 3-4 minutes for SMPTE image to install. (When complete the following message will display: SMPTE and QA images have been successfully copied)
- 4.) Press **ENTER** to exit the Service Desktop.



- 5.) Click the IMAGEWORKS icon.



- 6.) Display the SMPTE pattern. Use the browser to select Exam 1000, which contains the SMPTE pattern, and enlarge the image to full screen display.
- 7.) Select Viewer.
- 8.) Select 1:1 format.
- 9.) Increase the monitor's contrast to maximum.

Note: Adjust monitor contrast until the operator sees the anatomical structure (window raster)

- 10.) Increase the Brightness to maximum.
- 11.) Decrease the Brightness, until the raster just fades into, and matches, the monitor screen background. At this point, the 5% and 95% patches should be just visible.
 - If additional tweaking is required to attempt to match the monitor image to the filmed image, use only the brightness control.
 - If the CRT image exhibits any tearing or smearing of the alphanumeric characters, then reduce the contrast setting slightly until the tearing/smearing is just eliminated. The optimum setting for contrast is the highest setting that does not cause tearing/smearing of the alphanumeric characters.

- You should always finish up by displaying and filming images of anatomy (typical heads and bodies), and asking the technologist to compare the CRT image to the film image.
- 12.) To set up the text (left) monitor:
- Swap the right and left monitor's video cables to view the SMPTE pattern on the left monitor.
 - Repeat steps 2-12

2.7 Shut Down Application

Tools Required: None

Standard Level

If Applications is currently running, you must shutdown system applications.



- 1.) Click on the SERVICE DESKTOP icon.
- 2.) On the desktop toolbar select UTILITIES icon.
- 3.) Select APPLICATIONS SHUTDOWN (to bring down applications only).



Super User Level

- 1.) Open a UNIX Shell window.
- 2.) Type: `su -` ENTER at the prompt.
- 3.) Type the root (super user) password: `#bigguy`

2.8 Check/Set Date and Time

Tools Required: None

If date and time need to be corrected:

- 1.) Type/enter `setdate` ENTER.
- 2.) Follow the instructions to set the correct time and date.

2.9 Reconfig the OC

Tools Required: None

Note: The document collector box that arrived with your system contains the *Software Installation Procedures* manual, which documents the reconfiguration procedure in more detail.

2.9.1 Overview

On the following screens, you should make the changes necessary, pressing the corresponding button at the top of the screen to move from screen to screen. When you are done, you can either press the ACCEPT button to start the reconfiguration process, or press the QUIT button to exit without changing the system configuration.

While the reconfiguration is going on, messages are displayed in a shell window that closes when reconfiguration is complete. Should you later want to review the reconfiguration output, it is logged to the following file:

`/var/adm/install.log.YYYYMMDDWWHHMMSS`

Where

`YYYYMMDDWWHHMMSS` is the Date/Time that the reconfiguration was started.

To view the file, type: `more /var/adm/install.log.YYYYMMDDWWHHMMSS`

It is possible to abort the reconfiguration while entering information on the reconfiguration screens. Simply press the QUIT button at the top of the screen. There is NO safe way to abort the reconfiguration after pressing the ACCEPT button. If the entries made in the screens were incorrect, DO NOT try to stop the reconfiguration, instead wait for it to complete, and rerun reconfig, entering the correct parameters.

2.9.2 Procedure

- 1.) Change directory to scripts:
Type: `cd /usr/g/scripts` ENTER at the prompt.
- 2.) Launch the Install utility:
Type: `reconfig` ENTER at the prompt.
The OC displays the Install Utility Window as shown in [Figure 5-10](#).

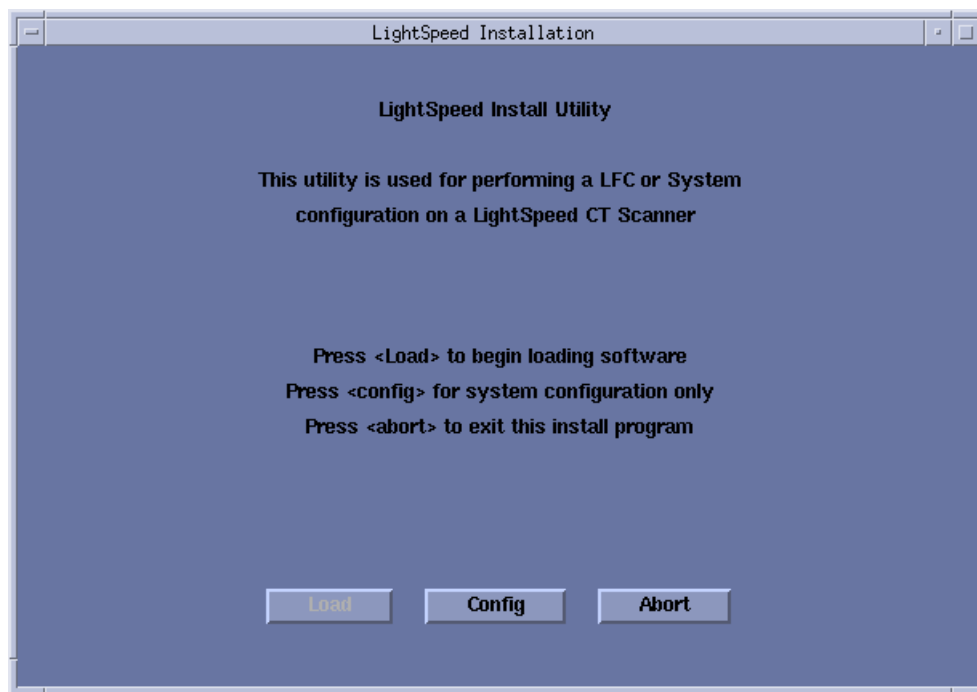


Figure 5-10 Install Utility Window

3.) Click on the CONFIG button.

The OC displays the System Configuration - System Settings Screen as shown in Figure 5-11.

Comment: The following pages show the screens that are used to change the configuration of the system. These screens are the same as those used for the Software Configuration during Load From Cold. The actual screens will vary depending on the current configuration of your system.

Figure 5-11 System Settings Screen

4.) Configure System Settings

- a.) Hospital Name (Figure 5-11, item 1) configures the name that will show up on images produced by this scanner. *Example:* ST MARYS HOSPITAL
- b.) Service ID (Figure 5-11, item 2) is issued by the Service organization.
Example: 262785CT2 (no spaces)
- c.) Select the Time Zone for the site.

Note:

Use the scrollbar at the bottom of the time-zone selection list to view the entire description of the time-zone you are about to select, to ensure that you are selecting the correct time-zone for your location.

If the time-zone of your location is not in the list above, select one of the universal times in the selection menu. In this case, automatic changes for daylight savings time will not take effect. See Load from Cold manual, if you require more information regarding time-zone setting & selection.

- d.) Next MOD # *Customer selected.*
- e.) Next Patient Exam # configures the next Exam number the scan user interface will use. At initial system installation, type: 1
- f.) Next Diagnostic Exam # *Customer Selected.*
- g.) Regenerate Database
- h.) Mobile System Select to tell the software if this CT is in a mobile environment or not.
- 5.) Select the PREFERENCES button to display the Preference Settings Screen as shown in Figure 5-12.
- 6.) Regenerate Database (refer to Figure 5-12) determines whether the Scan Database will be regenerated during reconfiguration. You should select NO.

Note:

If YES is selected, then all scan data will be deleted on the Scan Data Disk. To avoid accidental regen, NO is selected by default each time reconfig is run. Regenerating the Scan Database destroys all the scan and calibration files on the Scan Data Disk. Make sure that all images have been reconstructed before regenerating the Scan Database.

This procedure is normally only needed when a Scan Data Disk is replaced or reformatted. See Load from Cold manual, if you require more information regarding Scan Database Regeneration.

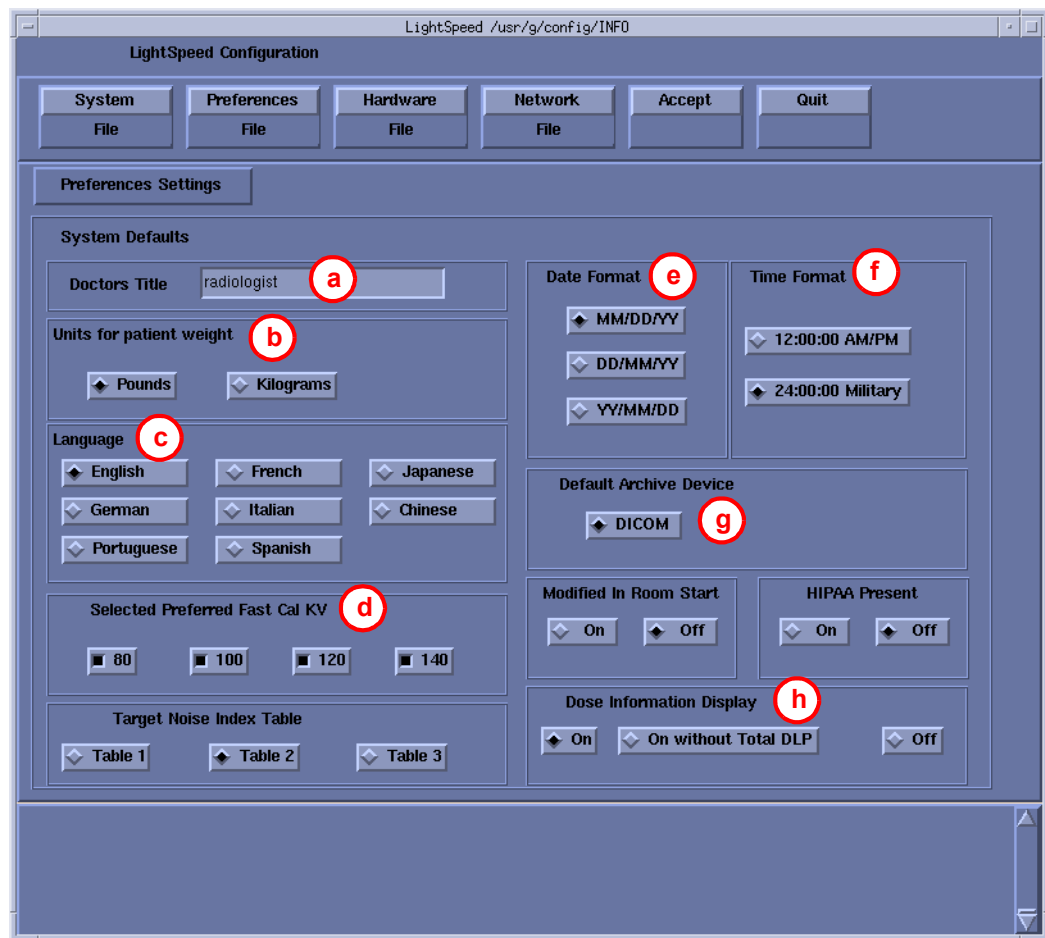


Figure 5-12 Preferences Setup Screen

7.) Configure Preferences Settings

- a.) Doctors Title Enter the title for the Doctor. (eg. radiologist)
- b.) Units for Patient Weight Tells the software whether pounds or kilograms are being used.
- c.) Language configures the language to be displayed on the Application screens (ENGLISH, FRENCH, GERMAN, ITALIAN, PORTUGUESE, SPANISH).
- d.) Preferred Fast Cal KV configures the preferred kV that the Fast Cal Routine will calibrate (80, 100, 120, 140 in the Selected Preferred Fast Cal KV field). The default selections are 120 and 140.
- e.) Date Format configure the format in which the date will be displayed on the images.
- f.) Time Format configure the format in which the time will be displayed on the images
- g.) Select the site preferred Dose Information Display option for the site to use in monitoring calculated Patient Dose:
 - * Select ON (full CTDIw Display)
 - * Select ON WITHOUT TOTAL DLP (no Dose Length Product Display), or
 - * Select OFF (no CTDIw Display)
- h.) Dose Information Display selects

Comment:

These kVs should include all kVs that the site uses for patient scanning. Deselecting All Preferred FastCal KVs is the same as selecting ALL the Preferred FastCal KVs

- 8.) Select the HARDWARE button to display the Hardware Settings Screen (Figure 5-13).

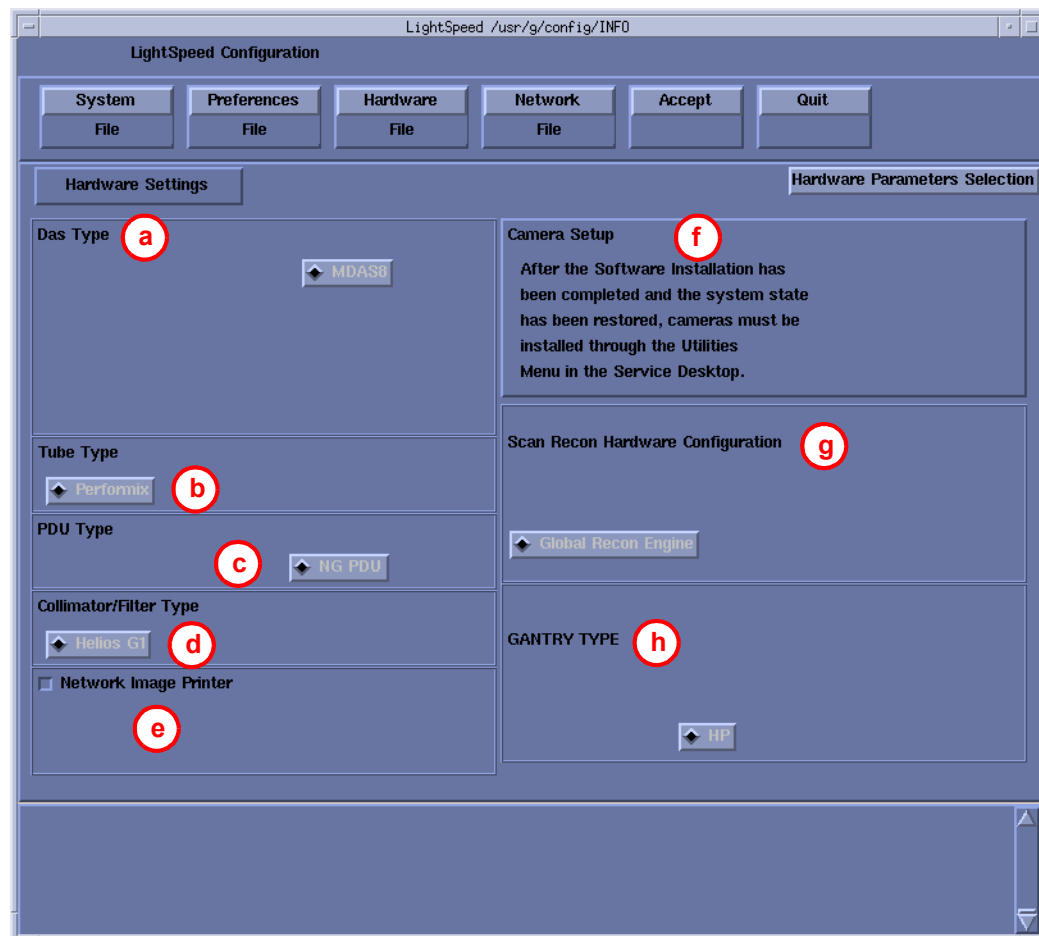


Figure 5-13 Hardware Settings Screen (example only - actual screen may vary)

- 9.) Configure Hardware Settings
- a.) DAS Type configures which DAS the software should expect.
 - b.) Tube Type configures which X-Ray Tube the software should expect.
 - c.) PDU Type configures which PDU the software should expect.
 - d.) Collimator/Filter Type configures which Collimator/Filter the software should expect. Presently, LightSpeed 5.X systems only support HELIOS G1.
 - e.) Select Network Image Printer only if the site plans to use a network connected postscript printer (such as Codonics printers) with this system.

Note: A "PRINT SERVER" IS NOT SUPPORTED (i.e., a computer with a printer attached); a Network Image Printer (which is supported) is a printer attached directly to the network.

- Comment:**
- f.) Camera Setup:
Camera Setup is not performed using Reconfig, but rather is performed from the Service Desktop once Applications have been started. See "DICOM Filming Devices Setup" on page 85.
 - g.) Scan Recon Hardware Configuration selects whether the 02 is present.
 - h.) Gantry Type selects the type of Gantry connected to this console

10.) Select the **NETWORK** button to display the Network Settings Screen as shown in [Figure 5-14](#).

Figure 5-14 Network Settings Screen

11.) Configure Network Settings

Comment: This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

See Chapter 7 for more information and complete details of setting the Hospital/System Network Configuration.

a.) Enter the Suite Name.

The Suite Name must start with a letter, followed by three alphanumeric characters. Total must be four characters long. The name of the OC interface will be <Suite Name>, within the scanner's subnet.

Typically, you should use su01 or ct01 ("su" and "ct" must be lowercase), unless the customer prefers a different suite name.

b.) Enter the hospital provided Host Name.

The Host Name identifies the network hostname and AE Title of the CT system to the hospital's network. The Host Name:

- * **MUST NOT** exceed 16 Characters.
- * **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _.

Comment: The Host Name is typically stmary or ct01.

c.) Enter the hospital provided IP Address for the system.

d.) Enter the hospital provided Net Mask (if the CT system is on a subnet).

- e.) Enter the Broadcast Address.

Comment:

The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1's or 0's depending on the configuration of the network. The standard default is 1's, but older Sun OS machines used 0's.

For example:

If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the Broadcast Address.

If the network contains genesis based scanners or other Sun OS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.

- f.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- g.) Select NIS (a.k.a. Yellow Pages database) Advanced Option—only if requested by the hospital network administrator—as follows:
- * Select ADVANCED OPTIONS button on the Network Settings Screen.
 - * Select USE NIS button.
 - * Enter the hospital provided Domain Name (from the hospital network administrator).
 - * Review all pages to be sure the information entered is correct before proceeding to the next step.

- 12.) Select ACCEPT on the System Configuration Screen.

Comment:

The system loads the CT Application Software, OS patches, kernel changes and configures the system on the OC.

The loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window that closes when the loading process is complete. All the window output is logged to a file named: _____

- 13.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES. (See [Figure 5-15](#).)

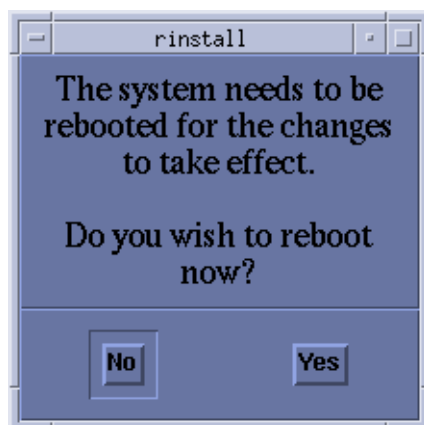


Figure 5-15 Reboot Screen

- 14.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.

2.10 Applications Start-Up

Open the Console shell window, and type: st ENTER. The applications desktop appears on the monitor.

2.11 GRE Console Boot-up Flow Chart

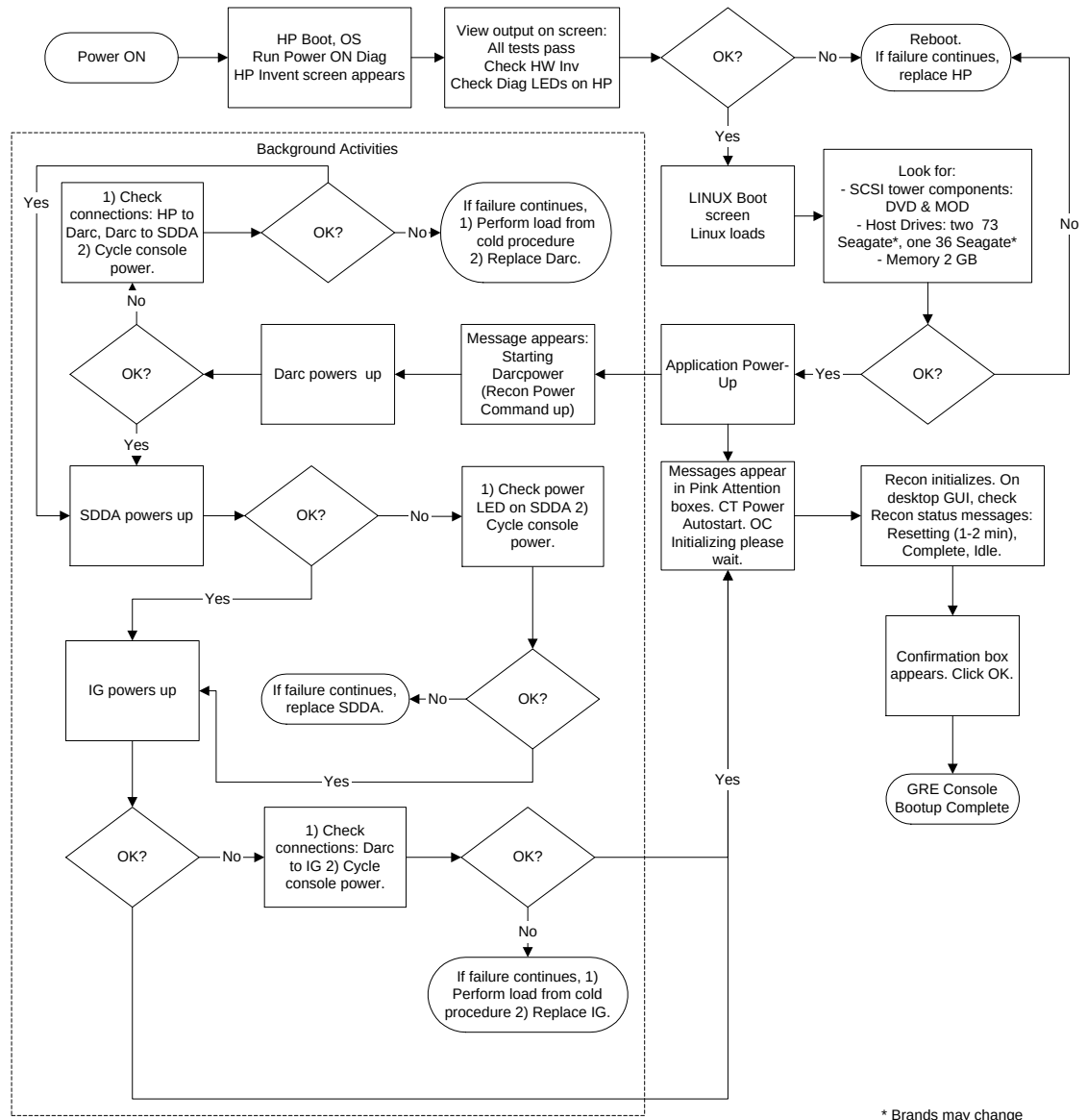


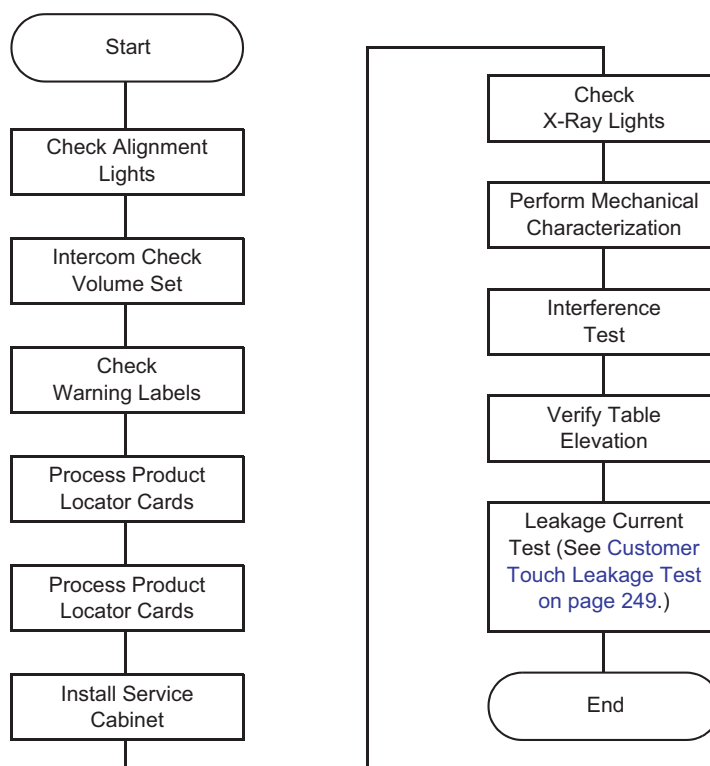
Figure 5-16 GRE Console Boot-up Flow Chart

Section 3.0 Table Gantry Integration

3.1 Introduction

Use these procedures to functionally check every part of the table/gantry subsystem.

Figure 5-17 Table Gantry Integration Process Overview



Required Tool

- Multimeter

3.2 Check Alignment Lights

3.2.1 Room Light Adjustment

Adjust the scan room lights to normal customer operating levels.

3.2.2 Turning the Alignment Lights ON



CAUTION Verify all personnel have cleared the system. The Gantry rotates during this check.

- 1.) Turn ON the AXIAL DRIVE ENABLE and HVDC ENABLE switches (located on the service switch panel).
- 2.) Turn on the alignment light switch on the gantry service panel. The gantry will rotate and the alignment lights will turn ON.



CAUTION LASER EYE INJURY!
NEVER STARE DIRECTLY INTO THE LASER BEAMS WHEN YOU OPERATE THE ALIGNMENT LIGHTS. STARING INTO THE BEAMS CAN CAUSE PERMANENT EYE DAMAGE.

3.2.3 Internal Axial Lights

- 1.) Place a sheet of plain white paper over the output port of each light.
- 2.) Verify that the two laser lines coincide and appear as a single line.

Note: GE designed the internal axial lasers on the current CT system to shine *down* on the collimator. Do NOT adjust the internal alignment lights at this time. The tomographic plane tests use the QA phantom to check the internal axial lasers alignment to the collimator.

3.2.4 External Axial to Internal Axial Distance

Note: Ensure that cradle is level.

- 1.) Raise the table to its highest elevation.
- 2.) Extend the cradle until you see both the internal and external laser lights shining on the cradle.
- 3.) Place a metric rule on the right edge of the cradle, and measure the distance from the internal axial laser line to the external axial line. Verify this distance equals 240.0 mm +1.0 mm.
- 4.) Place the rule on the left edge of the cradle, and measure again.
- 5.) Leave the cradle in its current position, and lower the table to the minimum elevation.
- 6.) Measure the distance between the internal and external lights on both edges of the cradle, as above. Verify the distance remains equal to 240.0 mm +1.0 mm.

3.2.5 Coronal Lights

- 1.) Place a sheet of plain white paper at the left side of the patient opening, in front of the coronal laser light. Verify the two coronal lines coincide.
- 2.) Move the paper to the right side of the patient opening. Verify the two coronal lines coincide.
- 3.) Place the paper in the center of the gantry opening. Use a level to verify that the coronal lines are horizontal.

3.2.6 Turn Lights OFF

Press the alignment light button on the gantry control panel, again, to turn the lights OFF.

3.3 Autovoice/Intercom Check

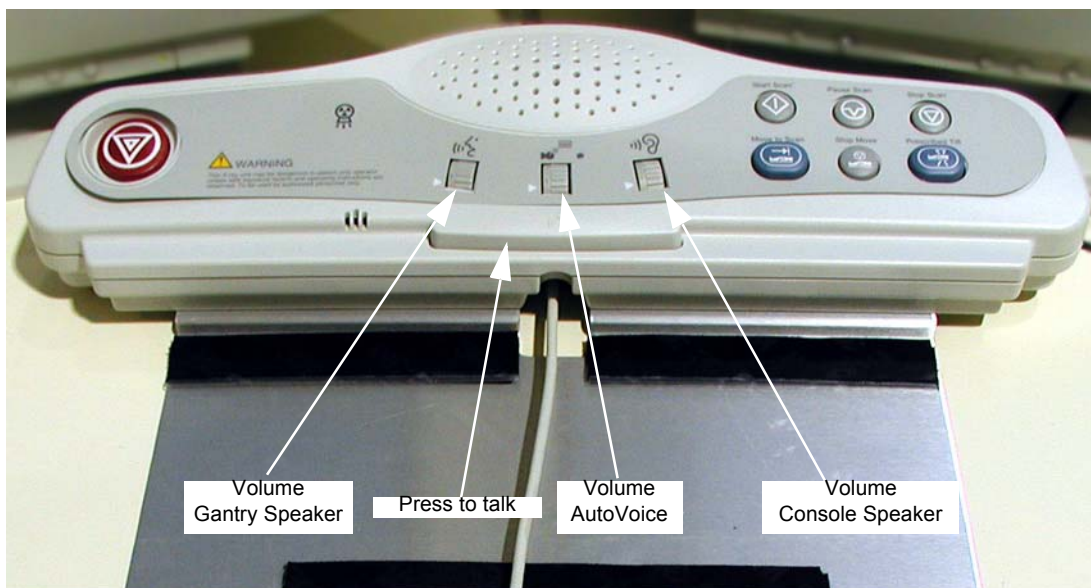


Figure 5-18 SCIM Volume Controls

3.3.1 Requirements

Two people are required to complete this procedure.

3.3.2 Patient Speaker

- 1.) To adjust the volume of the patient speaker in the table, adjust the left-most volume thumb wheel on the SCIM while speaking into the console microphone. (Press the bar on the SCIM to talk; release the bar to listen.)
- 2.) The patient should be able to clearly hear the operator.

3.3.3 Operator Console Speaker

To adjust the console speaker volume:

- 1.) Have an assistant speak into the gantry microphone.
- 2.) Adjust the SCIM console volume knob until you can clearly hear the patient.

3.3.4 Autovoice Volume

- 1.) On the Scan Desktop, select PROTOCOL MANAGEMENT.
- 2.) Select AUTO VOICE RECORD.
- 3.) Click on the 3.4 button, to the right of “FF2. Inspiration”.
- 4.) Click on the PLAY button, to play the Inspiration AutoVoice message.
- 5.) Adjust the center volume thumb wheel while Autovoice is playing, to set the volume for the gantry speaker.
- 6.) Repeat steps 4 and 5, as necessary to achieve satisfactory volume.
- 7.) Select DONE, then Select QUIT.

Note: If a satisfactory volume can not be achieved, refer to the *System Service Manual* and review the intercom module setup procedure.

3.4 CT System X-Ray ON Indicators, Cautions & Warning Labels

3.4.1 Check And Install System Warning Labels

All labels are installed in English and present on PDU, Console, Table, Gantry and Accessories. Replace the labels listed below ([Section 3.4.2](#) through [Section 3.4.7](#)) with the appropriate language labels for the country installed. Additionally, apply any other warning labels if present, on equipment where appropriate.

3.4.2 Console

3.4.2.1 On The SCIM

- Replace with the appropriate language overlay
- Replace with the appropriate language X-Ray warning label

3.4.2.2 On The Keyboard

- Replace with the appropriate language keyboard warning label
- Replace with appropriate language function key overlay label
- Replace with the appropriate language keyboard back warning label

3.4.3 Gantry

3.4.3.1 Scan Window Laser Labels

Replace with the appropriate language laser warning labels

3.4.3.2 Laser Window Labels

Replace with the appropriate language laser warning labels

3.4.3.3 Front Cover Laser Warning

Replace with the appropriate language laser warning labels

3.4.3.4 Front cover Information Labels

Replace with the appropriate language labels

3.4.3.5 Gantry System GIB (Rating) label

Replace with the appropriate language system GIB label

3.4.4 Table - 2000mm

3.4.4.1 Front Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.4.2 Back Cradle Pan - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5 Table - 1700mm

3.4.5.1 Front Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5.2 Rear Side Cover - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.5.3 Back Cradle Pan - Pinch Hazard Warning Label

Replace with the appropriate language caution label to each side of the cover

3.4.6 NGPDU

3.4.6.1 Front Cover - Emergency Off label

Replace with the appropriate language caution label

3.4.6.2 Front Cover - Gantry Enable Label

Replace with the appropriate language caution label

3.4.6.3 Front Cover - Power On Label

Replace with the appropriate language caution label

3.4.7 Accessories

3.4.7.1 Table Foot Extender Label

Replace with the appropriate language warning label

3.4.7.2 Coronal Head Holder Label

Replace with the appropriate language warning label

3.4.7.3 Sagittal Head Holder Label

Replace with the appropriate language warning label

3.4.7.4 Accessory Tray Label

Replace with the appropriate language warning label

3.4.7.5 IV Pole Label

Replace with the appropriate language caution label

3.4.8 Documentation - Verification

When finished update GE4879 (US Only) that all appropriate language labels were installed and present.

3.5 Check Warning Labels

3.5.1 On SCIM

- 1.) Make sure the X-Ray warning label appears in the correct location on the SCIM.
- 2.) Record this information on Form 4879 located in Chapter 9 of this book.

3.5.2 On Gantry

- 1.) Check that all laser warning labels are present on the gantry near the laser opening.
- 2.) There should also be warning labels on the lower right side of the gantry front cover.
- 3.) Record this information on Form 4879 located in Chapter 9 of this book.

3.5.3 On Laser

- 1.) Make sure all laser warning labels appear in the correct location on the outside of the gantry.
- 2.) Obtain and install replacements for any missing labels.

Figure 5-19 Laser Warnings and Precautions



3.6 Process Product Locator Cards

- 1.) Collect the product locator cards shipped with the system. There should be approximately 28 product locator cards with the average system.
- 2.) Update the online product locator web site with the required hospital information.
- 3.) Confirm that the serial numbers on the cards shipped with the system match those found on the web site for that GON number.
- 4.) Update as required. Place the cards in a plastic bag, then place them in the service cabinet.

3.7 Install Service Cabinet

The service cabinet is shipped disassembled. The cabinet assembly process takes about 1.5 hours to complete.

- 1.) Assemble the cabinet following the instructions located in the cabinet's shipping box.
- 2.) When completed, place the cabinet in the location shown on the site print.
- 3.) Place all of the service materials shipped with the system in the service cabinet.

3.8 Check X-Ray Lights

Perform several scans following the steps below. Verify that the X-ray ON lights are ON during the scans. When done, check the boxes in [Table 5-2](#).

- 1.) Make sure the axial drive enable and HVDC enable switches are ON.

- 2.) If you are not on the Service Desktop, click on the Service Desktop icon.
- 3.) Select DIAGNOSTICS.
- 4.) Select DIAGNOSTIC DATA COLLECTION.
- 5.) Set the scan time to 2.00.
- 6.) Set the kV to 80.
- 7.) Set the mA to 40.
- 8.) Press ACCEPT RX.
- 9.) Press START SCAN button when flashing.
- 10.) Record the above information on Form 4879 located in Chapter 9 of this book.

Light On?	Warning Light Locations
☐	SCIM
☐	Gantry Front
☐	Gantry Back (Use a mirror to view)
☐	Room Light (outside of the room)

Table 5-2 X-ray Light Chart

3.9 Mechanical Characterization

The relationship of table height to ISO center and internal-to-external landmarks must be characterized for proper interference matrix functionality.

Note: Do NOT perform tilt characterization.

3.9.1 Alignment Light Characterization

- 1.) Start the Mechanical Characterization tool from the Calibration tab on the Common Service Desktop.
- 2.) Select the CHARACTERIZE ALIGNMENT LIGHTS button from the interface.
- 3.) Follow the on-screen instructions.

3.9.2 Alignment Light Characterization

- 1.) Select the CHARACTERIZE TABLE HEIGHT button from the interface.
- 2.) Follow the on-screen instructions.

Note: If the table height is less than 21mm or greater than 25mm, relative to ISO, you must adjust the table height using the table leveling pad and adjusters. Raise or lower all four adjusters equally to achieve desired results.

3.10 Interference Test

PREREQUISITES

- Be sure that the System State was restored from **DVD** per Section 2.0.
- Reset the hardware to download the new characterization values before performing the table/gantry interference tests in this section.

CONFIGURATION

- 1.) Table flashed with latest SW
- 2.) Table must have elevation and cradle characterized
- 3.) Table must be mechanically aligned to gantry
- 4.) Table must have had the table-gantry characterization completed
- 5.) Verify the Table extender is installed.

TEST OUTLINE

The following tests verify the proper tilt and table interference matrix on the H-Power gantry.

- [Verify Table Elevation – Section 3.10.1](#)
- [Position Tilt, Move Table to Interference Limit – Section 3.10.2](#)
- [Position Table, Move Tilt to Interference Limit – Section 3.10.3](#)
- [Tilt Limits When Table Below Scan Plane Lower Limit – Section 3.10.4](#)

REQUIREMENTS

The following requirements are tested in this series of tests.

- 1.) No motion shall cause the table to hit the gantry (or gantry to hit the table)
- 2.) Requirement #1 shall include the use of the table extender.
- 3.) No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.

LIMITATIONS

These requirements will only be met when the table is NOT in service mode.

INTERPRETING TEST RESULTS

If test results indicate that elevation and/or tilt display readings fail to meet specifications, DO NOT adjust the limit switches. Instead, re-characterize and/or adjust tilt speed. Refer to Appendix C. Perform elevation and cradle first, then repeat the test. If it still fails, perform tilt.

3.10.1 Verify Table Elevation

#	TEST	EXPECTED RESULTS
3.10.1-1	Move the cradle to home position. Push the table down gantry push-button to lower the table to the minimum height.	Elevation Display should read 500.0 ± 3 mm
3.10.1-2	Push the table up gantry push button to raise the table to the maximum height. If the mechanical alignment of the table/gantry is not correct, as is often the case during manufacturing staging, this value may be out of range. Most of the following tests will still be valid: Those that might have some variation are indicated in bold type .	Elevation Display should read 25.0 ± 3 mm. If mechanical alignment of the table/gantry is not correct, this value can be as low as 0 mm and as high as 40 mm.

Table 5-3 Table Elevation Tests

3.10.2 Position Tilt, Move Table to Interference Limit

The following tests verify the table interference limits at different tilt locations.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table. Also make sure that the Gantry Tilting frame covers do not touch the stationary base covers, for all tilt angles used in this test.

#	TEST	EXPECTED RESULTS
3.10.2-1	Move cradle to home position and set internal landmark.	Cradle position on display should read 0.0
3.10.2-2	Raise table height to maximum height, set the internal landmark, move cradle into gantry 1m.	Table elevation on display should read 25.0 ± 3mm. (This value will be the same as in 3.10.1-2 above.) Cradle position on display should read 1000.0 mm.
3.10.2-3	Tilt the gantry to I30.0, then lower table until motion stops.	Table elevation on display should read 100 ± 3mm. Tilt display should read I30.
3.10.2-4	Tilt the gantry to I23.0, then lower table until motion stops.	Table elevation on display should read 155 ± 3mm. Tilt display should read I23.
3.10.2-5	Tilt the gantry to I20.0, then lower table until motion stops.	Table elevation on display should read 170 ± 3mm. Tilt display should read I20.
3.10.2-6	Raise the table elevation to maximum height.	Table elevation on display should read 25.0 ± 3mm. (This value will be the same as in 3.10.1-2 above.)
3.10.2-7	Tilt the gantry to S30.0, then lower table until motion stops.	Table elevation on display should read 156 ± 3mm. Tilt display should read S30.
3.10.2-8	Tilt the gantry to S23.0, then lower table until motion stops.	Table elevation on display should read 182 ± 3mm. Tilt display should read S23.
3.10.2-9	Tilt the gantry to S20.0, then lower table until motion stops.	Table elevation on display should read 192 ± 3mm. Tilt display should read S20.
3.10.2-10	Raise the table to 154 mm.	Table elevation on display should read 154mm.
3.10.2-11	Tilt gantry to S30 and verify the table height can be adjusted from 154 to 25 mm. (This value will be the same as in 3.10.1-2 above.)	Tilt display should read S30. Table lower limit should be 154 ± 3mm. Upper table limit should be 25 ± 3 mm. (This value will be the same as in 3.10.1-2 above.)
3.10.2-12	Set the table height to 98 mm.	Table elevation on display should read 98mm.
3.10.2-13	Tilt gantry to I30 and verify the table height can be adjusted from 98 to 25 mm. (This value will be the same as in 3.10.1-2 above.)	Tilt display should read I30. Table lower limit should be 98 ± 3mm. Upper table limit should be 25 ± 3 mm. (This value will be the same as in 3.10.1-2 above)

Table 5-4 Position Tilt, Move Table to Interference Limit Tests

3.10.3 Position Table, Move Tilt to Interference Limit

The following tests verify the tilt interference limits at different table heights.

- Note:
- “I” means top of gantry tilts toward the table base
 - “S” means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.10.3-1	Move cradle to home position and set internal landmark. Set gantry tilt to zero.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.
3.10.3-2	Raise table height to maximum height, set the internal landmark, move cradle into gantry 1m.	Table elevation on display should read 25.0 ± 3mm. (This value will be the same as in 3.10.1-2 above.) Cradle position on display should read 1000.0 mm.
3.10.3-3	Lower table until height is 115mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 115 mm. Tilt display should read S30 ±0.5°.
3.10.3-4	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 115 mm. Tilt display should read I28 ±0.5°.
3.10.3-5	Tilt the gantry to 0. Lower table until height is 200mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 200 mm. Tilt display should read S17 ±0.5°.
3.10.3-6	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 200 mm. Tilt display should read I14.5 ±0.5°.
3.10.3-7	Tilt the gantry to 0. Lower table until height is 210 mm. Tilt the gantry top away from the table ("S") until it stops.	Table elevation on display should read 210 mm. Tilt display should read S13.5 ±0.5°.
3.10.3-8	Tilt the gantry top toward the table ("I") until it stops.	Table elevation on display should read 210 mm. Tilt display should read I12 ±0.5°.

Table 5-5 Position Table, Move Tilt to Interference Limit Tests

3.10.4 Tilt Limits When Table Below Scan Plane Lower Limit

The following tests verify the table and tilt interference limits when the table height is below the scan plane.

- Note:
- "I" means top of gantry tilts toward the table base
 - "S" means top of gantry tilts away from the table base.

Important: For all tests, make sure there is 2.5 cm of clearance between the gantry and table.

#	TEST	EXPECTED RESULTS
3.10.4-1	Set gantry tilt to zero. Move cradle to home position, lower the table all the way, and set the internal landmark.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0. Table height should read 500 ± 3 mm. (This value will be the offset from 500 mm by the same amount as the upper table limit varies from 25 mm in 3.10.1-2 above.)
3.10.4-2	Tilt the gantry forward and backwards and verify the following tilt limits: S4.0 and I30.	Gantry tilt on display should read S4.0 ± 0.5°. Gantry tilt on display should read I30.0 ± 0.5°.

Table 5-6 Tilt Limits When Table Below Scan Plane Lower Limit Tests

#	TEST	EXPECTED RESULTS
3.10.4-3	Set Gantry tilt to 0. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at 58.5 mm $\pm$ 6 mm from the home position. (This value will not be valid if the table/gantry mechanical characterization is not correct as described in 3.10.1-2 above. Insure that the front of the table is at least 25 mm from the gantry.)
3.10.4-4	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0.
3.10.4-5	Raise the table to a height of 386 mm and verify the following tilt limits: S7.0 and I30.0.	Table height should read 386 mm. Gantry tilt on display should read S7.0 $\pm$ 0.5°. Gantry tilt on display should read I30.0 $\pm$ 0.5°.
3.10.4-6	Set Gantry tilt to 0. Set the internal landmark. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should stop at 91 mm $\pm$ 6 mm from the home position.
3.10.4-7	Set gantry tilt to zero. Move cradle to home position.	Cradle position on display should read 0.0. Gantry tilt on display should read 0.0.
3.10.4-8	Raise the table to a height of 242 mm and verify the following tilt limits: S12.0 and I30.0.	Table height should read 242 mm. Gantry tilt on display should read S12.0 $\pm$ 0.5°. Gantry tilt on display should read I30.0 $\pm$ 0.5°.
3.10.4-9	Set Gantry tilt to 0. Raise the table to 210 mm. Then, using the gantry push-buttons, move the cradle in towards the gantry until it is stopped.	Gantry tilt on display should read 0.0. Cradle should go all the way through the gantry bore to the full-extended position.
3.10.4-10	Set gantry tilt to zero. Move cradle to home position, lower the table all the way and set the internal landmark.	Cradle position on display should read 0.0 Gantry tilt on display should read 0.0. Table height should read 500 $\pm$ 3 mm. (This value will be the offset from 500 mm by the same amount as the upper table limit varies from 25 mm in 3.10.1-2 above.)
3.10.4-11	Tilt the gantry top toward the table to a tilt of 30 degrees	Display should read I30.
3.10.4-12	With the table down all the way, move the cradle in until it stops.	Cradle position should be 377.3 $\pm$ 3 mm.
3.10.4-13	Move the table to the home position, raise the table to a height of 386 mm, set the internal landmark, and then move the cradle in until it stops.	Cradle position should be 350 $\pm$ 50 mm.
3.10.4-14	Raise the table to a height of 100 mm, then move the cradle in.	Cradle should go all the way through the gantry bore to the full-extended position.

Table 5-6 Tilt Limits When Table Below Scan Plane Lower Limit Tests

Chapter 6

Image Quality



NOTICE
Potential for
Data Loss and/
or Equipment
Damage

To prevent potential data loss, please do the following:

- Record data collected from procedures in this chapter into Form F4879 when directed, located in [Chapter 9](#) of this book.
- Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0

Introduction

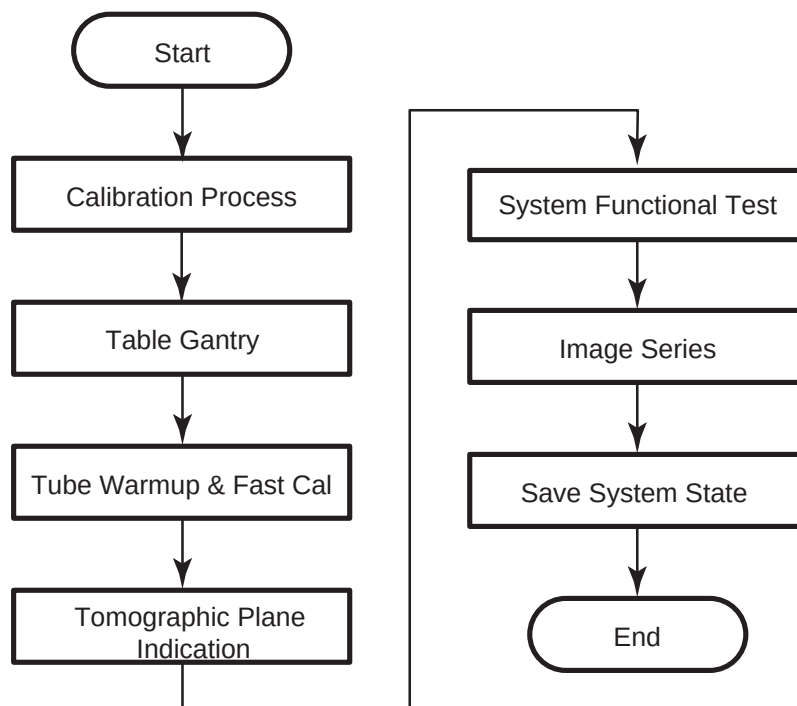


Figure 6-1 System Test Overview

Section 2.0

Calibration Process

2.1 Reference Procedures

Do NOT perform these procedures ("[Scanning with Service Protocols on page 50](#)" or "[Center Phantom on page 50](#)") until instructed to do so in other sections of this chapter

2.1.1 Scanning with Service Protocols

Locate the Manufacturing and Installation protocols under Infant area 20.

Note: **Manufacturing and Service share this Protocol list. Different product option offerings also use this list. Carefully follow the scan section instructions, and verify you acquired the images with the correct technique before filling out the data sheet. Otherwise you may troubleshoot an image problem that only exists because you used the wrong technique.**

- 1.) Select the NEW PATIENT icon on the left monitor
- 2.) Enter a Patient ID (e.g., getest).
- 3.) Click on the box labeled PEDIATRIC.
- 4.) Select a service protocol from the list, to display the corresponding view edit screen.

Optional Method: Enter the protocol number into the Protocol Number Field on the Exam Rx Screen.

2.1.2 Center Phantom

- 1.) Select SCANNER UTILITIES icon on the left monitor
- 2.) Select CENTER PHANTOM.
- 3.) Follow the on-screen procedures.
- 4.) The phantom center spec is $\pm 5\text{mm}$.
- 5.) Select QUIT, when the phantom is within specification.
- 6.) Level the phantom both front to back and side to side. (use a 6" level)

2.2 Prepare the QA Phantom

Note: **The QA phantom is shipped water filled.**

- 1.) Locate the multi-language sticker packet in the QA phantom shipping box.
- 2.) Attach the sticker with the customer's language to the face of the phantom hanger bracket.

2.3 Calibration Process Introduction

If your system has a factory supplied state MOD, you used it to load the system calibration files during the Restore System State ([2.3 on page 23](#) of this manual).

Section 3.0

Table/Gantry Alignment Procedure

3.1 Conditions

This procedure may require the removal of the table bottom, side and rail covers.

This procedure requires that mechanical final alignment was completed and passed.

3.2 Procedure

- 1.) If this is a Non-Fixed Site, check the Van level. Correct, if necessary.
- 2.) Check Table level. Correct, if necessary.
- 3.) Drive the table to its highest elevation (phantom holder removed).
- 4.) Drive the table cradle in and out five times to seat the rollers.
- 5.) Turn on the alignment lights. Check the scan window for proper installation.
- 6.) Advance the end of the cradle (black dot on cradle) to the internal light.
- 7.) Complete [Section 2.1.1, "Scanning with Service Protocols"](#) now.
- 8.) Select the service protocol, PERPENDICULAR ALIGNMENT. Protocol should appear as shown in [Table 6-1](#).

Scan Type	Direction	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm
Axial	Head First	120	40	Large	5mm/1i	2sec	S0	S0	Bone
Axial	Head First	120	40	Large	5mm/1i	2sec	I1000	I1000	Bone
Scout		120	40			90°	S0	I1000	
Scout		120	40			0°	S0	I1000	

Table 6-1 Perpendicular Alignment Scan Parameters

Set Internal Landmark. The red boxes will disappear from the screen.

- 9.) Click on CONFIRM, then press the START SCAN button when lighted, to perform the scan.
- 10.) Use over/under multi-image display to view both images at the same time:
 - a.) Click on IMAGE WORKS
 - b.) Highlight the scans just performed and click on VIEWER.
 - c.) Click on FORMAT, and select the "over/under" view (two boxes aligned vertically).
- 11.) Use the upper image (cradle front) to verify that table is centered:
 - a.) Click on the "grid" icon.
 - b.) Click on MEASURE, and select the straight (angled) line.
 - c.) Using the mouse, position one end of the line on the vertical axis.
 - d.) Position the other end at one outside edge of the cradle, forming a perfectly horizontal line (angle = 90). Read the distance displayed on screen.
 - e.) Repeat step d to measure the distance from the vertical axis to the other edge.
- 12.) Repeat step 11 for the lower image (cradle at 1000 mm).

- 13.) If the edges are off by greater than 1 mm (front to back), perform the following:
 - a.) Loosen the table anchors at locations #5, #6 and #7.
 - b.) Move the table half the measured distance.
 - c.) Return to Step 8. Repeat the procedure until the table edges line up (± 1 mm). If needed a scout can be completed to check table horizontal presentation
Check the two scout scans
 - i.) Check the tube at zero scan for table gantry perpendicularity. Make table adjustments as needed
 - ii.) Check the tube at 90 scan for cradle level and visual irregularity with the white side molding. Molding strips should be parallel within ± 3 mm. Cradle should be level within ± 3 mm.
 - d.) Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.) for fixed sites only. (75 ft.-lb.) for non-fixed systems.
 - e.) Reinstall all table parts removed to access the anchors, including covers, ground straps, elevation motor leads and the center support bar.
- 14.) Install all table and gantry covers.
- 15.) Use ERASE ALL to remove lines and grids from the images.
- 16.) Click on END EXAM to exit the program.

Section 4.0

Tube Warm Up and Fast Cal



- 1.) Select  to warm up the tube.
- 2.) Select FAST CALIBRATION from the Daily Prep menu.

Note:

Use the default Fast Cal selections determined by the system configuration. (The system defaults to all four kV stations, but you can choose kV stations to calibrate during reconfig.)

- 3.) Run the selected air calibrations.
- 4.) When the calibration process completes, click on QUIT.

Section 5.0

Tomographic Plane Indication

- 1.) Place the QA phantom on the phantom holder.
- 2.) Center the Phantom (refer to procedure [2.1.2 on page 50](#))
- 3.) Turn ON the internal alignment lights, and drive the phantom into the gantry opening, until the line on the phantom lines up with the internal laser lights.
- 4.) Verify that BOTH internal axial lasers line up along the line on the QA phantom. If not, check table/gantry, cradle, and/or laser alignment.
- 5.) Center the phantom in the scan plane with the calibration program. (See [2.1.2 on page 50](#), for details on the phantom centering procedure.)
- 6.) Select the service protocol, TOMO PLANE INDICATION. (See [2.1.1 on page 50](#), for details on scanning with service protocols.)

or

Manually select the scan parameters in [Table 6-2](#).

Scan Type	kV	mA	SFOV	Thickness	Scan Time	Start Loc.	End Loc.	Algorithm	Interval
Helical	120	200	Small	1.25HQ	1.0sec	I3.0	S3.0	Bone detail	0.2

Table 6-2 Tomographic Plane Indication Scan Parameters

- 7.) Display the image series, and locate the scan plane indicator, the longest bar in the bar pattern on the right side of the phantom. The right side of the phantom corresponds to the side of the image labeled **L** on the display screen.
- 8.) On the HHS Data Sheet, record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 9.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator. If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal $S0.0 \pm 1.0\text{mm}$. If necessary, adjust the internal alignment light position to meet the $S0.0 \pm 1.0\text{mm}$ requirement.
- 10.) Repeat the Tomographic Plane Indication test with the external alignment lights.
 - a.) Use the external alignment light, and press the external landmark.
 - b.) Verify the external light lines up along the black line on BOTH the left and right sides of the QA phantom.
 - c.) The scan plane indication must fall within the $S0.0 \pm 1.0\text{mm}$ specification.
- 11.) Initial below.

Section 6.0

Image Series

IMPORTANT: Run ALL Image Series Tests in Auto Mode. Manual procedures are provided in this section as REFERENCE ONLY.

6.1 Scan Protocol

The person who acquires the image series has the responsibility to review the images and verify they meet the specifications listed on data sheets. Responsibilities also include means and standard deviation measurements and keeping a record of failures that occur during the image series.

Unless otherwise stated, use the following scan parameters during the image series acquisition:

- Scan FOV equal to display FOV (Field of View)
- 512x512 matrix size

Note: **Consider any image series scan that does not meet specifications as failing.**

For means and standard deviations, 90% of the slices must pass. Any failure on a particular technique requires at least ten additional slices to evaluate effectively.

Systems with metal-free cradles have a phantom holder with a perpendicular adjustment (Z-axis) knob on it. **Each time you change phantoms**, make sure you use a bubble level, and the Z-axis knob on the phantom holder, to level the phantom.

6.2 Data Recording: Means and Standard Deviation

Any failure on a particular technique requires at least a ten additional slices to evaluate effectively. For means and standard deviations, 90% of the slices must pass.

- Record means to two decimal places, and round to the nearest one-tenth, (one decimal place) when you compare the resulting values to the spec.
- Record standard deviations to two decimal places, then round off to one decimal place, to compare it to the spec.
- Average standard deviations: Use two decimal places to average the values, then round off to one place.

Before you record the means and standard deviations, check the image data sheets to determine whether to average the means and standard deviations, or record them slice by slice. Make sure you record all the required image data on the HHS data sheets.

6.3 Term Definitions and Screens

Xc - Mean CT number for the specified center coordinates of the phantom image.

AvXc - Average Mean CT number for the center of the phantom image: Average the mean CT value for all specified center coordinates of all slices in an exam.

Xo - Mean CT number for the outside of the phantom image: Average the mean CT value for all specified outside coordinates of one slice.

AvXo - Average outside mean CT number for the number of slices in an exam.

AvSDc - Average image noise about the center image coordinate (measured as the standard deviation) of all slices in an exam.

AvSDo - Average image noise (standard deviation) for the outside of a phantom: Average of all outside coordinates of all the slices in an exam.

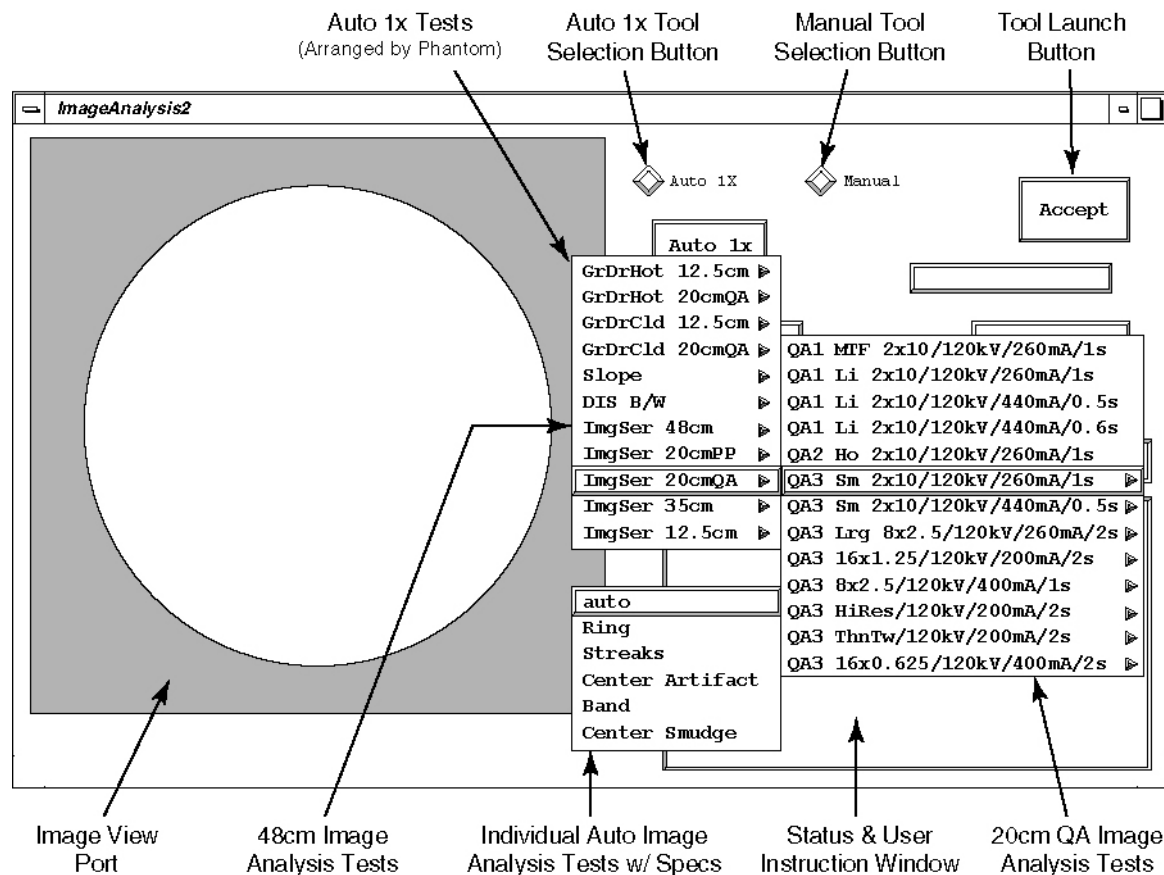


Figure 6-2 Image Analysis Tool User Interface - Auto 1x Test Pull Down Menu

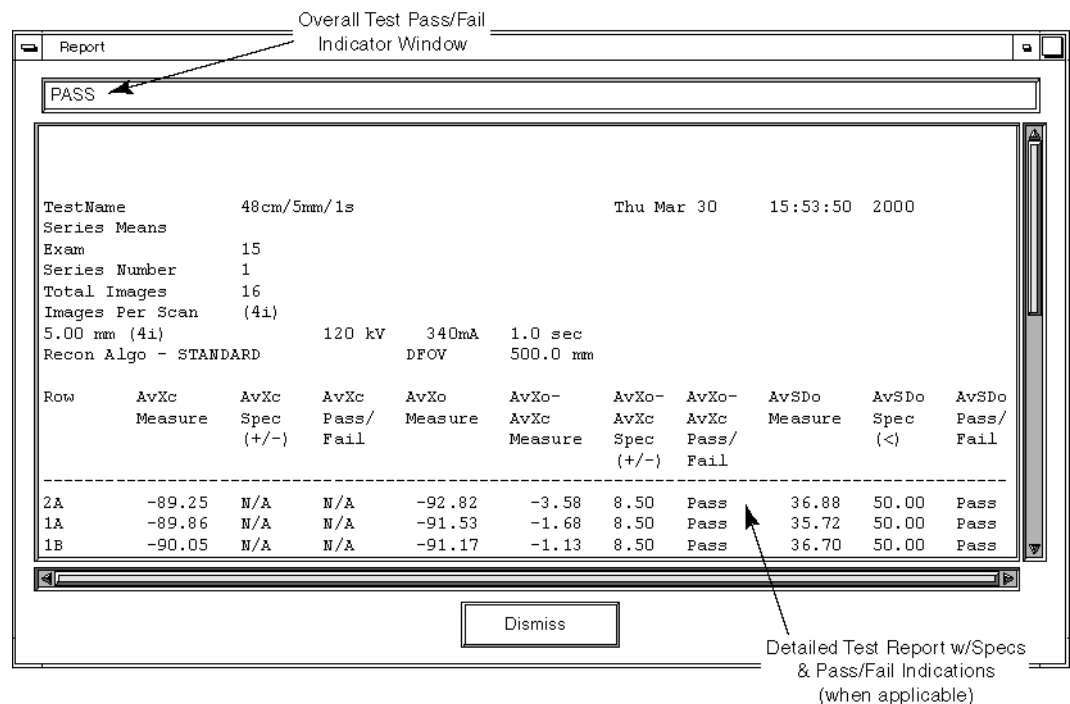


Figure 6-3 Image Analysis Tool User Interface - Test Results Report Window

6.4 48cm Phantom Image Series Image Performance Verification

6.4.1 Acquiring the 48cm Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 48cm Phantom on the Phantom Holder.
- 3.) Align, level, & center the 48cm Phantom.
 - Align phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan a single, 64 image, 48cm Phantom series.

MANUAL SCAN PROTOCOL SETUP

Refer to [Table6-3](#) and set-up an Axial scan with the parameters shown.

System	Series Description	Scan Type	Start Loc.	End Loc.	Total # of Images	Thick Speed	Interval (mm)	Gantry Tilt	SFOV	kV	mA	Total Exposure Time	DFOV (cm)	Recon Type
4 Slice	48 Lg Series 1	Axial Full 1.0 sec.	17.50	57.50	16	5.0 4i	0.00	S0.0	Large	120	340	4.0 sec.	50.0	Std

Table 6-3 48cm Phantom Image Series Scan Parameters

AUTO SCAN PROTOCOL SETUP

- a.) On the Exam Rx desktop, select NEW PATIENT.
- b.) Type the following entries in the two listed Patient Information fields:
 - * Patient ID: **Service**
 - * Name: **48cm Phantom Image Series**
- c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
- d.) On the Service Protocol Selection window, click on MANUFACTURING.
- e.) On the Service Category Protocol List, click on 45.7 IMGSER 48CM MDAS 16 16.
- f.) On the ExamRx protocol parameter display, skip to the 3rd Series.
(Series Description: 48 4x5.0/120kV/340mA/1s)
- g.) Set internal Landmark.
- 5.) Acquire a single, 64 image, 4 scan, image series of the 48 cm Phantom.

6.4.2 Image Performance Verification

6.4.2.1 Series Means (AvXo-AvXc and AvSDo) & Center Smudge

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on the IMAGE QUALITY icon.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.

Note: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a

- particular Auto 1x analysis test). Use **ALT/F1** to bring a window to the front. Use **ALT/F3** to send a window to the back.
- 2.) Using the **AUTO 1X** Tool on the Image Analysis screen, generate a Series Means and Center Smudge report for the 1st series of the 48cm Phantom acquisition.
 - a.) From the Service Browser window, select the 48cm Phantom series acquired in Section 6.4.1.
 - b.) On the Image Analysis Tool window, click on the **AUTO 1X** diamond shaped button. (See [Figure 6-2, on page 55.](#))
 - c.) Click on the **AUTO 1X** rectangular button, select the **IMGSER 48CM** test from the pull-down menu, slide cursor to the right and select **16X1.25/120KV/400MA/2S** from the test pull-down menu, and continue to slide cursor to the right to select **AUTO**.
 - d.) Click on the **ACCEPT** button.
 - 3.) Review the Image Analysis Report data and record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table 6-4](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. ([Table6-3](#))
 - b.) Verify the scan parameter values displayed in the report match those in [Table6-3](#).
 - c.) Record the Series Means (AvXo-AvXc, and AvSDc) data and the Center Smudge data for each row in [Table6-4](#).

6.4.2.2 Band and Streak Artifact

- 1.) Review the 48cm Phantom images acquired in Section 6.4.1 for Bands and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the **IMAGE WORKS** Desktop.
 - b.) From the Image Works Browser window, select the 48cm Phantom series acquired in Section 6.4.1.
 - c.) From the Image Works Browser window, click on the **VIEWER** button.
 - d.) While paging through the 64 images in the series, analyze images for any Band or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Band or Streak artifact.
 - 2.) Verify any images noted above (step 1e) meet 48cm Phantom Band or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 48cm Phantom series acquired in Section 6.4.1 that appeared to have a Band or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the **AUTO 1X** rectangular button, select the **IMGSER 48CM** test from the pull-down menu, slide cursor to the right and select **16X1.25/120KV/400MA/2S** from the test pull-down menu, and continue to slide cursor to the right to select **BAND** or **STREAK**.
 - d.) Click on the **ACCEPT** button.
- Note: If required, adjust Window/Level using the center mouse button.
- Note: Position and size the Band or Streak ROI using the left cursor button.
- Note: To ensure specifications are properly applied, do not adjust the Reference ROI's for the Band and Streak artifact tests. The Reference ROI's are adjustable after the 1st Accept Modification click.
- f.) Click on **ACCEPT MODIFICATION** twice, to generate a report.
 - g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.

h.) Repeat steps b through g, for each image noted in Step 1e.

Box Size = 1600 mm² to 2500 mm²
 45 mm (+/- 5 mm) x 45 mm (+/- 5 mm)
 45 (+/- 4 pixels) x 45 (+/- 4 pixels)

Box Positions: Box 1 = 0 mm x 0 mm
 Box 2 = 0 mm x -190 mm
 Box 3 = 190 mm x 0 mm
 Box 4 = 0 mm x 190 mm
 Box 5 = -190 mm x 0 mm

Row	Images	Brightness Uniformity ($\Delta v_{XO} - \Delta v_{XC}$)	Noise (Δv_{SDO})	Center Smudge Row Values	Comments
2A	1, 5, 9, 13				
1A	2, 6, 10, 14				
1B	3, 7, 11, 15				
2B	4, 8, 12, 16				
Specifications		< +/- 8.5	< 50	< +/- 14	

Table 6-4 48cm Phantom CT# Brightness Uniformity, Noise & Center Smudge Row Performance Data Sheet

6.4.3 Failure Recommended Actions

6.4.3.1 Series Means ($\Delta v_{XO}-\Delta v_{XC}$ or Δv_{SDO}) Failure Recovery

Specifications

Each Row of the series must pass 48cm Series Means (for the first series scan parameters) specifications:

- $\Delta v_{XO} - \Delta v_{XC}$: < +/- 8.5
- Δv_{SDO} : < 70.0

Recommended Recovery

- 1.) Repeat Detailed Cal for the 48cm Phantom.
- 2.) Repeat Sections 6.4.1 & 6.4.2, to verify Image Performance.

6.4.3.2 Center Smudge Failure Recovery

Specifications

80% the Rows must pass Center Smudge specifications when re-scanning:

Center Smudge: Smudge Factor < +/- 14.0

Recommended Recovery

- 1.) If one image of one Row fails (Row fails once):
 - Re-scan the 48cm Phantom acquiring two series of 64 images using the same protocol

- settings as listed in [Table6-3](#).
- Analyze the two 64 re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 128 re-scanned images do not fail Series Means (ΔV_{XO} - ΔV_{XC} and ΔV_{SDO}) or Center Smudge.
- 2.) If more than one image of one Row fails (Row fails more than once) or multiple rows are failing:
- Repeat Detailed Cal for the 48cm Phantom.
 - Re-scan the 48cm Phantom acquiring three series of 64 images using the same protocol settings as listed in ([Figure 6-3, on page 56](#)).
 - Analyze the three re-scanned image series using the Image Analysis IMGSER 48CM -> 16X1.25/120KV/400MA -> AUTO tool.
 - Verify that all the 192 re-scanned images do not fail Series Means (ΔV_{XO} - ΔV_{XC} and ΔV_{SDO}) or Center Smudge.

6.4.3.3 Band or Streak Artifact Failure Recovery

Specifications

- Band Artifact: Band Factor < +/- 8.0
- Streak Artifact: Streak Factor < +/- 4.0

Recommended Recovery

- 1.) Troubleshoot Band or Streak artifact failures, as described in the *System Service Manual*.
- 2.) Repeat Detailed Cal for the 48cm Phantom.
- 3.) Repeat Sections 6.4.1 & 6.4.2, to verify Image Performance.

6.5 20cm QA Phantom Image Series Image Performance Verification

6.5.1 Acquiring the 20cm QA Phantom Image Series

- 1.) Mount the Phantom Holder on the head-end of the table.
- 2.) Mount the 20cm QA Phantom on the Phantom Holder.
- 3.) Align, level, & center the 20cm QA Phantom.
 - Align black line on phantom using the internal laser lights.
 - Level phantom using bubble level and the Z Axis knob on the Phantom Holder.
 - Center phantom using the CENTER PHANTOM procedure in the left head SCANNER UTILITIES selection and the X and Y Axis knobs on the Phantom Holder.
- 4.) Set up the system to scan three 20cm QA Phantom image series with a Recon of the 1st series.
 - a.) On the Exam Rx desktop, select NEW PATIENT.
 - b.) Type the following entries in the listed Patient Information following fields:
 - * Patient ID: **Service**
 - * Name: **20cm QA Phantom Image Series**
 - c.) From the Protocol Selection display, click on the Service Tab under Anatomical Selector.
 - d.) On the Service Protocol Selection window, click on MANUFACTURING.

- e.) On the Service Category Protocol List, click on the appropriate Image Series selection as follow:

Protocol - First	Protocol - Second
Image Series QA	(none)

- f.) On the ExamRx protocol parameter display, select the 1st Series (Series Description: QA1 MTF 2x10/120kV/260mA/1s).
- g.) Set internal Landmark.
- 5.) Acquire the 1st 20cm QA Phantom image series and one Recon series (MTF and Contrast, Visible Lines).
- 6.) Acquire the 3rd 20cm QA Phantom image series (Series Description: QA2 Ho 2x10/120kV/260mA/1s).
- 7.) Acquire the last 20cm QA Phantom image series (Series Description: QA3 Sm 2x10/120kV/550mA/0.4s).

6.5.2 20cm QA Phantom Image Series Image Performance Verification

1st QA Phantom Image Series (4 Image MTF Average and Contrast Scale) Image Performance Verification

- 1.) Launch the Image Analysis Tool.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Desktop upper navigational bar, click on IMAGE QUALITY TESTS.
 - c.) In the left navigational area directory, click on IMAGE ANALYSIS.
- Note: During the Image Analysis process, you will need to switch between the Service Browser screen (to select the image series to analyze) and the Image Analysis screen (to run a particular Auto 1x analysis test). Use ALT/F1 to bring a window to the front. Use ALT/F3 to send a window to the back.
- 2.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#1 MTF and QA#1 Contrast Scale report for the 1st series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st series of the 20cm QA Phantom series acquired in section 6.5.1.
 - b.) On the Image Analysis Tool window, click on the AUTO 1X diamond shaped button. (See [Figure 6-2, on page 55.](#))
 - c.) Click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, slide cursor to the right to select QA1 MTF 2X10/120KV/260MA/1S from the test pull-down menu.
 - d.) Click on the ACCEPT button.
 - 3.) Review the Image Analysis Report data and record the Per Image MTF, 4 Image MTF Average, and Per Image Contrast Scale data for each row in Table 10-7.
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (See [Figure 6-3, on page 55.](#))
 - b.) Record Per Image MTF and Per Image Contrast Scale data for each image in [Table 6-5](#).

- c.) Record the 4 Image MTF Average in [Table 6-5](#).

Image	MTF	MTF 4-slice average	Contrast Scale	Comments
1		N.A.		
2		N.A.		
3		N.A.		
4		N.A.		
Specifications	N.A.	0.65 to 1.0	110.0 to 130.0	N.A.

Table 6-5 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (MTF and Contrast Scale)

2nd QA Phantom Image Series (4 Bone Retro Image Visible Lines) Image Perf. Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#1 Lines report for the 2nd series of the 20cm QA Phantom acquisition.
 - a.) From the Service Browser window, select the 1st image of the 2nd series of the 20cm QA Phantom series acquired in section 6.5.1 (on [page 59](#)).
 - b.) Click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, and slide cursor to the right to select QA1 LI 2X10/120KV/260MA/1S.
 - c.) Click on the ACCEPT button. The tool displays a Visible Lines pop-up window.
 - d.) On the Visible Lines pop-up window, click on the VISIBLE LINE button.
- Note: The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.
- e.) From the Visible Line pull-down menu, select the letter (A through F) that matches the smallest line pair pattern that you can discern in the Image View Port.
- f.) On the Visible Lines pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#1 Visible Lines data for the 1st image in [Table 6-6](#).
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 6-3](#), on [page 55](#).)
 - b.) Record the Per Image QA#1 Visible Lines data for the 1st image in [Table 6-6](#).
- 3.) Repeat Steps 1 and 2, for each of the remaining three images in the 2nd 20cm QA Image Series.

Image	Line Patterns Visible	Comments
1		
2		
3		
4		
Specifications	B, C, D, E, F	N.A.

Table 6-6 20cm QA#1 Phantom High Contrast Spatial Resolution Image Performance (Visible Lines)

3rd QA Phantom Image Series (4 Image Visible Holes) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a Per Image QA#2 Holes report for the 3rd series of the 20cm QA Phantom acquisition.

- a.) From the Service Browser window, select the 1st image of the 3rd series of the 20cm QA Phantom series acquired in section 6.5.1.
- b.) Click on the AUTO 1X rectangular button, select IMGSER 20CMQA from the pull-down menu, and slide cursor to the right to select QA2 HO 2X10/120KV/260MA/1S.
- c.) Click on the ACCEPT button. The tool displays a Visible Hole pop-up window.
- d.) On the Visible Hole pop-up window, click on the VISIBLE HOLE button.
- Note: The Image Analysis Tool automatically sets the Window/Level for optimal viewing. If required, adjust Window/Level using the center mouse button.
- e.) From the Visible Hole pull-down menu, select the number (1 through 5) that matches the smallest hole that you can discern in the pattern centered in the image displayed in the Image View Port.
- f.) On the Visible Hole pop-up window, click on the OK button.
- 2.) Review the Image Analysis Report data and record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6-7](#).
- a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to [Figure 6-3, on page 55](#).)
- b.) Record the Per Image QA#2 Visible Holes data for the 1st image in [Table 6-7](#).
- 3.) Repeat Steps 1 and 2 for the 3rd, 5th, and 7th images in the 3rd 20cm QA Image Series.

Image	Visible Holes Viewable at Window 20	Contrast Factor	Comments
1			
3			
5			
7			
Specifications	See Table 6-9	2.0 to 12.0	N.A.

Table 6-7 20cm QA#2 Phantom Low Contrast Detectability Image Performance (Visible Holes)

Contrast Factor Range (Box 1 Means - Box 2 Means)	Visible Number of Holes		Smallest Visible Hole Size
	Lower Limit *	Upper Limit *	
2.00 to 3.99	2	5	7.5 mm
4.00 to 7.99	3	5	5.0 mm
8.00 to 12.00	4	5	3.0 mm

* Required Number of Visible Holes depends on the Contrast Factor

Table 6-8 20cm QA#2 Phantom Visible Hole Specifications

Last QA Phantom Image Series (QA#3 Small) Image Performance Verification

- 1.) Using the Auto 1x Tool on the Image Analysis screen, generate a QA#3 Small report for the last series of the 20cm QA Phantom acquisition.
- a.) From the Service Browser window, select the last series of the 20cm QA Phantom series acquired in section 6.5.1 (on [page 59](#)).
- b.) Click on the AUTO 1X rectangular button, select IMGSER 20CMQA from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select AUTO. (Refer to

Figure 6-3, on page 55.)

- c.) Click on the ACCEPT button.
- 2.) Review the Image Analysis Report data and record the QA3 Small data for the last 20cm QA Image Series in Table 6-9.
 - a.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS. (Refer to Figure 6-3, on page 55.)
 - b.) Record the QA3 Small data (Row 2A1A and Row 2B1B Series Means $AvXc$ and $AvXo - AvXc$), QA3 ($AvSDc$), and the Center Smudge Row data in Table 6-9.

Box Size = 196 mm² to 256 mm²

15 mm (+/- 1 mm) x 15 mm (+/- 1 mm)

31 (+/- 2 pixels) x 31 (+/- 2 pixels)

Box Positions:

Box 1	=	0 mm x 0 mm
Box 2	=	0 mm x -80 mm
Box 3	=	80 mm x 0 mm
Box 4	=	0 mm x 80 mm
Box 5	=	-80 mm x 0 mm

Row	Images	AvXc	AvXo	AvXo - AvXc	AvSDo	AvSDc	Center Smudge Row Value	Comments
2A1A	1, 3, 5, 7				N.A.			
1B2B	2, 4, 6, 8				N.A.			
Specifications		+3.0 to -3.0	N.A.	+3.0 to -3.0	N.A.	+ 2.9 to +3.5	+2.2 to -2.2	

Table 6-9 20cm QA#3 Phantom CT#, Brightness Uniformity, Center Noise & Center Smudge Row Performance Data Sheet

- 3.) Review the last 20cm QA Phantom images acquired in Section 6.5.1 for Rings, Bands, and Streaks using the Image Works Desktop.
 - a.) From the Global Control Palette, click on the IMAGE WORKS DESKTOP.
 - b.) From the Image Works Browser window, select the last 20cm QA Phantom series acquired in section 6.5.1.
 - c.) From the Image Works Browser window, click on the VIEWER button.
 - d.) While paging through the eight images in the series, analyze images for any Ring, Band, or Streak artifacts. Adjust Window/Level using the center mouse button, as required.
 - e.) Note any image that appears to have a Ring, Band, or Streak artifact.
- 4.) Verify any images noted above (step 3e) meet 20cm QA Phantom Ring, Band, or Streak Image Performance requirements.
 - a.) From the Global Control Palette, click on the Service Desktop.
 - b.) From the Service Browser window, select an image of the 4th 20cm QA Phantom series acquired in section 6.5.1 that appeared to have a Ring, Band, or Streak while reviewing the series in Image Works.
 - c.) On the Image Analysis Tool window, click on the AUTO 1X rectangular button, select the IMGSER 20CMQA test from the pull-down menu, slide cursor to the right to select QA3 SM 2X10/120KV/260MA/1S from the test pull-down menu, and continue to slide cursor to the right to select RING, BAND, or STREAK.
 - d.) Click on the ACCEPT button.
 - e.) Position and size the Ring, Band, or Streak ROI, using the left cursor button.

Note: If required, adjust Window/Level using the center mouse button.

Note: To ensure specifications are properly applied, do not adjust the Reference ROIs for the Ring, Band, and Streak artifact tests. The Reference ROIs are adjustable after the 1st Accept Modification click.

- f.) Click on ACCEPT MODIFICATION twice to generate a report.
- g.) Verify the Overall Test Pass/Fail Indicator Window of the Report display indicates PASS.
- h.) Repeat steps b through g for each image noted in Step 3e.

6.5.3 Failure Recommended Actions

6.5.3.1 1st Image Series (4 Image MTF Average) Failure Recovery

Specifications

The 4 Image MTF Average must pass specifications:

4 Image MTF Average: 0.65 to 1.00

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.2 2nd Image Series (Visible Lines) Failure Recovery

Specifications

The Largest 5 line pair patterns (coded F, E, D, C, and B) must be visible for each of the four images in this series.

Recommended Recovery

Note: The most common failure for this test is that the phantom has air bubbles that are obscuring the line pair patterns.

- 1.) Carefully inspect the 20cm QA Phantom for air bubbles. If required, refill the phantom with water, to eliminate all air bubbles.
- 2.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.3 3rd Image Series (Visible Holes) Failure Recovery

Specifications

The number of visible holes in this image series must pass specifications as follows:

- Contrast Factor = 2.0 to 3.99:
Minimum of 2 holes must be visible (10mm & 7.5mm)
- Contrast Factor = 4.0 to 7.99:
Minimum of 3 holes must be visible (10mm, 7.5mm & 5mm)
- Contrast Factor = 8.0 to 12.0:
Minimum of 4 holes must be visible (10mm, 7.5mm, 5mm, & 3mm)

Recommended Recovery

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

6.5.3.4 4th Image Series Failure Recovery

Series Means

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 Series Means specifications:

- AvXc: $< \pm 3.0$
- $\text{AvXo} - \text{AvXc:}$ $< \pm 3.0$

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

QA#3 (AvSDc)

SPECIFICATIONS

Both Rows (2A1A, 1B2B) of the series must pass QA#3 AvSDc specifications:

- $\text{AvSDc} (< 5\text{K Scans}):$ 2.90 to 3.50
- $\text{AvSDc} (> 5\text{K Scans}):$ 2.80 to 3.60

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

Center Artifact or Center Smudge

SPECIFICATIONS

All eight images of the series must pass 20cm QA#3 Center Artifact and Center Smudge specifications:

- Center Artifact Factor: ± 3.5
- Center Smudge Factor: ± 2.2

RECOMMENDED RECOVERY

- 1.) Repeat Detailed Cal for the 20cm QA Phantom.

-
- 2.) Repeat Auto CT# Adjust for the 20cm QA Phantom.
 - 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.
-

Ring, Band, or Streak Artifact Failure Recovery

SPECIFICATIONS

- Ring Artifact: Ring Factor < +/- 4.8
- Band Artifact: Band Factor +/- 2.8 from 0cm to 8.5cm
< +/- 2.8 > 8.5cm
- Streak Artifact: Streak Factor < +/- 4.0

RECOMMENDED RECOVERY

- 1.) Troubleshoot Ring, Band or Streak artifact failures as described in the Service Manual.
- 2.) Repeat Detailed Cal for the 20cm QA Phantom.
- 3.) Repeat Sections 6.5.1 & 6.5.2, to verify Image Performance.

Section 7.0

System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer. Although the means, standard deviation, and resolution specifications do not apply during system functional tests, treat any artifact or image anomaly as a failure.

If you encounter a failure during the system tests:

- Record any evidence of artifacts, such as rings, streaks, shading, cupping, noise, or center artifacts.
- Correct artifacts, system test, or image series failures when they occur. Any delay in repairs could increase the number of retests.

- 1.) Place the QA phantom on the cradle.
 - Drive the table to an elevation of 100.
 - Align the line on the phantom with the internal laser lights.

Never scan above 50mA without first placing a phantom in the field of view. Levels in excess of 50mA can cause temporary radiation damage to the detector that lasts several hours. If you acquire image series cals with a radiation damaged detector, the cals may cause artifacts in subsequent image series scans.

- 2.) Select the service protocol, SYSTEM SCAN/CUST QUAL RELI.

Note: **Stop the service protocol after the second helical series. It is not necessary to proceed past that point in the protocol.**

- 3.) Using Diagnostic Data Analysis, review the MSD Plots for each AIR scan.
- 4.) Complete the scans.

Section 8.0

Save System State

Use the following commands to create the System State DVD.

1.) Load a DVD into the mod drive on the front of the console.

2.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon, .

3.) Click on the PM icon.



4.) Select SYSTEM STATE to open the System State Save/Restore menu.

5.) Select ALL

6.) Select SAVE

7.) When the save operation completes, select FILE and QUIT from the pull down menu.

8.) Remove the MOD from the drive.

Chapter 7

Customer Options Installation & Verification

Note: Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

Section 1.0 CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
 - * Direct 3D
 - * SmartScore Pro
 - * CT Perfusion
- ☐ Teleradiology Connections (Refer to [Section 10.0 on page 98](#))
- ☐ Auxiliary Monitor
 - * Flat Panel
 - * CRT

Section 2.0

Install Options

Refer to:

- GE Prints and schematics for mechanical (physical) location of option
- FDO shipment for identification of items
- Installation Specialist for installation instructions if they differ from print

2.1 5 inch MOD

Refer to documentation shipped with 5" MOD.

- 5" MOD Drive Option 5120693

2.2 Camera (Filming Device)

Refer to documentation shipped with camera

- Chapter 3 of this manual
- DICOM 2210573 GE Document
- DICOM Print 2152913

2.3 Injector

Refer to documentation shipped with the injector.

Complete a Functional test of the injector.

- The position of the ceiling mount assembly must allow the ceiling stand movement and gantry to tilt, typically 20 degrees, without interference.

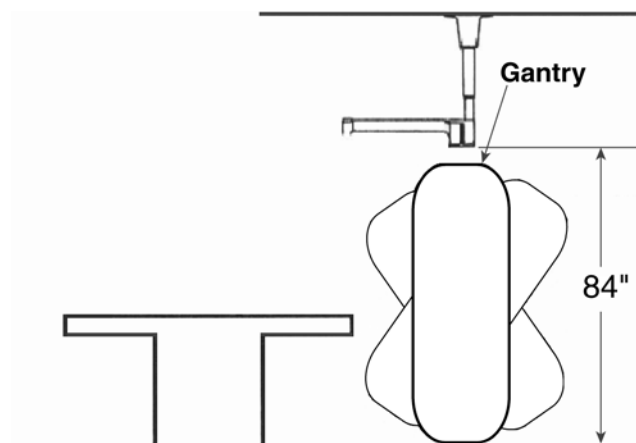


Figure 7-1 Ceiling Mount Position

2.4 Advantage Windows Workstation (AWW)

Refer to the directions provided:

- Pre-install 5117715
- Service 5117807

2.5 Uninterruptible Power System (UPS)

TBD

2.6 Other Options

Refer to CT-2211224-100 *Lightspeed Options* for all other options.

Section 3.0

DICOM Network Introduction

LightSpeed systems support two basic Networking Protocols:

- Advantage NET (IC, Signa 4.X, CT-HLA, CT/I ...)
- DICOM (CT/I, CT Synergy, Advantage Workstations, ...)

DICOM NETWORK INTRODUCTION

DICOM networks basically operate on the **tasks** or services that various devices on the network use or provide. These services are labeled as Application Entity Titles (AE Titles). The CT scanner system is a **user** of six DICOM Network Services and is a **provider** of two DICOM Services:

AS A DICOM SERVICE USER:

- Send or **Push** images to another network device.
- Send or **Push** images to a DICOM Printer.
- Review image database on another device and retrieve or **Pull** selected images from that device (Query/Retrieve User).
- Send or **Push** images to a an image storage device and obtain confirmation that the images have been archived (Storage Commitment).
- Obtain Patient Worklist Information from the Hospital HIS/RIS System.
- Store images on MOD media.

AS A DICOM SERVICE PROVIDER:

- Receive **Pushed** images from another network device
- Allow another network device to review the image database and to retrieve or **Pull** selected images (Query/Retrieve Provider)

For each DICOM Service that the CT system will be a **User** (except for storing images on MOD media), you must **declare** this device on the CT system using three menu selections. For some devices, you must declare not only the device, but each service (AE Title) that the device provides. For example, you may be required to declare a PACS System twice on the CT system: once as a destination to **push** images and, second, as destination that provides storage commitment capability after images have been **pushed**.

For each DICOM Service that the CT system will be a **Provider**, you must **declare** the CT system on the network device that will be using these services.

Information required to complete configuring a hospital DICOM network is provided by the hospital network administrator (hostnames, IP Addresses) and the DICOM Conformance Statement document (AE Titles, Port Numbers) provided with each DICOM compatible network device on the network.

Section 4.0

Before You Start

4.1 Network Physical Requirements

Before setting up the CT scanner system on the hospital network, verify the following physical items are complete:

- Scanner console, monitor, keyboard, and mouse are installed and connected.
- CT system power is ON
- Hospital Ethernet network RJ45 Class IV twisted pair cable is connected to the scanner console network receptacle.
- Hospital network connection is operational and is running 10baseT or 100baseT.

4.2 Network Identity Information

To declare the CT system on the network, ensure the following network identity information is available from the Hospital Network Administrator:

- Hostname (No more than 16 Characters)
- Internet Protocol (IP) Address
- Subnet Net Mask IP Address (if applicable)
- Broadcast Address (if applicable)

4.3 Scanner to DICOM Remote Hosts Network Information

To declare DICOM remote hosts (PACS systems, archival devices, review workstations) on the CT system, ensure the following information is available for each remote host:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address
- Network Protocol (DICOM for CT Systems)

From the Remote Host Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

4.4 Scanner to DICOM HIS/RIS Interface Network Information

To declare DICOM Hospital HIS/RIS Interface devices (Mitra and others) on the CT System, ensure the following information is available:

From the Hospital Network Administrator: *Internet Protocol (IP) Address*

From the HIS/RIS Interface Device DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM Service that the HIS/RIS interface provides)
- DICOM Listening Port Number

4.5 Scanner to DICOM Printers Network Information

To declare DICOM on the CT System, ensure the following information is available for each printer:

From the Hospital Network Administrator:

- Hostname
- Internet Protocol (IP) Address

From the Printer DICOM Conformance Statement Document:

- DICOM Application Entity Title or AE Title (DICOM service that remote host provides or uses)
- DICOM Listening Port Number

Section 5.0

Declaring the System on the Hospital Network

5.1 Enter Configuration Routine

- 1.) On the OC, open a Unix shell window.
 - 2.) Enter root as a superuser:
Type: `su -` **ENTER** at the prompt.
Type: `#bigguy` **ENTER** at the password prompt.
 - 3.) Change directory to scripts:
Type: `cd /user/g/scripts` **ENTER** at the root prompt.
 - 4.) Launch the Install Utility:
Type: `reconfig` **ENTER** at the prompt.
- The OC displays the Install Utility Window as shown in [Figure 7-2](#).

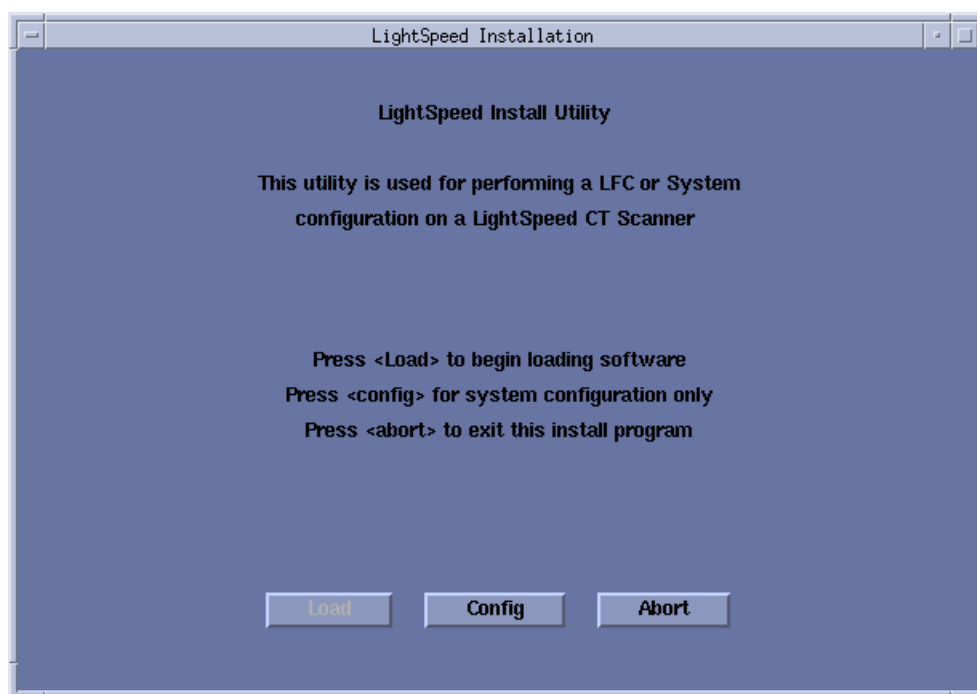


Figure 7-2 Install Utility Window

5.) Enter the Configuration Routine:

Using the mouse, click on the CONFIG button.

The OC displays the System Configuration - System Settings screen, as shown in [Figure 7-3](#).

Figure 7-3 System Settings Screen

5.2 Configure Network Settings

This screen provides the ability to declare the CT system on a hospital network. Key information such as Host Name, IP Address, Net Mask (for CT systems on a subnet) must be obtained from the hospital network administrator.

- 1.) Select the NETWORK button to display the Network Settings screen as shown in [Figure 7-4](#).
- 2.) Enter the Suite Name.

The Suite Name is a means of identifying this particular CT system as a part of a group of CT Systems in a suite configuration. This Suite Name will appear on all image headers.

The Suite Name must start with a letter, followed by three alphanumeric characters (total MUST be four characters long). The name of the OC interface will be <Suite Name>_oc and the SBC interface will be <Suite Name>_sbc.

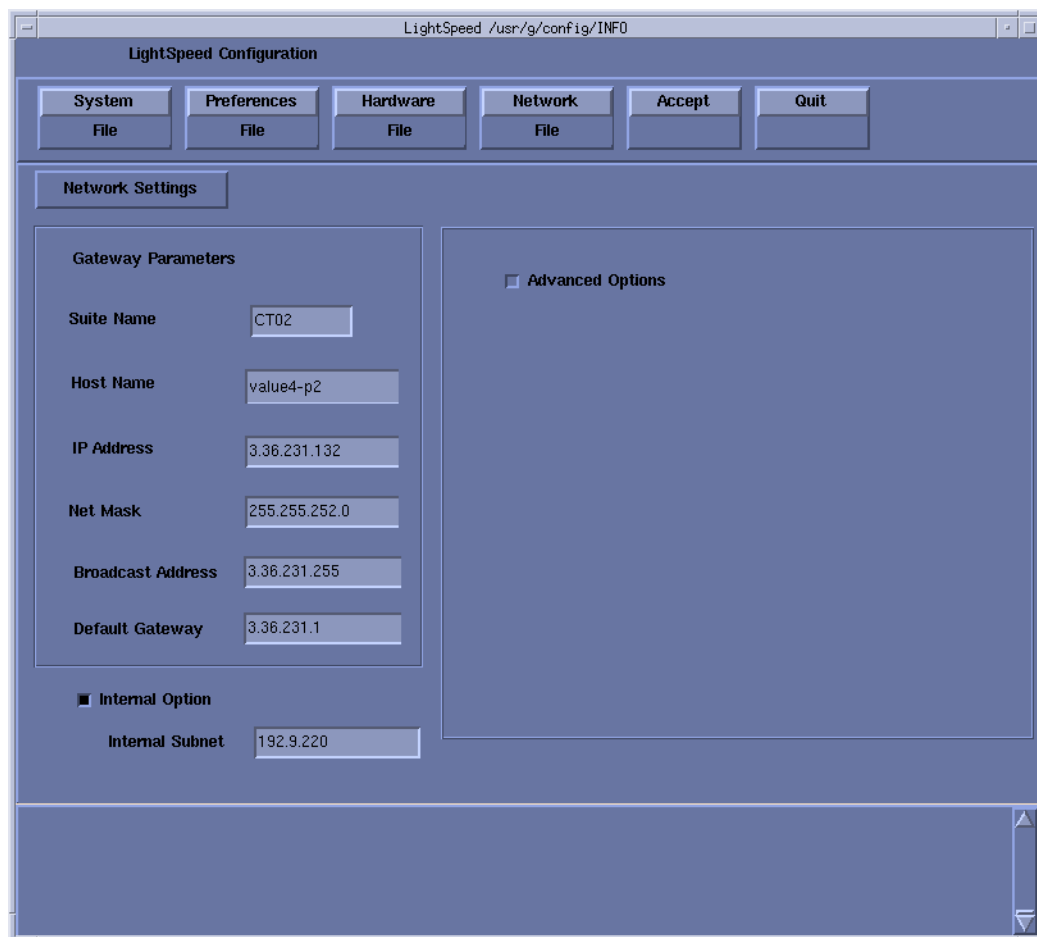


Figure 7-4 Networks Settings Screen

- 3.) Enter the hospital provided Host Name.
The Host Name identifies the network hostname and AE Title of the CT system.
The Host Name:
 - **MUST NOT** be <Suite Name>_oc or <SUITE NAME>_OC.
 - **MUST NOT** exceed 16 Characters.
 - **MUST** only contain the following characters: A through Z, a through z, 0 through 9, - and _
- 4.) Enter the hospital provided IP Address.
- 5.) Enter the hospital provided Net Mask (if the CT system is on a subnet).
- 6.) Enter the Broadcast Address:
The Broadcast Address should be the same as the IP Address except for the bits of the host id portion (last digit group) set to 1s or 0s depending on the configuration of the network. The standard default is 1s but older SunOS machines used 0s.

Example: If the IP Address is 192.100.9.17, the Broadcast Address should be 192.100.9.255 if the network is configured to use 1's to specify the broadcast address.
If the network contains genesis based scanners or other SunOS 3.5 or 4.1 computers, the Broadcast Address should be 192.100.9.0.
- 7.) Enter the hospital provided Default Gateway IP Address in the Default Gateway field (if applicable). If the site network does not use a default gateway, leave the field blank.
- 8.) Select NIS (Yellow Pages database) Advanced Option only if requested by the hospital

network administrator as follows:

- a.) Select ADVANCED OPTIONS button on the Network Settings screen.
 - b.) Select Use NIS? button.
 - c.) Enter the hospital provided Domain Name.
- 9.) Record all the Network parameters in the *Software Installation Procedures* Document, or on the worksheet in 2.2 on page 23.

5.3 Initiate System Reconfiguration

- 1.) Select ACCEPT on the System Configuration Screen.
The system loads the application software, OS patches, and kernel changes, and configures the system on both the OC and the SBC.
This loading process takes approximately 15 minutes. While the load is going on, the results are displayed in a shell window, which closes when the loading process is complete. All the window output is logged to a file named:
`/var/adm/install.log.YYYYMMDDWWHHMMSS.`
(Where *YYYYMMDDWWHHMMSS* is the Date/Time that the loading process was started.)
- 2.) When the loading process and configuration changes are complete, the system displays a prompt to reboot. Click on YES.
- 3.) The system will automatically login as ctuser after the reboot. Select OK on the Autostart Disabled popup message.
- 4.) To startup Applications, in the console shell window, type `startup` ENTER.

Section 6.0

Declaring Remote Hosts on the CT System

6.1 Enter Remote Host Configuration Screen

- 1.) On the OC, select the IMAGE WORKS icon.
- 2.) Select NETWORK.



6.2 Declaring Advantage NET Remote Hosts on the Scanner

Use Advantage NET Protocol networks to communicate with older CT or MR Systems (MR Signa 4.x, CT-HLA, CT/I Systems, and Workstations that support the Advantage NET protocol). Advantage NET Protocol does not offer full compatibility with LightSpeed DICOM formats.

Repeat the following procedure for each Advantage NET Remote Host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter Screen as shown in [Figure 7-5](#).

Figure 7-5 Advantage Net Network Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).

- 4.) Select ADVANTAGE NET as the Network Protocol.
The systems automatically de-highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM protocol parameters and do not apply to Advantage NET type devices.
- 5.) Select SAVE to store the parameter settings of the remote host.

6.3 Declaring DICOM Remote Hosts on the CT Scanner

Use DICOM protocol networks to communicate to DICOM devices such as CT/i, CT Synergy, DLX, MR Lx, and third party hosts.

Repeat the following procedure for each DICOM remote host device that the customer expects to have this CT system communicating with.

- 1.) Select REMOTE HOSTS from the pull down menu. The system displays the Remote Host Parameter screen as shown in [Figure 7-6](#).

Figure 7-6 DICOM Network Setting Protocol Parameter Settings

- 2.) Enter the hospital provided Host name.
- 3.) Enter the hospital provided Network Address (IP Address).
- 4.) Select DICOM as the Network Protocol.
The system automatically highlights the remaining parameter fields on the Remote Host parameter selection screen. These are dedicated DICOM Protocol parameters.
- 5.) Enter the TCP/IP Listening Port Number from the DICOM Conformance Statement provided with the device.

- 6.) Enter the AE Title from the DICOM Conformance Statement provided with the device.
Application Entity Titles (also known as ACR-Nema or Dicom Name) refer to the DICOM Network Services that a device provides to the CT System. For most devices, the AE Title is the same as the hostname (CT systems are equipped with this feature).
However, some devices such as PACS systems may have separate AE Titles and port numbers for each of the services that the PACS system provides. In these cases, you must enter a separate remote host (same hostname and IP Address) for each of the independent AE Title Services that the host provides (one host as an image **push-to** destination, another host as a **query/retrieve** provider, and another host as a **storage/commitment** provider).
Be sure to review the DICOM Conformance Statement for each device that will provide a remote host network service for the CT system (image **push-to** or store destination, Query/Retrieve, and Storage Commitment) to ensure that each service is correctly configured.
- 7.) Select the correct Archive Node choice for the device. The Archive Node selection field defines the ability of the remote host to act as a DICOM Storage/Commitment provider and indicate to the operator that a study/series/image was archived. Select:
 - AUTO to have the CT system automatically check to see if the designated remote host is a DICOM Storage/Commitment Provider.
 - YES if the device is the hospital designated DICOM Storage/Commitment Provider. During an Application Study Archive process, the local browser screen will indicate Archive Status = Y to the operator.
 - NO if the device is not a DICOM Storage/Commitment Provider.
- 8.) Select the correct Access to local host: settings. These two selections allow you to selectively block the remote host from using the LightSpeed DICOM services as a provider (image **push-to** destination, and a **Query/Retrieve** provider).
 - Send Images? Set to YES if the customer wants the CT system to be able to have images **pushed** to the system from the applicable remote host. Set to NO if the customer wants to block an image **push** from the applicable remote host.
 - Query/retrieve images? Set to YES if the customer wants the remote host to be able to review the image database (query) and **pull** selected images from the database. Set to NO if the customer does not want the remote host to have this ability.
- 9.) Select the correct Custom search? setting. This selection allows the CT scanner to selectively search through the remote host's image database when the operator is using remote browser screen to **query** the remote host. The search parameters that the CT system allows the customer to use are: last name contains, patient ID, exam number, accession number, and exam date.
 - Select ON if the device supports custom searches as part of the devices **Query/Retrieve** DICOM Provider service.
 - Select OFF if the device does not support custom searches.
- 10.) Record all the remote host network parameters for each remote host in the *Software Installation Procedures* Document.
- 11.) Select SAVE to store the parameter settings of the remote host.

Section 7.0

Declaring the CT System on Remote Hosts

7.1 Declaring the Scanner on Advantage NET Protocol Devices/Systems

Refer to the appropriate service manual provided with the Advantage NET Protocol device or system to find instructions how to declare the CT System as an Advantage NET remote host.

7.2 Declaring the Scanner on DICOM Protocol Devices/Systems

Refer to the appropriate Service Manual provided with the DICOM protocol device or system to find instructions how to declare the CT System as a DICOM remote host.

The CT System provides two DICOM Services as a provider to remote hosts:

- A remote host can **push** images to the CT image database.
- A remote host can review the CT image database (query) and **pull** selected images (retrieve).

Use the following parameter information to configure the DICOM device/system to either **push** images to the CT scanner and/or perform a **Query/Retrieve** operation:

- **Hostname:** Provided by the Hospital Network Administrator. Exactly the same scanner assigned hostname entered in Network Configuration Screen.
- **Application Entity Title:** Exactly the same entry as the Hostname.
- **Network Address:** Provided by the Hospital Network Administrator. Exactly the same scanner assigned IP Address entered in Network Configuration Screen.
- **Network Protocol:** DICOM 3.0.
- **Port Number:** For all DICOM service that the CT System provides, use 4006.
- **Provider Type:** This field concerns the LightSpeed DICOM **Query/Retrieve** provider capability. All CT systems are wstudy root systems, which allow queries at the exam, series, and image level.
- **Support Worklist:** This field concerns whether a DICOM **Query/Retrieve** provider capable device or system supports a filter search of the image database. All CT systems support a filtered search of the image database as part of the LightSpeed DICOM **Query/Retrieve** provider capability.

Section 8.0 DICOM HIS/RIS Setup

8.1 Prerequisites

Most hospital HIS/RIS systems are not DICOM compatible and require a DICOM HIS/RIS Worklist Interface to provide patient scheduling information to the CT system. Contact your local HNS support engineer to determine exactly what DICOM HIS/RIS Interface is appropriate for the customer.

In addition, the CT system must have the ConnectPRO software option installed to utilize the DICOM Protocol Worklist capability.

8.2 Loading ConnectPRO Software Option on the CT System

- 1.) If you are not on the Service Desktop, click on the SERVICE DESKTOP icon.



- 2.) Click on the CONFIGURATION icon.



- 3.) Click on INSTALL OPTIONS.

- 4.) Select INSTALL and click START. The console displays the Software Options window as shown in [Figure 7-7](#).

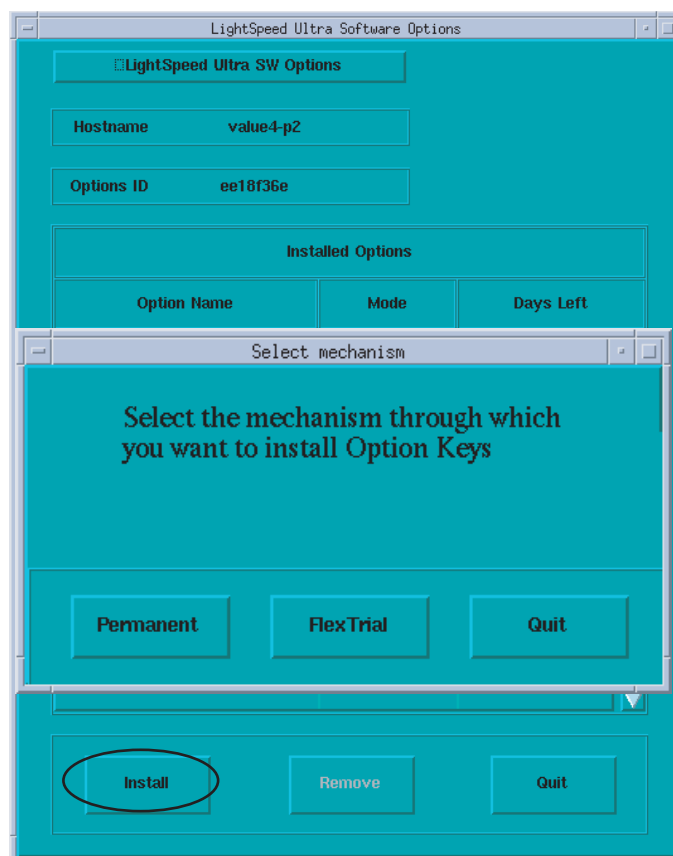


Figure 7-7 Options Window when First Selected

- 5.) Insert the ConnectPRO Options MOD into the MOD drive.
- 6.) Click on OK to continue.
- 7.) On the Software Option window, select the CONNECTPRO option in the Available Options window.
- 8.) Click on the INSTALL button.

The console may display a message box during the software loading operation. When the system has completed loading the software:

- the Installed Options window displays the ConnectPRO option, and
- the console displays the ConnectPro Setup window as shown in [Figure 7-8](#).

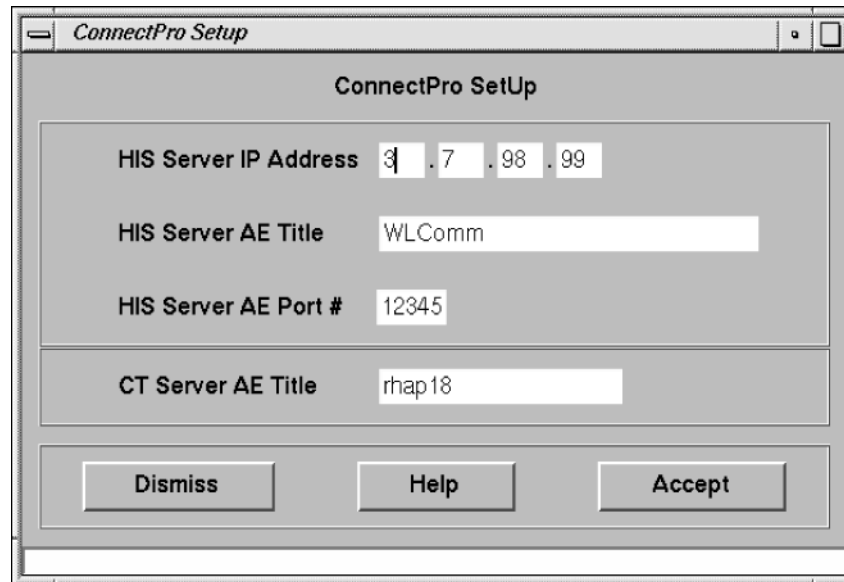


Figure 7-8 ConnectPro Setup Screen

- 9.) Enter the hospital provided HIS Server IP Address.
- 10.) Enter the HIS Server AE Title from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 11.) Enter the HIS Server AE Port # from the DICOM HIS/RIS Interface device DICOM Conformance Statement document.
- 12.) Record all the ConnectPRO HIS/RIS Network parameters in the *Software Installation Procedures* document.
The system will automatically load the LightSpeed CT Server AE Title (Hostname of the system).
- 13.) Click on the ACCEPT button. The system will check valid setup entries, execute a perl script that makes changes to the configuration files for HIS/RIS, and return to the Software Option window shown in [Figure 7-7](#).
- 14.) Click on the QUIT button and the subsequent OK button in the message window.
- 15.) Remove the MOD from the MOD drive, and write protect the side containing the ConnectPRO option.

Section 9.0 DICOM Filming Devices Setup

9.1 Prerequisites

Before configuring DICOM filming devices (cameras, printers) on the CT System, ensure the following are complete:

- Filming Device Service Representative to assist in camera/printer setup for best image quality film presentation.
- Hospital DICOM network is operational.
- Filming device is connected to the DICOM network with the correct filmer DICOM interface.
- Filming device is DICOM protocol compatible.
- Filming device DICOM Conformance Statement document is available.

Note: **Filmer DICOM Application Entity Titles may be site specific. Make sure that you check with the Filming Device Service Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination filming device.**

9.2 Declaring DICOM Filming Devices on the CT System



NOTICE
Potential For
Data Loss

Empty all filming queues before modifying camera parameters.

This section contains procedures for recording important Camera setup information. Use the table(s) at the end of this section to record information from the setup screens.



- 1.) Click on the SERVICE DESKTOP icon.
- 2.) On the Desktop Toolbar select CONFIGURATION -> INSTALL CAMERA. The Install Camera window appears, along with a warning message pop-up box, to remind you that all filming queues must be empty before you begin to update or delete a camera.
- 3.) The Graphical User Interface displayed shows a list of cameras installed (See Figure 7-10). First, you must click OK in the warning message box. See Figure 7-9.

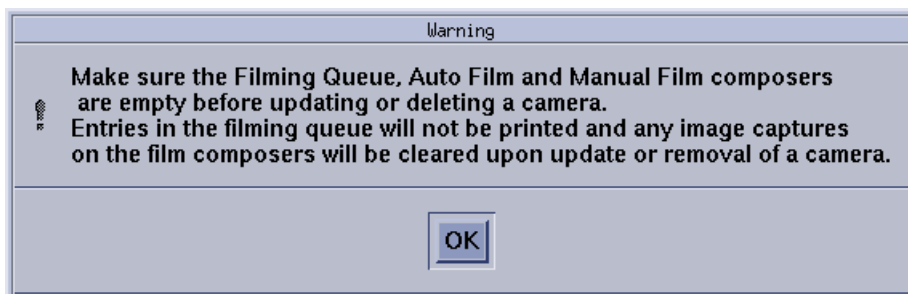


Figure 7-9 Warning Screen

- 4.) Now you are asked a series of questions.
- a.) To add a new camera, click the ADD button (See Figure 7-10).

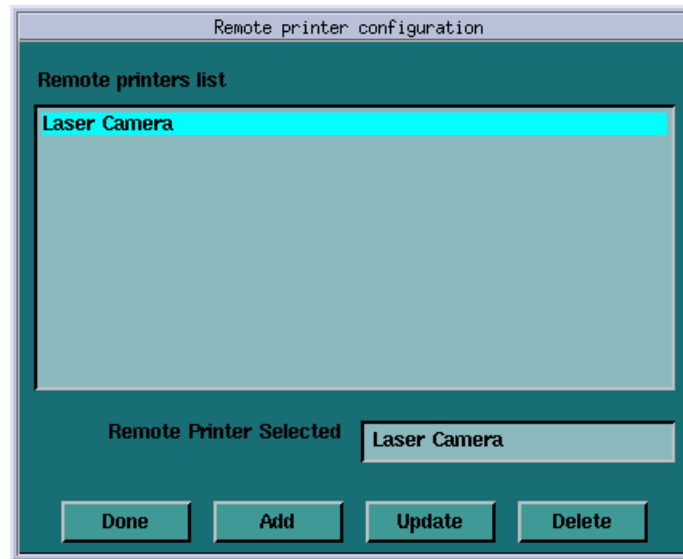


Figure 7-10 Printer Configuration GUI

- b.) A dialog window for the camera type (DASM/DICOM) appears. If no DASM is detected during the OC boot, the DASM button will be disabled (Figure 7-11). If a DASM is present and has not been detected, reboot the OC and run the camera configuration tool again.

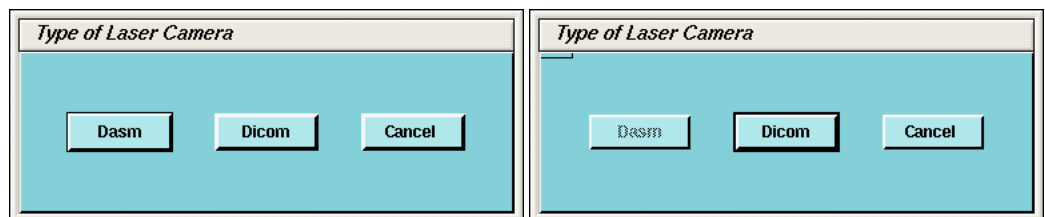


Figure 7-11 Dialog Box for camera Type

- 5.) To add a new laser camera, click DASM in the camera type dialog box. This brings up a list of available camera models. Select the appropriate model from the list and click SELECT (See Figure 7-12). Now you must configure it.

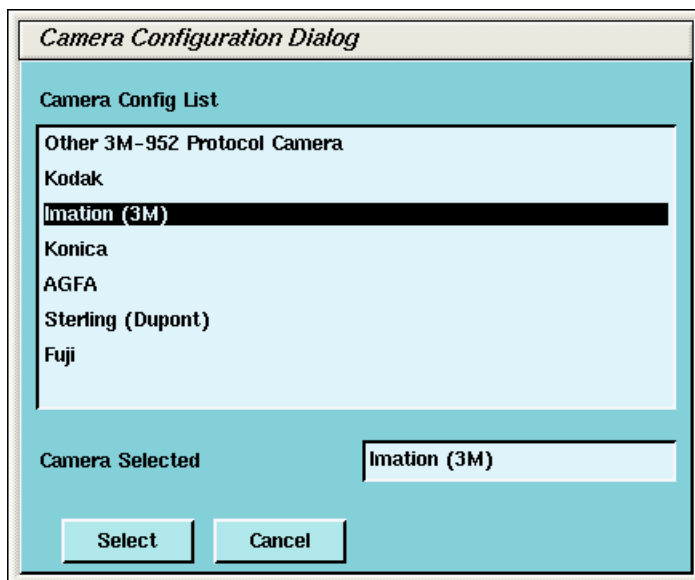


Figure 7-12 Camera Model Dialog (with DASM)

- a.) DASM Interface is automatically detected as either Analogue or Digital
- b.) Two Laser Options are available for laser cameras: SLIDES and ZOOM. Set this option only if the camera being installed supports slides and zoom. Setting the option allows it to be enabled or disabled at the application level.
- c.) Camera manufactures provide two (2) Magnification Type options for cameras. The SMOOTH resolution blurs the image, while the SHARP resolution makes the image pixels more pronounced. The default is smooth.

Comment:

To film good images, and have them look like images filmed by other GE CT products, the following camera settings are suggested:

Kodak: SMOOTH

Dupont/Sterling: SMOOTH

3M/Imation (Laser Camera): SHARP

3M/Imation (Dry View): SMOOTH

Agfa: SMOOTH

- d.) Select the appropriate File Format. Select ON from the drop down list boxes on the menu. Valid film formats are determined by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).

When finished setting parameters, click on DONE and proceed to step 8.

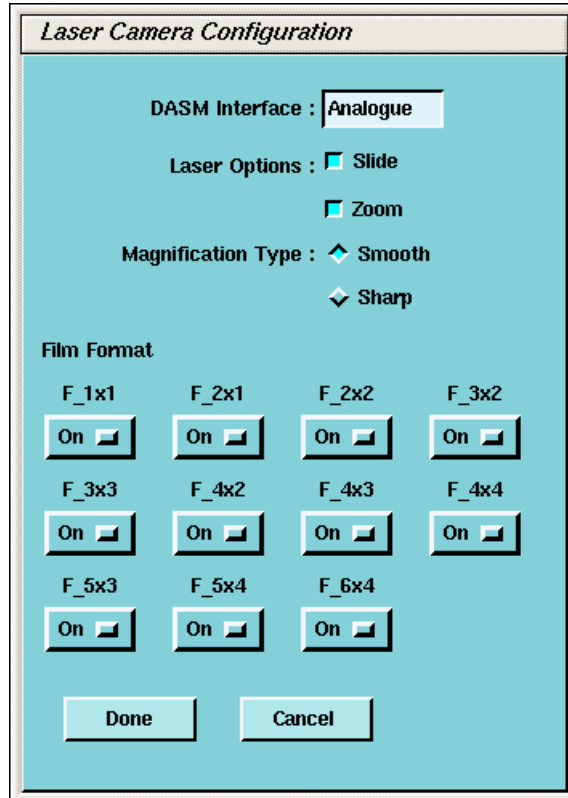


Figure 7-13 Laser Camera Configuration

- 6.) To add a new DICOM camera, click on ADD and then DICOM in the dialog box that appears.
 - a.) A list of camera models appears (See Figure 7-14). Select the appropriate model from the list and click SELECT. Clicking SELECT presets all the parameters to that models except the Network parameters.

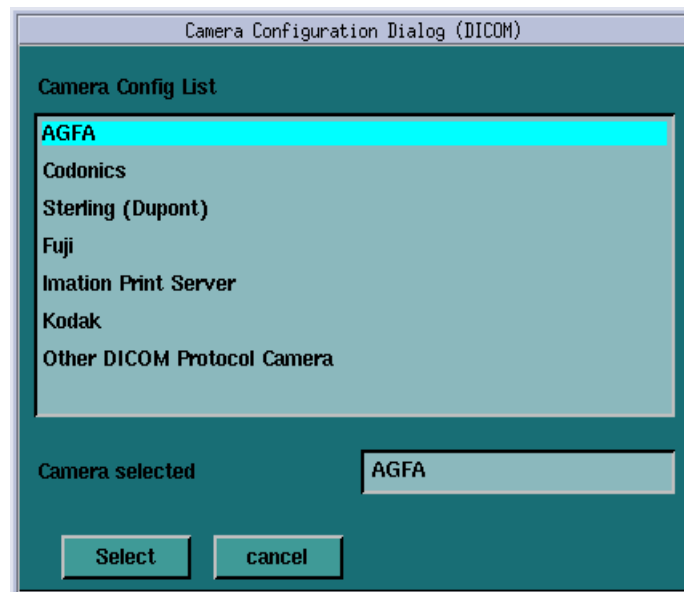


Figure 7-14 Camera Model Dialog (DICOM)

*Comment:
It's advised to
recheck the
preset
information with
the camera
manufacturer's
representative.*

Selection of a different camera model clears the Image Quality parameters, because these are camera manufacture dependent.

b.) Enter the Network Parameters (See Figure 7-15)

- > Device Name A unique name used to identify the camera.
- > Host Name DICOM print server host name, as defined by the hospital.
- > IP Address DICOM print server IP address, as defined by the hospital.
- > AE Title DICOM print server application entity title, as defined by the print server. *You should consult the manufacturers DICOM Conformance Statement.*

Note: The Application Entity Title for the Camera may be site specific. Make sure that you check with the Camera Manufacturer's Representative and the hospital network administrator to ensure you are using the correct AE Title for the destination DICOM Print Camera.

- > TCP/IP Listen Port DICOM print server TCP/IP listen port, as defined by the server. *You should consult the manufacturers DICOM Conformance Statement.*
- > Comments Optional comments used by the DICOM print server.

Figure 7-15 DICOM Camera Configuration

- c.) Medium Type specifies the type of film being used. Currently, only BLUE FILM and

CLEAR FILM are supported.

- d.) Set Destination to the final location for film output: either MAGAZINE or PROCESSOR.
- e.) Orientation selects film orientation. Only PORTRAIT is currently supported.
- f.) Set the Magnification Type. This parameter selects the algorithm used to interpolate pixels for proper film resolution. Set this parameter after consulting the camera manufacture to ensure optimal image quality. Choices are describe below:
 - > None No interpolation. This option is not suported by all cameras.
 - > Replicate Adjacent pixels are interpolated. This can result in "pixelized" images. *This algorithm is not normal preferred.*
 - > Bilinear A 1st order interpolation of pixels. Results in images usuaully described as blurred. *This algorithm is not normal preferred.*
 - > Cubic A 3rd order interpolation. Used with a large number of possible formulations. Camera manufactures define parameters called "smoothing type" to set coefficients used in this algorithm. The implementation of these "smoothing soefficients" is camera dependent.
- g.) Select the Standard Film Formats. Select the film format by choosing ON in the pull-down menu box located below each selection. See Figure 7-15. Valid film formats are set by the camera manufacture. IMATION for example, doesn't support 4x4, 2x4 or 1x2 and AGFA does not support 2x4) The DICOM print convention designates film formats by column and row (e.g. 12 on 1 film is 3x4).
- h.) After the camera has been configured, click the ADVANCED button. This creates the camera device file for you automatically and pops up the Advanced Parameters screen. See Figure 7-16.

Comment: For most Camera Manufacturers, the preferred selection is CUBIC.

Figure 7-16 Advanced Parameters (Camera Attribute Dialog)

- 7.) Advanced camera parameters control the image quality of films.

Note: For more information on the proper settings for these parameters, please see the Camera's DICOM Print Device Conformance Statement or the Camera Manufacturer Representative.

You may need to refer to a copy of the Conformance Statement as you are working with the Camera Manufacturer's Representative, to correctly set up the DICOM Print Camera I/Q and Time-out Settings.

- a.) **Configuration** - This parameter is camera manufacturer dependent as is typically used to specify the image contrast. The Configuration field may be up to 1024 characters long. The field will scroll automatically as text is entered. To review your entry, simply click and hold the middle mouse button, while the cursor is in the field, and drag the mouse towards the right (or left) as needed.

Note: Recommended Configuration Setting Values:

Agfa Drystar (MG3000) **PERCEPTION_LUT=KANAMORI (100)**
Imation Dryview (8700) **LUT=0,7**
Kodak Laser Printer 190 **CS434\CN0\PD1.20**

- b.) **Smoothing Type** - Set Smoothing Type to ON, and input the selected value. This parameter is used when the magnification type is CUBIC. It represents the coefficient for the image resolution algorithm. This parameter is camera manufacturer dependent, and should be re-verified with your radiology department.

Note: **Recommended Smoothing Type Starting Values and Ranges:**

Agfa DryStar (MG3000) Start Value: **140** Range: **137-150**
Imation Dryview (8700) Start Value: **3** Range: **3-13**
Kodak Laser Printer 190 Start Value: **Enhanced** Range: **Normal**

- c.) **Minimum and Maximum Density** - Used to set brightness of the images on film. The range of values is 0-4095, although the valid range for a specific camera is manufacture dependent. For **Maximum Density**, input the correct value into the text box. For **Minimum Density**, set it to ON and input the correct value in the text box.

Note: **Recommended Minimum and Maximum Density Starting Values:**

Agfa Drystar (MG3000) Min.: **20** or **23** Max: **300**
Imation Dryview (8700) Min.: **(Blank)** Max: **300**
Kodak Laser Printer 190 Min.: **20** Max: **300**

- d.) **Empty Image Density** - This parameter sets the density for empty film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- e.) **Border Density** - This sets the density for the border used around the film viewports. Typically, BLACK is used but WHITE is an available option. The minimum and maximum density values are used as the representation.
- f.) **Film Size** - Allows the system to specify a particular film size, if selected.
- g.) **Trim** - YES produces a white (clear) box surrounding each image.
- h.) **Priority** - This sets the print priority.
- i.) If you have completed entry of advanced parameters, click DONE.

CAMERA DATA TABLES: TO LOCATE INSTALL CAMERA INFORMATION: CLICK ON THE SERVICE DESKTOP BUTTON. ON THE DESKTOP TOOLBAR SELECT UTILITIES -> INSTALL -> INSTALL CAMERA. THE INSTALL CAMERA WINDOW APPEARS. SELECT EACH OF THE CAMERAS THAT ARE INSTALLED FROM THE LIST OF INSTALLED CAMERAS, AND CLICK ON UPDATE TO VIEW THE CAMERA'S SETTINGS. RECORD THE VALUES USED TO SET UP EACH CAMERA IN THE TABLES THAT FOLLOW. EXTRA TABLES

ARE PROVIDED FOR MULTIPLE CAMERAS

Note: You can determine this information by looking at the contents of the following files:

- For a DASM Camera: `/usr/g/ctuser/app-defaults/devices/camera.dev`
- For a DICOM Print Camera: `/usr/g/ctuser/app-defaults/devices/name.cfg`
where, *name.cfg* is the camera device name from the printer configuration GUI.

Example: `more <filename from above> ENTER`

DASM CAMERA #1 SETUP

GUI SETTING	SELECTIONS	VALUE
Camera Type	Model Type of Camera	
DASM Type	Digital or Analog	☐ Digital ☐ Analog
Options	Slides or Zoom	☐ Slides ☐ Analog
Film	Smooth or Sharp	☐ Smooth ☐ Sharp
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Film Format Default	1x1, 2x1, 2x2, 3x2, etc.	

Table 7-1

DICOM CAMERA #1

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-2 DICOM Camera #1

GUI SETTING	SELECTIONS	VALUES
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-2 DICOM Camera #1

DICOM CAMERA #2

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-3 DICOM Camera #2

GUI SETTING	SELECTIONS	VALUES
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-3 DICOM Camera #2

DICOM CAMERA #3

GUI SETTING	SELECTIONS	VALUES
DICOM Camera Type	Model Type of Camera	
Film Format Available	1x1, 2x1, 2x2, 3x2, etc.	
Network Parameters	Host Name	
	IP Address	
	AE Title	
	TCP Listen Port	
	Comments	
	Destination	☐ Magazine ☐ Processor
Special Set Up	Orientation	☐ Portrait ☐ Landscape
	Medium Type	☐ Blue ☐ Clear
	Magnification Type	☐ None ☐ Replicate ☐ Bilinear ☐ Cubic
*Advanced Parameters - IQ	Smoothing Type	☐ ON ☐ OFF Value:
	Configuration	
	Minimum Density	☐ ON ☐ OFF Value:
	Maximum Density	
	Empty Density (Black/White)	☐ Black ☐ White
	Border Density (Black/White)	☐ Black ☐ White
	TRIM	☐ YES ☐ NO
	Priority	☐ HI ☐ MED ☐ LOW
	Film Size	

NOTES

*To view Advanced DICOM Camera Settings, you must click on ADVANCED.

Table 7-4 DICOM Camera #3

9.3 Troubleshooting Tips

The following is a summary of troubleshooting information for DICOM print that was gathered during software testing and validation of the DICOM print feature.

There is also a significant amount of additional troubleshooting procedures, and the theory of the DICOM print feature in the System Service Manual. Should you have problems installing a DICOM print camera, first read the information in System Service Manual.

ERROR BRINGING UP THE CAMERA INSTALLATION/FILMING APPLICATION

- **Symptom:** After creating/modifying the DICOM print camera the install camera interface does not come up and the filming application indicates it cannot bring up the film composer.
Cause: The configuration field contains invalid information such as a backslash (\) as the final character in the field or brackets ({}).
- **Solution:** The camera.dev file in ~ctuser/app-defaults/devices must be manually edited to remove the offending characters in the set configuration line. Invalid characters include \{ }

CONFIGURATION INFORMATION FIELD

- **Symptom:** Cannot view the entire configuration field (> 25 characters)
- **Solution:** Hold down the middle mouse button and move the field contents

NEED TO SET DICOM PRINT ATTRIBUTES NOT SUPPORTED BY SOFTWARE

- **Symptom:** User wants the white border around each image box ON/OFF permanently for this system and it cannot be set as the default for the camera.
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
For Trim Off - set TRIM NO
For Trim On - set TRIM YES
- **Symptom:** DICOM print camera supports multiple film sizes and the user only wants to print if the film size is correct for LightSpeed 5.X (14x17). [Otherwise the camera will queue the films or return an error causing the queue to pause (based upon the DICOM print camera specifications).]
- **Solution:** Using your favorite editor, add the following line to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
To force a 14x17 film size - set filmSize 14INX17IN

NEED TO PREVENT DICOM PRINT ATTRIBUTES FROM BEING SENT TO DICOM PRINT CAMERA

- **Symptom:** Some DICOM print attributes are optional, and may result in fatal errors. For example, the Fuji camera does not support the Empty Image Density parameter for the film box.
- **Solution:** Using your favorite editor, add the following line(s) to the camera.dev file located in ~ctuser/app-defaults/devices after the DICOM print device has been otherwise configured.
- To prevent sending the Smoothing Parameter set FB_Smooth FALSE

- | | |
|--|-----------------|
| - To prevent sending the Border Density set | FB_Border FALSE |
| - To prevent sending the Empty Image Density set | FB_EID FALSE |
| - To prevent sending the Minimum Density set | FB_MinD FALSE |
| - To prevent sending the Trim Parameter set | FB_Trim FALSE |

ERROR TRYING TO CONNECT TO THE DICOM PRINT CAMERA

- **Symptom:** DICOM print server can be reached (ping), but Application error indicates "Unable to start filming interface" and the help message talks about running the install.dasm (Association Error)
- **Solution:** The system is unable to complete the association. Check the AE Title and the Port number of the DICOM print server and correct them through the Install Camera procedure.

FILM COMPOSER ERROR NOT USABLE

- **Symptom:** Film composer error says "unrecognized status - code 0"
- **Solution:** Review the log file, the attention and status windows. These areas have the correct filming status (for example, **film jam** and **supply empty**).

DEBUGGING CONNECTION ISSUES DIFFICULT

- **Symptom:** The timeouts for the DICOM print are very long, which means one needs to wait a long time before you know the application is not working.
- **Solution:** The timeouts for the DICOM print were setup to ensure that the system would work regardless of whether the DICOM print camera was on a LAN or a WAN halfway around the world. The DICOM print timeouts for the association and DIMSE classes (for example, **N-GET**, **N-DELETE**) can be modified within the DICOM print camera installation. They can be reduced down to 90 seconds.

DICOM PRINT ERROR ON N-GET TIMEOUT CONFUSING

- **Symptom:** When the N-GET timeout goes off, the error message in the prslog file will be "Could not get printer status, invalid command sequence for N-GET".
- **Solution:** When the user sees the above error they may want to consider that the issue may be an inactivity timer on the N-GET DIMSE service.

DICOM PRINT CAMERA SLIDE SUPPORT

- **Symptom:** Current implementation of DICOM print does not allow selection of slide format.
- **Solution:** Feature not currently supported. Possibly in future releases.

CONFUSION ON FILM FORMAT NOTATION

- **Symptom:** GEMS Laser Camera and DICOM Print film format notations are opposite.
- **Solution:**
 - GEMS Laser Camera film format notation has always been row x col (for example, 12 on 1 = 4x3)
 - DICOM Print Standard film format notation is col x row (for example, 12 on 1 = 3x4)

Section 10.0

Teleradiology (Framegrabber Type) Systems & Aux. Monitors

10.1 Introduction

Some systems may have teleradiology (TR) systems that framegrab the video. LightSpeed 5.X DOES NOT directly support this type of TR. The CT scanner's RGB color display video is a much larger format at a much higher pixel frequency. GE does not promise any direct compatibility with framegrabbing TR systems (DICOM 3.0 TR systems may work depending on the DICOM implementation but GEMS does not and cannot validate all the various TRs.)

In the framegrabber case, a high quality (300Mhz bandwidth) video splitter/amplifier (as listed above) is needed to intercept and re-drive the display CRT RGB video. Composite grey-scale would then be available on Green #2 (1280x1024 pixels at 72Hz). Any framegrabber hardware attempting to capture this signal must be capable of a 140Mhz pixel rate. This also involves TR system configuration parameters. The TR capture software may also need upgrading to deal with 1280x1024 and/or **crop** the signal. The TR remote display software may need upgrading to view the larger format. The image transmission times to the remote TR may be up to 4 times as long. GEMS will supply all technical information necessary to assist TR suppliers in making their systems work with LightSpeed 5.X, but GEMS cannot be responsible for this third party TR equipment, software, or compatibility with LightSpeed 5.X.

The following common parts are available from GE Medical Systems:

PART NUMBER	DESCRIPTION
2237018-2	Four-way Video Splitter (BNC Converter)
2154425	BNC to BNC 7' Splitter Cable
2142221	BNC to Host Computer 7' video Cable
2256482	DB15 to Host Computer 7' Video Cable
2256485	DB15 to BNC 7' Adapter Cable

Table 7-5 Common Parts Available From GEMs

10.2 Auxiliary Monitors Setup

Figure 7-17 illustrates the **only approved GE CT configuration** for remote monitors, including the part numbers for the only approved components. The use of other configurations and/or components should be strictly avoided.

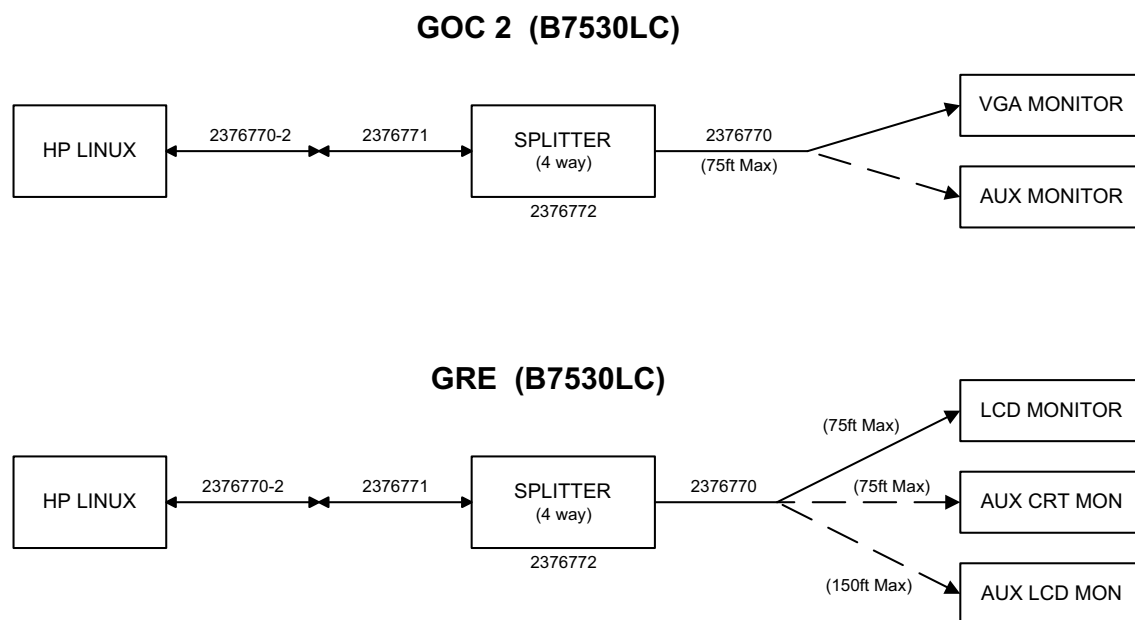


Figure 7-17 Video Splitter / Monitor Cabling

10.3 GEMS CT Analog Filming Interface Specs (Video & Serial)

Note: The analog (VDB) video output specs and serial interface were specified from, and are the same as, the Genesis interface output specs. This does NOT apply to the digital (LCAM) filming interface.

OUTPUT	VALUE
amplitude	1 volt peak-to-peak
video	0.643V $\pm$ 10%
setup	0.071V $\pm$ 10%
sync	0.286V $\pm$ 10%
DAC resolution	8 bits
diff. linearity	$\pm$ 1 LSB max
glitch area	80 picovolt seconds max, for any step size
rise/fall times	> 10 nsec, $\pm$ 10-90%
FS settling time	7.5 nsec typical to 1 LSB
transfer function	guaranteed monotonic
noise level	> 5.0 millivolt peak-to-peak, combined sync/async noise
DC offset	$\pm$ 1 VDC referenced to ground

Table 7-6 Video Output (Measured Into 75 Ω at BNC Output)

OUTPUT	VALUE
logic family	F series TTL
output low level	0.8VDC max
output high level	2.0VDC min
output period	41.336 nsecs $\pm$ 10%
transition times	10 nsec max, $\pm$ 10-90%

Table 7-7 Pixel Clock Output

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
pixel frequency	24.192 Mhz	24.192 Mhz
pixel period	41.336 nsec	41.336 nsec
horizontal line frequency	33.6 Khz	33.6 khz
horizontal line width	720 pixels	720 pixels
horizontal active	544 pixels	544 pixels
horizontal blanking	176 pixels	176 pixels
horizontal front porch	26 pixels	26 pixels
horizontal sync	76 pixels	76 pixels
horizontal back porch	74 pixels	74 pixels

Table 7-8 Video Timing Characteristics

VIDEO TIMING CHARACTERISTIC	60HZ	50 HZ
vertical frame frequency	60Hz	50Hz
vertical frame time	560 lines	672 lines
vertical active	524 lines	524 lines
vertical blanking	36 lines	148 lines
vertical sync	3 lines	3 lines
vertical back porch	30 lines	86 lines
vertical front porch	3 lines	59 lines
scanning format	non-interlaced	non-interlaced

Table 7-8 Video Timing Characteristics (Continued)

FORMAT	VALUE
visible field	544 pixels by 524 lines
image field	512 pixels by 512 lines
greyscale field	32 pixels by 16 level grey bar on left side of image
greyscale	software selectable on/off
greyscale off value	0 (black)
initial greyscale	255 (white) at upper left corner
border field	12 lines at bottom of visible field
border field value	any 8bit value, software programmable

Table 7-9 Video Display Format

ATTRIBUTE	VALUE
interface	R5422
25D conn pinout	pin 8 (RX+), pin 21 (RX-), pin 9 (TX+), pin 22 (TX-), pin 7 (GND)
baud rate	1200 baud
word length	8 bit, 1 start bit, 1 stop bit
parity	even
type	asynchronous

Table 7-10 Host Communications/Control Serial Port (analog interface only)

Note: The digital DASM/LCAM serial control has standard R5232 on pins 2, 3, & 7.

Section 11.0

Network Connections

BROAD-BAND

Broad-band is considered the standard network connection for LightSpeed 5.X. (A dial-up modem is optional.) Broad-band connections should use one of the following Category 5 patch cables:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

The CT system is connected to the network through the Console.

- A patch cable (not to exceed 10 feet) should be provided by the customer, and it is used to connect the console to a wall box. (See Notes on [Figure 7-18](#))
- Some customer-site units may require cable duct-work or conduit to route connecting network cables to the workstation, camera and console.
- The run from the hospital switch to the CT wall outlet must not exceed 290 ft. (88m). Bandwidth performance is degraded when the length reaches 300 ft. (91m) or greater.
- For the optional modem: **Two phone lines should be provided by the facility.** One line is for use with a modem and must be an analog line. The second line is a voice only line.

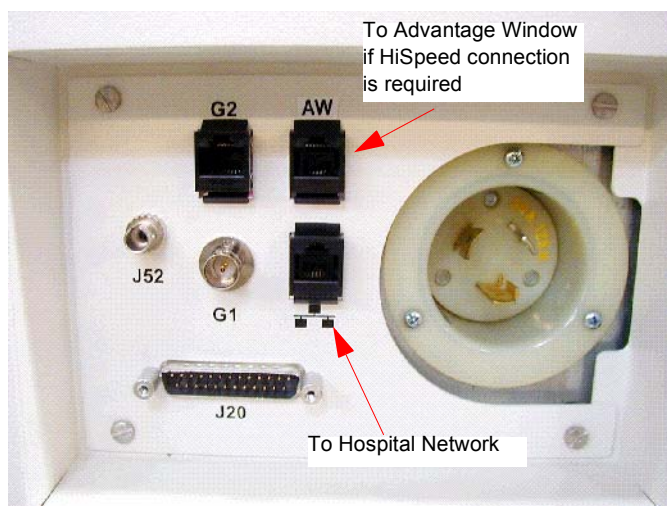


Figure 7-18 Console Rear Bulkhead

US PROCESS OVERVIEW

The United States network connectivity requirement for this product is broad-band. The US process relies on the Install Specialist to select a Customer Champion and identify an IT contact for the site. Together, those individuals then complete a site assessment to gauge what tasks are needed to fulfill the connection.

Anyone can contact the GE Connectivity team at 800.321.7937, Option #3, with questions.

CUSTOMER BROAD-BAND RESPONSIBILITIES

Provide GEMS Installation Specialist with an accurate site address, telephone number, contact name, and e-mail address for the:

- Customer Champion
 - Co-ordinate VPN activities between Radiology/Cardiology and the Information Technology (IT) departments
 - Act as a focal point in assuring site broad-band infrastructure meets GEMS requirements for connection as determined by a mutual assessment with the GEMS Connectivity team.
- IT Contact
 - Complete an equipment assessment with GEMS Connectivity team to determine site readiness for broad-band
 - Work with the Customer Champion to complete any identified infrastructure changes
 - Provide IP addresses for new CT equipment
 - Provide a VPN compatible appliance that will support the IPSec tunneling protocol and 3DES data encryption
 - To utilize an Internet Service Provider that supports static routing



GE Medical Systems

Remote Service Broadband - Customer Site Assessment

Site Name: _____ FE Name: _____
City, State: _____ FE Phone: _____
Date: _____ FE Email: _____

- | | Yes | No |
|---|---|--------------------------|
| 1. Does your site currently have a persistent (24x7) Internet connection? | ☐ | ☐ |
| 2. Is the GEMS Diagnostic Imaging equipment on the Local Area Network and will it be accessible to the Internet? | ☐ | ☐ |
| 3. Does your site have a VPN device today? | ☐ | ☐ |
| 4. Is the VPN device one of the models below? <i>If Yes, please select the model from the options below.</i> | ☐ | ☐ |
| ☐ a) Cisco Pix Firewalls
☐ b) Cisco Routers
☐ c) Cisco 3000 Series (Altiga acquisition)
☐ d) Checkpoint Firewalls Software Version 4.1 and higher
☐ e) Nortel Contivity Software Version 3.2 or higher
☐ f) Redcreek | ☐ g) Symantec (Raptor) firewalls
☐ h) Firebox
☐ i) Linux S/WAN
☐ j) Sidewinder
☐ k) Netscreen
☐ l) None
☐ m) Other _____ | |

**If No, the GEMS Connectivity Support Team can help determine device compatibility.*

5. Does your VPN device support "triple DES" Encryption? ☐ Yes ☐ No
6. Has approval been given to install this VPN connection? ☐ Yes ☐ No
- Site Approver's Name _____
7. Provide your VPN Installer information, this is the person who will be contacted to schedule the VPN install.

Customer Installer Name: _____
Installer Telephone Number: _____
Installer e-mail address: _____

Notes: _____

Field Engineer needs to provide compatible system information:

System ID	IP Address	Gateway Address
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____
_____	_____	_____

[Additional System and IP Address Spaces Available on Page 2](#)

If you have questions or need assessment support contact your Zone Champ or:

Joe Gracz - HQ Support 1-262-524-5261
Joseph.Gracz@med.ge.com

Once you have completed both pages of this form, please send it to:

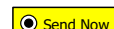
a) Judy Heyer judy.heyer@med.ge.com
b) Judy Heyer Fax# 414-918-4707

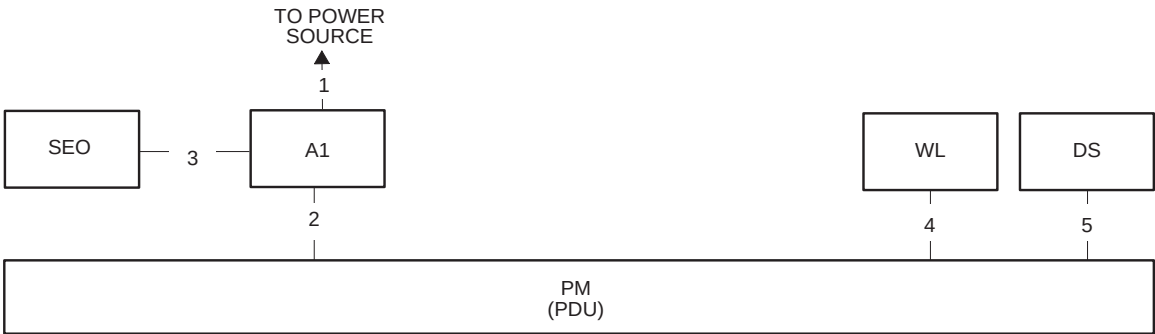
[Use the send button on page 2](#)



Once you have completed this form, please send it to:

judy.heyer@med.ge.com
414-918-4707





NOTES:

- 1) Used for remote diagnostics - Option
- 2) Refer to the appropriate Pre-installation / Installation documents for the Laser Camera
- 3) Category 5 cable. Use one of the following patch cords:

CAT Num	GE Part Num	Length
K9000WB	2215028-10	20 m
K9000KP	2215028-5	10 m
K9000JR	2215028-4	5 m
K9000WA	2215028-9	3 m

Only one phone connection is required for the system.

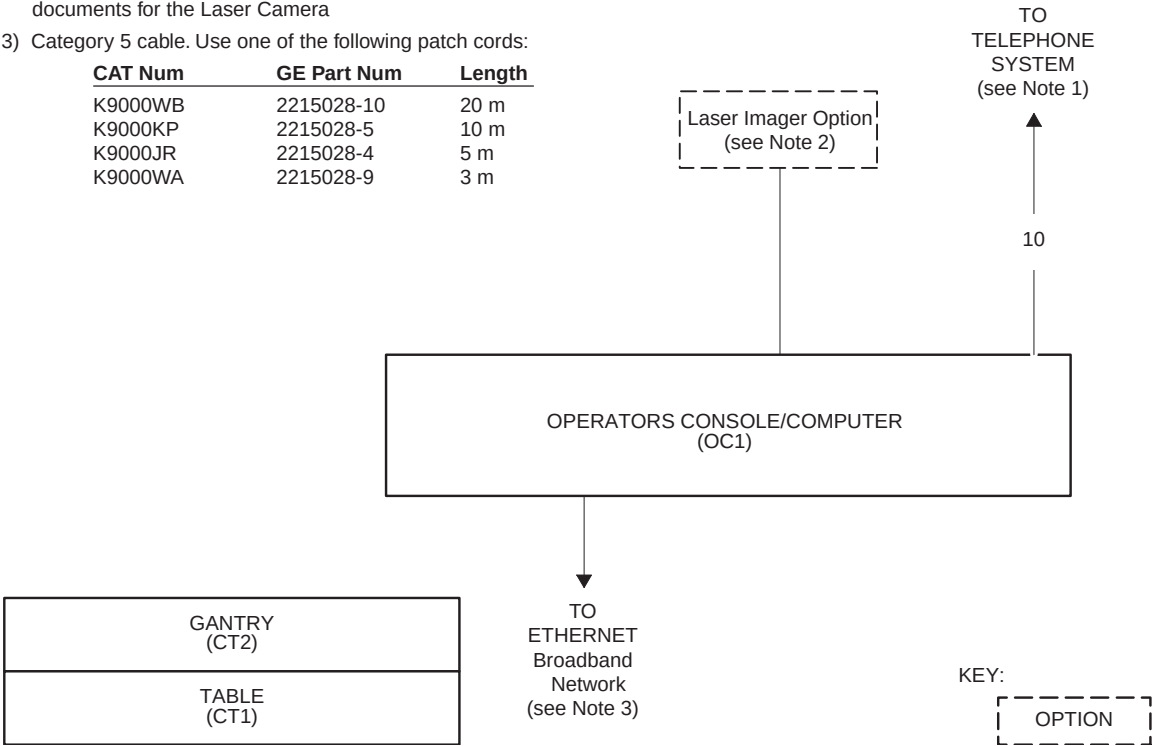


Figure 7-19 LightSpeed System Interconnection Runs

Chapter 8

Customer Touch Leakage Test

Section 1.0 Personnel Requirements

REQUIRED PERSONS	PRELIMINARY REQS	PROCEDURE	FINALIZATION
1	mins	1 hrs	mins

Section 2.0 Overview

This procedure must be completed after all options have been installed. It covers three safety and leakage current checks:

- Patient Touch current test (completed after installation)
- System Ground resistance measurement (completed during installation)
- Ground current typical (completed after installation - optional)

Section 3.0 Preliminary Requirements

3.1 Tools and Test Equipment

ITEM	QTY	PART#
Standard service tool kit	1	
Fluke 87	1	
AMP Clamp-on AC Probe	1	2338920
Dale 600 Leakage Current Meter	1	46-328406G1
Dale 600 Leakage Current Meter Ultra Sound Kit	1	46-285652G1
Dale 600 Leakage Current Meter European 220 VA	1	46-328406G2

3.2 Consumables

No consumables required.

3.3 Replacement Parts

No replacement parts required.

3.4 Safety

WARNING POTENTIAL FOR SHOCK
GROUND WIRES WILL HAVE GROUND CURRENT PRESENT WITH POWER "ON".
FOLLOW APPROPRIATE SAFETY PROCEDURES FOR WORKING WITH AN
ENERGIZED SYSTEM.

3.5 Required Conditions

No required conditions.

Section 4.0 Procedure

4.1 Patient Touch Leakage Current Test Procedure

- 1.) See [Figure 8-1](#) for the usage of the Dale 600 meter to measure leakage current.

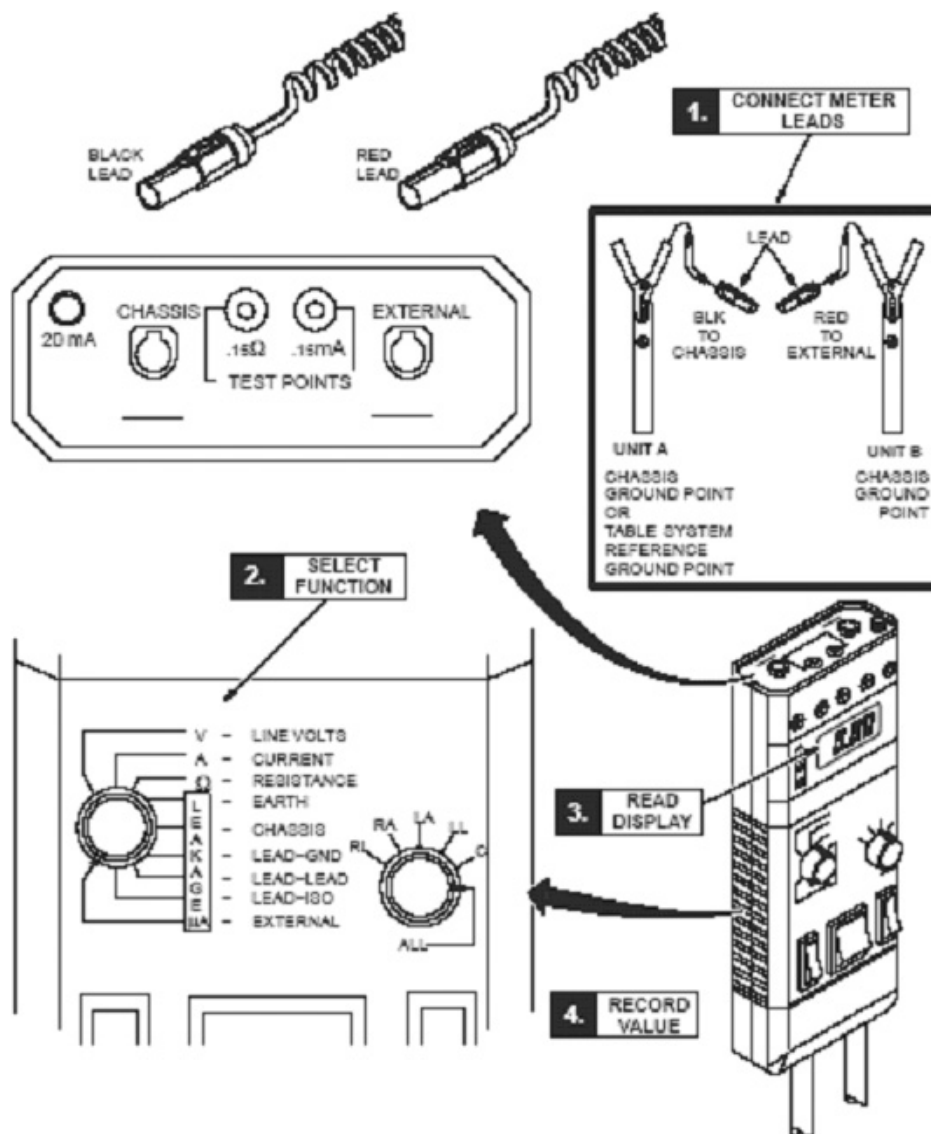


Figure 8-1 Using Dale 600 to Measure Leakage Current

- 2.) Plug the Dale 600 Meter into the outlet nearest the Table. (Table base covers must be removed to use table outlet.) Confirm the outlet is correctly wired per the three LED indicators on the Meter.
- 3.) Connect the black lead to the chassis plug and the red lead to the external plug on the top of the Dale 600 meter. Set the function switch to external and display readings will be in micro-amps.

The black lead will be connected to the table base ground buss and the red lead is then connected to the devices (components) under test.

- 4.) Leakage current cannot exceed the following and is tested with power on:
 - a.) **Critical care areas (invasive) - 10 μ A**
 - b.) General care areas - 20 μ A
 - c.) Not intended for patient area - 50 μ A
- 5.) Testing must be completed between the system reference ground point (table base) to unit reference ground points (i.e. gantry and table, see illustration 1).
- 6.) Test all conductive surfaces and components within patient reach or within 6 feet or 1.9m around table and 7.5 feet or 2.3m above. Measure at table maximum travels. On some site wall outlet cover plates and sinks may become an issue.
- 7.) Test all Optional components such as in room monitors, injector, overhead monitor suspensions, and table options.

Record these results in the data sheet. To be completed at install and during PM inspections.

Gnd Bus to	Install <10 μ A	6-month PM <10 μ A	6-month PM <10 μ A
Table gnd points			
Gantry gnd points			
Injector gnd points			
Boom in Room			
Monitors			
Sinks			

- 8.) Test conditions: Table max elevation Min - Max travel; Patient face up and face down, and head first feet first. Table min elevation 7.2' patient access to all conductive surfaces Including: IV poles, any smart step monitors and stands, table bearing rails and all other conductive surfaces. Be aware of Static discharge: from scan window, keypads, display, and touch pads or other plastic surfaces.
- 9.) Procedure Hints:
 - a.) Look for items that have abnormal measurements, high or low.
 - b.) These could indicate mis-wiring, loose connections, or poor connections due to corrosion, painted surfaces, etc.
 - c.) High leakage could indicate a wiring error such as a neutral connected to the ground.
 - d.) Fluctuating ground currents could indicate a short, poor connection, or facilities ground problem causing leakage currents from other areas of the facility to flow through our system grounds.

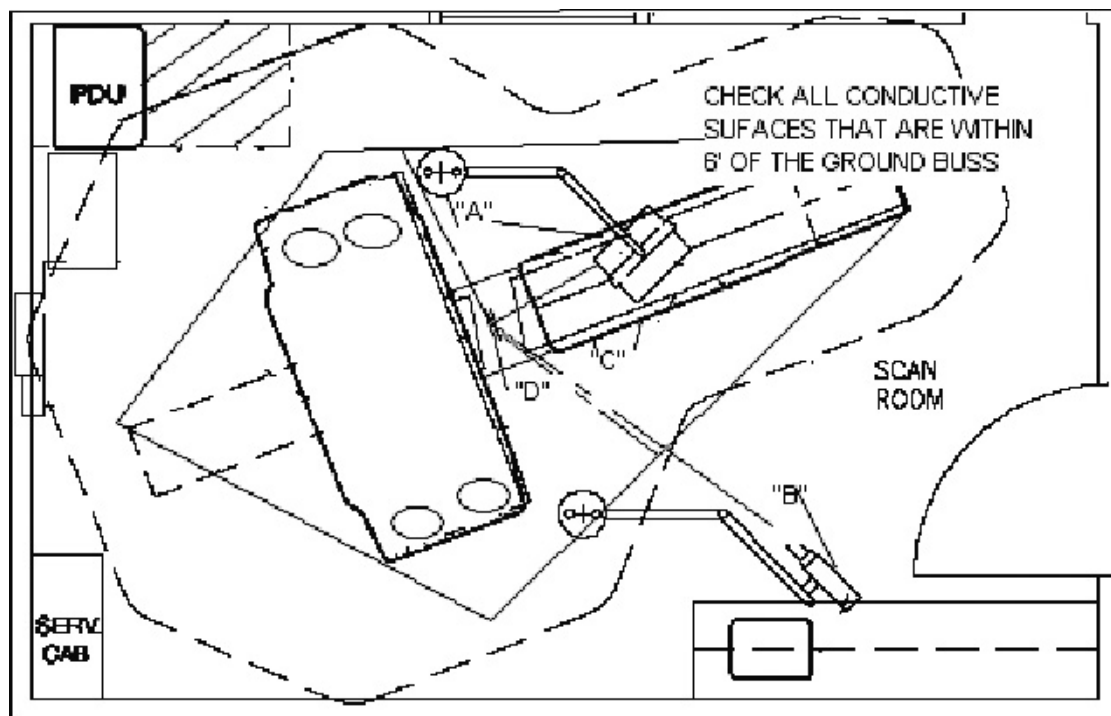


Figure 8-2 CT Leakage Current Map

4.2 Ground Resistance Checks Procedure (done during mechanical install)

Use Fluke 87 to measure ground resistance.

- 1.) Use a Fluke 87 or equivalent RMS Meter to verify the presence of less than 0.5 ohm of resistance between each of the following points:

Table 8-1 Resistance Verification - Site Ground

FROM	TO	
PDU Ground Bus	Vault Ground	☐ Check box when complete
PDU Ground Bus	Table/Gantry raceway ground point	☐ Check box when complete
Table/Gantry raceway ground point	Gantry Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Table Chassis	☐ Check box when complete
Table/Gantry raceway ground point	Operator Console Chassis	☐ Check box when complete
All Display or Computing Options (if any)	Operator Console Chassis	☐ Check box when complete

- 2.) Connect the meter leads to the Fluke 87 meter.
- 3.) Set the function switch to resistance ohms.
- 4.) The black and red leads will be connected per Table 8-1.

- 5.) All ground resistance testing is done with power off. Ground current will cause errors in the resistance readings.
- 6.) Use lockout/tag out procedures.
- 7.) Maximum ground resistance is 0.5 ohms impedance.
- 8.) Using the table footswitch ground bar as a reference. Check the ground resistance to all 5 system grounds.

Record the measurements in the datasheet:

SYSTEM GROUND BUS	TYPICAL READING	SITE DATA
Gantry Ground Stud	0.5 Ohms	
Console Ground Stud	0.5 Ohms	
PDU Ground Stud	0.5 Ohms	
Table Ground Stud	0.5 Ohms	

- Hints:
- Look for undersized ground wires and repair.

- Ground resistance readings should not fluctuate. If so this indicates an intermittent

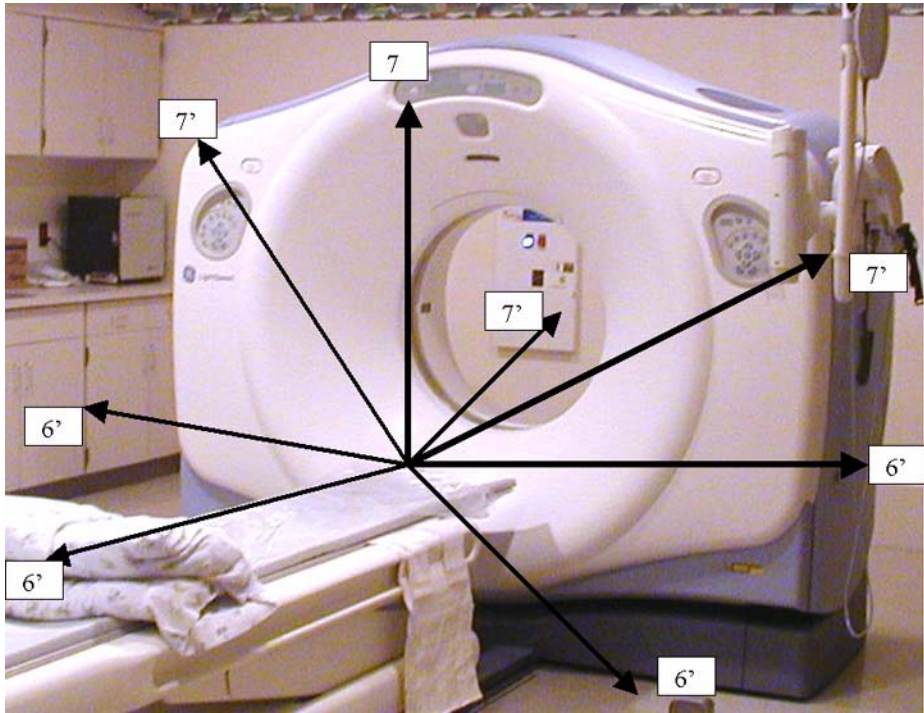


Figure 8-3 Measurement Points

Section 5.0

Finalization

No finalization required.

Chapter 9

Forms and Checklists

This chapter includes the following sections:

- [Section 1.0 - Mechanical Vendor Installation List](#)
- [Section 2.0 - Electrical Calibration Installation Checklist](#)
- [Section 3.0 - GE Form 4879 Installation Data Verification](#)
- [Section 4.0 - Install Completion Checklist](#)
- [Section 5.0 - Electronic Forms](#)

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Section 1.0

Mechanical Vendor Installation List

Vendor Company Name Install Trained

- ☐ GE FE ☐ Yes
- ☐ Mid-West ☐ No
- ☐ Jentree ☐ Somewhat
- ☐ Revels
- ☐ AMS Install Start Date _____
- ☐ AK Associates
- ☐ All Others Install Completion Date _____

Hours Spent _____ Number of Trips _____ FDO _____

CLEAN UP

- ☐ Move Customer boxes to one corner or counter.
- ☐ Place Field Engineer Material (phantoms, hoist, kits, tools) into the service cabinet.
- ☐ Put all documentation (including software, HHS doc) in the purple box and leave by the console.

REMOVE TRASH

- ☐ Remove and Recycle cardboard as required.
- ☐ Place skids and wood into the dumpster.

RETURN DOLLIES (USA ONLY)

- ☐ Check the packing list found in the purple box for dolly return location.
DO NOT return the console shipping skid to Milwaukee.

INSTALL COVERS

- ☐ Recheck Gantry and Table anchor for 75ft-lb.
- ☐ Install PDU covers.
- ☐ Install table foot-switch assembly and covers as requires in your area.
- ☐ Install console covers and align as needed for proper cover fit.
- ☐ Install gantry base covers, and rear cable box cover as requires.

CLEAN THE EQUIPMENT

- ☐ Use the cleaning supplies in the installation support kit to clean the equipment.

TOUCH UP NICKS

- ☐ Use the paint supplies found in the installation support kit to touch up nicks in the equipment.

COMPLETE PAPERWORK

- ☐ Collect all Product Locators and store in purple box.
- ☐ Complete the Field Engineer sign-off procedure with FE.
- ☐ Complete and sign this list.

SIGNATURE

Lead Mechanical Installer _____

Field Engineer _____

Mechanical Vendor installation comments _____

Short shipped? ☐ Yes ☐ No

Correct Documentation? ☐ Yes ☐ No ☐ Missing

Field Engineer comments _____

Return to: george.farrington@med.ge.comCT Service Engineering
Suhail.altal@med.ge.comProject Leader, Installation Services

ELECTRONIC VERSION OF THIS FORM

An electronic version of this form is available. Use of the form requires Microsoft® Word. To complete and submit this form electronically, see [Section 5.0 - Electronic Forms](#).

Section 2.0

Electrical Calibration Installation Checklist

FE Name _____ FDO _____ System ID _____

Install Start Date _____ Install Completion Date _____

Hours Spent _____ Number of Trips: ☐ 1 trip ☐ 2 trips ☐ 3 or more trips

CT SYSTEM

- ☐ Completed functional tests.
- ☐ Complete power and ground audit.
If not: ☐ TBD Later ☐ Not Required ☐ Scheduled by PG Leader
- ☐ Install seismic kit (if required in your area).
- ☐ Completed all required calibration.

CT OPTIONS

Table 9-1 CT Options

- | | | |
|---|--|---------------------------------------|
| • Complete injector functional tests. | ☐ Install Completed | ☐ Not Required |
| • Complete AWW functional tests. | ☐ Install Completed | ☐ Not Required |
| • Complete Filming/Camera/DASM functional tests. | ☐ Install Completed | ☐ Not Required |
| • Complete UPS functional tests. | ☐ Install Completed | ☐ Not Required |
| • Complete Network items installed and functional tests. | ☐ Install Completed | ☐ Not Required |
| • Verify that all customer software options are installed and functional. | ☐ Install Completed | ☐ Not Required |
| • Are Remote Video Connections Needed | ☐ Install Completed | ☐ Not Required |

SYSTEM CLEANUP

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

DOLLIES RETURN

- ☐ Confirm dollies are returned.

INSTALL COMPLETION MATERIAL CHECKS

- ☐ All HHS documentation completed.
- ☐ Write Install related CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

SIGNATURE

Field Engineer _____

Field Installation Comments / Issues _____

Alignment Calibration Required? ☐ Yes ☐ No

Correct Documentation Shipped? ☐ Yes ☐ No ☐ Missing

Return to: George.Farrington@med.ge.com CT Service Engineering
William.Gibson@med.ge.com HHS

ELECTRONIC VERSION OF THIS FORM

An electronic version of this form is available. Use of the form requires Microsoft® Word. To complete and submit this form electronically, see [Section 5.0 - Electronic Forms](#).

Section 3.0

GE Form 4879 Installation Data Verification

NOTICE Only use the Installation manual that arrives with your system for installation. Any other revisions of this manual may not exactly match your system.

3.1 Installation Site History

3.1.1 RECORDS KEEPING AND DOCUMENTATION - IMPORTANT!

- Locate the HHS data, sent from the factory, in the software collector box.
- Retain a copy of this HHS data and the data collected at the site, on site in the service cabinet.
- All sites that are required to submit an FDA 2579 form, must send a copy of all HHS test information to the HHS administrator along with a completed GE 4879 and FDA 2579 form. This applies to both GEE and Local FE's who complete VCT installations.
- A generator RECAL is not required on VCT installations unless a certified component is replaced.
- If any one of the calibration procedures fail, data collected is different, or you replace any
- Certified component in the High Voltage section, YOU MUST RECALIBRATE THE SYSTEM.
- To recalibrate the system, refer to procedures in the System Service manual.
- Complete all installation check list items, as proof the procedure was completed.

3.1.2 Component Tractability - IMPORTANT!

- Complete the Product Locator Report and mail it. Retain a copy on site with the HHS data records.
- Retain any additional information, per your local LCT procedures.

Completed By: GE Energy Healthcare FE Other

Date: _____

Name: _____

3.1.3 Installation Procedures Traceability - IMPORTANT!

PROCEDURE	COMPLETED BY (CIRCLE ONE)	TEST DATE	NAME (PRINT)
Mechanical Procedures Book 1	GE Energy Healthcare FE		
Electrical Procedures Book 2	GE Energy Healthcare FE		
IQ & Quality Procedures Book 2	GE Energy Healthcare FE		
GE 4879	GE Energy Healthcare FE		
GE 4879	GE Energy Healthcare FE		

PROCEDURE	COMPLETED BY (CIRCLE ONE)	TEST DATE	NAME (PRINT)
FDA e-2579	GE Energy Healthcare FE		
FDA e-2579	GE Energy Healthcare FE		

3.1.4 Installation Test & Tools Traceability - IMPORTANT!

Where Installed	
City	State
Customer's Telephone Number	Room No.
System ID #	GON #
Master Control - Console	
Model #	Serial #
Gantry	
Model #	Serial #
Installation Completion Date	
Test Equipment History	
DVM Manufacturer	DVM Model #
DVM Serial #	Cal Due Date
Leakage Meter Manufacturer	Model #
Serial #	Cal Due Date
Amp Meter Manufacturer	Model #
Serial #	Cal Due Date
Other - Manufacturer	Model #
Serial #	Cal Due Date

3.2 System X-Ray Indicators and Warning Labels

3.2.1 X-Ray On Indicators

- 1.) Perform several scans.
- 2.) Verify that the X-ray ON lights are on during the scans.
- 3.) Use a mirror to check the x-ray indicator light on the gantry rear cover

3.2.2 Operator Indicators - HHS Requirement

LIGHTS ON	LIGHT LOCATIONS
☐	SCIM
☐	Gantry Front
☐	Gantry Rear
☐	Room Warning Light

Initials: _____ Date: _____

X-Ray Indicators on the SCIM, Front and Back of Gantry Illuminate During X-Ray.

3.3 Warning Labels

3.3.1 Laser Warning labels

- 1.) Laser warning labels appear on the gantry lower right corner near the floor, and near each of the laser window opening on the gantry cover
- 2.) Obtain and install all warning labels in the appropriate language for the country installed.



Initials: _____ Date: _____

All laser warning labels are present.

3.3.2 Operator Use Warning Label - SCIM

Make sure the X-Ray warning label appears in the correct location on the SCIM.



Initials: _____ Date: _____

Warning labels are present.

3.3.3 Appropriate Language Warning Labels Verification

Check that all indicators, cautions and warning labels have been installed per the procedure in CT installation book 2, Chapter 5 Section 3.4.

Initials: _____ Date: _____

All Appropriate Language indicators, cautions and warning labels are installed.

3.4 Tomographic Plane Indication Accuracy

- 1.) Record the scan location (shown on the image annotation) of the image with the darkest scan plane indicator (darkest long bar).
- 2.) If your system meets all the installation and alignment specifications, the image at scan location zero (S0.0) should contain the scan plane indicator.

Specification:

(Must be within +/- 1mm)

If scan location S1.0 or scan location I1.0 has the darkest bar, the system still meets the specification. The scan plane deviation should equal S0.0 +1.0mm.



Repair: If necessary, adjust the internal alignment light position to meet the S0.0 +1.0mm requirement.

LANDMARK	DARKEST BAR LOCATION	DATA
Internal		
External		

Initials: _____ Date: _____

Tomographic plane indicator accuracy passed.

3.5 Table/Gantry Alignment Procedure

Repeat this procedure until the table edges line up (± 1 mm). If needed, a scout can be completed to check table horizontal presentation.

Check the two scout scans.

Specifications:

Molding strips should be parallel within +/- 3 mm. Cradle should be level within ± 1 mm.

Repair:

Repeat table to gantry alignment, and repeat scan test.

CRADLE LEVEL	DATA	PASS
Across Cradle		☐
Length of Cradle		☐
CRADLE ALIGNMENT		
Cradle Front min length	Edge to edge	☐
Cradle Rear max length	Edge to edge	☐
Cradle Center	From cradle CL	☐

Tighten the anchors and torque to 74.6 N-m (55 ft.-lb.).

Gantry	Completed	Table	Completed
8" Hilti QB	☐	8" Hilti QB	☐

Contractor Anchored Systems

Contractor name		Anchor type		Meets

Initials: _____ Date: _____

Table cradle alignment accuracy passed.

3.6 Check Alignment Lights - Test

- ☐ Internal Axial Lights
- ☐ External Axial Lights
- ☐ Coronal Lights

Initials: _____ Date: _____

Table cradle alignment Light accuracy passed.

3.7 Interference Test

Interference Test Requirement:

No tilt motion shall cause the gantry tilting frame to touch the stationary base covers for any tilt angle.

Initials: _____ Date: _____

All interference tests were completed and all tests passed.

3.8 System Functional Test

Use the system tests in the following sections to exercise all aspects of the system and to ensure system integrity before releasing to the customer.

Initials: _____ Date: _____

All functional tests were completed and all tests passed.

3.9 IQ Section - 5.X Image Quality Series

Complete the image series listed in Chapter 6, and record the values in the site installation manual.

Note: IQ Sections may be slightly different for RT, Pro16, LS16, Ultra and Plus systems.

3.10 Documentation Section - System Paperwork & Product Locator

3.10.1 Installation Verification

Copy or submit e-copied of these documents to the HHS administrator. Leave a copy on site as a record of system installation and test completion.

☐

Mechanical Installation Completion Form

To be completed by the mechanical installation team leader GEE or FE

☐

Electrical Installation Completion Form

To be completed by the electrical installation team leader GEE or FE

☐

GE 4879 Data Sheets

Note:

More than one form may be submitted for the same installation by the different installation teams. (For GEE completed work and FE completed work)

To be completed by the installation team leader. All parts that was completed.

A separate GE 4879 will be completed when the Image Quality testing is completed. This document is sent to the HHS administrator.

☐

GE Manufacturing HHS Data Sheets

Complete and copy by the installation team leader These document are sent with this data form to the HHS administrator.

GE Shipped Documentation

All shipped documentation was received.

☐

Customer Documentation

☐

FE Documentation

3.10.2 Product Locator

☐

Cards

Collect and update product locator.

Leave a copy or one of the cards on site

3.10.3 Regulatory Information Distribution



Document Distribution

- 1.) A copy of all items from section 1 above
- 2.) A copy of all items from section 2 above
- 3.) Mail to HHS administrator



HHS FDA 2579

Note:

More than one form may be submitted for the same installation by the different installation teams. (GEE completed work and FE completed work)

- 1.) Complete the FDA e-2579
- 2.) When completed submit to HHS administration. Correct all errors. Do Not print this form. A printable version will be sent to you from HHS administration for your records.
- 3.) Some states require a State Registration Number to complete this form. If you have questions concerning your state, contact the HHS administrator.

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Section 4.0

Install Completion Checklist

4.1 CT System

- ☐ Functional tests completed.
- ☐ Power and ground audit completed.
- ☐ Seismic kit installed (if needed in your area).

4.2 CT Options

- ☐ Injector functional tests completed.
- ☐ AWW functional tests completed.
- ☐ Filming/Camera/DASM functional tests completed.
- ☐ UPS functional tests completed.
- ☐ Network items installed and functional tests completed.
- ☐ Verify that all customer software options are installed and functional.
- ☐ Teleradiology Connections (Refer to [Section 10.0 on page 98](#))

4.3 System Cleanup

- ☐ Clean system with supplies shipped with your system.
- ☐ Touch-up nicks and scratches (as needed).
- ☐ Install pads and stickers (as needed).
Replace the caution labels with appropriate language labels.
- ☐ Store customer materials appropriately.
- ☐ Store service documents, tools, and materials appropriately.

4.4 Dollies Return

- ☐ Console is shipped on a pallet. Do Not return pallet to GEMS.

4.5 Install Completion Material Checks

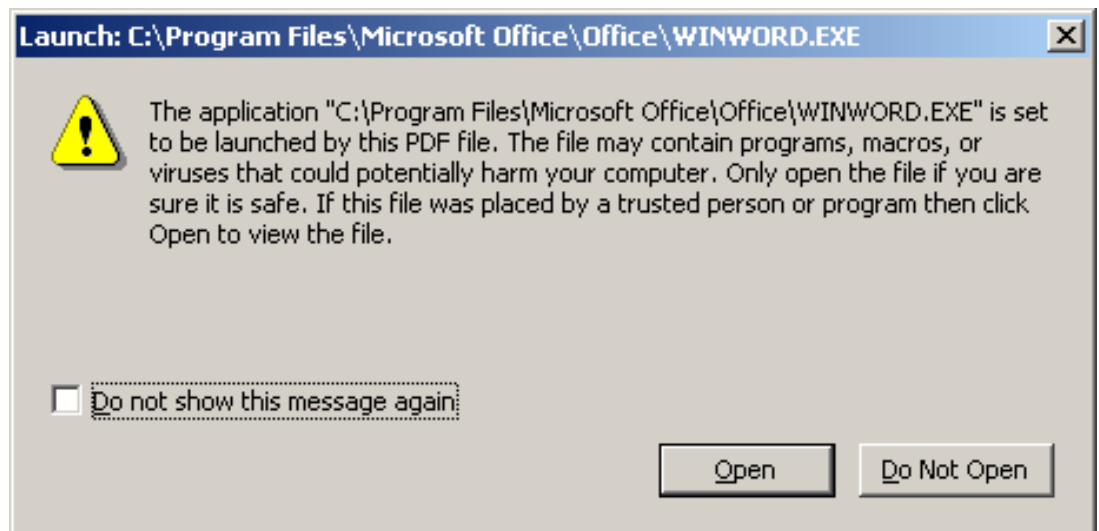
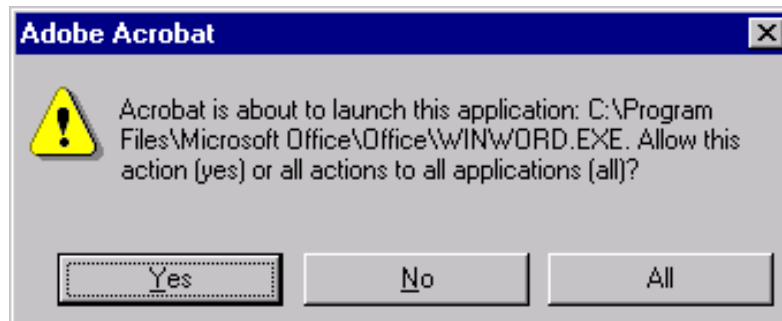
- ☐ Complete all HHS documentation. Complete Install CQAs as needed.
- ☐ Complete all dispatching activities (03-04-10 codes).
- ☐ Complete customer acceptance checks/test.
- ☐ Complete system transfer and notify appropriate GE personnel.
- ☐ Communicate with the customer and address installation issues.

Section 5.0

Electronic Forms

Several of the forms in this chapter may be completed and submitted electronically by following the procedure described below.

- 1.) To select and open the desired form, click on the appropriate link below. Either Microsoft® Word or Excel will launch, depending upon the type of form.
 - a.) Microsoft® Word documents:
 - * [Mechanical Vendor Installation List](#)
 - * [Electrical Calibration Installation Checklist](#)
 - b.) Microsoft® Excel spreadsheet:
 - * [HHS Data Sheet](#)
- 2.) A warning box similar to the samples below may appear. Click YES or OPEN as appropriate.



- 3.) Complete the form (TAB to move between fields).

NOTICE

Be certain to save a copy to a local hard drive.

- 4.) Submit the form, by choosing FILE > SEND TO > MAIL RECIPIENT. Enter the e-mail addresse(s) shown at the end of the signature section of the form in the Send To field.

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Appendix A

Additional Characterization Procedures

For information related to alignment, setup, and calibration procedures, please refer to the appropriate LightSpeed 5.X Plus System Service Manual.

AppendixB

Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5032	Alternating Current
	335-1	Three-phase Alternating Current
	335-1	Three-phase Alternating Current with neutral conductor
		Direct Current
	417-5019	Protective Earth (Ground)
	348	Attention, consult ACCOMPANYING DOCUMENTS
	417-5008	OFF (Power: disconnection from the mains)
	417-5007	ON (Power: connection to the mains)
		Warning, HIGH VOLTAGE
		Emergency Stop
		Type B

Table 9-2 Symbols

SYMBOL	PUBLICATION	DESCRIPTION
	417-5339	X-ray Source Assembly Emitting
	417-5009	Standby
		Start
		Table Set
		Abort
		Intercom
		(on Operator Console) Power On: light on Standby: light off

Table 9-2 Symbols



Imagination at work

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